Electrical Engineering Mathematics

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From fractions and equations to derivatives, integrals and phasors

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Electrical Engineering Mathematics
Text and figures: Erik Pihl.
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The C code in Chapter 18 may also be used under the MIT License, which is the licence for all code in the course repository. The repository also holds the lesson material and the worked solutions to the exercises, in Swedish.

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Preface

This book contains the mathematics used in electrical engineering. It is set from a course in electrical engineering mathematics, lesson by lesson, and covers:

- Arithmetic, algebra, and linear equations and systems of equations.
- Powers, roots, binary and hexadecimal numbers, quadratic equations, and exponential functions and logarithms, including decibels.
- Vectors, functions and trigonometric functions.
- Derivatives and integrals, with applications in circuit theory.
- Complex numbers in rectangular, polar and Euler form, and phasors for AC calculations.

The mathematics is applied to electrical problems throughout. The first chapter already uses fractions on resistors in parallel, and the book keeps returning to circuits:

- Resistors in series and in parallel, voltage dividers and power calculations.
- Charging and discharging capacitors in RC circuits.
- AC voltages: amplitude, frequency, phase and trigonometric equations.
- Impedance calculations with complex numbers, and phasor addition.
- Sampling sinusoidal signals, and computing derivatives, integrals and complex numbers in C, in the lab in Chapter 18.

What you will be able to do

Once you have worked through the book, you should be able to:

- Simplify algebraic expressions and solve the equations and systems of equations that arise in electrical engineering.
- Calculate with powers, roots, logarithms and decibels.
- Calculate and interpret derivatives and integrals of common functions.
- Calculate with complex numbers and phasors to analyse AC circuits.
- Implement the mathematics of the course in C with `math.h` and `complex.h`.

What you are expected to know

The book assumes compulsory-school mathematics, and even most of that is revised in the first chapters: number sets, the order of operations, fractions and percentages in Chapter 1, equations in Chapter 3 and powers in Chapter 5. If you feel unsure of the basics, simply start from the beginning.

The lab in Chapter 18 also assumes basic C programming: variables, functions, `printf()` and loops. Floating-point numbers, `math.h` and `complex.h` are covered in the chapter.

How the book is organised

Each chapter corresponds to a lesson in the course and has the same number: Chapter 5 is lesson 5, and its material is in the course repository under `lectures/L05`.

Chapter	Topic
1	Number sets and basic arithmetic
2	Algebra
3	Linear equations and inequalities
4	Systems of equations
5	Powers, roots and number systems
6	Quadratic equations, with practice test 1
7	Vectors
8	Functions
9	Trigonometric functions
10	Exponential functions, more logarithms and decibels
11	Review of Chapters 7–10, with practice test 2
12	Derivatives, part I
13	Derivatives, part II
14	Integrals
15	Complex numbers, part I
16	Complex numbers, part II
17	Complex numbers, part III
18	Mathematics in C: lab
19	Review and exam preparation, with practice exam

A chapter opens with what it contains and then goes through the theory, with examples that show every step. Then comes a summary with the chapter's formulas, what you should be able to do and a few questions under the heading *Test yourself*, and last the chapter's

exercises. Chapters 11 and 19 have no new theory; they review, and hold a practice test and a practice exam respectively.

The exercises

The exercises in each chapter are divided into parts and numbered as in the course, so that Exercise 2.3 is the third exercise in Part 2:

Part	What it contains
Part 1	Introductory exercises, or review of the previous chapter.
Part 2	New material: the chapter's own theory, often applied to a circuit.
Part 3	Extra exercises for further practice after the lesson.

In the course, Parts 1 and 2 are done during the lesson, first on your own and then together, while Part 3 is optional practice.

Answers and worked solutions

The answers to all the exercises are in the answer key at the back of the book. Every exercise has a small tag, ANSWER P. 231, giving the page its answer is on; in the PDF, both the tag and the exercise's number in the answer key are links between the exercise and its answer.

The answer key holds only the answers, not the solutions. An answer shows whether you got it right, but not how to get there. Full worked solutions, in Swedish, are published in the course repository after each lesson, under `lectures/LNN/appendix/c_solutions.md`. Always work an exercise yourself first, then compare with the answer key, and read the worked solution when your answer does not match or when you want to see another way.

What you need

A calculator, any calculator, is used throughout the book. Many exercises are still meant to be done without one, and they say so. The lab in Chapter 18 needs nothing but a web browser: the code is written and run in an online compiler, with nothing to install.

Notation

The book follows the notation common in electrical engineering, and that of the Swedish edition it is translated from:

- **Decimal point**: 4.7. The Swedish edition, the course material and the exams write a decimal comma, 4,7, for the same number.

- **Units** are written with a small space after the number: $4.7\,\mathrm{k\Omega}$, $50\,\mathrm{Hz}$, $30\,\%$.
- **Voltage** is written U, as in international standards and in the Swedish course; many English-language texts write V instead. Subscripts are English: U_{out} is the output voltage.
- **Resistance is written before current**, as in Ohm's law: $U = RI$ and $P = RI^2$.
- **Parallel connection** is written $R_1 \mathbin{/\!/} R_2$.
- **Logarithms**: log is the base-10 logarithm and ln the natural logarithm.
- **The imaginary unit** is written j, since i already stands for current.
- **Angles** are in radians unless otherwise stated; degrees are marked with $^\circ$.

CHAPTER 1

Number sets and basic arithmetic

The numbers we calculate with, the order we calculate in, and the first circuit.

IN THIS CHAPTER

- Number sets: natural numbers, integers, rational numbers and real numbers.
- The number line and absolute value.
- Basic arithmetic and the order of operations.
- Fractions, decimals and percentages.
- Resistors in series and in parallel, and Ohm's law.

This chapter lays the foundation for the rest of the book. It goes through the number sets that mathematics builds on, how numbers are placed on the number line and what the absolute value of a number is, the order in which operations are carried out, and how to calculate with fractions, decimals and percentages. Here you already meet the first application in electrical engineering: resistors in series and in parallel, and Ohm's law, where fractions and the order of operations are used to calculate resistance, current and voltage in a simple circuit.

§1.1 to §1.5 are the chapter's theory, with examples. §1.6 summarises the formulas and what you should be able to do after the chapter, and §1.7 holds the exercises. Most of them can be done in your head or with pencil and paper.

1.1 Number sets

Mathematics is built around different sets of numbers. Each set extends the one before it.

Set	Notation	Description	Examples
Natural numbers	$\mathbb{N}$	Positive integers and zero	$0, 1, 2, 3, \ldots$
Integers	$\mathbb{Z}$	Natural numbers and negative numbers	$\ldots, -2, -1, 0, 1, 2, \ldots$
Rational numbers	$\mathbb{Q}$	Numbers that can be written as $\frac{p}{q}$, where p and q are integers and $q \neq 0$	$\frac{1}{2}$, $-\frac{3}{4}$, 0.75
Irrational numbers	–	Numbers that *cannot* be written as $\frac{p}{q}$	π, $\sqrt{2}$, e
Real numbers	$\mathbb{R}$	All rational and irrational numbers	Every number on the number line

Electrical engineering uses all of these sets. Resistance values are often rational ($R = 4.7\,\text{k}\Omega$), while constants such as π appear in the formulas for AC systems.

1.2 The number line and absolute value

Every real number can be placed on a **number line**. The number's distance from the origin (zero) is called its **absolute value** and is written $|a|$:

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}$$

EXAMPLE

$$|5| = 5, \qquad |-3| = 3, \qquad |0| = 0$$

The absolute value is never negative: $|a| \geq 0$.

In electrical engineering, the absolute value is used, among other things, to give the amplitude of an AC voltage, whether the voltage is positive or negative at the moment.

1.3 Order of operations

When an expression contains several operations, they are carried out in this order:

1. **Brackets**: work out the innermost first and work outwards.
2. **Powers and roots**: work out exponents and roots.
3. **Multiplication and division**: from left to right.
4. **Addition and subtraction**: from left to right.

EXAMPLE

$$3 + 2 \times 4^2 - (1 + 5) = 3 + 2 \times 16 - 6 = 3 + 32 - 6 = 29$$

1.4 Fractions

1.4.1 Simplifying

A fraction is simplified by dividing the numerator and the denominator by their greatest common divisor (GCD):

$$\frac{12}{18} = \frac{12 \div 6}{18 \div 6} = \frac{2}{3}$$

1.4.2 Addition and subtraction

Fractions are added and subtracted over a **common denominator**:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$$

EXAMPLE

$$\frac{1}{3} + \frac{1}{4} = \frac{4}{12} + \frac{3}{12} = \frac{7}{12}$$

1.4.3 Multiplication

Numerator is multiplied by numerator, denominator by denominator:

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

EXAMPLE

$$\frac{2}{3} \times \frac{3}{5} = \frac{6}{15} = \frac{2}{5}$$

1.4.4 Division

Dividing by a fraction is the same as multiplying by the fraction turned upside down, its *reciprocal*:

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

EXAMPLE

$$\frac{3}{4} \div \frac{3}{8} = \frac{3}{4} \times \frac{8}{3} = \frac{24}{12} = 2$$

1.4.5 Resistors in parallel

In electrical engineering, fractions arise naturally in calculating resistors in parallel. The total resistance R_{TOT} of two resistors R_1 and R_2 in parallel is given by

$$\frac{1}{R_{\text{TOT}}} = \frac{1}{R_1} + \frac{1}{R_2}.$$

EXAMPLE With $R_1 = 6\,\Omega$ and $R_2 = 3\,\Omega$,

$$\frac{1}{R_{\text{TOT}}} = \frac{1}{6} + \frac{1}{3} = \frac{1}{6} + \frac{2}{6} = \frac{3}{6} = \frac{1}{2},$$

so $R_{\text{TOT}} = 2\,\Omega$.

1.4.6 Series and parallel connection

When several resistors sit in the same circuit, they can be replaced by a single resistor, a so-called **equivalent resistance**.

A series connection means that the resistors are connected one after the other. The equivalent resistance R_{s} is the sum of the resistances:

$$R_{\text{s}} = R_1 + R_2$$

A parallel connection means that the resistors are connected side by side, which is written $R_1 \mathbin{/\!/} R_2$. Besides the formula in §1.4.5, the parallel resistance R_{p} of *two* resistors can be calculated as product over sum, which is often quicker:

$$R_{\text{p}} = R_1 \mathbin{/\!/} R_2 = \frac{R_1 \times R_2}{R_1 + R_2}$$

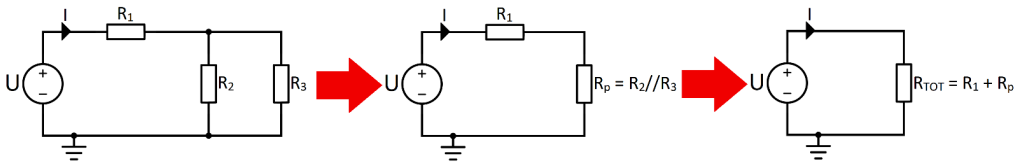
The parallel resistance is always smaller than the smallest of the resistors in it.

Ohm's law relates the voltage U, the resistance R and the current I:

$$U = R \times I \qquad \Leftrightarrow \qquad I = \frac{U}{R} \qquad \Leftrightarrow \qquad R = \frac{U}{I}$$

Tip: If the resistance is in kΩ and the voltage in V, the current comes out directly in mA. That saves keeping track of powers of ten.

EXAMPLE **Simplifying a circuit.** The circuit below is simplified step by step until only one resistor is left. First the parallel connection of R_2 and R_3 is replaced by R_{p}, then R_1 is added to give the total resistance R_{TOT}.



With $R_1 = 2\,\mathrm{k\Omega}$, $R_2 = 12\,\mathrm{k\Omega}$, $R_3 = 6\,\mathrm{k\Omega}$ and $U = 12\,\mathrm{V}$,

$$R_\mathrm{p} = \frac{12 \times 6}{12 + 6} = \frac{72}{18} = 4\,\mathrm{k\Omega},$$
$$R_\mathrm{TOT} = R_1 + R_\mathrm{p} = 2 + 4 = 6\,\mathrm{k\Omega},$$
$$I = \frac{U}{R_\mathrm{TOT}} = \frac{12}{6} = 2\,\mathrm{mA}.$$

The order is decisive: the parallel connection is a calculation of its own that must be finished before R_1 is added, just as brackets are worked out first in the order of operations.

1.5 Percentages and proportion

A **percentage** is another way of writing a fraction with the denominator 100:

$$p\,\% = \frac{p}{100}$$

A percentage of a value A is calculated as

$$x\,\% \text{ of } A = \frac{x}{100} \times A.$$

EXAMPLE What is 30 % of 200 V?

$$\frac{30}{100} \times 200 = 60\,\mathrm{V}$$

Efficiency is a common percentage measure in electrical engineering:

$$\eta = \frac{P_\mathrm{out}}{P_\mathrm{in}} \times 100\,\%$$

EXAMPLE A transformer takes in 500 W and delivers 470 W. The efficiency is

$$\eta = \frac{470}{500} \times 100\,\% = 94\,\%.$$

1.6 Summary

Concept	Definition or formula
Absolute value	$\lvert a\rvert \geq 0$, the distance to zero
Order of operations	Brackets, powers, $\times$ and $\div$, $+$ and $-$
Adding fractions	Common denominator
Multiplying fractions	Numerator $\times$ numerator, denominator $\times$ denominator
Percentage	$p\,\% = \frac{p}{100}$
Parallel resistance	$\frac{1}{R_{\text{TOT}}} = \frac{1}{R_1} + \frac{1}{R_2}$
Parallel resistance, two resistors	$R_{\text{p}} = \frac{R_1 \times R_2}{R_1 + R_2}$
Series resistance	$R_{\text{s}} = R_1 + R_2$
Ohm's law	$U = R \times I$
Efficiency	$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\,\%$

What you should be able to do

After this chapter, you should be able to:

- Know the different number sets and their properties.
- Apply the order of operations and the basic arithmetic operations.
- Calculate with fractions, decimals and percentages.
- Calculate resistance, current and voltage in simple series and parallel connections.

Test yourself

ANSWER P. 248

1. Which number set does $\sqrt{5}$ belong to? Justify your answer briefly.
2. Calculate the parallel resistance of $R_1 = 4\,\Omega$ and $R_2 = 12\,\Omega$.

Next chapter

Chapter 2 is about algebra: algebraic expressions, simplification and factorisation.

1.7 Exercises

How to work with the exercises. Parts 1 and 2 are done during the lesson: first on your own, a few minutes per exercise, then together. Part 3 is extra exercises for further practice after the lesson. Most of them can be done in your head or with pencil and paper.
Answers and worked solutions. The answer to every exercise is in the answer key at the back of the book, and its page is given next to the exercise. An answer shows whether you got it right, but not how. Full worked solutions, in Swedish, are published in the course repository after the lesson, in `lectures/L01/appendix/c_solutions.md`. Always try it yourself first.

PART 1
Introductory exercises

Exercise 1.1 Calculations with the order of operations

ANSWER P. 247

Calculate the following expressions **without** a calculator.

a) $3 + 2 \times 5$ **b)** $(3 + 2) \times 5$ **c)** $12 - 4 \times 2 + 1$ **d)** $\dfrac{10 + 2}{3} \times 4 - 1$

Exercise 1.2 Fractions

ANSWER P. 247

Calculate and simplify.

a) $\dfrac{1}{4} + \dfrac{1}{6}$ **b)** $\dfrac{3}{5} - \dfrac{1}{4}$ **c)** $\dfrac{2}{3} \times \dfrac{9}{4}$ **d)** $\dfrac{5}{6} \div \dfrac{5}{12}$

Exercise 1.3 Percentages

ANSWER P. 247

a) Calculate 40 % of 250.

b) What percentage of 72 is 18?

c) A resistor has the nominal value 100 Ω with a tolerance of ±5 %. Give the lowest and the highest permitted resistance.

PART 2

New material

Exercise 2.1 Resistors in parallel

ANSWER P. 247

Two resistors $R_1 = 12\,\Omega$ and $R_2 = 6\,\Omega$ are connected in parallel. The total resistance R_{TOT} is given by

$$\frac{1}{R_{TOT}} = \frac{1}{R_1} + \frac{1}{R_2}.$$

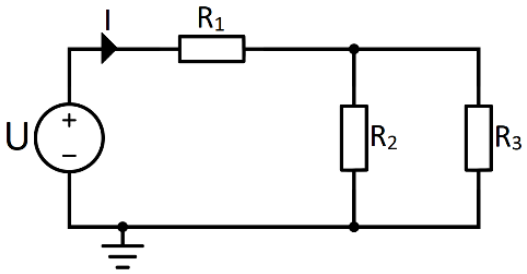
a) Calculate $\frac{1}{R_{TOT}}$.

b) Calculate R_{TOT}.

Exercise 2.2 Series and parallel connection

ANSWER P. 247

The circuit below consists of the resistor R_1 in series with the parallel connection of R_2 and R_3.



The component values are $R_1 = 4\,\text{k}\Omega$, $R_2 = 24\,\text{k}\Omega$ and $R_3 = 8\,\text{k}\Omega$. The parallel resistance $R_p = R_2 \,/\!/\, R_3$ can be calculated in two ways:

$$\frac{1}{R_p} = \frac{1}{R_2} + \frac{1}{R_3} \qquad \text{or} \qquad R_p = \frac{R_2 \times R_3}{R_2 + R_3}$$

a) Calculate R_p with both formulas and check that they give the same answer.

b) Calculate the circuit's total resistance $R_{TOT} = R_1 + R_p$.

c) Why must the parallel connection be calculated before R_1 is added? Relate your answer to the order of operations.

Exercise 2.3 Current and voltage in the circuit

ANSWER P. 247

The circuit in Exercise 2.2 is supplied with the voltage $U = 20\,\text{V}$. Ohm's law relates voltage, resistance and current:

$$U = R \times I$$

> **Tip:** If you calculate with the resistance in kΩ and the voltage in V, you get the current directly in mA.

a) Calculate the current I that the voltage source delivers.

b) Calculate the voltage U_1 across R_1.

c) Calculate the voltage U_p across the parallel connection.

d) Calculate the currents I_2 and I_3 through R_2 and R_3 respectively. Check that $I_2 + I_3 = I$.

Exercise 2.4 Absolute value

ANSWER P. 247

An AC voltage varies between $-8\,\text{V}$ and $+8\,\text{V}$.

a) What is the amplitude $|U|$ of the voltage?

b) At a certain instant, the voltage is $u(t) = -5.3\,\text{V}$. Calculate $|u(t)|$.

Exercise 2.5 Efficiency

ANSWER P. 247

A battery-powered motor takes in $P_\text{in} = 24\,\text{W}$ and delivers $P_\text{out} = 18\,\text{W}$.

a) Calculate the motor's efficiency η as a percentage.

b) If the efficiency is improved to 90 % at the same input power, what will the output power be?

Exercise 2.6 Number sets

ANSWER P. 247

Place each number in the *smallest* number set it belongs to: $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, the irrational numbers or $\mathbb{R}$.

a) 7 **b)** -3 **c)** $\frac{3}{4}$ **d)** $\sqrt{2}$ **e)** 0.25 **f)** π

PART 3

Extra exercises

The exercises below are extra practice, best done on your own after the lesson.

Exercise 3.1 Order of operations with negative numbers

ANSWER P. 247

Calculate **without** a calculator.

a) $-3 + 5 \times (-2)$ **b)** $(-4)^2 - 4^2$ **c)** $\dfrac{-12}{-3} + 2 \times (-5)$

d) $8 - 2(3 - 7)$ **e)** $\dfrac{(-2)^3 + 10}{2}$

Exercise 3.2 The number line and absolute value

ANSWER P. 247

a) Sort the numbers in order of size, smallest first: -2.5, $|-3|$, $\dfrac{7}{2}$, $-|4|$, 0.

b) Calculate $|-7| - |3|$.

c) Calculate $|3 - 8|$ and $|8 - 3|$. What does the result say about the distance between two numbers?

d) Two measured values are $U_1 = 4.8\,\text{V}$ and $U_2 = 5.1\,\text{V}$. Write the size of the difference with an absolute value and calculate it.

Exercise 3.3 Fractions, decimals and percentages

ANSWER P. 247

Fill in the empty cells. Give the fractions in their simplest form.

Fraction	Decimal	Percentage
$\frac{1}{4}$		
	0.6	
		12.5 %
$\frac{3}{8}$		

Exercise 3.4 Fractions in several steps

ANSWER P. 247

Calculate and simplify.

a) $\dfrac{2}{3} + \dfrac{1}{6} - \dfrac{1}{2}$ **b)** $\dfrac{3}{4} \times \dfrac{8}{9} \div \dfrac{2}{3}$ **c)** $\dfrac{\frac{1}{2} + \frac{1}{3}}{\frac{5}{6}}$ **d)** $\dfrac{1}{\frac{1}{4} + \frac{1}{4}}$

e) Which circuit calculation do you recognise in d)?

Exercise 3.5 Percentage change

ANSWER P. 247

a) A voltage rises from 12 V to 15 V. By how many per cent does it rise?

b) A current falls from 400 mA to 340 mA. By how many per cent does it fall?

c) A resistance first increases by 10 % and then decreases by 10 %. Is it back at its original value? Justify your answer.

d) A power of 60 W is increased by 25 %. What is the new power?

Exercise 3.6 Tolerance and measurement deviation

ANSWER P. 247

a) A resistor is marked $4.7\,\text{k}\Omega \pm 10\,\%$. Give the lowest and the highest permitted value.

b) The resistor measures 4.9 kΩ. Is it within tolerance?

c) By how many per cent does the measured value deviate from the nominal value?

d) Another resistor is marked $220\,\Omega \pm 5\,\%$ and measures 208 Ω. Does it pass?

Exercise 3.7 Number sets: true or false

ANSWER P. 248

Decide whether the statement is true or false, and justify your answer briefly.

a) Every integer is also a rational number.

b) Every rational number is also an integer.

c) $\sqrt{9}$ is an irrational number.

d) The sum of two irrational numbers is always irrational.

e) The decimal 0.333 … is a rational number.

f) Every real number can be written as a fraction of two integers.

Exercise 3.8 Equal resistors in parallel

ANSWER P. 248

a) Two resistors of 100 Ω each are connected in parallel. Calculate R_p.

b) Four resistors of 100 Ω each are connected in parallel. Calculate R_p.

c) Show in general that n equal resistors R in parallel give $R_\text{p} = \dfrac{R}{n}$.

d) How many 100 Ω resistors must be connected in parallel to give 20 Ω?

Exercise 3.9 Find the unknown resistor

ANSWER P. 248

a) $R_1 = 30\,\Omega$ is connected in parallel with R_2 and gives $R_p = 12\,\Omega$. Calculate R_2.

b) $R_1 = 220\,\Omega$ is to be supplemented with a resistor in series so that $R_{TOT} = 500\,\Omega$. Calculate that resistor.

c) Check in a) that R_p is smaller than the smallest of the resistors in it.

d) Can a parallel connection of $R_1 = 30\,\Omega$ and some other resistor give $R_p = 40\,\Omega$? Justify your answer.

Exercise 3.10 Ohm's law: fill in the table

ANSWER P. 248

Fill in the empty cells using Ohm's law $U = R \times I$.

> **Tip:** With the resistance in kΩ and the voltage in V, the current comes out directly in mA.

U	R	I
12 V	4 kΩ	
	2.2 kΩ	5 mA
9 V		3 mA
5 V	1 kΩ	

Exercise 3.11 Efficiency in two stages

ANSWER P. 248

A power supply consists of a transformer with the efficiency $\eta_1 = 95\,\%$ followed by a rectifier with the efficiency $\eta_2 = 80\,\%$. The input power is $P_{in} = 200\,\text{W}$.

a) Calculate the power after the transformer.

b) Calculate the output power after the rectifier.

c) Calculate the total efficiency η_{TOT}.

d) How much power is lost as heat in total?

CHAPTER 2

Algebra

Calculating with letters: simplifying, expanding and factorising.

IN THIS CHAPTER

- Algebraic expressions and variables.
- Simplifying algebraic expressions.
- The distributive law and expanding brackets.
- The squaring rules and the difference of two squares.
- Factorisation.
- Ohm's law and the power formulas in algebraic form.

In electrical engineering, almost every relationship is written with letters: Ohm's law $U = RI$, the power $P = UI$ and the resistance of a series connection $R_{\text{TOT}} = R_1 + R_2$. This chapter is about how to calculate with such expressions. It goes through what variables, terms, factors and coefficients are, how like terms are collected, how brackets are expanded with the distributive law, the squaring rules and the difference of two squares, and how an expression is factorised. The rules are then used on circuits, where the right rewriting often saves work. The chapter builds on the order of operations and the fractions of Chapter 1; powers, roots, units and prefixes come in Chapter 5, and equations and working with formulas continue in Chapter 3.

§2.1 to §2.7 are the chapter's theory, with examples. §2.8 summarises the rules and what you should be able to do after the chapter, and §2.9 holds the exercises. Most of them can be done in your head or with pencil and paper.

2.1 Variables and algebraic expressions

A **variable** is an unknown number represented by a letter, for example x, R, U or I.

An **algebraic expression** is a combination of variables, constants and operations, for example

$$3x + 2y - 5, \qquad \frac{U}{R}, \qquad 2R_1 + R_2.$$

2.1.1 Terms, factors and coefficients

In the expression $5x^2 - 3x + 7$:

- $5x^2$, $-3x$ and 7 are **terms**. The terms are separated by $+$ or $-$.
- 5 is the **coefficient** of x^2 and -3 is the coefficient of x, while 7 is a **constant**.
- In the product $5 \cdot x^2$, 5 and x^2 are **factors**.

2.2 Simplifying: collecting like terms

Terms with the *same variable part* are called **like terms** and can be added:

$$3x + 5x = 8x, \qquad 4R - R = 3R, \qquad 2x^2 + 3x - x^2 + x = x^2 + 4x$$

Terms with *different* variable parts can**not** be simplified further:

$$3x + 2y \quad \text{(cannot be simplified further)}$$

2.3 The distributive law

The distributive law is used to expand brackets:

$$a(b + c) = ab + ac$$

EXAMPLE

$$3(2x + 4) = 6x + 12$$
$$-2(R_1 + R_2 - 3) = -2R_1 - 2R_2 + 6$$

The distributive law is used in reverse when factorising; see §2.6.

2.4 The squaring rules

The squaring rules are used a great deal in electrical engineering, for example in calculating power and impedance.

2.4.1 Square of a sum

$$(a + b)^2 = a^2 + 2ab + b^2$$

EXAMPLE

$$(R + 2)^2 = R^2 + 4R + 4$$

2.4.2 Square of a difference

$$(a-b)^2 = a^2 - 2ab + b^2$$

EXAMPLE

$$(U-3)^2 = U^2 - 6U + 9$$

2.5 The difference of two squares

$$(a+b)(a-b) = a^2 - b^2$$

EXAMPLE

$$(x+5)(x-5) = x^2 - 25$$
$$(2R+1)(2R-1) = 4R^2 - 1$$

2.6 Factorisation

Factorisation means writing an expression as a **product of factors**, the reverse of expanding. There are several methods.

2.6.1 Common factor

Identify the factor that all the terms have in common and take it outside a bracket:

$$6x + 9 = 3(2x+3)$$
$$4R_1^2 + 8R_1 = 4R_1(R_1+2)$$
$$RI^2 - RI = RI(I-1)$$

2.6.2 The difference of two squares in reverse

An expression of the form $a^2 - b^2$ can be factorised:

$$a^2 - b^2 = (a+b)(a-b)$$

EXAMPLE

$$x^2 - 16 = (x+4)(x-4)$$
$$4U^2 - 9 = (2U+3)(2U-3)$$

2.6.3 The squaring rules in reverse

Expressions of the form $a^2 + 2ab + b^2$ or $a^2 - 2ab + b^2$ can be factorised:

$$x^2 + 6x + 9 = (x + 3)^2$$
$$R^2 - 4R + 4 = (R - 2)^2$$

2.7 Application in electrical engineering

Ohm's law in algebraic form is

$$U = R \cdot I.$$

The law can be rewritten depending on which quantity is sought:

$$R = \frac{U}{I}, \qquad I = \frac{U}{R}$$

The **power formula** gives rise to algebraic expressions:

$$P = U \cdot I = R \cdot I^2 = \frac{U^2}{R}$$

For the resistors R_1 and R_2 in series,

$$R_{\text{TOT}} = R_1 + R_2,$$

which is a linear algebraic expression.

2.8 Summary

Rule	Formula
Distributive law	$a(b + c) = ab + ac$
Square of a sum	$(a + b)^2 = a^2 + 2ab + b^2$
Square of a difference	$(a - b)^2 = a^2 - 2ab + b^2$
Difference of two squares	$(a + b)(a - b) = a^2 - b^2$
Common factor	$ab + ac = a(b + c)$
Power	$P = UI = RI^2 = \frac{U^2}{R}$

What you should be able to do

After this chapter, you should be able to:

- Understand what an algebraic expression is and how variables are used.
- Simplify algebraic expressions.
- Expand and factorise expressions.
- Use the rules to simplify and speed up calculations in electrical circuits.

Test yourself

ANSWER P. 249

1. Factorise $9R^2 - 4$.
2. Simplify $2(3I + 1) - (I - 5)$.
3. A heating element with $R = 20\,\Omega$ is supplied first with $U_1 = 52\,\text{V}$ and then with $U_2 = 48\,\text{V}$. Use the difference of two squares to calculate the difference in power

$$P_1 - P_2 = \frac{U_1^2 - U_2^2}{R}$$

without a calculator.

Next chapter

Chapter 3 is about linear equations and inequalities: how they are solved, how the solution of an inequality is drawn on the number line and how equations with absolute values are handled.

2.9 Exercises

How to work with the exercises. Parts 1 and 2 are done during the lesson: first on your own, a few minutes per exercise, then together. Part 3 is extra exercises for further practice after the lesson. Most of them can be done in your head or with pencil and paper.
Answers and worked solutions. The answer to every exercise is in the answer key at the back of the book, and its page is given next to the exercise. An answer shows whether you got it right, but not how. Full worked solutions, in Swedish, are published in the course repository after the lesson, in `lectures/L02/appendix/c_solutions.md`. Always try it yourself first.

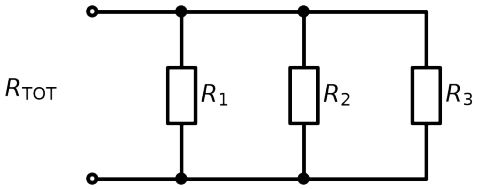
PART 1

Review exercises

Exercise 1.1 Three resistors in parallel

ANSWER P. 248

Three resistors $R_1 = 6\,\Omega$, $R_2 = 3\,\Omega$ and $R_3 = 2\,\Omega$ are connected in parallel.



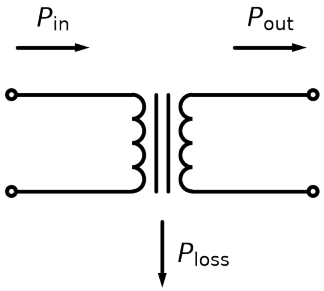
Calculate the total resistance R_{TOT}, which is given by

$$\frac{1}{R_{TOT}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}.$$

Exercise 1.2 Efficiency and power

ANSWER P. 248

A transformer has the efficiency $\eta = 92\,\%$ and the input power $P_{in} = 500\,\text{W}$.



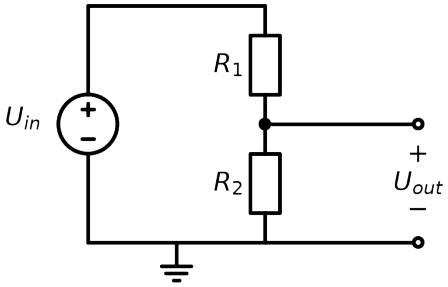
a) Calculate the output power P_{out}.

b) How much power P_{loss} is lost as heat?

Exercise 1.3 Order of operations in a circuit formula

ANSWER P. 248

Calculate the output voltage U_{out} of the voltage divider with the formula below when $U_{\text{in}} = 12\,\text{V}$, $R_1 = 8\,\Omega$ and $R_2 = 4\,\Omega$.



$$U_{\text{out}} = U_{\text{in}} \times \frac{R_2}{R_1 + R_2}$$

PART 2

New material

Exercise 2.1 Simplifying algebraic expressions

ANSWER P. 248

Simplify the following expressions.

a) $5I_1 + 3I_2 - 2I_1 + 7I_2$

b) $4R_1 - 3R_2 + R_1 + 5R_2 - 2R_1$

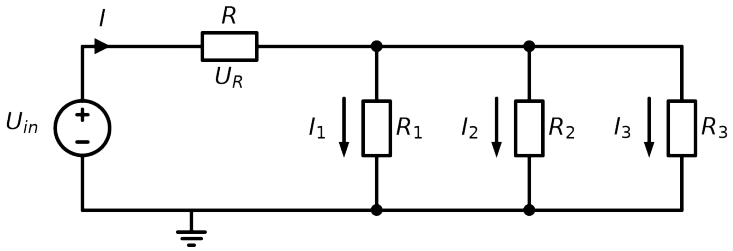
c) $3I^2 - 2I + 4I^2 + 5I - 1$

d) Substitute $I = 2$ into both the original and the simplified expression in c) and check that they give the same value.

Exercise 2.2 The distributive law in circuits

ANSWER P. 248

The resistor R in the circuit below carries the sum of the three branch currents I_1, I_2 and I_3.



The voltage drop across R is therefore given by

$$U_R = RI = R(I_1 + I_2 + I_3).$$

a) Expand the brackets and give each term with its unit.

b) Calculate U_R with the original form when $R = 100\,\Omega$, $I_1 = 20\,\text{mA}$, $I_2 = 30\,\text{mA}$ and $I_3 = 50\,\text{mA}$.

c) Calculate U_R once more with the expanded expression and check that the answers agree.

Exercise 2.3 Factorisation

ANSWER P. 248

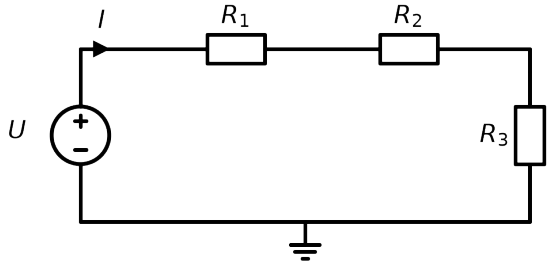
Factorise the following expressions.

a) $6I + 9RI$ **b)** $P_1^2 - P_2^2$ **c)** $U^2 - 10U + 25$ **d)** $4R^2 - 1$

Exercise 2.4 Common factor: total power in a series circuit

ANSWER P. 248

Three resistors $R_1 = 120\,\Omega$, $R_2 = 180\,\Omega$ and $R_3 = 200\,\Omega$ are connected in series and so carry the same current $I = 50\,\text{mA}$.



The power in each resistor is given by $P_k = R_k I^2$.

a) Write an expression for the total power $P_{\text{TOT}} = P_1 + P_2 + P_3$ and take out the common factor.

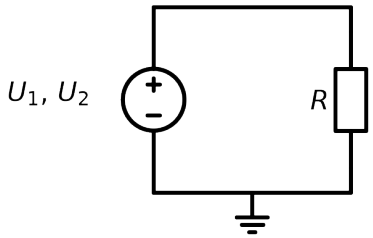
b) Calculate P_{TOT} with the factorised expression.

c) Calculate P_1, P_2 and P_3 separately and check that the sum is the same. Which way took the least work?

Exercise 2.5 The difference of two squares: difference in power

ANSWER P. 248

A heating element with the resistance R is supplied first with the voltage U_1 and then with U_2.



The power is given by $P = \dfrac{U^2}{R}$.

a) Show that the difference in power can be written

$$P_1 - P_2 = \frac{(U_1 + U_2)(U_1 - U_2)}{R}.$$

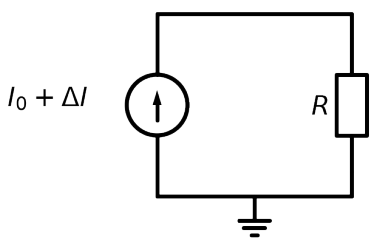
b) Calculate $P_1 - P_2$ **without** a calculator when $U_1 = 24\,\text{V}$, $U_2 = 20\,\text{V}$ and $R = 8\,\Omega$.

c) Check your answer by calculating P_1 and P_2 separately.

Exercise 2.6 The squaring rule: power after a change in current

ANSWER P. 248

The current $I_0 = 2\,\text{A}$ flows through a resistor $R = 10\,\Omega$. The current increases by $\Delta I = 0.1\,\text{A}$, so that the new current is $I_0 + \Delta I$.



The power is given by $P = RI^2$.

a) Expand $P = R(I_0 + \Delta I)^2$ with the squaring rule.

b) Calculate the new power with the expanded expression.

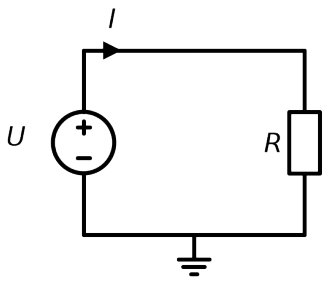
c) How large is the increase in power ΔP compared with $P_0 = RI_0^2$?

d) How large is the error if you ignore the term $R\,\Delta I^2$? Give the error as a percentage of ΔP.

Exercise 2.7 Power expressions and Ohm's law

ANSWER P. 248

The power dissipated in the resistor below can be written $P = RI^2$.



a) Show that $P = RI^2$ can be factorised as $P = (RI) \cdot I$, and identify which quantity RI represents.

b) Substitute $I = \dfrac{U}{R}$ into $P = RI^2$ and show that the expression simplifies to $P = \dfrac{U^2}{R}$.

c) A resistor $R = 6\,\Omega$ is supplied with $U = 12\,\text{V}$. Calculate the power with all three forms $P = UI$, $P = RI^2$ and $P = \dfrac{U^2}{R}$, and check that they give the same answer.

PART 3

Extra exercises

The exercises below are extra practice, best done on your own after the lesson.

Exercise 3.1 Terms, factors and coefficients

ANSWER P. 248

Consider the expression $7R^2 - 4R + 9$.

a) How many terms does the expression consist of?

b) Give the coefficient of R^2 and of R.

c) Give the constant term.

d) Which factors make up the term $7R^2$?

e) Calculate the value of the expression for $R = 2$.

Exercise 3.2 Simplifying with brackets

ANSWER P. 248

Simplify.

a) $2(3I + 4) - 3(I - 2)$ **b)** $-(U - 5) + 2(U + 1)$

c) $5R_1 - [2R_1 - (R_1 + 3)]$ **d)** $3(2P - 1) - 2(3P - 4)$

Exercise 3.3 Expanding

ANSWER P. 248

Expand and simplify.

a) $(R + 3)(R + 5)$ **b)** $(2I - 1)(I + 4)$ **c)** $(U + 2)(U - 2)$

d) $(3R - 2)(3R - 2)$ **e)** $2I(I + 3) - I(2I - 1)$

Exercise 3.4 The squaring rules

ANSWER P. 248

Expand.

a) $(I + 4)^2$ **b)** $(R - 6)^2$ **c)** $(2U + 3)^2$ **d)** $(5 - I)^2$

e) Show with $a = 3$ and $b = 4$ that $(a + b)^2 \neq a^2 + b^2$.

Exercise 3.5 The difference of two squares as a mental-arithmetic trick

ANSWER P. 248

Use $(a + b)(a - b) = a^2 - b^2$ and calculate **without** a calculator.

a) 102×98 **b)** 45×35 **c)** 2.1×1.9 **d)** $51^2 - 49^2$

e) The voltage across a resistor $R = 5\,\Omega$ changes from $U_1 = 25\,\text{V}$ to $U_2 = 15\,\text{V}$. Calculate $P_1 - P_2 = \dfrac{U_1^2 - U_2^2}{R}$ without squaring any number.

Exercise 3.6 Factorise

ANSWER P. 248

Factorise as far as possible.

a) $8R + 12$ **b)** $RI^2 + RI$ **c)** $U^2 - 49$

d) $I^2 + 12I + 36$ **e)** $9R^2 - 16$ **f)** $2U^2 - 8$

Exercise 3.7 Simplifying algebraic fractions

ANSWER P. 248

Factorise and cancel. Assume that the denominator is not zero.

a) $\dfrac{6RI}{3R}$ **b)** $\dfrac{R^2 - 9}{R + 3}$ **c)** $\dfrac{2U^2 + 4U}{2U}$

d) $\dfrac{I^2 - 10I + 25}{I - 5}$ **e)** $\dfrac{4R^2 - 1}{2R - 1}$

Exercise 3.8 Substituting into the power formulas

ANSWER P. 248

A resistor $R = 8\,\Omega$ carries the current $I = 1.5\,\text{A}$.

a) Calculate the voltage $U = RI$.

b) Calculate the power with $P = RI^2$.

c) Calculate the power with $P = \dfrac{U^2}{R}$ and check that the answer agrees.

d) The current doubles. By what factor does the power change? Justify your answer algebraically.

Exercise 3.9 The algebra of the voltage divider

ANSWER P. 248

For two resistors in series,

$$U_1 = U_{\text{in}} \times \frac{R_1}{R_1 + R_2} \qquad \text{and} \qquad U_2 = U_{\text{in}} \times \frac{R_2}{R_1 + R_2}.$$

a) Show algebraically that $U_1 + U_2 = U_{\text{in}}$.

b) Show that $\dfrac{U_1}{U_2} = \dfrac{R_1}{R_2}$.

c) What is U_1 if $R_1 = R_2$?

d) Calculate U_1 and U_2 when $U_{\text{in}} = 15\,\text{V}$, $R_1 = 1\,\text{k}\Omega$ and $R_2 = 4\,\text{k}\Omega$.

Exercise 3.10 Find the error

ANSWER P. 248

Each equality below contains a common algebra mistake. Explain what is wrong and write the correct right-hand side.

a) $(R + 2)^2 = R^2 + 4$

b) $3(I - 2) = 3I - 2$

c) $\dfrac{R + 4}{4} = R$

d) $-(U - 3) = -U - 3$

e) $2R \times 3R = 6R$

f) $(2I)^2 = 2I^2$

CHAPTER 3

Linear equations and inequalities

Finding the unknown: equations, inequalities and absolute values.

IN THIS CHAPTER

- Linear equations and the principles for solving them.
- Proportions.
- Inequalities and solution intervals on the number line.
- Equations with absolute values.

Chapter 2 simplified and factorised expressions. This chapter sets two expressions equal to each other and solves the equation that results. A linear equation is solved with two principles, the addition principle and the multiplication principle, and the same principles solve inequalities, with one important exception: the inequality sign reverses when you multiply or divide by a negative number. The chapter also covers proportions and equations with absolute values. The applications recur throughout the course: solving Ohm's law for an unknown resistance, finding the safe operating area of a component and expressing a tolerance with an absolute value.

§3.1 to §3.6 are the chapter's theory, with examples. §3.7 summarises the rules and what you should be able to do after the chapter, and §3.8 holds the exercises. Most of them can be done in your head or with pencil and paper.

3.1 Equations

An **equation** is a statement that two expressions are equal:

$$\text{left-hand side} = \text{right-hand side}$$

A **linear equation** in one unknown x is an equation in which x appears to at most the first power:

$$ax + b = c, \quad a \neq 0$$

To **solve an equation** is to find the value or values of x that satisfy it.

3.2 The principles for solving equations

The solution of an equation does not change if you perform the *same* operation on both sides.

Principle	Description
Addition principle	Add or subtract the same number on both sides
Multiplication principle	Multiply or divide both sides by the same number ($\neq 0$)

Strategy: Move every term with the unknown to one side and the constants to the other.

3.3 Solving linear equations step by step

EXAMPLE Solve $3x - 5 = 7$.

$$\begin{aligned} 3x - 5 &= 7 \\ 3x &= 12 && \text{add 5 to both sides} \\ x &= 4 && \text{divide both sides by 3} \end{aligned}$$

Check: $3 \cdot 4 - 5 = 12 - 5 = 7$, which is correct.

EXAMPLE **Ohm's law.** A circuit has the voltage $U = 12\,\text{V}$ and the current $I = 0.5\,\text{A}$. Calculate the resistance R. Ohm's law gives

$$U = R \cdot I \quad \Leftrightarrow \quad 12 = R \cdot 0.5.$$

Divide by 0.5:

$$R = \frac{12}{0.5} = 24\,\Omega$$

EXAMPLE **With brackets.** Solve $2(3x + 1) = 14$.

$$\begin{aligned} 6x + 2 &= 14 && \text{expand the brackets} \\ 6x &= 12 && \text{subtract 2} \\ x &= 2 && \text{divide by 6} \end{aligned}$$

3.4 Proportions

A proportion is an equation of the form

$$\frac{a}{b} = \frac{c}{d}$$

and is solved by **cross-multiplication**, that is, by multiplying diagonally:

$$ad = bc$$

EXAMPLE Solve $\dfrac{x}{6} = \dfrac{4}{3}$.

$$3x = 24 \quad \Rightarrow \quad x = 8$$

3.5 Inequalities

An **inequality** states that one expression is greater or less than another:

Symbol	Read as
$a < b$	a is less than b
$a > b$	a is greater than b
$a \leq b$	a is less than or equal to b
$a \geq b$	a is greater than or equal to b

3.5.1 Solving inequalities

Inequalities are solved exactly like equations, with one important exception:

Note: If you multiply or divide by a **negative number**, the inequality sign reverses.

EXAMPLE Solve $2x + 3 > 9$.

$$2x > 6 \quad \Rightarrow \quad x > 3$$

The solution is every x greater than 3, that is, $x \in (3, \infty)$.

EXAMPLE **Negative number.** Solve $-3x \leq 12$. Divide by -3; the sign reverses:

$$x \geq -4$$

3.5.2 Practical application: safe operating area

A component can take at most $P_{\max} = 2\,\text{W}$, and the power is given by $P = U \cdot I$ with $I = 0.5\,\text{A}$. The permitted voltage range follows from

$$U \cdot 0.5 \leq 2 \quad \Rightarrow \quad U \leq 4\,\text{V}.$$

The voltage can be at most 4 V without damaging the component.

3.6 Absolute values in equations

Equations with absolute values generally have two solutions. The equation $|ax + b| = c$, with $c \geq 0$, gives

$$ax + b = c \quad \text{or} \quad ax + b = -c.$$

EXAMPLE Solve $|2x - 3| = 5$.

$$\begin{aligned} \text{Case 1:} \quad & 2x - 3 = 5 \quad \Rightarrow \quad x = 4 \\ \text{Case 2:} \quad & 2x - 3 = -5 \quad \Rightarrow \quad x = -1 \end{aligned}$$

The solution is $x = 4$ or $x = -1$.

3.7 Summary

Concept	Key rule
Linear equation	Isolate the unknown with the addition and multiplication principles
Proportion	Cross-multiplication: $\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$
Inequality	The same rules as for an equation, but the sign reverses when you multiply or divide by a negative number
Absolute value	$\lvert ax + b \rvert = c \Rightarrow$ two cases: $ax + b = c$ or $ax + b = -c$

What you should be able to do

After this chapter, you should be able to:

- Solve linear equations algebraically.
- Understand and solve inequalities.
- Understand and calculate the absolute value of a number or an expression.
- Set up and solve equations and inequalities from electrical relationships.

Test yourself

ANSWER P. 249

1. Solve the equation $4(x - 2) = 2x + 8$ and check the answer.
2. The voltage is $U = 3.6\,\text{V}$ and the resistance $R = 120\,\Omega$. Calculate the current I with Ohm's law.
3. The terminal voltage of a loaded battery is $U = 9 - 0.5I$. Find the currents that satisfy $U \geq 6\,\text{V}$.

Next chapter

Chapter 4 is about systems of equations: two equations in two unknowns, solved by the substitution method or the elimination method.

3.8 Exercises

> **How to work with the exercises.** Parts 1 and 2 are done during the lesson: first on your own, a few minutes per exercise, then together. Part 3 is extra exercises for further practice after the lesson. Most of them can be done in your head or with pencil and paper.
> **Answers and worked solutions.** The answer to every exercise is in the answer key at the back of the book, and its page is given next to the exercise. An answer shows whether you got it right, but not how. Full worked solutions, in Swedish, are published in the course repository after the lesson, in `lectures/L03/appendix/c_solutions.md`. Always try it yourself first.

PART 1

Review exercises

Exercise 1.1 Simplification

ANSWER P. 249

Simplify the following expressions.

a) $3(2R + 4) - 2(R - 1)$ **b)** $(U + 3)^2 - 9$ **c)** $\dfrac{6I^2 + 4I}{2I}$, assume $I \neq 0$

Exercise 1.2 Factorisation

ANSWER P. 249

Factorise.

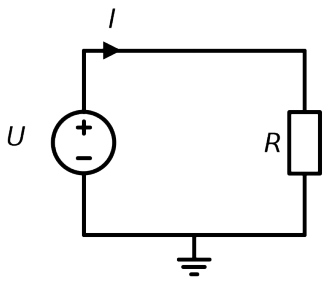
a) $R^2 - 25$ **b)** $3U^2 + 6U$ **c)** $I^2 - 8I + 16$

PART 2
New material

Exercise 2.1 Ohm's law: find the resistance

ANSWER P. 249

The voltage across a resistor is $U = 15\,\text{V}$ and the current is $I = 0.3\,\text{A}$.



Ohm's law states that $U = R \cdot I$.

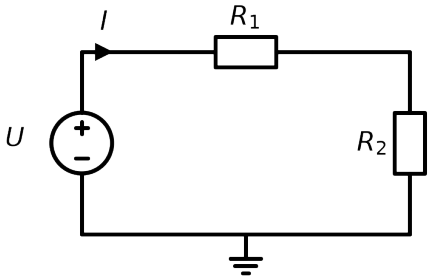
a) Set up the equation and solve for R.

b) Check your answer.

Exercise 2.2 Series circuit: an equation with brackets

ANSWER P. 249

Two resistors are connected in series. The circuit is supplied with $U = 24\,\text{V}$ and the current is $I = 0.4\,\text{A}$. One resistor is $R_2 = 40\,\Omega$; the other is unknown.



Ohm's law for the whole circuit gives

$$U = (R_1 + R_2) \cdot I.$$

a) Substitute the values and solve the equation for R_1 **without** expanding the brackets.

b) Solve the equation again, but expand the brackets first. Check that you get the same answer.

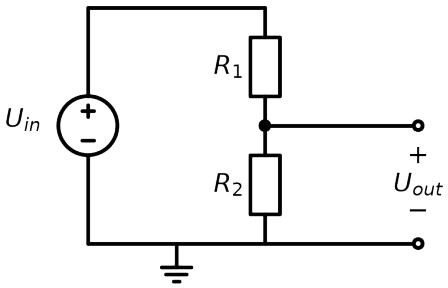
c) Check the solution by calculating the current $I = \dfrac{U}{R_1 + R_2}$.

Exercise 2.3 Voltage division

ANSWER P. 249

In a voltage divider, the output voltage is given by

$$U_{\text{out}} = \frac{R_2}{R_1 + R_2} \cdot U_{\text{in}}.$$



Assume that $U_{\text{in}} = 9\,\text{V}$, $R_2 = 3\,\Omega$ and $U_{\text{out}} = 3\,\text{V}$.

a) Set up the equation for U_{out} and solve for R_1.

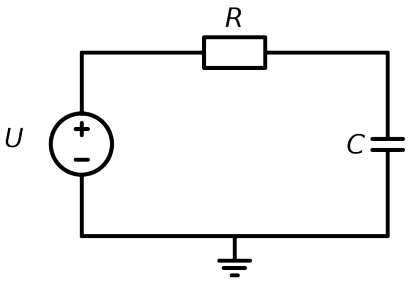
b) Check your answer.

Exercise 2.4 Time constant of an RC circuit

ANSWER P. 249

The time constant τ (tau) of an RC circuit is

$$\tau = R \cdot C.$$

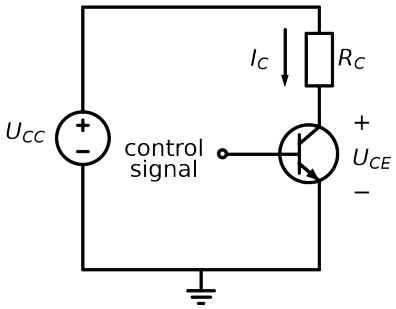


The capacitor's capacitance is $C = 100\,\mu\text{F} = 100 \times 10^{-6}\,\text{F}$ and the time constant is $\tau = 0.05\,\text{s}$. Calculate the resistance R.

Exercise 2.5 Safe operating area

ANSWER P. 249

The power dissipation of a transistor is calculated as $P = U_{CE} \cdot I_C$.



The transistor can take at most $P_{max} = 500\,\text{mW}$, and the collector current is $I_C = 50\,\text{mA}$.

a) Set up an inequality to find the largest permitted voltage drop U_{CE}.

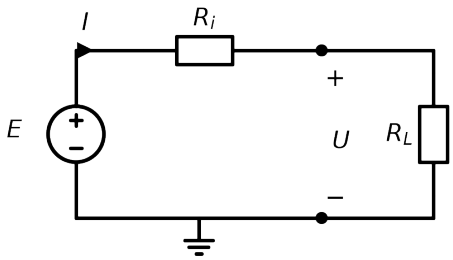
b) Solve the inequality and give the largest permitted U_{CE} in volts.

Exercise 2.6 Loaded battery: an inequality where the sign reverses

ANSWER P. 249

A battery has the open-circuit voltage $E = 12\,\text{V}$ and the internal resistance $R_i = 0.8\,\Omega$. When the battery is loaded with the current I, the terminal voltage is

$$U = E - R_i \cdot I.$$



a) Calculate the terminal voltage U when $I = 2\,\text{A}$.

b) The connected device needs at least 9 V to work. Set up the inequality $U \geq 9$ and solve it for I.

> **Tip:** Think about what happens to the inequality sign when you divide by a negative number.

c) Another battery has $E = 13\,\text{V}$ and $R_\text{i} = 1.0\,\Omega$. At what current do the two batteries give the same terminal voltage? Set up an equation with the unknown on both sides and solve it.

d) Which of the batteries gives the higher terminal voltage at $I = 8\,\text{A}$?

Exercise 2.7 Absolute value in signal analysis

ANSWER P. 249

A signal can deviate positively or negatively from a reference voltage. The deviation δ satisfies $|\delta| \le 0.5\,\text{V}$.

a) Rewrite the absolute-value inequality as a double inequality without the absolute value.

b) If the reference voltage is $5\,\text{V}$, give the permitted range of the actual voltage.

Exercise 2.8 Current regulation: an absolute-value equation

ANSWER P. 249

A current regulator compares the measured current I (in amperes) with its setpoint and gives the error voltage

$$u_\text{e} = 0.4I - 2.$$

a) At what current is the error voltage zero, that is, what is the setpoint?

b) The regulator raises an alarm when the error voltage has the magnitude $1.2\,\text{V}$. Solve the equation $|0.4I - 2| = 1.2$ as two cases and give the two currents at which the alarm is triggered.

c) Which range of currents satisfies $|0.4I - 2| < 1.2$, that is, when does the regulator *not* raise an alarm?

PART 3

Extra exercises

The exercises below are extra practice, best done on your own after the lesson.

Exercise 3.1 Simple linear equations

ANSWER P. 249

Solve the equations.

a) $x + 7 = 12$ **b)** $5x = 45$ **c)** $3x - 4 = 11$

d) $\dfrac{x}{4} = 6$ **e)** $8 - 2x = 0$ **f)** $-3x = 18$

Exercise 3.2 The unknown on both sides

ANSWER P. 249

Solve the equations and check your answers.

a) $5x - 3 = 2x + 9$ **b)** $7 - 2x = x + 1$ **c)** $4(x - 1) = 2(x + 3)$

d) $3(2x + 1) = 6x + 3$ **e)** $2(x + 4) = 2x + 5$

> **Tip:** Two of the equations behave differently from the others. Think about what that means.

Exercise 3.3 Equations with fractions

ANSWER P. 249

Solve the equations.

a) $\frac{x}{3} + \frac{x}{6} = 3$ **b)** $\frac{2x - 1}{5} = 3$

c) $\frac{x}{2} - \frac{x}{5} = 3$ **d)** $\frac{12}{x} = 4$, assume $x \neq 0$

Exercise 3.4 Proportions

ANSWER P. 249

a) Solve $\frac{x}{8} = \frac{3}{4}$.

b) Solve $\frac{5}{x} = \frac{2}{6}$.

c) In a voltage divider, $\frac{U_1}{U_2} = \frac{R_1}{R_2}$. Calculate R_1 when $U_1 = 4\,\text{V}$, $U_2 = 8\,\text{V}$ and $R_2 = 2\,\text{k}\Omega$.

d) A resistor has 6 V across it at the current 0.2 A. What voltage is needed for the current 0.5 A? Set up a proportion and solve it.

Exercise 3.5 Solving a formula for a variable

ANSWER P. 249

Solve for the variable given.

a) $U = RI$, solve for I. **b)** $P = RI^2$, solve for R.

c) $R_{\text{TOT}} = R_1 + R_2$, solve for R_2. **d)** $\tau = RC$, solve for C.

e) $U = E - R_i I$, solve for R_i. **f)** $\eta = \frac{P_{\text{out}}}{P_{\text{in}}}$, solve for P_{in}.

Exercise 3.6 Inequalities

ANSWER P. 249

Solve the inequalities and describe each solution as an interval on the number line.

a) $x + 3 < 8$ **b)** $2x - 1 \geq 7$ **c)** $-x > 4$

d) $-2x + 6 \leq 0$ **e)** $5 - 3x > 14$

Exercise 3.7 Double inequalities

ANSWER P. 249

a) Solve $2 < x + 1 < 7$.

b) Solve $-3 \leq 2x - 1 \leq 5$.

c) A component is supplied with $U = 5\,\text{V}$ and can take at most $P_{\max} = 0.25\,\text{W}$. Set up and solve an inequality for the permitted current I.

d) An input is read as a logic one when its voltage lies in the range $2.0\,\text{V} \leq u \leq 3.3\,\text{V}$. The input voltage is $u = 0.5x$ volts. Which values of x give a logic one?

Exercise 3.8 Absolute values in equations

ANSWER P. 249

Solve the equations.

a) $|x| = 6$ **b)** $|x - 3| = 5$ **c)** $|2x + 1| = 7$

d) $|x + 4| = 0$ **e)** $|3x - 2| = -5$

Exercise 3.9 Absolute values in inequalities and tolerances

ANSWER P. 249

a) Write $|x - 5| \leq 2$ without the absolute value and solve the inequality.

b) Solve $|2x - 6| < 4$.

c) A supply voltage is to be $5\,\text{V}$ with a tolerance of $\pm 2\,\%$. Write the requirement as an absolute-value inequality and give the permitted voltage range.

d) A resistor is marked $1\,\text{k}\Omega \pm 1\,\%$ and measures $1\,015\,\Omega$. Does the resistor meet the requirement $|R - 1000| \leq 10$?

Exercise 3.10 Temperature-dependent resistance

ANSWER P. 249

The resistance of a resistor varies with temperature as

$$R = R_0(1 + \alpha \cdot \Delta T).$$

Here $R_0 = 100\,\Omega$ is the resistance at $20\,^\circ\text{C}$, $\alpha = 0.004\,^\circ\text{C}^{-1}$ and ΔT is the change in temperature from $20\,^\circ\text{C}$.

a) Calculate R at $70\,^\circ\text{C}$.

b) At what temperature is $R = 110\,\Omega$?

c) The circuit works as long as $R \leq 130\,\Omega$. What temperature range is permitted?

d) Solve the formula for ΔT in general.

CHAPTER 4

Systems of equations

Several conditions that must hold at once, two ways to solve them, and Kirchhoff's laws.

IN THIS CHAPTER

- Linear systems of equations in two unknowns.
- The substitution method.
- The elimination method (addition method).
- Graphical interpretation of systems of equations.
- Kirchhoff's laws, which give systems of equations for the currents and voltages in a circuit.

In Chapter 3 you solved equations in one unknown. A circuit often has several unknown currents and voltages, and every relationship between them becomes an equation of its own; they must all hold at once. This chapter is about such systems of equations, primarily two linear equations in two unknowns. You learn two methods of solving them, the substitution method and the elimination method, and see what the solution means graphically: the point where two straight lines intersect, which also decides whether the system has one solution, none at all or infinitely many. Finally, Kirchhoff's laws are used to set up a system of equations for the currents in a circuit.

§4.1 to §4.4 are the chapter's theory, with examples. §4.5 summarises the methods and what you should be able to do after the chapter, and §4.6 holds the exercises. Most of them can be done in your head or with pencil and paper.

4.1 What is a system of equations?

A **linear system of equations** consists of several equations in several unknowns that must be satisfied *simultaneously*. The most common is a system of **two equations in two unknowns**:

$$\begin{cases} a_1 x + b_1 y = c_1 \\ a_2 x + b_2 y = c_2 \end{cases}$$

The **solution** is the pair (x, y) that satisfies both equations. Graphically, it is the **point of intersection** of the two straight lines.

4.1.1 Number of solutions

Situation	Number of solutions
The lines intersect	Exactly one solution
The lines are parallel (but do not coincide)	No solution
The lines coincide	Infinitely many solutions

4.2 The substitution method

The substitution method works like this:

1. Choose one equation and express one unknown in terms of the other.
2. Substitute the expression into the other equation.
3. Solve the resulting equation, which has one unknown.
4. Calculate the other unknown.
5. Check in *both* of the original equations.

EXAMPLE Solve the system

$$\begin{cases} x + 2y = 7 \\ 3x - y = 7 \end{cases}$$

Step 1: Equation (1) gives $x = 7 - 2y$.
Step 2: Substitute into equation (2):

$$3(7 - 2y) - y = 7$$

Step 3: Simplify and solve:

$$21 - 6y - y = 7 \quad \Rightarrow \quad 21 - 7y = 7 \quad \Rightarrow \quad y = 2$$

Step 4: Calculate x:

$$x = 7 - 2 \cdot 2 = 3$$

Step 5: Check: $3 + 4 = 7$ ✓ and $9 - 2 = 7$ ✓
The solution is $(x, y) = (3, 2)$.

4.3 The elimination method (addition method)

Idea: Multiply the equations by suitable numbers so that one unknown gets coefficients that are *equal but opposite*, then add the equations to eliminate that unknown.

1. Multiply one or both equations by suitable constants.
2. Add the equations line by line, so that one unknown disappears.
3. Solve the remaining equation.
4. Calculate the eliminated unknown from one of the original equations.

EXAMPLE Solve the system

$$\begin{cases} 2x + 3y = 12 \\ 4x - y = 10 \end{cases}$$

Step 1: Multiply equation (2) by 3:

$$\begin{cases} 2x + 3y = 12 \\ 12x - 3y = 30 \end{cases}$$

Step 2: Add the equations:

$$14x = 42 \quad \Rightarrow \quad x = 3$$

Step 3: Substitute $x = 3$ into equation (1):

$$6 + 3y = 12 \quad \Rightarrow \quad y = 2$$

The solution is $(x, y) = (3, 2)$.

4.4 Application: Kirchhoff's laws

Systems of equations are fundamental to circuit theory. With Kirchhoff's laws, the distribution of currents and voltages in a circuit can be calculated.

4.4.1 Kirchhoff's voltage law (KVL)

The sum of all the voltages around a closed loop is zero:

$$\sum U = 0$$

4.4.2 Kirchhoff's current law (KCL)

The sum of the currents into a node is zero, that is, the currents in equal the currents out:

$$\sum I = 0$$

EXAMPLE Consider a circuit with two loops and the currents I_1 and I_2. KVL gives

$$\begin{cases} 10 = 2I_1 + 4(I_1 - I_2) \\ 0 = 4(I_2 - I_1) + 6I_2 \end{cases}$$

which simplifies to

$$\begin{cases} 6I_1 - 4I_2 = 10 \\ -4I_1 + 10I_2 = 0 \end{cases}$$

Equation (2) gives $I_1 = 2.5I_2$. Substituting this into equation (1) gives

$$6 \cdot 2.5I_2 - 4I_2 = 10 \quad \Rightarrow \quad I_2 = \frac{10}{11} \approx 0.91\,\text{A},$$

$$I_1 = 2.5 \cdot \frac{10}{11} = \frac{25}{11} \approx 2.27\,\text{A}.$$

4.5 Summary

Method	Best when
Substitution method	One unknown is easy to express in terms of the other
Elimination method	The coefficients are integers and the elimination is straightforward

The steps are the same in both methods:

1. Set the system up clearly.
2. Solve for one unknown.
3. Calculate the other unknown.
4. Check in *both* equations.

What you should be able to do

After this chapter, you should be able to:

- Understand what a system of equations is and when it is used.
- Solve linear systems of equations by the substitution method.
- Solve linear systems of equations by the elimination method.

Test yourself

ANSWER P. 250

1. Solve the system $x + 2y = 7$ and $3x - y = 7$ by a method of your choice.
2. In a circuit, KCL gives $I_1 + I_2 = 4\,\text{A}$, and also $I_1 = 3I_2$. Find I_1 and I_2.

Next chapter

Chapter 5 is about powers and roots, scientific notation and significant figures, and binary and hexadecimal numbers.

4.6 Exercises

> **How to work with the exercises.** Parts 1 and 2 are done during the lesson: first on your own, a few minutes per exercise, then together. Part 3 is extra exercises for further practice after the lesson. Most of them can be done in your head or with pencil and paper.
> **Answers and worked solutions.** The answer to every exercise is in the answer key at the back of the book, and its page is given next to the exercise. An answer shows whether you got it right, but not how. Full worked solutions, in Swedish, are published in the course repository after the lesson, in `lectures/L04/appendix/c_solutions.md`. Always try it yourself first.

PART 1

Review exercises

Exercise 1.1 Ohm's law: find the voltage

ANSWER P. 250

A circuit has $R = 47\,\Omega$ and $I = 200\,\text{mA} = 0.2\,\text{A}$.

a) Calculate the voltage U.

b) The power in the resistor is given by $P = U \cdot I$. Calculate the power.

Exercise 1.2 A linear equation in an RC circuit

ANSWER P. 250

The time constant τ of an RC circuit is given by $\tau = R \cdot C$. In the circuit, $\tau = 20\,\text{ms}$ and $C = 10\,\mu\text{F}$.

a) Set up the equation and solve for R.

b) Give R in $\text{k}\Omega$.

Exercise 1.3 Inequality

ANSWER P. 250

An amplifier delivers the output voltage $U_{\text{out}} = 5 \cdot U_{\text{in}}$. The output voltage must not exceed 15 V. Set up and solve an inequality for U_{in}.

PART 2

New material

Exercise 2.1 An algebraic system of equations

ANSWER P. 250

Solve the system of equations

$$\begin{cases} 3x + y = 11 \\ x - 2y = -2 \end{cases}$$

Exercise 2.2 Currents from Kirchhoff's current law

ANSWER P. 250

At a node, KCL holds: the sum of the currents in equals the sum of the currents out. The currents I_1 and I_2 flow into a node, and the current $I_3 = 5$ A flows out. We also know that $I_1 = 2I_2$.

a) Set up a system of equations for I_1 and I_2.

b) Solve the system and give the values of the currents.

Exercise 2.3 Voltage and current via KVL

ANSWER P. 250

A circuit consists of a voltage source $E = 12$ V and two resistors R_1 and R_2 in series. KVL gives $E = U_1 + U_2$, that is, $U_1 + U_2 = 12$. The voltage drop across R_2 is also three times that across R_1: $U_2 = 3U_1$.

a) Set up the system of equations.

b) Solve the system and calculate U_1 and U_2.

c) The current is $I = 2$ A. Calculate R_1 and R_2 with Ohm's law.

PART 3

Extra exercises

The exercises below are extra practice, best done on your own after the lesson.

Exercise 3.1 The substitution method

ANSWER P. 250

Solve the systems by the substitution method.

a) $\begin{cases} y = 2x + 1 \\ 3x + y = 11 \end{cases}$ **b)** $\begin{cases} x = 3y \\ x + 2y = 15 \end{cases}$ **c)** $\begin{cases} x + y = 10 \\ x - y = 4 \end{cases}$

Exercise 3.2 The elimination method

ANSWER P. 250

Solve the systems by the elimination method.

a) $\begin{cases} 2x + y = 7 \\ x - y = 2 \end{cases}$ **b)** $\begin{cases} 3x + 2y = 16 \\ x - 2y = 0 \end{cases}$ **c)** $\begin{cases} 5x + 3y = 21 \\ 2x - y = 4 \end{cases}$ **d)** $\begin{cases} 3x + 4y = 10 \\ 2x - 3y = 1 \end{cases}$

Exercise 3.3 Number of solutions

ANSWER P. 250

Decide whether the system has exactly one solution, no solution or infinitely many. Justify your answer by how the lines lie.

a) $\begin{cases} x + y = 4 \\ 2x + 2y = 8 \end{cases}$ **b)** $\begin{cases} x + y = 4 \\ x + y = 6 \end{cases}$ **c)** $\begin{cases} x + y = 4 \\ x - y = 0 \end{cases}$ **d)** $\begin{cases} 2x - 4y = 6 \\ -x + 2y = -3 \end{cases}$

Exercise 3.4 Graphical interpretation

ANSWER P. 250

Consider the system

$$\begin{cases} y = 2x - 1 \\ y = -x + 5 \end{cases}$$

a) Make a table of values for both lines with $x = 0, 1, 2, 3, 4$.

b) At what x do the two expressions give the same y? Read the point of intersection from the table.

c) Solve the system algebraically and check your reading.

d) What would the table look like if the system had no solution?

Exercise 3.5 Systems with fractions and decimals

ANSWER P. 250

Solve the systems.

a) $\begin{cases} 0.5x + y = 4 \\ x - y = 2 \end{cases}$

b) $\begin{cases} \frac{x}{2} + \frac{y}{3} = 4 \\ x - y = 3 \end{cases}$

> **Tip:** Clear the denominators before you choose a method.

Exercise 3.6 KCL at a node

ANSWER P. 250

The currents I_1 and I_2 flow into a node, while $I_3 = 12\,\text{A}$ flows out. A measurement also shows that I_1 is 4 A greater than I_2.

a) Set up a system of equations for I_1 and I_2.

b) Solve the system.

c) Check the solution in both equations.

Exercise 3.7 Current division in a parallel connection

ANSWER P. 250

Two resistors R_1 and R_2 are connected in parallel across the voltage $U = 24\,\text{V}$. The total current is $I = 3\,\text{A}$, and the branch current through R_1 is twice that through R_2.

a) Set up a system of equations for the branch currents I_1 and I_2.

b) Solve the system.

c) Calculate R_1 and R_2 with Ohm's law.

d) Check that the parallel resistance $R_1 \mathbin{/\!/} R_2$ agrees with $\frac{U}{I}$.

Exercise 3.8 KVL in a series circuit

ANSWER P. 250

A series circuit with two resistors is supplied with $E = 15\,\text{V}$. The voltage drops satisfy KVL, and U_1 is 3 V greater than U_2.

a) Set up the system of equations.

b) Solve the system.

c) The current is $I = 0.3\,\text{A}$. Calculate R_1 and R_2.

d) Check that $R_1 + R_2 = \frac{E}{I}$.

Exercise 3.9 Operating point: source and load

ANSWER P. 250

A voltage source with internal resistance gives the terminal voltage $U = 12 - 2I$, where I is the current in amperes. The source is loaded with a $4\,\Omega$ resistor, which by Ohm's law gives $U = 4I$.

a) Set up the system of equations for U and I.

b) Solve the system.

c) Check the solution in both equations.

d) What is the point called where the characteristics of the source and the load intersect?

Exercise 3.10 Designing a voltage divider

ANSWER P. 250

Two resistors are connected in series, and the total resistance is $R_1 + R_2 = 900\,\Omega$. When the pair is supplied with $U = 9\,\text{V}$, the voltage across R_1 is $6\,\text{V}$.

a) Use the voltage division $\dfrac{U_1}{U} = \dfrac{R_1}{R_1 + R_2}$ to get a second equation.

b) Solve the system and give R_1 and R_2.

c) Check your answer with the voltage divider formula.

Exercise 3.11 A system with three unknowns

ANSWER P. 250

In a circuit, the following relationships hold between three branch currents (in amperes):

$$\begin{cases} I_1 + I_2 + I_3 = 9 \\ I_1 - I_2 = 1 \\ I_3 = 4 \end{cases}$$

a) Solve the system.

b) Check the solution in all three equations.

c) Why is the third equation especially convenient to start with?

Exercise 3.12 Setting up the system from a text

ANSWER P. 250

A circuit board is built with two types of resistor, A and B. Five resistors of type A together with three of type B give $4\,390\,\Omega$ in series. Two of type A together with four of type B give $3\,660\,\Omega$.

a) Introduce the notation R_A and R_B and set up the system of equations.

b) Solve the system.

c) Check your answer in both equations.

Exercise 3.13 Checking a proposed solution

ANSWER P. 250

Decide, **without** solving the system, whether the given pair of numbers is a solution.

a) $(x, y) = (2, 3)$ for $x + y = 5$ and $2x - y = 1$.

b) $(x, y) = (1, 4)$ for $3x + y = 7$ and $x - y = -3$.

c) $(x, y) = (5, 1)$ for $x - 2y = 3$ and $x + y = 7$.

d) Why is it not enough to check in only one of the equations?

Exercise 3.14 Mixed systems

ANSWER P. 250

Solve by the method you think suits best.

a) $\begin{cases} 4x - y = 5 \\ 2x + 3y = 13 \end{cases}$ **b)** $\begin{cases} x + 3y = 1 \\ 2x - y = 9 \end{cases}$ **c)** $\begin{cases} 6x + 2y = 10 \\ 3x - y = 7 \end{cases}$

d) Briefly justify why you chose each method.

CHAPTER 5

Powers, roots and number systems

Repeated multiplication, roots, powers of ten, how many digits an answer should have, and numbers in base two and sixteen.

In this chapter

- Powers and the laws of exponents.
- Square roots and nth roots.
- Scientific notation (standard form) and SI prefixes.
- Significant figures and rounding.
- Calculating electrical power with powers and powers of ten.
- Binary and hexadecimal numbers.

In electrical engineering, most values are either very large or very small: a resistance of a few megohms and a capacitance of a few picofarads can sit in the same circuit. This chapter goes through powers and the rules that apply to them, square roots and nth roots, and how roots can be written as powers with fractional exponents. Numbers are then written in scientific notation and with SI prefixes, and you learn to calculate with powers of ten and to give a result with a sensible number of significant figures. Next comes calculating electrical power, where both squares and powers of ten are needed. The chapter ends with binary and hexadecimal numbers: the same positional number system we use every day, but with base two or sixteen instead of ten.

§5.1 to §5.8 are the chapter's theory, with examples. §5.9 summarises the rules and what you should be able to do after the chapter, and §5.10 holds the exercises. Most of them can be done in your head or with pencil and paper.

5.1 Powers

A **power** is a number multiplied by itself repeatedly:

$$a^n = \underbrace{a \times a \times \cdots \times a}_{n \text{ factors}}$$

Here a is called the **base** and n the **exponent**.

EXAMPLE

$$2^4 = 2 \times 2 \times 2 \times 2 = 16, \qquad 10^3 = 1\,000$$

5.2 Laws of exponents

The following seven rules hold for all $a \neq 0$ and $b \neq 0$.

No.	Rule	Formula
1	Product rule	$a^m \cdot a^n = a^{m+n}$
2	Quotient rule	$\dfrac{a^m}{a^n} = a^{m-n}$
3	Power of a power	$(a^m)^n = a^{m \cdot n}$
4	Power of a product	$(a \cdot b)^n = a^n \cdot b^n$
5	Power of a quotient	$\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$
6	Zero exponent	$a^0 = 1$
7	Negative exponent	$a^{-n} = \dfrac{1}{a^n}$

EXAMPLE **Rule 1.**

$$10^3 \cdot 10^4 = 10^{3+4} = 10^7$$

EXAMPLE **Rule 7.**

$$10^{-3} = \frac{1}{10^3} = \frac{1}{1000} = 0.001$$

5.3 Roots

The **square root** of a $(a \geq 0)$ is the non-negative number whose square is a:

$$\sqrt{a} = a^{1/2} \quad \text{since} \quad \left(\sqrt{a}\right)^2 = a$$

The n**th root** of a is the number whose nth power is a:

$$\sqrt[n]{a} = a^{1/n}$$

5.3.1 Fractional exponents

Roots and powers are combined with fractional exponents:

$$a^{m/n} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m$$

EXAMPLE

$$8^{2/3} = \left(\sqrt[3]{8}\right)^2 = 2^2 = 4$$

5.3.2 Rules for roots

$$\sqrt{a \cdot b} = \sqrt{a} \cdot \sqrt{b}, \qquad \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

Note: $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$. This is a common mistake!

5.4 Scientific notation (standard form)

Electrical engineering involves extremely large and extremely small numbers, for example $R = 1\,000\,000\,\Omega$ and $C = 0.000\,000\,001$ F.

Scientific notation writes the number as $a \times 10^n$, where $1 \leq a < 10$:

$$1\,000\,000 = 1.0 \times 10^6, \qquad 0.000\,000\,001 = 1.0 \times 10^{-9}$$

5.4.1 SI prefixes

Prefix	Symbol	Factor
Giga	G	10^9
Mega	M	10^6
Kilo	k	10^3
Milli	m	10^{-3}
Micro	μ	10^{-6}
Nano	n	10^{-9}
Pico	p	10^{-12}

EXAMPLE $4.7\,\text{k}\Omega = 4.7 \times 10^3\,\Omega = 4\,700\,\Omega$

EXAMPLE $33\,\text{nF} = 33 \times 10^{-9}\,\text{F} = 3.3 \times 10^{-8}\,\text{F}$

5.5 Calculating in scientific notation

Multiplication:

$$(3 \times 10^4) \times (2 \times 10^3) = 6 \times 10^7$$

Division:

$$\frac{6 \times 10^6}{2 \times 10^2} = 3 \times 10^4$$

Addition (the terms must have the same exponent):

$$3.5 \times 10^3 + 2.0 \times 10^3 = 5.5 \times 10^3$$

5.6 Significant figures and rounding

Significant figures are the digits that carry meaningful information in a measured value.

- $4\,700\,\Omega$ can have 2, 3 or 4 significant figures; there is no telling without context.
- $4.70 \times 10^3\,\Omega$ clearly has **3 significant figures**.

The result of a calculation should normally be given with as many significant figures as the least precise input.

Rounding rule: If the digit after the rounding position is 5 or more, round up; otherwise round down.

5.7 Application: electrical power

The power dissipated in a resistor R at the voltage U is

$$P = \frac{U^2}{R}.$$

EXAMPLE With $U = 6\,\text{V}$ and $R = 50\,\Omega$,

$$P = \frac{6^2}{50} = \frac{36}{50} = 0.72\,\text{W}.$$

At a given current I, the power is

$$P = RI^2.$$

EXAMPLE With $I = 20\,\text{mA} = 20 \times 10^{-3}\,\text{A}$ and $R = 1\,\text{k}\Omega = 10^3\,\Omega$,

$$P = 10^3 \times (20 \times 10^{-3})^2 = 10^3 \times 400 \times 10^{-6} = 0.4\,\text{W}.$$

5.8 Binary and hexadecimal numbers

Our everyday number system is a **positional number system** with base 10: the value of a digit depends on where in the number it stands, and each position corresponds to a power of ten. The rightmost position has the value $10^0 = 1$:

$$4\,703 = 4 \cdot 10^3 + 7 \cdot 10^2 + 0 \cdot 10^1 + 3 \cdot 10^0$$

The same idea works with any base. In digital electronics and programming, the **binary** and **hexadecimal** number systems are used as well as the decimal one. The table shows the number 45 in all three:

Number system	Base	Digits	Written	In C
Decimal	10	0–9	45	`45`
Binary	2	0 and 1	101101_2	`0b101101`
Hexadecimal	16	0–9 and A–F	$2D_{16}$	`0x2D`

The hexadecimal digits A–F stand for 10–15. The small number after a number gives its base; without one, the number is decimal.

A binary digit is called a **bit**, and eight bits are called a **byte**. With n bits, 2^n different values can be stored, from 0 to $2^n - 1$. A byte can therefore hold $2^8 = 256$ different values, 0–255.

5.8.1 From binary or hexadecimal to decimal

Multiply each digit by the value of its position and add. The positions are numbered from the right, starting at 0, so position n has the value 2^n in a binary number and 16^n in a hexadecimal number:

Position n	7	6	5	4	3	2	1	0
2^n	128	64	32	16	8	4	2	1

For hexadecimal numbers, $16^0 = 1$, $16^1 = 16$, $16^2 = 256$ and $16^3 = 4\,096$ are usually enough.

EXAMPLE

$$1011_2 = 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 8 + 0 + 2 + 1 = 11$$

EXAMPLE F stands for 15:

$$2F_{16} = 2 \cdot 16^1 + 15 \cdot 16^0 = 32 + 15 = 47$$

5.8.2 From decimal to binary or hexadecimal

Divide the number by the base repeatedly and note the remainder each time, until the quotient is 0. The remainders, read **from bottom to top**, are the number in the new base.

EXAMPLE **From decimal to binary.** Convert 13 to binary.

Division	Quotient	Remainder
13/2	6	1
6/2	3	0
3/2	1	1
1/2	0	1

The remainders from bottom to top give $13 = 1101_2$. Check: $8 + 4 + 0 + 1 = 13$ ✓

EXAMPLE **From decimal to hexadecimal.** Convert 470 to hexadecimal. A remainder between 10 and 15 is written as a letter.

Division	Quotient	Remainder
470/16	29	6
29/16	1	13 = D
1/16	0	1

The remainders from bottom to top give $470 = 1D6_{16}$. Check: $1 \cdot 256 + 13 \cdot 16 + 6 = 470$ ✓

Tip: A calculator gives the quotient as a decimal number. $470/16 = 29.375$ means the quotient 29 and the remainder $0.375 \cdot 16 = 6$.

5.8.3 Between binary and hexadecimal

Since $16 = 2^4$, each hexadecimal digit corresponds to exactly four bits:

Hex	Binary	Hex	Binary	Hex	Binary	Hex	Binary
0	0000	4	0100	8	1000	C	1100
1	0001	5	0101	9	1001	D	1101
2	0010	6	0110	A	1010	E	1110
3	0011	7	0111	B	1011	F	1111

From binary to hexadecimal: split the bits into groups of four, **from the right**, pad with zeros on the left if needed, and replace each group with its hexadecimal digit:

$$1011\,0110_2 = \underbrace{1011}_{\mathrm{B}}\ \underbrace{0110}_{6} = \mathrm{B6}_{16}$$

From hexadecimal to binary: replace each digit with its four bits:

$$4\mathrm{D}_{16} = \underbrace{0100}_{4}\ \underbrace{1101}_{\mathrm{D}} = 0100\,1101_2$$

A byte therefore corresponds to exactly two hexadecimal digits, from `0x00` to `0xFF`. This is why, for example, the contents of a microcontroller's registers are often written in hexadecimal: `0xB6` is easier to read than `0b10110110`.

> **Note:** In computing, kilo is sometimes used for $2^{10} = 1\,024$ instead of 1 000. The unambiguous name is **kibi** (Ki): 1 KiB = 1 024 bytes.

5.9 Summary

Concept	Key formula
Product rule	$a^m \cdot a^n = a^{m+n}$
Quotient rule	$a^m / a^n = a^{m-n}$
Power of a power	$(a^m)^n = a^{mn}$
Negative exponent	$a^{-n} = 1/a^n$
Square root	$\sqrt{a} = a^{1/2}$
Fractional exponent	$a^{m/n} = \sqrt[n]{a^m}$
Scientific notation	$a \times 10^n$, $1 \le a < 10$
Electrical power	$P = U^2/R = RI^2$
Positional number system	$1011_2 = 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 11$
Binary and hexadecimal	One hexadecimal digit corresponds to four bits
Number of values with n bits	2^n, from 0 to $2^n - 1$

What you should be able to do

After this chapter, you should be able to:

- Know and apply the laws of exponents.
- Calculate and simplify expressions with roots.
- Write numbers in scientific notation and handle significant figures.
- Convert numbers between decimal, binary and hexadecimal form.

Test yourself

ANSWER P. 251

1. Simplify $\dfrac{(3x^2)^3}{9x^4}$ with the laws of exponents.
2. A capacitance is $C = 220\,\text{nF}$. Give the value in farads in scientific notation.
3. Convert $1101\,0010_2$ to hexadecimal and decimal form.

Next chapter

Chapter 6 is about quadratic equations: the PQ-formula and completing the square.

5.10 Exercises

How to work with the exercises. Parts 1 and 2 are done during the lesson: first on your own, a few minutes per exercise, then together. Part 3 is extra exercises for further practice after the lesson. Most of them can be done in your head or with pencil and paper.
Answers and worked solutions. The answer to every exercise is in the answer key at the back of the book, and its page is given next to the exercise. An answer shows whether you got it right, but not how. Full worked solutions, in Swedish, are published in the course repository after the lesson, in `lectures/L05/appendix/c_solutions.md`. Always try it yourself first.

PART 1

Review exercises

Exercise 1.1 System of equations: voltage and current

ANSWER P. 251

Solve the following system of equations (currents in amperes):

$$\begin{cases} 2I_1 + 3I_2 = 13 \\ 4I_1 - I_2 = 5 \end{cases}$$

Exercise 1.2 Circuit analysis with KVL and KCL

ANSWER P. 251

A circuit has three resistors in a network. KVL and KCL give the following system for U_A and U_B (node voltages in volts):

$$\begin{cases} 3U_A - U_B = 12 \\ -U_A + 4U_B = 6 \end{cases}$$

Solve the system and give U_A and U_B.

PART 2

New material

Exercise 2.1 Laws of exponents

ANSWER P. 251

Simplify with the laws of exponents. Leave the answer with positive exponents.

a) $x^3 \cdot x^5$ **b)** $\dfrac{R^6}{R^2}$ **c)** $(I^2)^3$ **d)** $U^{-2} \cdot U^5$ **e)** $\left(\dfrac{2}{R}\right)^3$

Exercise 2.2 Scientific notation for component values

ANSWER P. 251

Write the following values in scientific notation, $a \times 10^n$ with $1 \leq a < 10$.

a) 47 000 Ω **b)** 0.000 022 F **c)** 3 300 000 Hz **d)** 0.000 000 015 A

Exercise 2.3 Calculating power

ANSWER P. 251

Calculate the power P dissipated in a resistor using $P = \dfrac{U^2}{R}$ and $P = RI^2$.

a) $U = 5\,\text{V}$ and $R = 100\,\Omega$. Calculate P in mW.

b) $I = 40\,\text{mA}$ and $R = 50\,\Omega$. Calculate P in mW.

Exercise 2.4 From binary and hexadecimal to decimal

ANSWER P. 251

Convert to decimal form.

a) 1010_2 **b)** 11001_2 **c)** $3A_{16}$ **d)** `0xC8`

Exercise 2.5 From decimal to binary and hexadecimal

ANSWER P. 251

a) Convert 22 to binary.

b) Convert 100 to binary.

c) Write the binary number from b) in hexadecimal by splitting the bits into groups of four.

d) Write `0x5A` in binary.

PART 3

Extra exercises

The exercises below are extra practice, best done on your own after the lesson.

Exercise 3.1 Evaluating powers

ANSWER P. 251

Calculate **without** a calculator.

a) 3^4 **b)** $(-2)^5$ **c)** 10^{-2} **d)** 5^0 **e)** 2^{-3} **f)** 0.1^3

Exercise 3.2 Laws of exponents

ANSWER P. 251

Simplify. The answer should have positive exponents.

a) $a^7 \cdot a^{-3}$ **b)** $\dfrac{U^5}{U^8}$ **c)** $(2R^3)^2$ **d)** $\dfrac{(I^2)^4}{I^3}$ **e)** $\left(\dfrac{R^2}{3}\right)^3$ **f)** $\dfrac{6x^5}{2x^2}$

Exercise 3.3 Powers of ten

ANSWER P. 251

Write as a single power of ten.

a) $10^4 \cdot 10^{-7}$ **b)** $\dfrac{10^{-2}}{10^{-5}}$ **c)** $\left(10^3\right)^{-2}$ **d)** $\dfrac{10^6 \cdot 10^{-2}}{10^{-3}}$ **e)** $\sqrt{10^{-6}}$

Exercise 3.4 Roots

ANSWER P. 251

a) Calculate $\sqrt{81}$.

b) Calculate $\sqrt{0.25}$.

c) Calculate $\sqrt[3]{27}$.

d) Calculate $\sqrt{16 \times 10^{-4}}$.

e) Simplify $\sqrt{50}$ to exact form.

f) Show with $a = 9$ and $b = 16$ that $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$.

Exercise 3.5 Fractional exponents

ANSWER P. 251

Calculate.

a) $16^{1/2}$ **b)** $27^{1/3}$ **c)** $9^{3/2}$ **d)** $8^{-1/3}$ **e)** $\left(\frac{1}{4}\right)^{-1/2}$

Exercise 3.6 From prefix to base unit

ANSWER P. 251

Rewrite in the base unit, in scientific notation.

a) 2.2 kΩ **b)** 470 μF **c)** 15 pF **d)** 250 mA **e)** 1.8 MHz **f)** 6.8 nH

Exercise 3.7 From scientific notation to prefix

ANSWER P. 251

Give the value with the prefix that gives a number between 1 and 1000.

a) 3.3×10^{-6} F **b)** $4.7 \times 10^{5}\,\Omega$ **c)** 2.5×10^{-4} A

d) 8×10^{7} Hz **e)** 1.2×10^{-11} F

Exercise 3.8 Calculating in scientific notation

ANSWER P. 251

Calculate and give the answer in scientific notation.

a) $\left(4 \times 10^{5}\right)\left(2 \times 10^{-8}\right)$ **b)** $\dfrac{9 \times 10^{-3}}{3 \times 10^{2}}$ **c)** $\left(5 \times 10^{3}\right)^{2}$

d) $2.5 \times 10^{-3} + 7.5 \times 10^{-4}$ **e)** $\dfrac{\left(6 \times 10^{-2}\right)^{2}}{4 \times 10^{-5}}$

Exercise 3.9 Significant figures and rounding

ANSWER P. 251

a) How many significant figures does 4.70×10^{3} have?

b) Round 0.024 68 to two significant figures.

c) Round 12 345 Ω to three significant figures and write the answer in scientific notation.

d) A voltage is measured as $U = 5.0$ V and a resistance as $R = 100\,\Omega$ (three significant figures). With how many significant figures should the current $I = \dfrac{U}{R}$ be given? Calculate the current.

Exercise 3.10 Ohm's law with powers of ten

ANSWER P. 251

Calculate and give the answer with a suitable prefix.

a) $U = 3.3\,\text{V}$ and $R = 220\,\Omega$. Calculate I.

b) $I = 250\,\mu\text{A}$ and $R = 47\,\text{k}\Omega$. Calculate U.

c) $U = 12\,\text{V}$ and $I = 4\,\text{mA}$. Calculate R.

Exercise 3.11 Power with powers of ten

ANSWER P. 251

a) $R = 1.5\,\text{k}\Omega$ and $I = 20\,\text{mA}$. Calculate $P = RI^2$.

b) $U = 5\,\text{V}$ and $R = 2.2\,\text{k}\Omega$. Calculate $P = \dfrac{U^2}{R}$ in mW.

c) A resistor is rated $470\,\Omega$ and $0.25\,\text{W}$. Calculate the largest permitted current.

d) Calculate the largest permitted voltage across the resistor in c).

Exercise 3.12 Time constant with prefixes

ANSWER P. 251

The time constant of an RC circuit is $\tau = RC$.

a) $R = 10\,\text{k}\Omega$ and $C = 4.7\,\mu\text{F}$. Calculate τ in ms.

b) $\tau = 1\,\text{ms}$ and $C = 100\,\text{nF}$. Calculate R.

c) $R = 2.2\,\text{M}\Omega$ and $\tau = 1.1\,\text{s}$. Calculate C and give the answer in nF.

Exercise 3.13 Peak value, RMS value and power

ANSWER P. 251

For a sine voltage, $U_{\text{RMS}} = \dfrac{|U|}{\sqrt{2}}$.

a) Calculate U_{RMS} when the amplitude is $|U| = 10\,\text{V}$.

b) Calculate the amplitude $|U|$ when $U_{\text{RMS}} = 230\,\text{V}$.

c) The power in a resistor is calculated from the RMS value, $P = \dfrac{U_{\text{RMS}}^2}{R}$. Show that this can be written $P = \dfrac{|U|^2}{2R}$.

d) Calculate P when $|U| = 12\,\text{V}$ and $R = 8\,\Omega$.

Exercise 3.14 Find the error

ANSWER P. 251

Each line contains a common mistake with powers. Explain the mistake and write the correct answer.

a) $2^3 \cdot 2^4 = 2^{12}$ **b)** $(a^2)^3 = a^5$ **c)** $\frac{10^6}{10^2} = 10^3$

d) $(2a)^3 = 2a^3$ **e)** $\sqrt{9+16} = 3+4$ **f)** $a^{-2} = -a^2$

Exercise 3.15 Binary numbers

ANSWER P. 251

Convert the binary numbers in a)–c) to decimal and the decimal numbers in d)–f) to binary.

a) 111_2 **b)** 10000_2 **c)** 101010_2

d) 37 **e)** 64 **f)** 255

Exercise 3.16 Hexadecimal numbers

ANSWER P. 251

Convert the numbers in a)–c) to decimal and the numbers in d)–f) to hexadecimal.

a) $1F_{16}$ **b)** `0x80` **c)** ABC_{16}

d) 175 **e)** 4 096 **f)** 1 000

Exercise 3.17 Between binary and hexadecimal

ANSWER P. 251

Write the binary numbers in a)–c) in hexadecimal and the hexadecimal numbers in d)–f) in binary.

a) $1001\,1110_2$ **b)** $1111\,0000_2$ **c)** 101011_2

d) `0x3C` **e)** `0xA5` **f)** `0x7F`

Exercise 3.18 Bits, bytes and registers

ANSWER P. 251

The bits in a binary number are numbered from the right, starting at 0.

a) How many different values can be stored with 4 bits? What is the largest value?

b) An ADC (analogue-to-digital converter) with a resolution of 10 bits converts a voltage to an integer. How many different values can it give, and what is the largest?

c) An 8-bit register has the value `0x24`. Which bits are ones?

d) In an 8-bit register, bit 0 and bit 7 are to be ones and the other bits zeros. Give the value of the register in binary, hexadecimal and decimal.

CHAPTER 6

Quadratic equations

When the unknown is squared: two roots, one or none.

IN THIS CHAPTER

- Quadratic equations and their solutions.
- The factorisation method and the PQ-formula.
- Completing the square.
- Current from power, and resistances from their sum and product.
- The ABC-formula, as optional extra material.
- Practice test 1, a review of Chapters 1–6.

So far the unknown has appeared to the first power, and the equations could be solved by isolating the unknown step by step. In a quadratic equation the unknown is squared, and that is no longer enough. This chapter shows how such an equation is solved: by factorisation, with the PQ-formula and by completing the square, which is how the formula is derived. The expression under the root sign in the PQ-formula decides whether the equation has two, one or no real roots. In electrical engineering, quadratic equations arise when, for example, the current is to be calculated from the power $P = RI^2$, or two resistors are to be found from their sum and product. The chapter's theory closes with the ABC-formula, which is optional extra material.

§6.1 to §6.4 are the chapter's theory, with examples. §6.5 summarises the methods and what you should be able to do after the chapter, and §6.6 holds the exercises. The chapter closes with practice test 1 in §6.7, which reviews Chapters 1–6 before the first test.

6.1 Definition

A **quadratic equation** (an equation of the second degree) is an equation of the form

$$ax^2 + bx + c = 0, \quad a \neq 0.$$

The coefficients a, b and c are real numbers. The equation can have **0, 1 or 2 real roots**.

EXAMPLE $2x^2 - 3x - 5 = 0$ is a quadratic equation with $a = 2$, $b = -3$ and $c = -5$.

6.2 Methods of solution

6.2.1 The factorisation method

If $ax^2 + bx + c$ can be factorised as $a(x - r_1)(x - r_2)$, the roots are $x = r_1$ and $x = r_2$.

EXAMPLE Solve $x^2 - 5x + 6 = 0$. Factorise:

$$(x - 2)(x - 3) = 0$$

The roots are given by $x - 2 = 0$ or $x - 3 = 0$:

$$x_1 = 2, \quad x_2 = 3$$

EXAMPLE **Difference of two squares.** Solve $x^2 - 9 = 0$.

$$(x + 3)(x - 3) = 0 \quad \Rightarrow \quad x = -3 \text{ or } x = 3$$

6.2.2 The PQ-formula

The PQ-formula, the name Swedish schools give the quadratic formula in this form, applies to equations in *normalised* form $x^2 + px + q = 0$, that is, with $a = 1$:

$$x = -\frac{p}{2} \pm \sqrt{\left(\frac{p}{2}\right)^2 - q}$$

EXAMPLE Solve $x^2 - 4x + 3 = 0$. Here $p = -4$ and $q = 3$:

$$x = \frac{4}{2} \pm \sqrt{\left(\frac{4}{2}\right)^2 - 3} = 2 \pm \sqrt{4 - 3} = 2 \pm 1$$
$$x_1 = 3, \quad x_2 = 1$$

The expression under the root sign is called the **discriminant** and decides the number of real roots: if it is positive there are two roots, if it is zero they coincide in a double root, and if it is negative there are no real roots.

The PQ-formula can also be used by moving everything except x^2 to the right-hand side, so that the equation has the form $x^2 = px + q$. Then p and q go into the formula with the signs they have on the right-hand side:

$$x^2 = px + q \quad \Rightarrow \quad x = \frac{p}{2} \pm \sqrt{\left(\frac{p}{2}\right)^2 + q}$$

EXAMPLE **Alternative form.** Solve $x^2 + 2x + 2 = 26$. Move everything except x^2 to the right-hand side:

$$x^2 = -2x + 24$$

Here $p = -2$ and $q = 24$:

$$x = \frac{-2}{2} \pm \sqrt{\left(\frac{-2}{2}\right)^2 + 24} = -1 \pm \sqrt{1 + 24} = -1 \pm 5$$

$$x_1 = 4, \quad x_2 = -6$$

The two forms give the same calculation; p and q simply have opposite signs. Choose the form you prefer and stick to it.

If $a \neq 1$, first divide every term by a, so that the equation is normalised.

EXAMPLE Solve $2x^2 + 5x - 3 = 0$. Divide by 2:

$$x^2 + 2.5x - 1.5 = 0$$

Here $p = 2.5$ and $q = -1.5$:

$$x = -1.25 \pm \sqrt{1.25^2 + 1.5} = -1.25 \pm \sqrt{3.0625} = -1.25 \pm 1.75$$

$$x_1 = 0.5, \quad x_2 = -3$$

6.2.3 Completing the square

Every quadratic equation can be written in the form $(x + k)^2 = m$ by 'completing the square'. For $x^2 + bx + c = 0$ it goes like this:

1. Move the constant: $x^2 + bx = -c$.
2. Add $\left(\frac{b}{2}\right)^2$ to both sides.
3. The left-hand side becomes a perfect square.

EXAMPLE Solve $x^2 + 6x + 5 = 0$.

$$x^2 + 6x = -5$$

Add $\left(\frac{6}{2}\right)^2 = 9$:

$$x^2 + 6x + 9 = -5 + 9 = 4$$

$$(x + 3)^2 = 4 \quad \Rightarrow \quad x + 3 = \pm 2 \quad \Rightarrow \quad x_1 = -1, \quad x_2 = -5$$

6.3 Applications in electrical engineering

6.3.1 Current from power and resistance

The power in a resistor R is given by $P = RI^2$. Given P and R, the current I can be calculated:

$$RI^2 = P \quad \Rightarrow \quad I^2 = \frac{P}{R} \quad \Rightarrow \quad I = \sqrt{\frac{P}{R}}$$

This is a *pure* quadratic equation, with no linear term.

EXAMPLE With $P = 8\,\text{W}$ and $R = 50\,\Omega$,

$$I = \sqrt{\frac{8}{50}} = \sqrt{0.16} = 0.4\,\text{A}.$$

6.3.2 Resistance from a series connection

Two resistors R_1 and R_2 are connected in series with $R_1 + R_2 = 10\,\Omega$ and $R_1 \cdot R_2 = 24\,\Omega^2$. Solve the first equation for $R_2 = 10 - R_1$ and substitute it into the second:

$$R_1(10 - R_1) = 24 \quad \Rightarrow \quad R_1^2 - 10R_1 + 24 = 0$$

The PQ-formula gives

$$R_1 = 5 \pm \sqrt{25 - 24} = 5 \pm 1.$$

The root $R_1 = 6\,\Omega$ gives $R_2 = 10 - 6 = 4\,\Omega$. The other root, $R_1 = 4\,\Omega$, gives the same pair in the opposite order, that is,

$$R_1 = 6\,\Omega, \quad R_2 = 4\,\Omega.$$

6.4 Extra: the ABC-formula

The ABC-formula (the quadratic formula) is an alternative to the PQ-formula, included as optional extra material. It gives the same roots, but can be used directly on $ax^2 + bx + c = 0$ without first normalising the equation:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

EXAMPLE Solve $2x^2 + 5x - 3 = 0$. Here $a = 2$, $b = 5$ and $c = -3$:

$$x = \frac{-5 \pm \sqrt{5^2 - 4 \cdot 2 \cdot (-3)}}{2 \cdot 2} = \frac{-5 \pm \sqrt{49}}{4} = \frac{-5 \pm 7}{4} \quad \Rightarrow \quad x_1 = 0.5, \quad x_2 = -3$$

The roots are the same as the PQ-formula gave for the same equation in §6.2.2.

6.5 Summary

Method	Best when ...
Factorisation	Integer roots can be spotted quickly
PQ-formula	General use; normalise first if $a \neq 1$
Completing the square	Deriving formulas and understanding the process

$$\boxed{x = -\frac{p}{2} \pm \sqrt{\left(\frac{p}{2}\right)^2 - q}}$$

What you should be able to do

After this chapter, you should be able to:

- Understand when an equation is of the second degree.
- Solve quadratic equations with the PQ-formula.
- Solve quadratic equations by completing the square.

Test yourself

ANSWER P. 252

1. Solve $2x^2 - 8x = 0$ by the factorisation method.
2. The current I satisfies $8I^2 = 2$. Calculate I and justify your choice of root.

Next chapter

Chapter 7 is about vectors: concepts, notation and calculations.

6.6 Exercises

How to work with the exercises. Parts 1 and 2 are done during the lesson: first on your own, a few minutes per exercise, then together. Part 3 is extra exercises for further practice after the lesson. Most of them can be done in your head or with pencil and paper.
Answers and worked solutions. The answer to every exercise is in the answer key at the back of the book, and its page is given next to the exercise. An answer shows whether you got it right, but not how. Full worked solutions, in Swedish, are published in the course repository after the lesson, in `lectures/L06/appendix/c_solutions.md`. Always try it yourself first.

PART 1

Review exercises

Exercise 1.1 Laws of exponents

ANSWER P. 252

Simplify, with positive exponents in the answer.

a) $\dfrac{10^6 \cdot 10^{-3}}{10^2}$ **b)** $(2 \cdot 10^3)^2$ **c)** $\sqrt{4 \times 10^6}$

Exercise 1.2 Scientific notation in a circuit calculation

ANSWER P. 252

A circuit has $U = 3.3\,\text{V}$ and $R = 4.7\,\text{k}\Omega = 4.7 \times 10^3\,\Omega$.

a) Calculate the current $I = \dfrac{U}{R}$ and give the answer in mA in scientific notation.

b) Calculate the power $P = \dfrac{U^2}{R}$ and give the answer in mW.

Exercise 1.3 Square root

ANSWER P. 252

a) Calculate $\sqrt{144}$.

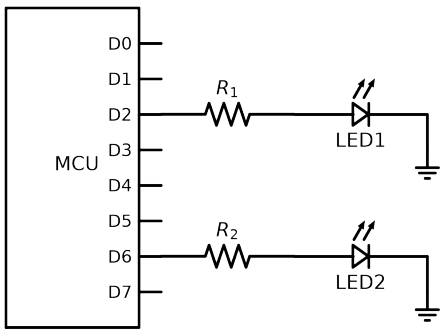
b) Calculate $\sqrt{0.0049}$.

c) A resistor dissipates $P = 2\,\text{W}$ at the voltage U, and $R = 200\,\Omega$. Calculate U using $P = \dfrac{U^2}{R}$.

Exercise 1.4 LEDs on a microcontroller

ANSWER P. 252

A microcontroller (MCU) has eight digital pins D0–D7. Two red LEDs, LED1 and LED2, are connected to D2 and D6 respectively, each through its own resistor to ground.



The pins are controlled with two 8-bit registers, in which each bit belongs to one pin: bit 0 (furthest to the right) belongs to D0, bit 1 to D1 and so on up to bit 7, which belongs to D7.

Register	Bit = 1	Bit = 0
`DDRD`	The pin is an output	The pin is an input
`PORTD`	The output is high (5.0 V)	The output is low (0 V)

Both registers have the value `0x00` at the start.

a) The LEDs' pins are to be configured as outputs. Which value must be written to `DDRD` to make the bits D2 and D6 equal to 1? Give the value in binary and in hexadecimal.

b) LED1 is to be lit. Which value must be written to `PORTD` to make the bit D2 equal to 1? Give the value in binary and in hexadecimal.

c) Now LED2 is to be lit as well, **without** turning LED1 off. In C this is done with `PORTD |= x`, which here works like PORTD = PORTD + x in mathematics. Which value must x have, and which value does `PORTD` have afterwards? Give the values in binary and in hexadecimal.

d) The supply voltage is $V_{CC} = 5.0\,\text{V}$ and a red LED has the forward voltage $U_F = 2.0\,\text{V}$, so the voltage across the resistor is $V_{CC} - U_F$ when the LED is lit. Dimension the resistors so that the current through each LED is about 10 mA. Choose a value from the E12 series:

10 12 15 18 22 27 33 39 47 56 68 82

The values can be multiplied by any power of ten, for example 47 Ω, 470 Ω and 4.7 kΩ.

e) Calculate the power consumption of each LED circuit (resistor and LED together) when the LED is lit, with the resistor value from d).

PART 2

New material

Exercise 2.1 A pure quadratic equation

ANSWER P. 252

The current I (in amperes) satisfies

$$50I^2 = 8.$$

a) Solve the equation for I.

b) Explain why the negative root should be rejected in this electrical context.

Exercise 2.2 Quadratic equations with the PQ-formula

ANSWER P. 252

Solve the following equations with the PQ-formula.

a) $x^2 - 5x + 6 = 0$ **b)** $x^2 + 3x - 10 = 0$ **c)** $x^2 - 6x + 9 = 0$

Exercise 2.3 Quadratic equations that are not normalised

ANSWER P. 252

First divide so that the coefficient of x^2 is 1, then solve with the PQ-formula.

a) $2x^2 - 7x + 3 = 0$ **b)** $3x^2 + 5x - 2 = 0$

Exercise 2.4 Resistors in series

ANSWER P. 252

Two resistors R_1 and R_2 in series satisfy

$$R_1 + R_2 = 13\,\Omega \qquad \text{and} \qquad R_1 \cdot R_2 = 36\,\Omega^2.$$

a) Solve the first equation for R_2 and substitute it into the second, to get a quadratic equation with R_1 as the unknown.

b) Solve the equation and give both resistances.

PART 3

Extra exercises

The exercises below are extra practice, best done on your own after the lesson.

Exercise 3.1 Pure quadratic equations

ANSWER P. 252

Solve the equations.

a) $x^2 = 49$ **b)** $3x^2 = 27$ **c)** $x^2 - 5 = 20$

d) $2x^2 + 8 = 0$ **e)** $\dfrac{x^2}{4} = 9$

Exercise 3.2 The factorisation method

ANSWER P. 252

Solve by factorising and using the zero-product rule.

a) $x^2 - 7x = 0$ **b)** $x^2 - 16 = 0$ **c)** $x^2 + 5x + 6 = 0$

d) $2x^2 - 8x = 0$ **e)** $x^2 - 4x + 4 = 0$

Exercise 3.3 The PQ-formula

ANSWER P. 252

Solve with the PQ-formula.

a) $x^2 + 2x - 8 = 0$ **b)** $x^2 - 8x + 12 = 0$

c) $x^2 + x - 12 = 0$ **d)** $x^2 - 2x - 15 = 0$

Exercise 3.4 Normalise first

ANSWER P. 252

The equations below are not normalised. First divide so that $a = 1$, then solve with the PQ-formula.

a) $2x^2 - 10x + 12 = 0$ **b)** $3x^2 + 12x - 15 = 0$ **c)** $-x^2 + 4x - 3 = 0$

Exercise 3.5 The number of real roots

ANSWER P. 252

Give the number of real roots **without** solving the equation completely, by examining the sign of the expression under the root sign in the PQ-formula. Normalise first where needed.

a) $x^2 - 6x + 5 = 0$ **b)** $x^2 + 4x + 4 = 0$ **c)** $x^2 + x + 3 = 0$

d) $2x^2 - 4x + 2 = 0$ **e)** $3x^2 + 2x - 1 = 0$

Exercise 3.6 Completing the square

ANSWER P. 252

Solve by completing the square.

a) $x^2 + 4x - 5 = 0$ **b)** $x^2 - 10x + 21 = 0$ **c)** $x^2 + 6x + 4 = 0$

Exercise 3.7 Current from power

ANSWER P. 252

The power in a resistor is given by $P = RI^2$.

a) $P = 12.5\,\text{W}$ and $R = 50\,\Omega$. Calculate I.

b) $P = 0.18\,\text{W}$ and $R = 200\,\Omega$. Calculate I in mA.

c) A resistor is marked $470\,\Omega$ and $0.5\,\text{W}$. Calculate the largest permitted current.

d) Why is the negative root rejected in every part?

Exercise 3.8 Voltage from power

ANSWER P. 252

The power can also be written $P = \dfrac{U^2}{R}$.

a) $P = 5\,\text{W}$ and $R = 20\,\Omega$. Calculate U.

b) $P = 0.125\,\text{W}$ and $R = 2\,000\,\Omega$. Calculate U.

c) A heating element is to give off $P = 1\,000\,\text{W}$ at $U = 230\,\text{V}$. What resistance is needed?

Exercise 3.9 Two resistors from their sum and product

ANSWER P. 252

a) Two resistors satisfy $R_1 + R_2 = 20\,\Omega$ and $R_1 R_2 = 96\,\Omega^2$. Find the resistances.

b) Two resistors satisfy $R_1 + R_2 = 15\,\Omega$ and $R_1 R_2 = 56\,\Omega^2$. Find the resistances.

c) Can two resistors have the sum $10\,\Omega$ and the product $30\,\Omega^2$? Justify your answer with the PQ-formula.

Exercise 3.10 Series and parallel resistance at once

ANSWER P. 252

Two resistors together give $10\,\Omega$ in series and $2.4\,\Omega$ in parallel.

a) Show that $R_1 R_2 = 24\,\Omega^2$.

b) Set up and solve the quadratic equation for the resistances.

c) Check both the series and the parallel resistance.

Exercise 3.11 Power in a load

ANSWER P. 252

An unknown resistor R is connected in series with $R_1 = 4\,\Omega$ and supplied with $E = 12\,\text{V}$. The power developed in R is to be $P = 8\,\text{W}$. The current in the circuit is $I = \dfrac{E}{R_1 + R}$ and the power in R is $P = RI^2$.

a) Show that the requirement leads to the equation $R^2 - 10R + 16 = 0$.

b) Solve the equation.

c) Check **both** solutions by calculating the current and the power.

d) Calculate the power in R when $R = 4\,\Omega$ and compare with the two solutions.

Exercise 3.12 True or false

ANSWER P. 252

Decide whether the statement is true or false and justify your answer briefly.

a) Every quadratic equation has two real roots.

b) The equation $x^2 = 16$ has the solution $x = 4$.

c) The PQ-formula can be used directly on $2x^2 + 4x - 6 = 0$.

d) If the expression under the root sign in the PQ-formula is zero, the two roots coincide.

e) If a product of two factors is zero, at least one of the factors must be zero.

f) The equation $x^2 + 4 = 0$ has no real solutions.

6.7 Practice test 1

The practice test covers Chapters 1–6 and prepares you for test 1, which is set out the same way and has the same rules. The answers are in the answer key, and full worked solutions, in Swedish, in `lectures/L06/appendix/practice_test/solutions.md`.

Permitted aids. Writing materials and any calculator, and one A4 sheet (both sides), handwritten or typed, with formulas, examples and notes of your own choice.

Other information. The test does not affect the grade or the examination in the course, but it can give 0.5 bonus points on the regular exam. The bonus requires at least 3.5 of the maximum 5.0 points. Every question requires a solution with a stated answer, including the unit where needed. Good luck!

> **Note:** Mobile phones may not be used while the test is in progress and must be left in the place indicated.

Question 1

ANSWER P. 253 1.0 POINTS

The resistance of two resistors R_1 and R_2 in parallel can be calculated with the formula

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}.$$

Calculate the resistance R_2 if $R_1 = 10\,\text{k}\Omega$ and the parallel resistance is $R = 4.5\,\text{k}\Omega$. Give the answer in $\text{k}\Omega$ to one significant figure.

> **Note:** Ω is the unit of resistance and is read 'ohm'.

Question 2

ANSWER P. 253 1.0 POINTS

Simplify the following expression as far as possible:

$$\frac{\dfrac{c^8 - c^4}{c^2}}{c(c^2 - 1)}$$

Question 3

ANSWER P. 253 1.0 POINTS

Solve the system of equations below. Give the answers exactly, as fractions.

$$\begin{cases} 4y - 2x + 6 = 0 \\ 3y = 5x + 4 \end{cases}$$

Question 4

ANSWER P. 253 0.5 POINTS

An 8-bit register in a microcontroller has the value `0x9C`. Write the value in binary and in decimal.

Question 5

ANSWER P. 253 0.5 POINTS

A resistor with the resistance $R = 4.7\,\mathrm{k\Omega}$ carries the current $I = 2.5\,\mathrm{mA}$. Calculate the power P using powers of ten:

$$P = RI^2$$

Give the answer in watts in scientific notation, to two significant figures.

Question 6

ANSWER P. 253 1.0 POINTS

Solve the following equation:

$$8(x+2)^2 - 5(x-3)^2 = 4 - 20x$$

Give the answer to two decimal places.

CHAPTER 7

Vectors

Quantities with both magnitude and direction, and how to calculate with them.

IN THIS CHAPTER

- Distances and angles in a coordinate system.
- Vectors: concepts and notation.
- Magnitude and angle, and the angle between two vectors.
- Calculations with vectors: addition, subtraction and multiplication by a number.

A number such as a resistance or a temperature has only a size. Many quantities also have a direction: a force pulls one particular way, and in AC circuits voltages and currents are described by a magnitude and a phase angle. Such quantities are described by vectors. This chapter begins with distances and angles in a coordinate system and then goes through vectors: how they are written with components, how their length and angle are calculated, and how they are added, subtracted and multiplied by a number. The angle between two vectors follows from the vectors' own angles. Vectors return later in the book, when voltages and currents are handled as phasors and complex numbers.

§7.1 to §7.5 are the chapter's theory, and §7.6 shows it in a worked example. §7.7 summarises the formulas and what you should be able to do after the chapter, and §7.8 holds the exercises.

7.1 Coordinate systems and distances

A **coordinate system** (the xy-plane) has a horizontal x-axis and a vertical y-axis that meet at the origin $(0, 0)$.

The distance d between two points (x_1, y_1) and (x_2, y_2) is given by Pythagoras' theorem:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

The angle that the line segment makes with the positive x-axis is

$$\theta = \arctan\left(\frac{y_2 - y_1}{x_2 - x_1}\right).$$

> **Note:** Always check which quadrant the point is in and correct the angle if needed: $+180^\circ$ for quadrants II and III.

7.2 Vectors – concepts and notation

A **vector** represents a quantity with both magnitude and direction. In a plane, the vector $\mathbf{u}$ is written with its components:

$$\mathbf{u} = (u_x, u_y)$$

Unlike a scalar, a vector carries information about direction. In electrical engineering, vectors are used to represent voltages and currents in AC systems.

7.3 Magnitude and angle

The **magnitude** (length) of $\mathbf{u} = (u_x, u_y)$ is

$$|\mathbf{u}| = \sqrt{u_x^2 + u_y^2}.$$

The **angle** v_u with the positive x-axis is

$$v_u = \arctan\left(\frac{u_y}{u_x}\right) + \text{quadrant correction},$$

where the quadrant correction is given by the table.

Quadrant	u_x	u_y	Correction
I	+	+	none
II	−	+	$+180^\circ$
III	−	−	$+180^\circ$
IV	+	−	none (negative angle)

7.4 The angle between two vectors

The angle α between $\mathbf{u}$ and $\mathbf{v}$ is the difference between the vectors' angles with the positive x-axis. So first calculate v_u and v_v separately, and then the difference:

$$\alpha = v_v - v_u$$

The angle between two vectors is given as a value between 0° and 180°. If the difference is negative, change its sign, and if it is greater than 180°, subtract it from 360°.

7.5 Calculating with vectors

7.5.1 Addition and subtraction

$$\mathbf{u} \pm \mathbf{v} = (u_x \pm v_x,\ u_y \pm v_y)$$

7.5.2 Multiplication by a number

When a vector is multiplied by a number k, both components are multiplied by the number:

$$k\mathbf{u} = (k \cdot u_x,\ k \cdot u_y)$$

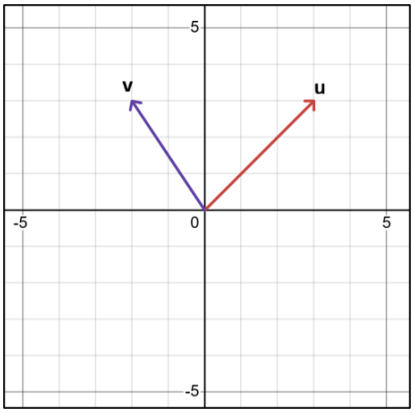
For example, for $\mathbf{U} = (3, 4)$,

$$2\mathbf{U} = (2 \cdot 3,\ 2 \cdot 4) = (6, 8).$$

A negative number also reverses the direction of the vector.

7.6 Worked example

EXAMPLE **Vectors in a coordinate system.** Two vectors **u** and **v** are shown in the figure below.



a) Find the coordinates of the vectors.

b) Calculate the magnitudes $|\mathbf{u}|$ and $|\mathbf{v}|$ of the vectors.

c) Calculate the angles v_u and v_v of the vectors.

d) Calculate the angle between the vectors **u** and **v**.

e) Calculate a third vector $\mathbf{w} = \mathbf{u} + 2\mathbf{v}$.

f) Determine a vector of length 5 that points in the opposite direction to **u**.

a) Looking at the figure, we see that the vector **u** has the x-coordinate 3 and the y-coordinate 3. So

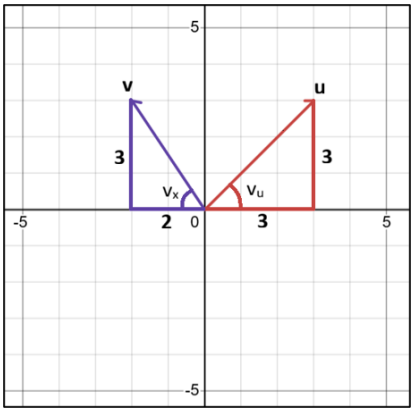
$$\mathbf{u} = (3, 3)$$

We also see that the vector **v** has the x-coordinate -2 and the y-coordinate 3. So

$$\mathbf{v} = (-2, 3)$$

b) The magnitude of each vector equals its hypotenuse and is therefore easily calculated with Pythagoras' theorem.

This can be visualised by drawing each vector as a triangle, see the figure below.



The magnitudes $|\mathbf{u}|$ and $|\mathbf{v}|$ are calculated as follows:

$$|\mathbf{u}| = \sqrt{3^2 + 3^2} = \sqrt{18} \approx 4.24$$
$$|\mathbf{v}| = \sqrt{3^2 + (-2)^2} = \sqrt{13} \approx 3.61$$

c) To calculate the angles of the vectors we can use the tangent. Once again we see the vectors as triangles.

For the vector $\mathbf{u} = (3, 3)$, the opposite and the adjacent side are both equal to 3. So

$$\tan(v_u) = \frac{3}{3} = 1,$$

which can be transformed into

$$v_u = \arctan(1) = 45^\circ$$

For the vector **v**, which has a negative x-value, the corresponding triangle can be drawn as shown in the figure above. Note that in this case we have marked an angle v_x, which is the supplementary angle of v_v:

$$v_v + v_x = 180^\circ,$$

which means that

$$v_v = 180° - v_x$$

If we calculate the angle v_x, we can therefore easily calculate the angle v_v. The angle v_x is easily calculated with the tangent, since the corresponding triangle has an opposite side equal to 3 and an adjacent side equal to 2. So

$$\tan(v_x) = \frac{3}{2} = 1.5 \quad \Rightarrow \quad v_x = \arctan(1.5) \approx 56.3°$$

We can then easily calculate the angle v_v:

$$v_v = 180° - v_x = 180° - 56.3° = 123.7°$$

d) To calculate the angle α between **v** and **u**, we calculate the difference between their respective angles v_v and v_u:

$$\alpha = v_v - v_u$$

We insert the values into the formula and see that the angle between the vectors is about 78.7°, since

$$\alpha = v_v - v_u = 123.7° - 45° \approx 78.7°$$

e) We calculate the vector **w** algebraically by inserting the values of the vectors $\mathbf{u} = (3, 3)$ and $\mathbf{v} = (-2, 3)$ into the given formula:

$$\mathbf{w} = \mathbf{u} + 2\mathbf{v} = (3, 3) + 2(-2, 3),$$

where

$$2(-2, 3) = (2 \cdot (-2),\ 2 \cdot 3) = (-4, 6)$$

So

$$\mathbf{w} = (3, 3) + (-4, 6)$$

To complete the calculation we add the x- and y-values and see that $\mathbf{w} = (-1, 9)$, since

$$\mathbf{w} = (3 - 4,\ 3 + 6) = (-1, 9)$$

f) We can draw a vector of length 5 in the opposite direction to **u**. Let us call this vector **a**. That the vector points the opposite way means that it is turned 180°, that is half a turn, from **u**. For its angle v_a we therefore have

$$v_a = v_u + 180°$$

Since the angle v_u was calculated earlier to be 45°, the angle v_a of the vector **a** is 225°, since

$$v_a = 45° + 180° = 225°$$

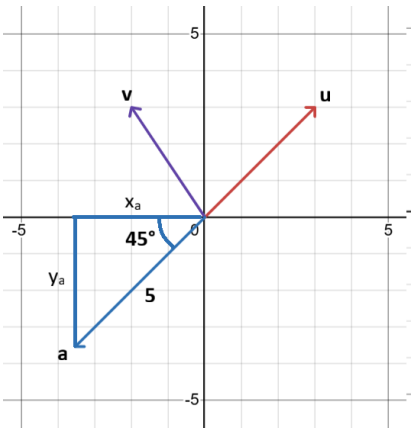
The length of a given vector equals its magnitude. So the magnitude $|\mathbf{a}|$ of the vector **a** is 5:

$$|\mathbf{a}| = 5$$

We can then calculate the vector **a** by two methods, graphically or algebraically.

Method 1 – Graphical solution. As we saw earlier, the vector can be drawn as a triangle whose hypotenuse equals its magnitude. Since this vector is turned half a turn from the vector **u**, we can draw the vector **a** as a triangle with the angle 45° seen from 180°, that is, turned half a turn.

With this information we can draw the vector **a** as shown below.



We can calculate the magnitude $|x_a|$ of the adjacent side x_a of the vector **a** with the cosine:

$$\cos 45° = \frac{|x_a|}{5},$$

which means that

$$|x_a| = 5 \cdot \cos 45° \approx 3.54$$

Since x_a lies to the left of the origin, x_a is negative. So

$$x_a \approx -3.54$$

We can also calculate the magnitude $|y_a|$ of the opposite side y_a of the vector **a** with the sine:

$$\sin 45° = \frac{|y_a|}{5},$$

which means that

$$|y_a| = 5 \cdot \sin 45° \approx 3.54$$

Since y_a lies below the origin, y_a is negative. So

$$y_a \approx -3.54$$

From the calculated values the vector **a** can then be determined:

$$\mathbf{a} \approx (-3.54,\ -3.54)$$

Method 2 – Algebraic solution. We saw earlier that the angle v_a of the vector **a** is 225°, since it is turned half a turn (180°) from the vector **u**, whose angle v_u was calculated earlier to be 45°:

$$v_a = v_u + 180^\circ = 45^\circ + 180^\circ = 225^\circ$$

The magnitude $|\mathbf{a}|$ of the vector, which equals its length, was also specified as 5 in the problem:

$$|\mathbf{a}| = 5$$

We can easily calculate the x- and y-values of the vector with the cosine and sine. We begin by calculating the x-value x_a of the vector:

$$\cos v_a = \frac{x_a}{|\mathbf{a}|},$$

which is equivalent to

$$x_a = |\mathbf{a}| \cdot \cos v_a$$

By inserting values into the formula above we see that the x-value x_a of the vector is -3.54, since

$$x_a = 5 \cdot \cos 225^\circ \approx -3.54$$

We can then calculate the y-value y_a of the vector **a** with the sine:

$$\sin v_a = \frac{y_a}{|\mathbf{a}|},$$

which is equivalent to

$$y_a = |\mathbf{a}| \cdot \sin v_a$$

By inserting values into the formula above we see that the y-value y_a of the vector **a** is also -3.54, since

$$y_a = 5 \cdot \sin 225^\circ \approx -3.54$$

So

$$\mathbf{a} \approx (-3.54,\ -3.54)$$

7.7 Summary

Concept	Formula
Magnitude	$\lvert\mathbf{u}\rvert = \sqrt{u_x^2 + u_y^2}$
Angle with the x-axis	$v = \arctan(u_y/u_x)$ + quadrant correction
Angle between vectors	$\alpha = v_v - v_u$
Addition	$\mathbf{u} + \mathbf{v} = (u_x + v_x, u_y + v_y)$
Multiplication by a number	$k\mathbf{u} = (ku_x, ku_y)$

What you should be able to do

After this chapter, you should be able to:

- Understand the concept of a vector and how a vector is represented mathematically.
- Carry out simple calculations with vectors.

Test yourself

ANSWER P. 254

1. A vector is $\mathbf{u} = (3, -4)$. Calculate the magnitude $\lvert\mathbf{u}\rvert$ and the angle with the x-axis.
2. Calculate $\mathbf{w} = 2\mathbf{u} - \mathbf{v}$ for $\mathbf{u} = (2, 1)$ and $\mathbf{v} = (-1, 3)$.

Next chapter

Chapter 8 is about functions.

7.8 Exercises

How to work with the exercises. Parts 1 and 2 are done during the lesson: first on your own, a few minutes per exercise, then together. Part 3 is extra exercises for further practice after the lesson. Most of them can be done in your head or with pencil and paper.
Answers and worked solutions. The answer to every exercise is in the answer key at the back of the book, and its page is given next to the exercise. An answer shows whether you got it right, but not how. Full worked solutions, in Swedish, are published in the course repository after the lesson, in `lectures/L07/appendix/c_solutions.md`. Always try it yourself first.

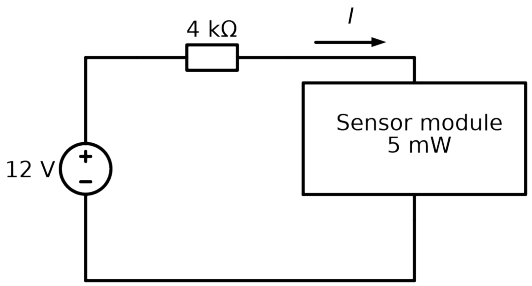
PART 1

Review exercises

Exercise 1.1 Sensor module on a railway wagon

ANSWER P. 253

A sensor module on a railway wagon is supplied from the wagon's 12 V system. To limit the current in the event of a short circuit, a protective resistor of 4 kΩ sits in series with the module. A measurement shows that the sensor module consumes 5 mW.



You want to find out how large the current I through the circuit is, but have no ammeter to hand, only your knowledge of circuit theory.

a) The voltage across the protective resistor is $4I$ (with I in mA, $\text{k}\Omega \cdot \text{mA} = \text{V}$). Show that the voltage across the sensor module is $12 - 4I$ and that the power relation leads to the equation

$$4I^2 - 12I + 5 = 0.$$

b) Solve the equation for the current I.

c) The equation gives two solutions. Calculate the voltage across the sensor module and the power loss in the protective resistor in both cases. Which operating point is the most reasonable for a sensor module, and why?

PART 2

New material

In AC circuits, voltages and currents are represented by vectors, known as **phasors**: the length of the vector gives the magnitude, and its angle with the positive x-axis gives the phase angle. Two phasors with the same direction are *in phase*, and two phasors with opposite directions are *in antiphase*.

Exercise 2.1 Adding voltages

ANSWER P. 253

Two AC voltages are represented by the phasors $\mathbf{U}_1 = (3, 2)$ V and $\mathbf{U}_2 = (4, 6)$ V. Find the coordinates of:

a) $\mathbf{U}_1 + \mathbf{U}_2$ **b)** $2\mathbf{U}_1 + 3\mathbf{U}_2$ **c)** $\mathbf{U}_1 - \mathbf{U}_2$ **d)** $-2\mathbf{U}_1 - 4\mathbf{U}_2$

Exercise 2.2 Calculating with three currents

ANSWER P. 253

Three alternating currents are represented by the phasors $\mathbf{I}_a = (1, 4)$ A, $\mathbf{I}_b = (-4, 2)$ A and $\mathbf{I}_c = (3, -4)$ A. Find the coordinates of:

a) $\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c$ **b)** $\mathbf{I}_a - \mathbf{I}_b - \mathbf{I}_c$ **c)** $2\mathbf{I}_a + 4\mathbf{I}_b$ **d)** $2\mathbf{I}_a - 4\mathbf{I}_b$

Exercise 2.3 Magnitudes of voltages

ANSWER P. 253

The voltages $\mathbf{U}_a = (2, 1)$ V and $\mathbf{U}_b = (1, -2)$ V are given. Find, in exact form, the magnitude of:

a) $\mathbf{U}_a$ **b)** $3\mathbf{U}_a + \mathbf{U}_b$ **c)** $\mathbf{U}_a - \mathbf{U}_b$ **d)** $3\mathbf{U}_a - 4\mathbf{U}_b$

Exercise 2.4 Solve the equations

ANSWER P. 253

Find x and y:

a) $(x, y) = 5(3, 1) + 2(4, -5)$ **b)** $(15, 12) = (x, 10) + (23, y)$

Exercise 2.5 Currents in phase and in antiphase

ANSWER P. 253

If $\mathbf{I}_1 = k \cdot \mathbf{I}_2$, the phasors $\mathbf{I}_1$ and $\mathbf{I}_2$ are parallel: in phase if $k > 0$ and in antiphase if $k < 0$. Which of the following currents are parallel to $\mathbf{I} = (2, -3)$ A?

a) $(6, -9)$ A **b)** $(-8, 16)$ A **c)** $(10, 15)$ A **d)** $(-12, 18)$ A

Exercise 2.6 Finding unknown coefficients

ANSWER P. 253

The voltages $\mathbf{U}_1 = (-2, 3)$ V and $\mathbf{U}_2 = (3, -1)$ V are given. Find the numbers a and b such that

$$a\mathbf{U}_1 + \mathbf{U}_2 = (0, b) \text{ V}.$$

Exercise 2.7 Phase shift between voltage and current

ANSWER P. 253

The voltage across a load is $\mathbf{U} = (4, 4)$ V and the current through it is $\mathbf{I} = (-1, 2)$ A.

a) Calculate the magnitudes $|\mathbf{U}|$ and $|\mathbf{I}|$.

b) Calculate the phase angles v_U and v_I with the positive x-axis.

c) Calculate the phase shift, that is, the angle α between $\mathbf{U}$ and $\mathbf{I}$.

PART 3

Extra exercises

The exercises below are extra practice, best done on your own after the lesson.

Exercise 3.1 Distance between points

ANSWER P. 253

Calculate the distance between the points:

a) $(0, 0)$ and $(3, 4)$ **b)** $(1, 2)$ and $(4, 6)$ **c)** $(-2, 1)$ and $(2, -2)$

d) $(5, -1)$ and $(5, 7)$ **e)** $(-3, 2)$ and $(1, 2)$

Exercise 3.2 The vector between two points

ANSWER P. 253

The vector from the point A to the point B is given by $\overrightarrow{AB} = (x_B - x_A; y_B - y_A)$.

a) $A = (1, 2)$ and $B = (5, 5)$. Find $\overrightarrow{AB}$ and $|\overrightarrow{AB}|$.

b) $A = (-1, 4)$ and $B = (2, 0)$. Find $\overrightarrow{AB}$ and $|\overrightarrow{AB}|$.

c) Find $\overrightarrow{BA}$ in a) and compare it with $\overrightarrow{AB}$.

d) What holds in general between $\overrightarrow{AB}$ and $\overrightarrow{BA}$?

Exercise 3.3 The magnitude of a voltage

ANSWER P. 253

Find the magnitude $|\mathbf{U}|$ of the voltage in exact form and as an approximation to two decimal places.

a) $\mathbf{U} = (6, 8)$ V **b)** $\mathbf{U} = (-5, 12)$ V **c)** $\mathbf{U} = (1, 1)$ V

d) $\mathbf{U} = (-3, -3)$ V **e)** $\mathbf{U} = (0, -7)$ V

Exercise 3.4 Phase angle

ANSWER P. 253

Find the phase angle v of the current with the positive x-axis. State which quadrant the phasor lies in and whether a quadrant correction is needed.

a) $\mathbf{I} = (1, 1)$ A **b)** $\mathbf{I} = (-1, 1)$ A **c)** $\mathbf{I} = (-1, -1)$ A

d) $\mathbf{I} = (1, -1)$ A **e)** $\mathbf{I} = (4, 3)$ A

Exercise 3.5 From magnitude and phase angle to components

ANSWER P. 253

A voltage with the magnitude $|\mathbf{U}|$ and the phase angle v with the positive x-axis has the components

$$U_x = |\mathbf{U}| \cos v, \qquad U_y = |\mathbf{U}| \sin v.$$

Find the components:

a) $|\mathbf{U}| = 10\,\text{V}$, $v = 30°$

b) $|\mathbf{U}| = 5\,\text{V}$, $v = 90°$

c) $|\mathbf{U}| = 4\,\text{V}$, $v = 180°$

d) $|\mathbf{U}| = 2\,\text{V}$, $v = 225°$

e) Check in a) that $\sqrt{U_x^2 + U_y^2} = 10\,\text{V}$.

Exercise 3.6 Phase shift

ANSWER P. 253

First calculate the phase angles v_U and v_I with the positive x-axis, and then the phase shift α, that is, the angle between the voltage and the current.

a) $\mathbf{U} = (1, 0)\,\text{V}$, $\mathbf{I} = (1, 1)\,\text{A}$

b) $\mathbf{U} = (3, 0)\,\text{V}$, $\mathbf{I} = (0, 4)\,\text{A}$

c) $\mathbf{U} = (2, 1)\,\text{V}$, $\mathbf{I} = (-1, 3)\,\text{A}$

d) $\mathbf{U} = (2, 2)\,\text{V}$, $\mathbf{I} = (-3, -3)\,\text{A}$

Exercise 3.7 Resistor and inductor in series

ANSWER P. 253

Across a resistor and an inductor in series lie the voltages $\mathbf{U}_R = (12, 0)\,\text{V}$ and $\mathbf{U}_L = (0, 5)\,\text{V}$.

a) Calculate the total voltage $\mathbf{U} = \mathbf{U}_R + \mathbf{U}_L$.

b) Calculate the magnitude $|\mathbf{U}|$.

c) Calculate the phase angle with the positive x-axis.

d) Why is $|\mathbf{U}|$ not equal to $12 + 5 = 17\,\text{V}$?

Exercise 3.8 Resistor, inductor and capacitor in series

ANSWER P. 253

In a series circuit, the voltage $\mathbf{U}_R = (30, 0)\,\text{V}$ lies across the resistor, $\mathbf{U}_L = (0, 70)\,\text{V}$ across the inductor and $\mathbf{U}_C = (0, -30)\,\text{V}$ across the capacitor.

a) Calculate the total voltage $\mathbf{U} = \mathbf{U}_R + \mathbf{U}_L + \mathbf{U}_C$.

b) Calculate the magnitude $|\mathbf{U}|$.

c) Calculate the phase angle with the positive x-axis.

d) What is the total voltage if the capacitor voltage is instead $(0, -70)\,\text{V}$? What does the result mean?

Exercise 3.9 Unit vectors

ANSWER P. 253

A unit vector has length 1 and is obtained as $\hat{\mathbf{u}} = \dfrac{\mathbf{u}}{|\mathbf{u}|}$. The current $\mathbf{I} = (3, 4)$ A is given.

a) Find the unit vector in the direction of the current.

b) Check that the answer in a) really has length 1.

c) Find the unit vector in the direction $(-6, 8)$.

d) Find a current with the magnitude 15 A that is in phase with $\mathbf{I}$.

e) Find a current with the magnitude 15 A that is in antiphase with $\mathbf{I}$.

Exercise 3.10 Three-phase system

ANSWER P. 253

In a three-phase system the three phase voltages have the same magnitude but different phase angles: $\mathbf{U}_1 = (230, 0)$ V, $\mathbf{U}_2 = (-115, -199)$ V and $\mathbf{U}_3 = (-115, 199)$ V.

a) Calculate the sum $\mathbf{U}_1 + \mathbf{U}_2 + \mathbf{U}_3$.

b) What does the result mean?

c) Calculate the magnitude and the phase angle of $\mathbf{U}_2$.

d) Calculate the line voltage $\mathbf{U}_{12} = \mathbf{U}_1 - \mathbf{U}_2$ and its magnitude.

Exercise 3.11 In phase and in antiphase

ANSWER P. 253

The voltage $\mathbf{U} = (4, -6)$ V is given.

a) Which of $(2, -3)$ V, $(-8, 12)$ V, $(6, -4)$ V and $(-2;\ 3)$ V are parallel to $\mathbf{U}$?

b) Which of them are in antiphase with $\mathbf{U}$?

c) Find t such that $(t, 9)$ V is parallel to $\mathbf{U}$.

d) Can a phasor be parallel to itself? Justify your answer.

Exercise 3.12 Resistor and capacitor in parallel

ANSWER P. 253

Through a resistor and a capacitor connected in parallel flow the currents $\mathbf{I}_R = (4;\ 0)$ A and $\mathbf{I}_C = (0, 3)$ A.

a) Calculate the total current $\mathbf{I} = \mathbf{I}_R + \mathbf{I}_C$.

b) Calculate the magnitude $|\mathbf{I}|$.

c) Calculate the phase angle with the positive x-axis.

d) How large must the capacitor current be for the phase angle to become 45°?

Exercise 3.13 The power triangle

ANSWER P. 253

The power in an AC circuit can be described by the vector $\mathbf{S} = (P, Q)$, where P is the active power in watts (W) and Q is the reactive power in var. The magnitude $|\mathbf{S}|$ of the vector is the apparent power in volt-amperes (VA). A motor has $P = 800\,\text{W}$ and $Q = 600\,\text{var}$.

a) Calculate the apparent power $|\mathbf{S}|$.

b) Calculate the angle v of the vector with the positive x-axis.

c) The power factor is $\cos v$. Calculate it.

d) A capacitor is connected and contributes the vector $(0, -600)$. Find the new vector $\mathbf{S}$ and the new apparent power.

Exercise 3.14 Find the error

ANSWER P. 254

Each line contains a common vector error. Explain the error and give the correct answer.

a) $|(3, 4)| = 3 + 4 = 7$

b) $2(3, -1) = (6, -1)$

c) Two phasors have the angles 30° and 100° with the positive x-axis, so the angle between them is 130°.

d) The vector $(-3, 4)$ makes the angle $\arctan\left(\dfrac{4}{-3}\right) \approx -53.1°$ with the positive x-axis.

e) $|-2\mathbf{u}| = -2|\mathbf{u}|$

PART 4

Mathematics in C

The exercise below is shown during the lesson, and you can also do it yourself. The program is written and run in **OnlineGDB**, `https://www.onlinegdb.com/online_c_compiler`, with the language set to **C**.

The exercise uses **functions** in C. A function in C works like a mathematical function: it takes one or more values, called **parameters**, and **returns** a result with `return`. The function `double magnitude(double x, double y)`, for example, takes two floating-point numbers and returns one. It is written above `main()` and called as `magnitude(3.0, 4.0)`, just as $f(3)$ is written in mathematics. The functions `sqrt(x)` and `atan2(y, x)` and the constant `M_PI` are in `math.h`.

Exercise 4.1 Vectors in C

ANSWER P. 254

The vectors $\mathbf{u} = (3, 4)$ and $\mathbf{v} = (-4, 3)$ are given.

a) Find $|\mathbf{u}|$, $|\mathbf{v}|$, the angles v_u and v_v to the positive x-axis, the sum $\mathbf{u} + \mathbf{v}$ and the angle α between the vectors. Calculate by hand.

b) Write a program with two functions:

- `double magnitude(double x, double y)`, which returns the magnitude $\sqrt{x^2 + y^2}$.
- `double angle_deg(double x, double y)`, which returns the angle to the positive x-axis in degrees. Use `atan2(y, x)`, which gives the angle in radians, and convert to degrees.

Let the program print the magnitude and angle of $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{u} + \mathbf{v}$, and the angle between $\mathbf{u}$ and $\mathbf{v}$.

c) Compare with the hand calculation. `atan2()` needs no quadrant correction. Why does `atan(3.0 / -4.0)` give the wrong angle for $\mathbf{v}$?

CHAPTER 8

Functions

One value in and exactly one value out: the function, its graph and the most common types.

IN THIS CHAPTER

- The concept of a function: a rule that gives exactly one value for each permitted x.
- Domain and range.
- Common types of function: linear, power, exponential and polynomial functions.
- Reading a graph, and solving problems with functions and graphs.

This chapter is about functions, one of the most important tools in mathematics. A function is a rule that gives exactly one function value for every permitted value of x, and it can be described by a formula, a table or a graph. The chapter reviews the concepts from compulsory school and upper-secondary Mathematics 1: domain and range, the most common types of function and what can be read from a graph. In electrical engineering, functions describe how one quantity depends on another, for example how the resistance of a resistor depends on the temperature or how the power in a resistor depends on the current.

§8.1 to §8.5 are the chapter's theory, with examples. §8.6 summarises the concepts and what you should be able to do after the chapter, and §8.7 holds the exercises. Most of them can be done with pencil and paper, but a few need a calculator.

8.1 What is a function?

A **function** f is a rule that gives exactly one value $f(x)$ for every permitted value of x:

$$f : x \mapsto f(x)$$

Functions can be represented as algebraic expressions, tables or graphs.

EXAMPLE $f(x) = 2x + 1$ gives $f(3) = 7$ and $f(-1) = -1$.

8.2 Domain and range

The **domain** D_f is the set of all x-values for which $f(x)$ is defined.

The **range** V_f is the set of all possible function values $f(x)$.

Function	Domain	Range
$f(x) = 2x + 3$	$\mathbb{R}$	$\mathbb{R}$
$f(x) = x^2$	$\mathbb{R}$	$[0, \infty)$
$f(x) = \sqrt{x}$	$[0, \infty)$	$[0, \infty)$
$f(x) = \dfrac{1}{x}$	$\mathbb{R} \setminus \{0\}$	$\mathbb{R} \setminus \{0\}$

8.3 Common types of function

8.3.1 Linear function

$$f(x) = kx + m$$

where k is the slope and m is the y-intercept.

8.3.2 Power function

$$f(x) = x^n$$

8.3.3 Exponential function

$$f(x) = C \cdot a^x, \qquad a > 0,\ a \neq 1$$

- $a > 1$: an increasing function.
- $0 < a < 1$: a decreasing function.
- $f(0) = C$, the initial value.

8.3.4 Exponential equations

When the unknown x is in the exponent, for example $2^x = 8$, we solve the equation with logarithms. We use two logarithms, both of which are on the calculator:

- log is the base-10 logarithm, so $\log 10 = 1$.
- ln is the natural logarithm (base $e \approx 2.718$), so $\ln e = 1$.

The key is the law of logarithms that moves the exponent down in front of the logarithm:

$$\log(a^x) = x \cdot \log a, \qquad \ln(a^x) = x \cdot \ln a$$

EXAMPLE Solve the equation $2^x = 8$. Take the logarithm of both sides with the base-10 logarithm, move the exponent down and divide both sides by $\log 2$:

$$\log(2^x) = \log 8$$
$$x \cdot \log 2 = \log 8$$
$$x = \frac{\log 8}{\log 2} = 3$$

Check: $2^3 = 8$.

In general, then,

$$a^x = b \quad \Rightarrow \quad x = \frac{\log b}{\log a}.$$

Note! The same logarithm must be used on both sides. It does not matter which one: with ln the answer is the same, $x = \frac{\ln 8}{\ln 2} = 3$. Mixing them, however, does not work, for example $\frac{\log 8}{\ln 2}$.

Base e: When the base is e, ln is the simplest to use, since $\ln e = 1$:

$$e^x = 5 \quad \Rightarrow \quad \ln(e^x) = \ln 5 \quad \Rightarrow \quad x \cdot \ln e = \ln 5 \quad \Rightarrow \quad x = \ln 5 \approx 1.61$$

With log it works too, $x = \frac{\log 5}{\log e} \approx 1.61$, but it is more cumbersome.

8.3.5 Polynomial function

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$$

8.4 Reading a graph

From a graph you can read:

- the domain, that is, which x-values exist,
- the range, that is, which y-values are reached,
- the zeros, the x where $f(x) = 0$,
- the maximum and minimum points.

8.5 Worked examples

EXAMPLE **Temperature sensor.** A sensor gives the temperature $f(x)$ (in °C) as a function of the input voltage x (in V):

$$f(x) = 100x - 50, \qquad 0 \le x \le 5\,\text{V}$$

a) Find the domain and the range. The domain is $D_f = [0, 5]\,\text{V}$, and

$$f(0) = -50\,°\text{C}, \quad f(5) = 450\,°\text{C} \quad \Rightarrow \quad V_f = [-50, 450]\,°\text{C}.$$

b) Calculate the input voltage when the temperature is 30 °C.

$$100x - 50 = 30 \quad \Rightarrow \quad x = 0.8\,\text{V}$$

EXAMPLE **A linear function from two points.** A linear function passes through $(1, 3)$ and $(3, 7)$. Find the equation $f(x) = kx + m$ of the function.

The slope is

$$k = \frac{7 - 3}{3 - 1} = 2.$$

Substituting $(1, 3)$ gives $m = 3 - 2 \cdot 1 = 1$, so

$$f(x) = 2x + 1.$$

EXAMPLE **Exponential function and growth.** A data set starts with 6×10^6 data points and grows by 3 % a year:

$$f(x) = 6 \times 10^6 \cdot 1.03^x$$

a) Calculate $f(5)$.

$$f(5) = 6 \times 10^6 \cdot 1.03^5 \approx 6.96 \times 10^6$$

b) When has the data set doubled?

$$1.03^x = 2 \quad \Rightarrow \quad x = \frac{\log 2}{\log 1.03} \approx 23.5\,\text{years}$$

8.6 Summary

Concept	Explanation
Function	Every x gives exactly one $f(x)$
Domain	Permitted x-values
Range	Possible $f(x)$-values
Linear function	$f(x) = kx + m$
Exponential function	$f(x) = C \cdot a^x$
Exponential equation	$a^x = b \Rightarrow x = \frac{\log b}{\log a}$

What you should be able to do

After this chapter, you should be able to:

- Recall the key concepts of functions from compulsory school and Mathematics 1.
- Understand the concept of a function.
- Recognise the common types of function.
- Use functions and graphs to solve problems.

Test yourself

ANSWER P. 255

1. Give an example of a function with a restricted domain and explain why it is restricted.
2. Calculate $f(2)$ and $f(-1)$ for $f(x) = 2x^2 - x + 3$.

Next chapter

Chapter 9 is about trigonometric functions: amplitude, period, phase and angle measure.

8.7 Exercises

How to work with the exercises. Parts 1 and 2 are done during the lesson: first on your own, a few minutes per exercise, then together. Part 3 is extra exercises for further practice after the lesson. Most of them can be done in your head or with pencil and paper.
Answers and worked solutions. The answer to every exercise is in the answer key at the back of the book, and its page is given next to the exercise. An answer shows whether you got it right, but not how. Full worked solutions, in Swedish, are published in the course repository after the lesson, in `lectures/L08/appendix/c_solutions.md`. Always try it yourself first.

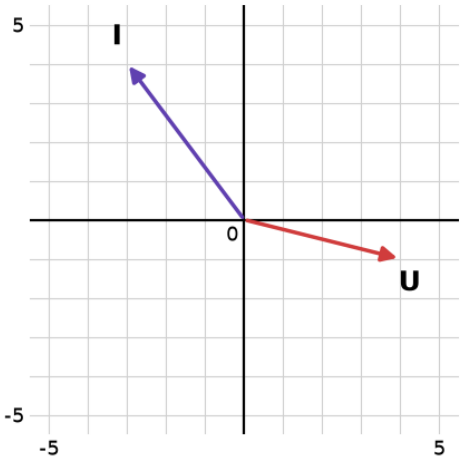
PART 1

Review exercises

Exercise 1.1 Voltage and current as phasors

ANSWER P. 254

An alternating voltage **U** and an alternating current **I** are represented by the phasors in the figure below. The voltage is given in V and the current in A.



a) Find the coordinates of the phasors.

b) Calculate the magnitudes (absolute values) $|\mathbf{U}|$ and $|\mathbf{I}|$.

c) Calculate the phase angles v_U and v_I with the positive x-axis.

d) Calculate the phase shift, that is, the angle α between **U** and **I**.

e) The current **I** meets a second current $\mathbf{I}_2 = (7, -1)$ A at a node. Calculate the total current $\mathbf{I}_{\text{tot}} = \mathbf{I} + \mathbf{I}_2$ and its magnitude.

f) Find a current with the magnitude 10 A that is in antiphase with **I**.

PART 2

New material

Exercise 2.1 Is the relation a function?

ANSWER P. 254

Decide whether the current I is a function of the voltage U. Justify your answer.

a) $I = \dfrac{U}{10}$ (Ohm's law for a $10\,\Omega$ resistor)

b) The measured pairs (U, I) with U in V and I in mA: $\{(2, 100),\ (4, 200),\ (6, 250),\ (8, 250)\}$

c) The measured pairs (U, I) with U in V and I in mA: $\{(1, 2),\ (1, 5),\ (2, 3)\}$

Exercise 2.2 Domain and range

ANSWER P. 254

Find the domain and the range.

a) The temperature sensor `TMP36` gives the output voltage $U(T) = 0.01T + 0.5$ (in V) at the temperature T (in °C). The sensor works for $-40 \leq T \leq 125$.

b) The voltage across a $100\,\Omega$ resistor that develops the power P (in W) is $U(P) = \sqrt{100P}$.

c) The power in a $10\,\Omega$ resistor is $P(U) = \dfrac{U^2}{10}$, where the voltage can be $-20 \leq U \leq 20$ (in V).

Exercise 2.3 Linear resistance of a temperature-dependent resistor

ANSWER P. 254

The resistance R (in Ω) of a certain temperature-dependent resistor depends linearly on the temperature T (in °C) as $R(T) = kT + m$. At $T = 0\,°\text{C}$, $R = 100\,\Omega$, and at $T = 50\,°\text{C}$, $R = 175\,\Omega$.

a) Find k and m.

b) Calculate the resistance at $T = 25\,°\text{C}$.

Exercise 2.4 An ageing battery

ANSWER P. 254

A new lithium-ion battery has the capacity 4000 mAh and loses 3 % of its capacity a year. After x years the capacity is

$$C(x) = 4000 \cdot 0.97^x \text{ mAh}.$$

a) Calculate $C(10)$ and interpret the answer.

b) A battery is usually considered worn out when its capacity has dropped to 80 %. After how many years does that happen?

Exercise 2.5 Power as a power function of the current

ANSWER P. 254

The power P (in W) developed in a resistor $R = 50\,\Omega$ depends on the current I (in A) as

$$P(I) = R \cdot I^2.$$

a) What type of function is $P(I)$?

b) Calculate $P(0.5)$ and $P(2)$.

c) A fuse in the circuit limits the current to $0 \le I \le 3$ A. Find the domain and range of $P(I)$ in this interval.

Exercise 2.6 Interpreting measurements

ANSWER P. 254

A voltage $U(t)$ is measured at several times during a discharge:

t (s)	0	1	2	3	4
$U(t)$ (V)	12	9	6	3	0

a) Show from the table that the relationship between t and $U(t)$ is linear.

b) Find a formula $U(t) = kt + m$ from the table.

c) If the pattern in the table continued, at what time t would $U(t) = -3$ V? Is this a reasonable value for a discharge voltage? Explain why the linear model cannot hold for all $t \ge 0$.

PART 3

Extra exercises

The exercises below are extra practice, best done on your own after the lesson.

Exercise 3.1 Calculating function values

ANSWER P. 254

A battery has the terminal voltage $U(I) = 12 - 2I$ (in V), a resistor develops the power $P(I) = 5I^2$ (in W), and the current through a load is $I(R) = \dfrac{24}{R}$ (in A). Calculate:

a) $U(1)$ **b)** $U(4)$ **c)** $P(0)$

d) $P(3)$ **e)** $I(8)$ **f)** $I(12)$

Exercise 3.2 Solving the equation $f(x) = c$

ANSWER P. 254

The same functions as in exercise 3.1.

a) At what current is the terminal voltage 9 V?

b) Find the zero of $U(I)$. What does it mean for the battery?

c) At what current does the resistor develop 20 W?

d) Can $P(I)$ be negative? Justify your answer.

e) Which resistance gives the current 0.5 A?

Exercise 3.3 Domain

ANSWER P. 254

Find the mathematical domain. Then state which values are physically reasonable.

a) $I(R) = \dfrac{10}{R}$

b) $I(R) = \dfrac{12}{R+2}$

c) $U(P) = \sqrt{50P}$

d) $X(f) = \dfrac{1000}{f}$

Exercise 3.4 Range

ANSWER P. 254

Find the range.

a) $P(I) = 10I^2$ (in W) for $0 \le I \le 2$ (in A)

b) $U(I) = 12 - 2I$ (in V) for $0 \le I \le 6$ (in A)

c) A rectifier gives the output voltage $u_{\text{out}} = |u_{\text{in}}|$ for $-10 \le u_{\text{in}} \le 10$ (in V).

d) $T(U) = 100U - 50$ (in °C) for $0.1 \le U \le 1.75$ (in V)

Exercise 3.5 A linear function from two points

ANSWER P. 254

Find the linear function.

a) A battery has the terminal voltage 9 V at 0 A and 8 V at 2 A. Find $U(I)$.

b) A temperature sensor gives 0.5 V at 0 °C and 1.5 V at 100 °C. Find $U(T)$.

c) A Pt100 element has the resistance 100 Ω at 0 °C and 138.5 Ω at 100 °C. Find $R(T)$.

d) A charging current is 2 A at $t = 0$ s and 2 A at $t = 5$ s. Find $I(t)$.

Exercise 3.6 Interpreting k and m

ANSWER P. 254

A loaded battery has the terminal voltage $U(I) = 12 - 0.5I$ (in V), where I is the current in A.

a) What does k mean graphically?

b) What does m mean graphically?

c) Is $U(I)$ increasing or decreasing? Justify your answer.

d) Find the zero of $U(I)$.

e) What do $k = -0.5$ and $m = 12$ represent physically?

Exercise 3.7 Power functions

ANSWER P. 255

The power in a $10\,\Omega$ resistor is $P(I) = 10I^2$ (in W).

a) Calculate $P(3)$ and $P(-3)$. Why is the power the same when the current changes direction?

b) How many times larger does the power become if the current is tripled?

c) How does the power change if the current is halved?

d) The power can also be written $P(U) = \dfrac{U^2}{10}$. Calculate $P(5)$ and $P(10)$. How does the power change when the voltage is doubled?

Exercise 3.8 Exponential functions

ANSWER P. 255

An amplifier chain doubles the signal in each stage, so that $u(n) = 5 \cdot 2^n$ (in mV) after n stages. An attenuator instead halves the signal in each stage, so that $g(n) = 100 \cdot 0.5^n$ (in mV).

a) Calculate $u(0)$, $u(1)$ and $u(3)$.

b) Calculate $g(0)$, $g(1)$ and $g(3)$.

c) Which of the functions is increasing and which is decreasing? What decides it?

d) What is C in each function, and what does it mean?

Exercise 3.9 Growth factor

ANSWER P. 255

a) The price of electricity increases by 5 % a year. What is the growth factor a?

b) The capacity of a battery decreases by 20 % a year. What is a?

c) A signal is attenuated to 70 % of its value for each stage it passes. The initial amplitude is 2 V. Write a function for the amplitude after n stages.

d) Calculate the amplitude after 3 stages.

e) After how many stages is the amplitude below 1 V?

Exercise 3.10 Linear or exponential?

ANSWER P. 255

Two quantities have been measured once a year: $f(x)$ is the number of charging points in a municipality (in hundreds) and $g(x)$ is the number of connected sensors in a network (in thousands), where x is the number of years.

x	0	1	2	3	4
$f(x)$	3	5	7	9	11
$g(x)$	3	6	12	24	48

a) Which of the functions is linear? Justify your answer from the table.

b) Which is exponential? Justify your answer.

c) Find a formula for $f(x)$.

d) Find a formula for $g(x)$.

e) Calculate $f(10)$ and $g(10)$.

Exercise 3.11 Zeros

ANSWER P. 255

Find the zeros and interpret them.

a) The terminal voltage $U(I) = 12 - 4I$ (in V) of a battery, where I is the current in A.

b) The power $P(I) = 12I - 2I^2$ (in W) that the same kind of battery, with the internal resistance 2 Ω, delivers to a load.

c) The voltage $U(t) = 6 - 0.5t$ (in V) during a discharge, where t is the time in s.

Exercise 3.12 A voltage divider as a function

ANSWER P. 255

A voltage divider consists of $R_1 = 1\,\text{k}\Omega$ and a potentiometer with the resistance x (in kΩ). The output voltage is given by

$$U(x) = 12 \cdot \frac{x}{1+x}\,\text{V}, \qquad 0 \le x \le 9.$$

a) Calculate $U(0)$, $U(1)$ and $U(9)$.

b) Find the range V_U for the given interval.

c) At what x does U become 8 V?

d) Can U ever reach 12 V? Justify your answer.

Exercise 3.13 Discharging a capacitor

ANSWER P. 255

The voltage across a discharging capacitor is given by

$$u(t) = 10 \cdot e^{-t/2}\,\text{V}, \qquad t \geq 0,$$

where t is in seconds.

a) Calculate $u(0)$.

b) Calculate $u(2)$ and $u(4)$.

c) Is the function increasing or decreasing?

d) Find the range V_u for $t \geq 0$.

e) Does $u(t)$ ever become exactly zero? Justify your answer.

Exercise 3.14 True or false

ANSWER P. 255

Decide whether the statement is true or false, and justify your answer briefly.

a) The power $P(I) = RI^2$ is negative if the current is negative.

b) The current $I(R) = \dfrac{12}{R}$ is defined for all R.

c) That $U(2) = 5$ means that U has a zero at $I = 2\,\text{A}$.

d) If the same voltage U has been measured with two different currents I, then I is not a function of U.

e) The voltage $u(t) = 3 \cdot 2^t$ (in V) has the initial value 3 V.

PART 4

Mathematics in C

The exercise below is shown during the lesson, and you can also do it yourself. The program is written and run in **OnlineGDB**, `https://www.onlinegdb.com/online_c_compiler`, with the language set to **C**. Functions in C are described in Exercise 4.1 of Chapter 7.

Exercise 4.1 Functions in C

ANSWER P. 255

The temperature sensor in §8.5 gives the temperature $f(x) = 100x - 50\,^\circ\text{C}$ at the input voltage x V, where $0 \leq x \leq 5$. A mathematical function and a function in C rest on the same idea: one value in, one value out.

a) Compute $f(0)$, $f(0.8)$ and $f(5)$ by hand.

b) Write a function `double temperature(double voltage)` that returns $f(x)$. Then print a table of values with $x = 0;\ 0.5;\ 1.0;\ \ldots;\ 5.0$ V and the temperature on each line. Hint: let an integer variable `i` go from 0 to 10 in a loop and compute $x = 0.5 \cdot i$.

c) Solve $f(x) = 100x - 50$ for x and write a function `double voltage(double temperature)` that works backwards, from temperature to voltage. Which voltage gives 30 °C? Also check that `voltage(temperature(0.8))` gives 0.8.

CHAPTER 9

Trigonometric functions

Angles in degrees and radians, the sine function and the equation of an AC voltage.

IN THIS CHAPTER

- Angle measure: degrees and radians.
- The sine function and its properties.
- The AC voltage equation: amplitude, frequency, period and phase.
- Reading the properties of an AC voltage from a graph.

This chapter is about the trigonometric functions, and above all the sine function, which describes the AC voltage of the mains and the signals in almost all electronics. The chapter begins with the angle measures degrees and radians and how to convert between them, then goes through the properties of the sine function and shows how a sinusoidal AC voltage is described by its amplitude, angular frequency and phase angle. Finally, you read the amplitude, period, frequency and phase of an AC voltage from a graph and write its equation.

§9.1 to §9.5 are the chapter's theory, with examples. §9.6 summarises the formulas and what you should be able to do after the chapter, and §9.7 holds the exercises. Part 1 of them reviews earlier chapters. Several of them need a calculator that can work in radians.

9.1 Angle measure: degrees and radians

Angles can be given in **degrees** (°) or **radians** (rad). The conversion formulas are

$$v_{\text{rad}} = v_{\text{deg}} \cdot \frac{\pi}{180^\circ} \qquad \text{and} \qquad v_{\text{deg}} = v_{\text{rad}} \cdot \frac{180^\circ}{\pi}.$$

Degrees	Radians
0°	0
30°	$\pi/6$
45°	$\pi/4$
60°	$\pi/3$
90°	$\pi/2$
180°	π
360°	2π

9.2 The sine function

The sine function $\sin(\theta)$ is defined by the unit circle and has the following properties:

- domain: $\mathbb{R}$,
- range: $[-1, 1]$,
- period: 2π, one full turn.

9.3 The AC voltage equation

A sinusoidal AC voltage is described by

$$u(t) = |U| \sin(\omega t + \delta).$$

Parameter	Symbol	Unit	Description
Amplitude	$\lvert U\rvert$	V	The peak value
Angular frequency	ω	rad/s	$\omega = 2\pi f$
Frequency	f	Hz	Number of periods per second
Period	T	s	$T = 1/f$
Phase angle	δ	rad	Shift in time

The quantities are related by

$$\omega = 2\pi f = \frac{2\pi}{T}.$$

Phase shift in time. A positive phase angle δ means that the voltage is early, that is, shifted to the left in the graph. A negative δ means that it is delayed.

9.4 Reading properties from a graph

From a sine graph you can read:

- the **amplitude** $|U|$: the distance from zero to the peak value,
- the **period** T: the length of one complete cycle,
- the **frequency** $f = 1/T$,
- the **phase angle** δ: compare the time of the peak with $T/4$, since with no phase shift the peak would have come at $t = T/4$.

9.5 Worked examples

EXAMPLE **Angle conversions.** Convert the following angles to radians: $45°$, $90°$, $270°$, $-60°$ and $135°$. With the formula $v_{\text{rad}} = v_{\text{deg}} \cdot \pi/180°$, this gives

$$45° \to \frac{\pi}{4}, \quad 90° \to \frac{\pi}{2}, \quad 270° \to \frac{3\pi}{2}, \quad -60° \to -\frac{\pi}{3}, \quad 135° \to \frac{3\pi}{4}.$$

EXAMPLE **Finding the properties from a graph.** An AC voltage has the amplitude $|U| = 4\,\text{V}$ and the period $T = 40\,\text{ms}$, and the peak value is reached at $t = 15\,\text{ms}$. The **angular frequency** is

$$f = \frac{1}{T} = \frac{1}{0.04} = 25\,\text{Hz}, \qquad \omega = 2\pi \cdot 25 = 50\pi\,\text{rad/s}.$$

The phase angle. With no phase shift, the peak comes at $T/4 = 10\,\text{ms}$. It comes at $15\,\text{ms}$, that is, $5\,\text{ms}$ late, which gives a negative phase:

$$\delta = -2\pi \cdot \frac{5\,\text{ms}}{40\,\text{ms}} = -\frac{\pi}{4}\,\text{rad}$$

The equation is

$$u(t) = 4\sin\left(50\pi t - \frac{\pi}{4}\right)\,\text{V}.$$

EXAMPLE **Writing the equation from given parameters.** An AC voltage has the amplitude $4\,\text{V}$, the frequency $50\,\text{Hz}$ and the phase $-30°$. Then

$$\delta = \frac{-30° \cdot \pi}{180°} = -\frac{\pi}{6}\,\text{rad}, \qquad \omega = 2\pi \cdot 50 = 100\pi\,\text{rad/s},$$

and the equation is

$$u(t) = 4\sin\left(100\pi t - \frac{\pi}{6}\right)\,\text{V}.$$

9.6 Summary

Concept	Formula
Degrees to radians	$v_{\text{rad}} = v_{\text{deg}} \cdot \pi/180°$
Angular frequency	$\omega = 2\pi f = 2\pi/T$
Period	$T = 1/f$
AC voltage equation	$u(t) = \lvert U \rvert \sin(\omega t + \delta)$

What you should be able to do

After this chapter, you should be able to:

- Recognise the trigonometric functions and their parameters.
- Use the angle measures degrees and radians and convert between them.

Test yourself

ANSWER P. 256

1. Convert 150° to radians and $\dfrac{5\pi}{6}$ to degrees.
2. An AC voltage has the amplitude 5 V, the frequency 50 Hz and the phase $\pi/4$ rad. Write the voltage's equation $u(t)$.

Next chapter

Chapter 10 is about exponential functions and more logarithms, and decibels.

9.7 Exercises

How to work with the exercises. Parts 1 and 2 are done during the lesson: first on your own, a few minutes per exercise, then together. Part 3 is extra exercises for further practice after the lesson. Most of them can be done in your head or with pencil and paper.
Answers and worked solutions. The answer to every exercise is in the answer key at the back of the book, and its page is given next to the exercise. An answer shows whether you got it right, but not how. Full worked solutions, in Swedish, are published in the course repository after the lesson, in `lectures/L09/appendix/c_solutions.md`. Always try it yourself first.

PART 1

Review exercises

Exercise 1.1 Charging a capacitor

ANSWER P. 255

A capacitor is charged in an RC circuit with the voltage

$$u_c(t) = 5\left(1 - e^{-t/2}\right),$$

where $u_c(t)$ is the voltage across the capacitor in volts and t is the time in seconds, defined for $0 \leq t \leq 10$.

a) Calculate the capacitor voltage $u_c(t)$ at $t = 0$.

b) Find the range of the voltage.

c) Find the time t at which the voltage $u_c(t)$ across the capacitor is 4 V.

Exercise 1.2 Motion of a servo motor

ANSWER P. 255

The angular position of a servo motor (in degrees) is given by

$$v(t) = -0.5t^2 + 4t,$$

where t is the time in seconds, defined for $0 \leq t \leq 8$.

a) Calculate the angle v at $t = 0$, 2, 4, 6 and 8 s.

b) Draw the graph and mark when the servo motor reaches its maximum angle.

c) Find the times at which the angle is 3.5°.

Exercise 1.3 Gain in an electronic system

ANSWER P. 255

In an amplifier system, the gain is described by the rational function

$$G(x) = \frac{x^2 + x - 12}{x^2 - 16},$$

where $G(x)$ is the gain factor and x a dimensionless variable.

a) Factorise both the numerator and the denominator.

b) Give the values of x that are not permitted.

c) Simplify the expression as far as possible.

PART 2

New material

Exercise 2.1 Angles to radians

ANSWER P. 255

Convert the following angles to radians.

a) $-30°$ **b)** $20°$ **c)** $225°$ **d)** $-45°$ **e)** $300°$

Exercise 2.2 Angles to degrees

ANSWER P. 255

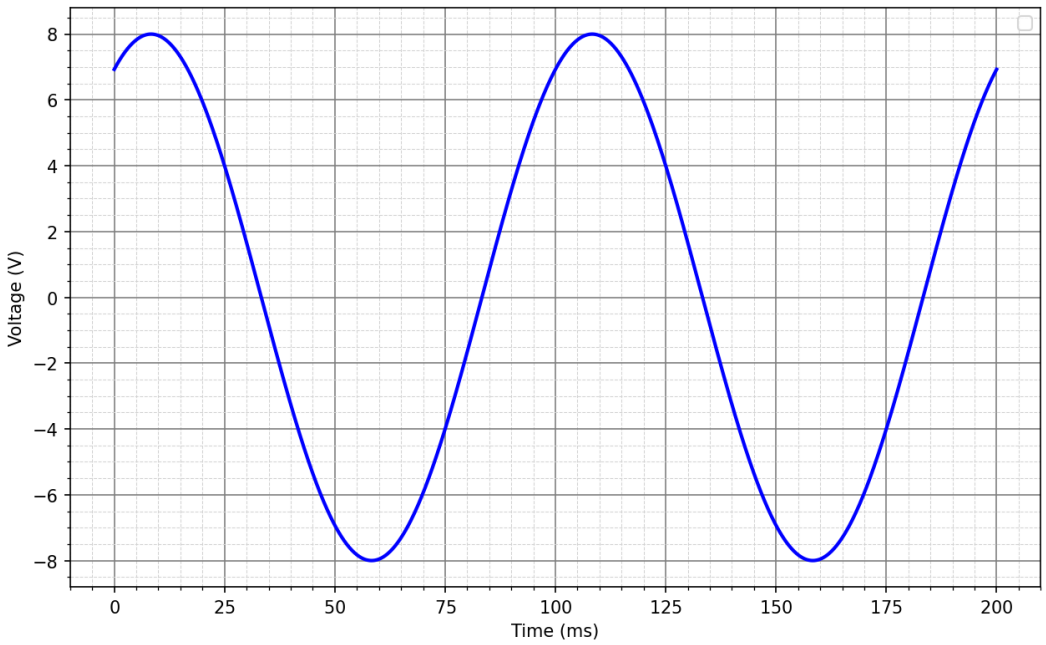
Convert the following angles to degrees.

a) $\pi/2$ **b)** $3\pi/4$ **c)** $-5\pi/6$ **d)** -1 **e)** 2.2

Exercise 2.3 Determining the properties of an AC voltage

ANSWER P. 255

An AC voltage is shown in the figure below.



Note: The peak value is reached after about 8.33 ms.

Find

- the voltage's amplitude $|U|$ in V,
- the voltage's frequency f in Hz,

- the voltage's angular frequency ω in rad/s,
- the voltage's phase δ in rad,
- the voltage's equation in the form $u(t) = |U| \sin(\omega t + \delta)$.

Exercise 2.4 An AC voltage with a given phase shift

ANSWER P. 256

An AC voltage has the amplitude 3 V, the frequency 25 Hz and the phase 45°. Find the equation $u(t)$ of the AC voltage with the phase in radians, and draw the sine curve over one period T.

Exercise 2.5 Calculating the phase of an AC voltage

ANSWER P. 256

The equation of an AC voltage is

$$u(t) = 5 \sin(60\pi t + \delta)\,\text{V}.$$

At the time $t = 10\,\text{ms}$, $u(t) = 1.25\,\text{V}$. Calculate the phase δ.

PART 3

Extra exercises

The exercises below are extra practice, best done on your own after the lesson.

Exercise 3.1 Degrees to radians

ANSWER P. 256

Convert to radians and give the answer in exact form.

a) 60° **b)** 120° **c)** −90° **d)** 210° **e)** 15° **f)** 360°

Exercise 3.2 Radians to degrees

ANSWER P. 256

Convert to degrees.

a) $\frac{\pi}{3}$ **b)** $\frac{5\pi}{4}$ **c)** $-\frac{\pi}{2}$ **d)** $\frac{7\pi}{6}$ **e)** 3 rad **f)** 0.5 rad

Exercise 3.3 Sine values

ANSWER P. 256

Give the value in exact form. Use the unit circle.

a) $\sin 0$ **b)** $\sin \frac{\pi}{2}$ **c)** $\sin \pi$ **d)** $\sin \frac{3\pi}{2}$ **e)** $\sin \frac{\pi}{6}$ **f)** $\sin \frac{\pi}{4}$

Exercise 3.4 Frequency, period and angular frequency

ANSWER P. 256

Fill in the empty cells using $T = \frac{1}{f}$ and $\omega = 2\pi f$.

f	T	ω
50 Hz		
	1 ms	
		400π rad/s
400 Hz		

Exercise 3.5 Reading properties from the equation

ANSWER P. 256

Give the amplitude, angular frequency, frequency, period and phase of

a) $u(t) = 5\sin(100\pi t)$

b) $u(t) = 12\sin\left(200\pi t + \frac{\pi}{2}\right)$

c) $u(t) = 3\sin\left(50\pi t - \frac{\pi}{6}\right)$

d) $u(t) = 8\sin(2000\pi t)$

Exercise 3.6 Writing the equation

ANSWER P. 256

Write the equation of the AC voltage in the form $u(t) = |U|\sin(\omega t + \delta)$ with the phase in radians.

a) $|U| = 10\,\text{V}, f = 50\,\text{Hz}, \delta = 0$

b) $|U| = 6\,\text{V}, f = 100\,\text{Hz}, \delta = 90°$

c) $|U| = 2\,\text{V}, T = 5\,\text{ms}, \delta = -60°$

d) $|U| = 325\,\text{V}, f = 50\,\text{Hz}, \delta = \frac{\pi}{4}$

Exercise 3.7 Instantaneous values

ANSWER P. 256

An AC voltage is given by $u(t) = 10\sin(100\pi t)$ volts.

a) Calculate $u(0)$.

b) Calculate $u(5\,\text{ms})$.

c) Calculate $u(10\,\text{ms})$.

d) Calculate $u(15\,\text{ms})$.

e) At what times during the first period is $u(t) = 0$?

Exercise 3.8 RMS value and power

ANSWER P. 256

For a sinusoidal voltage, $U_{\text{RMS}} = \dfrac{|U|}{\sqrt{2}}$, and the power in a resistor is $P = \dfrac{U_{\text{RMS}}^2}{R}$.

a) Calculate U_{RMS} for $u(t) = 12\sin(100\pi t)$.

b) An AC voltage has $U_{\text{RMS}} = 230\,\text{V}$ and $f = 50\,\text{Hz}$. Write $u(t)$ with $\delta = 0$.

c) The voltage in a) is applied across a resistor $R = 100\,\Omega$. Calculate the power.

Exercise 3.9 Phase shift in time

ANSWER P. 256

An AC voltage has the period $T = 20\,\text{ms}$.

a) How many milliseconds does the phase $\delta = \dfrac{\pi}{2}$ correspond to?

b) How many milliseconds does $\delta = \dfrac{\pi}{6}$ correspond to?

c) A curve reaches its peak value 2 ms **earlier** than a sine curve with no phase shift. Calculate δ.

d) Another curve reaches its peak 5 ms **later**. Calculate δ.

Exercise 3.10 Two voltages with a phase difference

ANSWER P. 256

Two AC voltages are given by

$$u_1(t) = 4\sin(100\pi t) \qquad \text{and} \qquad u_2(t) = 4\sin\left(100\pi t + \frac{\pi}{3}\right).$$

a) Which voltage leads the other, and by how many degrees?

b) How many milliseconds does the phase difference correspond to?

c) Calculate u_1 and u_2 at $t = 5\,\text{ms}$.

d) Do the voltages have the same amplitude and frequency?

Exercise 3.11 Solving a trigonometric equation

ANSWER P. 256

An AC voltage is given by $u(t) = 6\sin(100\pi t)$ volts.

a) At what time is $u(t) = 3\,\text{V}$ for the first time?

b) Give the next time at which $u(t) = 3\,\text{V}$.

c) At what time is the peak value first reached?

d) Why does the equation $u(t) = 3$ have infinitely many solutions?

Exercise 3.12 True or false

ANSWER P. 256

Decide whether the statement is true or false, and justify your answer briefly.

a) $\sin\theta$ can take values greater than 1.

b) If the frequency is doubled, the period is halved.

c) The angular frequency ω is measured in hertz.

d) A positive phase δ shifts the curve to the left, that is, earlier in time.

e) $180°$ corresponds to π radians.

f) The amplitude is the distance between the lowest and the highest value of the curve.

PART 4

Mathematics in C

The exercise below is shown during the lesson, and you can also do it yourself. The program is written and run in **OnlineGDB**, `https://www.onlinegdb.com/online_c_compiler`, with the language set to **C**. Functions in C are described in Exercise 4.1 of Chapter 7.

Exercise 4.1 Sine in C

ANSWER P. 256

The functions in `math.h` work with angles in **radians**.

a) Give $\sin 30°$ and convert $30°$ to radians. Calculate by hand.

b) Write a function `double deg_to_rad(double degrees)` that converts degrees to radians. Print both `sin(30.0)` and `sin(deg_to_rad(30.0))`. Why do they differ?

c) An AC voltage is given by $u(t) = 10\sin\left(100\pi t + \frac{\pi}{6}\right)$ V. Write a function `double voltage(double t)` that returns $u(t)$, and a table of values with $t = 0;\ 2;\ 4;\ \ldots;\ 20$ ms and the voltage on each line. Hint: let an integer variable `i` go from 0 to 10 and compute $t = 0.002 \cdot i$ s.

d) What is the period, and can it be seen in the table? What is the largest value in the table, and why is it smaller than the amplitude 10 V? At what time is $u(t) = 10$ V?

CHAPTER 10

Exponential functions, more logarithms and decibels

How to solve for an unknown exponent, and why gain is calculated in decibels.

IN THIS CHAPTER

- Exponential functions and exponential equations.
- Logarithmic functions: the base-10 logarithm and the natural logarithm.
- The laws of logarithms.
- Decibels.

When the unknown number sits in an exponent, as in $2^x = 64$, we solve the equation by taking the logarithm of both sides, as in §8.3.4: a logarithm turns the exponent into a factor, which can then be solved for. This chapter goes further with logarithms: how they are defined and which laws apply to them. The same tool describes quantities that decrease or grow by a fixed factor per unit of time, for example the charge in a battery. Last come decibels, the logarithmic scale electrical engineering uses for gain and attenuation, where the amplifier stages in a chain are added instead of multiplied.

§10.1 to §10.5 are the chapter's theory, with examples. §10.6 summarises the formulas and what you should be able to do after the chapter, and §10.7 holds the exercises, where Part 1 reviews the trigonometry of Chapter 9. For many of them you need a calculator with logarithms. You may like to read Wikipedia's article on the decibel, `https://en.wikipedia.org/wiki/Decibel`, before you start.

10.1 Exponential functions and equations

An equation with an **exponential function** $f(x) = a^x$ $(a > 0, a \neq 1)$ can be solved for x with logarithms.

Solving $a^x = b$: take the logarithm of both sides.

$$x \cdot \log a = \log b \quad \Rightarrow \quad x = \frac{\log b}{\log a}$$

10.2 Logarithms

The **base-10 logarithm** $\log_{10}$, usually written log, is defined by

$$\log_{10}(10^x) = x \quad \Leftrightarrow \quad 10^{\log x} = x.$$

The **natural logarithm** ln has the base $e \approx 2.718$:

$$\ln(e^x) = x \quad \Leftrightarrow \quad e^{\ln x} = x$$

Why e? The number $e = 2.71828\ldots$ may seem arbitrary, but it appears by itself whenever something grows in step with its own size. Suppose an amount grows by 100 % per year, but the growth is added in n equal parts during the year. The growth factor for the whole year is then $\left(1 + \frac{1}{n}\right)^n$:

n	$\left(1 + \frac{1}{n}\right)^n$
1	2
2	2.25
12	≈ 2.613
365	≈ 2.715
$n \to \infty$	$e \approx 2.718$

The more often the growth is added, the closer the factor gets to e, but it never exceeds it. Why e gives the simplest formulas, and is therefore the base of the natural logarithm, is shown with derivatives in §13.2.

10.3 The laws of logarithms

Logarithms obey the following laws:

$$\log(ab) = \log a + \log b$$
$$\log\left(\frac{a}{b}\right) = \log a - \log b$$
$$\log(a^n) = n \cdot \log a$$

Change of base:

$$\log_a b = \frac{\log b}{\log a} = \frac{\ln b}{\ln a}$$

10.4 Decibels (dB)

Decibels are used in electrical engineering to give gain and attenuation on a logarithmic scale.

Voltage gain:

$$G_{\mathrm{dB}} = 20 \log_{10}\left(\frac{U_{\mathrm{out}}}{U_{\mathrm{in}}}\right)$$

Conversely:

$$\frac{U_{\mathrm{out}}}{U_{\mathrm{in}}} = 10^{G_{\mathrm{dB}}/20}$$

Power gain:

$$G_{\mathrm{dB}} = 10 \log_{10}\left(\frac{P_{\mathrm{out}}}{P_{\mathrm{in}}}\right)$$

Level in dBV, decibels relative to 1 V:

$$U_{\mathrm{dBV}} = 20 \log_{10}\left(\frac{U_{\mathrm{RMS}}}{1\,\mathrm{V}}\right)$$

10.5 Worked examples

EXAMPLE **Solving logarithmic equations.** Solve **a)** $3^x = 81$, **b)** $2^{x-1} = 64$ and **c)** $e^{x-2} = 150$.

a) $3^x = 3^4 \Rightarrow x = 4$. Alternatively, $x = \frac{\log 81}{\log 3} = 4$.

b)

$$(x-1)\log 2 = \log 64 \quad \Rightarrow \quad x - 1 = \frac{\log 64}{\log 2} = 6 \quad \Rightarrow \quad x = 7$$

c)

$$x - 2 = \ln 150 \approx 5.01 \quad \Rightarrow \quad x \approx 7.01$$

EXAMPLE **Half-life of a battery charge.** A battery loses half its charge in 30 hours:

$$u(30) = U_0 \cdot a^{30} = 0.5\, U_0$$

Calculate the growth factor a:

$$a^{30} = 0.5 \quad \Rightarrow \quad a = 0.5^{1/30} \approx 0.977$$

When is 20 % left?

$$a^t = 0.2 \quad \Rightarrow \quad t = \frac{\log 0.2}{\log a} = \frac{\log 0.2}{\log 0.5^{1/30}} \approx 69.7\,\mathrm{h}$$

Calculate with the unrounded value of a; with 0.977 the answer becomes 69.2 h.

EXAMPLE **Linear voltage gain.** Two signals have the levels $L_1 = 20\,\text{dB}$ and $L_2 = 46\,\text{dB}$.

$$G_{\text{dB}} = L_2 - L_1 = 26\,\text{dB}$$
$$G_{\text{lin}} = 10^{26/20} = 10^{1.3} \approx 20$$

EXAMPLE **dBV to volts.** A sinusoidal voltage has the level 31.0 dBV.

$$U_{\text{RMS}} = 10^{31.0/20} \approx 35.5\,\text{V}$$
$$|U| = U_{\text{RMS}} \cdot \sqrt{2} \approx 35.5 \cdot 1.414 \approx 50.2\,\text{V}$$

10.6 Summary

Concept	Formula
Exponential equation	$a^x = b \;\Rightarrow\; x = \frac{\log b}{\log a}$
$\log(a^n)$	$n \cdot \log a$
Voltage gain (dB)	$G_{\text{dB}} = 20 \log\left(\frac{U_{\text{out}}}{U_{\text{in}}}\right)$
dB to linear gain	$G_{\text{lin}} = 10^{G_{\text{dB}}/20}$

What you should be able to do

After this chapter, you should be able to:

- Calculate with, and solve equations with, logarithms and exponential functions.
- Understand what decibels are.

Test yourself

ANSWER P. 257

1. Solve $4^x = 64$.
2. An amplifier has $G_{\text{dB}} = 40\,\text{dB}$. Calculate the linear voltage gain G_{lin}.

Next chapter

Chapter 11 reviews Chapters 7–10: vectors, functions, trigonometry, logarithms and decibels. The chapter ends with practice test 2.

10.7 Exercises

> **How to work with the exercises.** Parts 1 and 2 are done during the lesson: first on your own, a few minutes per exercise, then together. Part 1 reviews the trigonometry of Chapter 9, and Part 2 is the chapter's new material. Part 3 is extra exercises for further practice after the lesson.
> **Answers and worked solutions.** The answer to every exercise is in the answer key at the back of the book, and its page is given next to the exercise. An answer shows whether you got it right, but not how. Full worked solutions, in Swedish, are published in the course repository after the lesson, in `lectures/L10/appendix/c_solutions.md`. Always try it yourself first.

PART 1

Review exercises

Exercise 1.1 Angles to radians

ANSWER P. 256

Convert the following angles to radians.

a) $15°$ **b)** $-75°$

Exercise 1.2 Angles to degrees

ANSWER P. 256

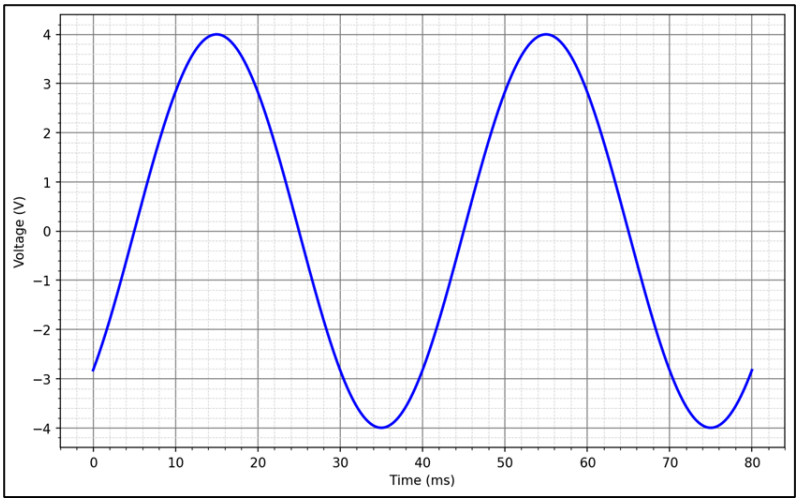
Convert the following angles to degrees.

a) $-\pi/6$ **b)** $3\pi/2$

Exercise 1.3 Determining the properties of an AC voltage

ANSWER P. 256

An AC voltage is shown in the figure below.



Find the voltage's equation in the form $u(t) = |U| \sin(\omega t + \delta)$.

Exercise 1.4 Equation and graph of an AC voltage

ANSWER P. 256

An AC voltage has the amplitude 4 V, the frequency 50 Hz and the phase $-30°$. Find the equation $u(t)$ of the AC voltage, with the phase in radians, and draw the sine curve over one period T.

Exercise 1.5 Calculating the phase of an alternating current

ANSWER P. 256

The equation of a given AC voltage is

$$u(t) = 6 \sin(80\pi t + \delta).$$

At the time $t = 15\,\text{ms}$, $u(t) = 3\,\text{V}$. Calculate the phase δ.

PART 2

New material

Exercise 2.1 Logarithmic equations

ANSWER P. 257

Solve the following logarithmic equations.

a) $5^x = 125$ **b)** $2^{x-3} = 128$ **c)** $e^{x+1} = 120$

Exercise 2.2 Half-life of the charge in a battery

ANSWER P. 257

The charge in a battery can be described by the corresponding voltage with the function

$$u(t) = U_0 \cdot a^t,$$

where

- $u(t)$ is the present charge,
- U_0 is the original charge,
- a is the growth factor,
- t is the number of hours that have passed.

A battery loses half its charge in 40 hours. Calculate after how long only 10 % of the charge remains.

Exercise 2.3 Linear gain

ANSWER P. 257

Two signals have the voltage levels $L_1 = 15\,\text{dB}$ and $L_2 = 49\,\text{dB}$. Calculate the linear voltage gain G_{lin} between these two levels. Use the formula

$$G_{\text{dB}} = 20 \log_{10}(G_{\text{lin}})$$

and remember that $G_{\text{dB}} = L_2 - L_1$.

Exercise 2.4 Calculating an RMS value in dBV

ANSWER P. 257

The relationship between the RMS value of a voltage and the corresponding value in dBV (decibels relative to one volt) is

$$U_{\text{dBV}} = 20 \log_{10}\left(\frac{U_{\text{RMS}}}{1\,\text{V}}\right),$$

where U_{RMS} is the RMS value of the voltage in V and U_{dBV} the voltage in dBV. For a sinusoidal voltage, the amplitude is also $|U| = U_{\text{RMS}} \cdot \sqrt{2}$. A sinusoidal voltage has the level 17.0 dBV. Find the amplitude in V.

PART 3

Extra exercises

The exercises below are extra practice, best done on your own after the lesson. Most of them can be done in your head or with pencil and paper.

Exercise 3.1 Logarithms without a calculator

ANSWER P. 257

Calculate.

a) $\log 100$ **b)** $\log 1000$ **c)** $\log 1$ **d)** $\log 0.01$ **e)** $\ln e$ **f)** $\ln 1$

Exercise 3.2 The laws of logarithms

ANSWER P. 257

Write as a single logarithm and simplify as far as possible.

a) $\log 3 + \log 4$ **b)** $\log 20 - \log 4$ **c)** $2 \log 5$

d) $\log 2 + \log 5$ **e)** $3 \log 2 - \log 4$

Exercise 3.3 Exponential equations

ANSWER P. 257

Solve the equations.

a) $2^x = 32$ **b)** $10^x = 0.001$ **c)** $5^x = 100$ **d)** $e^x = 20$ **e)** $4^{x+1} = 64$

Exercise 3.4 Logarithmic equations

ANSWER P. 257

Solve the equations.

a) $\log x = 2$ **b)** $\ln x = 0$ **c)** $\log(x + 5) = 1$

d) $2 \log x = 4$ **e)** $\ln(2x) = 1$

Exercise 3.5 Voltage gain in dB

ANSWER P. 257

Use

$$G_{\text{dB}} = 20 \log_{10} \left(\frac{U_{\text{out}}}{U_{\text{in}}} \right).$$

a) $U_{\text{in}} = 10\,\text{mV}$ and $U_{\text{out}} = 1\,\text{V}$. Calculate G_{dB}.

b) $U_{\text{in}} = 2\,\text{V}$ and $U_{\text{out}} = 2\,\text{V}$. Calculate G_{dB}.

c) $U_{\text{in}} = 5\,\text{V}$ and $U_{\text{out}} = 0.5\,\text{V}$. Calculate G_{dB}.

d) What does a negative value of G_{dB} mean?

Exercise 3.6 From dB to linear gain

ANSWER P. 257

Use

$$G_{\text{lin}} = 10^{G_{\text{dB}}/20}.$$

a) $G_{\text{dB}} = 20\,\text{dB}$ **b)** $G_{\text{dB}} = 60\,\text{dB}$ **c)** $G_{\text{dB}} = 6\,\text{dB}$ **d)** $G_{\text{dB}} = -3\,\text{dB}$

e) An amplifier with $G_{\text{dB}} = 26\,\text{dB}$ is fed with $U_{\text{in}} = 50\,\text{mV}$. Calculate U_{out}.

Exercise 3.7 Power gain in dB

ANSWER P. 257

Use

$$G_{\text{dB}} = 10\log_{10}\left(\frac{P_{\text{out}}}{P_{\text{in}}}\right).$$

a) $P_{\text{in}} = 1\,\text{W}$ and $P_{\text{out}} = 100\,\text{W}$. Calculate G_{dB}.

b) $P_{\text{in}} = 20\,\text{W}$ and $P_{\text{out}} = 10\,\text{W}$. Calculate G_{dB}.

c) Why is the factor 10 used for power but 20 for voltage?

d) How many dB does a doubling of the power correspond to? And a doubling of the voltage?

Exercise 3.8 Cascaded stages

ANSWER P. 257

A signal chain consists of two amplifier stages with $G_1 = 12\,\text{dB}$ and $G_2 = 18\,\text{dB}$, and a cable that attenuates 4 dB.

a) Calculate the total gain in dB.

b) Calculate the total linear gain.

c) Calculate each stage linearly and check that the product agrees with b).

d) Why are decibels convenient to calculate with when stages are cascaded?

Exercise 3.9 Level in dBV

ANSWER P. 257

Use

$$U_{\text{dBV}} = 20\log_{10}\left(\frac{U_{\text{RMS}}}{1\,\text{V}}\right)$$

and, for a sinusoidal voltage, $|U| = U_{\text{RMS}} \cdot \sqrt{2}$.

a) $U_{\text{RMS}} = 1\,\text{V}$. Calculate the level in dBV.

b) $U_{\text{RMS}} = 10\,\text{V}$. Calculate the level in dBV.

c) $U_{\text{RMS}} = 0.1\,\text{V}$. Calculate the level in dBV.

d) A signal is at 12 dBV. Calculate U_{RMS}.

e) Calculate the amplitude $|U|$ of the signal in d).

Exercise 3.10 Half-life and doubling time

ANSWER P. 257

A quantity is described by $f(t) = C \cdot a^t$.

a) A battery loses 2 % of its charge per hour. Find the growth factor a.

b) After how many hours is half the charge left?

c) A set of measurement data grows by 10 % per day. After how many days has it doubled?

d) Show in general that the doubling time is given by $t = \dfrac{\log 2}{\log a}$.

Exercise 3.11 Find the error

ANSWER P. 257

Each line contains a common mistake. Explain the mistake and give the correct answer.

a) $\log(a + b) = \log a + \log b$

b) $\dfrac{\log 8}{\log 2} = \log 4$

c) $\log(a^n) = (\log a)^n$

d) $G_{\text{dB}} = 0$ means that the signal disappears completely.

e) $10^{\log x} = 10x$

f) $\ln 0 = 1$

PART 4

Mathematics in C

The exercise below is shown during the lesson, and you can also do it yourself. The program is written and run in **OnlineGDB**, `https://www.onlinegdb.com/online_c_compiler`, with the language set to **C**. Functions in C are described in Exercise 4.1 of Chapter 7.

Exercise 4.1 Decibels in C

ANSWER P. 257

The voltage gain in decibels is $G_{\text{dB}} = 20 \log_{10}\left(\frac{U_{\text{out}}}{U_{\text{in}}}\right)$, and conversely $\frac{U_{\text{out}}}{U_{\text{in}}} = 10^{G_{\text{dB}}/20}$.

a) Compute the gain in dB for the ratios $\frac{U_{\text{out}}}{U_{\text{in}}} = 1$, 10 and 100, and the ratio for -3 dB. Calculate by hand.

b) Write two functions:

- `double gain_db(double ratio)`, which returns $20 \log_{10}(\text{ratio})$. Use `log10()`.
- `double ratio_from_db(double db)`, which returns $10^{\text{db}/20}$. Use `pow()`.

Print a table with the ratios 0.5, 1, 2, 10, 100 and 1000, their gain in dB, and the ratio that `ratio_from_db()` gives back. Put the ratios in an array and go through it with a loop.

c) How many dB are added when the ratio is doubled, and when it is multiplied by ten? What happens when the ratio is less than 1?

CHAPTER 11

Review: vectors, functions, trigonometry and logarithms

Four chapters in one sweep, and a test to measure yourself against.

IN THIS CHAPTER

- Review of Chapters 7–10: vectors, functions, trigonometry, logarithms and decibels.
- Review exercises, one part for each chapter.
- Practice test 2, with the answers in the answer key.

This chapter has no new theory. Instead, it gathers up what was covered from Chapter 7 to Chapter 10: vectors and phasors, functions and their domains and ranges, sinusoidal AC voltages, and logarithms and decibels. All of this is used in the chapters that follow, when the derivative is introduced, so it pays to sort out now whatever still feels uncertain. Revise the theory in Chapters 7–10 before you start, and preferably do the review exercises on your own.

§11.1 states what you should be able to do after the chapter. §11.2 holds the review exercises, arranged in one part for each chapter, and §11.3 practice test 2, which is set out like a real test. Take it on your own, with pencil, paper and a calculator, and then compare with the answer key.

11.1 Summary

What you should be able to do

After this chapter, you should:

- Have consolidated and reviewed the vectors, functions, trigonometry and logarithms of Chapters 7–10 before the derivative.
- Have identified the gaps in your own knowledge in good time.

Reflect

1. Which topic from Chapters 7–10 did you feel least sure of during the practice test?
2. What will you revise before you go on to the next chapter?

Next chapter

Chapter 12 introduces the derivative: its definition, rate of change and the tangent.

11.2 Exercises

How to work with the exercises. The exercises review Chapters 7–10 before the practice test and are best done on your own, one part for each chapter. Start with the parts you feel least sure of.
Answers and worked solutions. The answer to every exercise is in the answer key at the back of the book, and its page is given next to the exercise. An answer shows whether you got it right, but not how. Full worked solutions, in Swedish, are published in the course repository after the lesson, in `lectures/L11/appendix/c_solutions.md`. Always try it yourself first.

PART 1
Vectors

These exercises review Chapter 7.

Exercise 1.1 Calculating with vectors

ANSWER P. 257

The vectors $\mathbf{u} = (2, 5)$ and $\mathbf{v} = (-3, 1)$ are given. Find:

a) $\mathbf{u} + \mathbf{v}$ **b)** $\mathbf{u} - \mathbf{v}$ **c)** $3\mathbf{u}$ **d)** $2\mathbf{u} - 3\mathbf{v}$

Exercise 1.2 Magnitude and angle

ANSWER P. 257

a) Calculate $|(8, 6)|$.

b) Calculate $|(-5, -12)|$.

c) Find the angle to the positive x-axis for $(8, 6)$.

d) Find the angle for $(-5, -12)$. Remember the quadrant correction.

Exercise 1.3 The angle between two vectors

ANSWER P. 257

a) Calculate the angles with the positive x-axis for $\mathbf{u} = (1, 2)$ and $\mathbf{v} = (3, 1)$.

b) Calculate the angle between $\mathbf{u}$ and $\mathbf{v}$ in a).

c) Are $\mathbf{u} = (3, 1)$ and $\mathbf{v} = (2, -6)$ perpendicular? Justify your answer with the angles of the vectors.

Exercise 1.4 Unit vector and a given length

ANSWER P. 257

The vector $\mathbf{v} = (-8, 6)$ is given.

a) Find the unit vector in the direction of $\mathbf{v}$.

b) Find a vector of length 25 in the same direction as $\mathbf{v}$.

c) Find a vector of length 5 in the opposite direction to $\mathbf{v}$.

Exercise 1.5 Voltages as phasors

ANSWER P. 257

The voltages across a resistor and a capacitor in series are $\mathbf{U}_R = (9, 0)$ V and $\mathbf{U}_C = (0, -12)$ V.

a) Calculate the total voltage $\mathbf{U} = \mathbf{U}_R + \mathbf{U}_C$.

b) Calculate $|\mathbf{U}|$.

c) Calculate the angle to the positive x-axis.

d) Why is $|\mathbf{U}|$ not equal to $9 + 12 = 21$ V?

PART 2

Functions

These exercises review Chapter 8.

Exercise 2.1 Domain and range

ANSWER P. 257

Find the domain D_f and the range V_f of:

a) $f(x) = \sqrt{x - 2}$ **b)** $f(x) = \dfrac{1}{x + 1}$ **c)** $f(x) = -x^2 + 4$

Exercise 2.2 A linear model of a thermistor

ANSWER P. 257

The resistance of a thermistor depends linearly on the temperature as $R(T) = kT + m$. Measurements give $R(0°\text{C}) = 200\,\Omega$ and $R(40°\text{C}) = 120\,\Omega$.

a) Find k and m.

b) Calculate R at 25°C.

c) At what temperature is $R = 100\,\Omega$?

d) Is the function increasing or decreasing?

Exercise 2.3 An exponential model

ANSWER P. 257

The number of faults in a system decreases by 15 % per month. At the start there are 80 faults.

a) Write a function $f(x)$ for the number of faults after x months.

b) Calculate the number of faults after 6 months.

c) After how many months are there fewer than 20 faults?

Exercise 2.4 Power as a power function

ANSWER P. 257

The power in a resistor is given by $P(I) = 25I^2$ watts for $0 \leq I \leq 2$ A.

a) Calculate $P(0.4)$.

b) Calculate $P(2)$.

c) Find V_P for the given interval.

d) At what current is $P = 36$ W?

Exercise 2.5 A rational function

ANSWER P. 258

The function $f(x) = \dfrac{x^2 - 9}{x - 3}$ is given.

a) Factorise the numerator.

b) Which values of x are not permitted?

c) Simplify the expression.

d) Find the zero of the function.

PART 3

Trigonometry

These exercises review Chapter 9.

Exercise 3.1 Angle measure

ANSWER P. 258

a) Convert 135° to radians.

b) Convert −120° to radians.

c) Convert $\frac{5\pi}{6}$ to degrees.

d) Convert 2 rad to degrees.

Exercise 3.2 Properties from the equation

ANSWER P. 258

An AC voltage is given by $u(t) = 6\sin\left(400\pi t - \frac{\pi}{3}\right)$ volts.

a) Give the amplitude.

b) Give the angular frequency and the frequency.

c) Give the period.

d) How many milliseconds does the phase correspond to, and is the curve early or late?

Exercise 3.3 Writing the equation

ANSWER P. 258

An AC voltage has the amplitude 12 V, the period $T = 2$ ms and the phase 30°. Find $u(t)$ in the form $|U|\sin(\omega t + \delta)$ with the phase in radians.

Exercise 3.4 Instantaneous value, RMS value and power

ANSWER P. 258

An AC voltage is given by $u(t) = 8\sin(100\pi t)$ volts. For a sinusoidal voltage, $U_{\text{RMS}} = \frac{|U|}{\sqrt{2}}$, and the power in a resistor is $P = \frac{U_{\text{RMS}}^2}{R}$.

a) Calculate $u(5\,\text{ms})$.

b) Calculate U_{RMS}.

c) The voltage is applied across a resistor $R = 50\,\Omega$. Calculate the power.

Exercise 3.5 A trigonometric equation

ANSWER P. 258

An AC voltage is given by $u(t) = 10\sin(200\pi t)$ volts.

a) At what time is $u(t) = 5$ V for the first time?

b) Give the next time at which $u(t) = 5$ V.

c) At what time is the peak value first reached?

PART 4

Logarithms and decibels

These exercises review Chapter 10.

Exercise 4.1 Exponential and logarithmic equations

ANSWER P. 258

Solve the equations.

a) $2^x = 128$ **b)** $10^x = 500$ **c)** $e^{2x} = 5$ **d)** $\log(x-1) = 2$

Exercise 4.2 The laws of logarithms

ANSWER P. 258

Calculate or simplify.

a) $\log 4 + \log 25$ **b)** $\log 500 - \log 5$ **c)** $3\log 2$ **d)** $\ln e^3$

Exercise 4.3 Gain in dB

ANSWER P. 258

An amplifier has $U_{\text{in}} = 20\,\text{mV}$ and $U_{\text{out}} = 2\,\text{V}$.

a) Calculate the gain in dB.

b) After the amplifier comes a filter that attenuates 12 dB. Calculate the total gain in dB.

c) Calculate the total linear gain.

d) Calculate the output voltage after the filter.

Exercise 4.4 Power in dB and dBm

ANSWER P. 258

The power level in dBm is given relative to 1 mW:

$$P_{\text{dBm}} = 10\log_{10}\left(\frac{P}{1\,\text{mW}}\right)$$

a) An amplifier has $P_{\text{in}} = 2\,\text{W}$ and $P_{\text{out}} = 50\,\text{W}$. Calculate the gain in dB.

b) A transmitter has the output power 20 dBm. Give the power in mW.

c) How many dBm does 1 W correspond to?

Exercise 4.5 Discharging a capacitor

ANSWER P. 258

The voltage is given by $u(t) = 12e^{-t/(RC)}$ volts with $R = 22\,\text{k}\Omega$ and $C = 100\,\mu\text{F}$.

a) Calculate the time constant $\tau = RC$.

b) Calculate the voltage after 3 seconds.

c) After how long has the voltage halved?

d) After how long is the voltage down to 1 V?

11.3 Practice test 2

The practice test covers Chapters 7–10: vectors, functions, trigonometry, logarithms and decibels. It is set out like a real test. In the lesson it is done question by question: you get a few minutes for each question, and then the teacher goes through it with the whole class. If you work on your own, take it in one sitting before you look in the answer key, where the answers are.

> **How the test works.** Permitted aids are writing materials, any calculator and one A4 sheet (both sides), handwritten or typed, with formulas, examples and notes of your own choice. The test has five questions worth 1.0 point each, so at most 5.0 points. Every question requires a worked solution with the answer stated, including the unit where needed. The real test is set out the same way: it does not affect your grade, but at least 3.5 of 5.0 points gives 0.5 bonus points on the regular exam.

Question 1

ANSWER P. 258 1.0 POINTS

You have two vectors $\mathbf{u} = (1, 2)$ and $\mathbf{v} = (-3, 4)$.

a) Draw the vectors in a coordinate system.

b) Calculate the magnitudes $|\mathbf{u}|$ and $|\mathbf{v}|$ of the vectors.

c) Calculate the angles v_u and v_v of the vectors.

d) Calculate a third vector $\mathbf{w} = 2\mathbf{u} - \mathbf{v}$.

Question 2

ANSWER P. 258 1.0 POINTS

Simplify the following rational expression and state which values of x are not permitted:

$$f(x) = \frac{x^2 - x - 2}{x - 2}$$

Question 3

ANSWER P. 258 1.0 POINTS

When a capacitor is discharged through a resistor, the voltage decreases exponentially with time. The voltage $u(t)$ can be described by

$$u(t) = U_0 e^{-t/(RC)},$$

where

- $u(t)$ is the voltage across the capacitor at the time t,
- U_0 is the initial voltage in V,
- RC is the circuit's time constant in seconds,
- t is the time in seconds.

A capacitor with the capacitance 470 μF is connected to a 10 kΩ resistor. It is initially charged to 12 V and starts to discharge at $t = 0$.

a) Calculate the voltage after five seconds.

b) Find the function's domain and range for the time interval $0 \leq t \leq 20$ s.

c) Calculate after how long the voltage has fallen to half its original value.

Question 4

ANSWER P. 258 1.0 POINTS

An AC voltage has the amplitude 5 V, the frequency 100 Hz and the phase 90°.

a) Find the equation $u(t)$ of the AC voltage. Give the phase in radians.

b) Draw the sine curve of the AC voltage over one period T.

Question 5

ANSWER P. 258 1.0 POINTS

An audio amplifier has a gain of 32 dB. By what factor does it amplify the voltage?

CHAPTER 12

Derivatives, part I

How fast a quantity changes, the slope of the tangent, and the peaks and valleys of a curve.

IN THIS CHAPTER

- Rate of change and the tangent.
- The definition of the derivative.
- The derivatives of power and polynomial functions.
- Zeros of the derivative, maximum and minimum points.

The derivative describes how fast a quantity changes. In electrical engineering it turns up directly in circuit theory:

$$i_C(t) = C\frac{du_C}{dt}, \qquad u_L(t) = L\frac{di_L}{dt}$$

The current through a capacitor depends on how fast the voltage across it changes, and the voltage across an inductor depends on how fast the current through it changes. The derivative is therefore not abstract mathematics, but the tool that describes transients in RC and RL circuits. This chapter builds up the concept from the difference quotient to the slope of the tangent, gives the rules of differentiation for polynomials and shows how the derivative finds the maximum and minimum points of a function. A short introduction to the derivative as a concept is in the video https://www.youtube.com/watch?v=N2PpRnFqnqY.

§12.1 to §12.6 are the chapter's theory, with examples. §12.7 summarises the rules and what you should be able to do after the chapter, and §12.8 holds the exercises.

12.1 Rate of change and the difference quotient

The **difference quotient** (the average rate of change) of $f(x)$ over the interval $[x, x+h]$ is

$$\frac{f(x+h)-f(x)}{h}.$$

Geometrically, this is the slope of the **secant** through the points $(x, f(x))$ and $(x+h, f(x+h))$.

12.2 The definition of the derivative

The derivative $f'(x)$ is the limit of the difference quotient as $h \to 0$:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Geometrically, $f'(x)$ is the **slope of the tangent** to the curve $y = f(x)$ at the point x. The derivative is also called the **instantaneous rate of change**.

12.3 Rules of differentiation for polynomials

Function	Derivative
$f(x) = C$ (constant)	$f'(x) = 0$
$f(x) = x^n$	$f'(x) = nx^{n-1}$
$f(x) = Cx^n$	$f'(x) = Cnx^{n-1}$
$f(x) = u(x) + v(x)$	$f'(x) = u'(x) + v'(x)$

EXAMPLE

$$f(x) = 3x^4 - 2x^2 + 5 \quad \Rightarrow \quad f'(x) = 12x^3 - 4x$$

12.4 The equation of the tangent

The tangent to $y = f(x)$ at the point $(x_0, f(x_0))$ has the equation

$$y = f'(x_0)(x - x_0) + f(x_0).$$

12.5 Turning points – maximum and minimum

A **stationary point** satisfies $f'(x_0) = 0$. Whether it is a maximum or a minimum is decided with the second-derivative test:

- $f''(x_0) > 0$: **minimum** (the curve is concave up),
- $f''(x_0) < 0$: **maximum** (the curve is concave down).

12.6 Worked examples

EXAMPLE **Differentiating polynomial functions.** Differentiate

$$\textbf{a)}\quad f(x) = -2x^2 + 2x + 4,$$
$$\textbf{b)}\quad f(x) = 3x^3 - 6x^2 + \frac{3x}{4} - 5,$$
$$\textbf{c)}\quad f(x) = -x^4 + x^3 + \frac{2x^2}{3} - 3x + 2.$$

The rule for Cx^n is applied term by term:

$$\textbf{a)}\quad f'(x) = -4x + 2$$
$$\textbf{b)}\quad f'(x) = 9x^2 - 12x + \frac{3}{4}$$
$$\textbf{c)}\quad f'(x) = -4x^3 + 3x^2 + \frac{4x}{3} - 3$$

EXAMPLE **Analysing a parabola.** The function is $f(x) = -x^2 + 6x - 5$.
a) Find $f'(x)$.

$$f'(x) = -2x + 6$$

b) Solve $f'(x) = 0$.

$$-2x + 6 = 0 \quad \Rightarrow \quad x = 3$$

c) Determine the type with $f''(x)$.

$$f''(x) = -2 < 0 \quad \Rightarrow \quad \text{maximum}$$

d) Calculate the maximum value.

$$f(3) = -9 + 18 - 5 = 4$$

The maximum point is $(3, 4)$.

EXAMPLE **Current in an RC circuit.** The current is approximated by $i(t) = -0.4t^3 + 2.4t^2 - 3t + 1.2$ amperes, with the time t in seconds.
a) Calculate $i'(t)$.

$$i'(t) = -1.2t^2 + 4.8t - 3$$

b) Solve $i'(t) = 0$ with the PQ-formula.

$$t^2 - 4t + 2.5 = 0 \quad \Rightarrow \quad t = 2 \pm \sqrt{1.5}$$

$$t_1 \approx 0.78\,\text{s}, \qquad t_2 \approx 3.22\,\text{s}$$

c) The second derivative is $i''(t) = -2.4t + 4.8$, and

$$\begin{aligned} i''(0.78) &\approx 2.9 > 0 \quad &\Rightarrow \quad &\text{minimum at } t_1, \\ i''(3.22) &\approx -2.9 < 0 \quad &\Rightarrow \quad &\text{maximum at } t_2. \end{aligned}$$

d) The current at the stationary points is

$$i(0.78) \approx 0.13\,\text{A (minimum)}, \qquad i(3.22) \approx 3.07\,\text{A (maximum)}.$$

12.7 Summary

Concept	Formula or rule
Derivative of x^n	nx^{n-1}
Stationary point	$f'(x_0) = 0$
Maximum	$f'(x_0) = 0$ and $f''(x_0) < 0$
Minimum	$f'(x_0) = 0$ and $f''(x_0) > 0$
Equation of the tangent	$y = f'(x_0)(x - x_0) + f(x_0)$

What you should be able to do

After this chapter, you should be able to:

- Understand how the derivative relates to the tangent, the equation of a straight line, and maximum and minimum points.
- Differentiate polynomial functions.

Test yourself

ANSWER P. 259

1. Differentiate $f(x) = 4x^3 - 2x^2 + 7$.
2. Find the coordinates of the turning point of $f(x) = x^2 - 2x - 8$ and decide whether it is a maximum or a minimum.

Next chapter

Chapter 13 is about the derivatives of trigonometric, exponential and logarithmic functions.

12.8 Exercises

> **How to work with the exercises.** Part 1 is done during the lesson: first on your own, a few minutes per exercise, then together. Part 2 is extra exercises for further practice after the lesson; most of them can be done in your head or with pencil and paper.
> **Answers and worked solutions.** The answer to every exercise is in the answer key at the back of the book, and its page is given next to the exercise. An answer shows whether you got it right, but not how. Full worked solutions, in Swedish, are published in the course repository after the lesson, in `lectures/L12/appendix/c_solutions.md`. Always try it yourself first.

PART 1

Lesson exercises

Exercise 1.1 Differentiating functions

ANSWER P. 258

Differentiate the following functions.

a) $f(x) = -3x^2 + 5x + 2$

b) $f(x) = 2x^3 - 5x^2 + \frac{2x}{3} - 4$

c) $f(x) = 2x^4 - x^3 + \frac{3x^2}{5} + 4x - 1$

Exercise 1.2 Analysing a parabola

ANSWER P. 258

Consider the function

$$f(x) = -x^2 + 8x - 7.$$

a) Differentiate the function $f(x)$, that is, find the expression for $f'(x)$.

b) Find where the function is stationary, that is, solve the equation $f'(x) = 0$.

c) Use the second derivative to decide whether the point is a **maximum** or a **minimum**.

d) Calculate the greatest or least value of the function.

e) Plot the graph of $f(x)$ in GeoGebra, `https://www.geogebra.org/graphing?lang=en`, and check that your answers are right.

Exercise 1.3 Analysing the current in an RC circuit

ANSWER P. 258

The current through a certain RC circuit can be approximated by the polynomial function

$$i(t) = -0.2t^3 + 1.8t^2 - 3.6t + 2.5,$$

where $i(t)$ is the current through the circuit in A and t the time in seconds.

a) Differentiate the function to find the expression for the change in the current (the rate of change of the current) $i'(t)$.

b) Calculate the time (or times) at which the current is stationary, that is, when $i'(t) = 0$.

c) Use the **second derivative** to decide whether each stationary point is a **maximum** or a **minimum**.

d) Calculate the maximum and minimum values of the current, that is, the current $i(t)$ at the stationary points.

e) Plot the graph of $i(t)$ in GeoGebra, https://www.geogebra.org/graphing?lang=en, and check that your answers are right.

PART 2

Extra exercises

The exercises below are extra practice, best done on your own after the lesson.

Exercise 2.1 Differentiating polynomials

ANSWER P. 258

Differentiate the following functions.

a) $f(x) = x^5$ **b)** $f(x) = 4x^3$ **c)** $f(x) = 7$

d) $f(x) = 2x^2 - 5x + 1$ **e)** $f(x) = -3x^4 + x$ **f)** $f(x) = \dfrac{x^3}{3}$

Exercise 2.2 Difference quotient and the definition of the derivative

ANSWER P. 258

The function $f(x) = x^2$ is given.

a) Set up the difference quotient over the interval $[2, 2 + h]$ and simplify it.

b) What value does the quotient approach as $h \to 0$?

c) Check your answer with the rule $f'(x) = 2x$.

d) Calculate the difference quotient for $h = 0.1$ and $h = 0.01$.

Exercise 2.3 The derivative at a point

ANSWER P. 258

The function $f(x) = x^3 - 3x$ is given.

a) Find $f'(x)$. **b)** Calculate $f'(0)$. **c)** Calculate $f'(2)$.

d) At which points is the tangent horizontal?

Exercise 2.4 The equation of the tangent

ANSWER P. 258

The function $f(x) = x^2 - 4x + 5$ is given. Consider the point where $x_0 = 3$.

a) Calculate $f(3)$. **b)** Calculate $f'(3)$.

c) Find the equation of the tangent. **d)** Where does the tangent cross the x-axis?

Exercise 2.5 Stationary point of a parabola

ANSWER P. 259

The function $f(x) = x^2 - 6x + 8$ is given.

a) Find $f'(x)$.

b) Solve $f'(x) = 0$.

c) Use $f''(x)$ to decide whether the point is a maximum or a minimum.

d) Calculate the extreme value of the function.

Exercise 2.6 Stationary points of a cubic function

ANSWER P. 259

The function $f(x) = x^3 - 3x^2 - 9x + 5$ is given.

a) Find $f'(x)$ and factorise the expression.

b) Find the x-values of the stationary points.

c) Use $f''(x)$ to decide which point is the maximum and which the minimum.

d) Calculate the function values at both points.

Exercise 2.7 Increasing and decreasing

ANSWER P. 259

The function $f(x) = x^2 - 4x$ is given.

a) Find $f'(x)$. **b)** For which x is the function increasing?

c) For which x is it decreasing? **d)** What happens at $x = 2$?

Exercise 2.8 The second derivative

ANSWER P. 259

Find $f'(x)$ and $f''(x)$.

a) $f(x) = x^4 - 2x^2$ **b)** $f(x) = 5x^3 + 2x$ **c)** $f(x) = -x^2 + 7x - 1$

d) What does the sign of $f''(x)$ say about the shape of the curve?

Exercise 2.9 Current through a capacitor

ANSWER P. 259

The current through a capacitor is given by $i_C(t) = C\dfrac{du}{dt}$. A capacitor with $C = 100\,\mu\text{F}$ has the voltage $u(t) = 4t^2 + 2t$ volts.

a) Find $u'(t)$.

b) Find an expression for $i_C(t)$ in mA.

c) Calculate the current at $t = 1\,\text{s}$.

d) At what time is the current 2 mA?

Exercise 2.10 Voltage across an inductor

ANSWER P. 259

The voltage across an inductor is given by $u_L(t) = L\dfrac{di}{dt}$. An inductor with $L = 0.5\,\text{H}$ carries the current $i(t) = 2t^2 - 4t + 3$ amperes.

a) Find $i'(t)$.

b) Find an expression for $u_L(t)$.

c) At what time is $u_L = 0$?

d) What happens to the current at this time?

Exercise 2.11 Maximum power

ANSWER P. 259

The power that a generator delivers to a load varies with the current as $P(I) = 24I - 3I^2$ watts.

a) Find $P'(I)$.

b) At what current is the power stationary?

c) Use $P''(I)$ to decide whether it is a maximum or a minimum.

d) Calculate the maximum power.

Exercise 2.12 Rate of change of a signal

ANSWER P. 259

A voltage is given by $u(t) = -2t^2 + 12t$ volts for $0 \leq t \leq 6\,\text{s}$.

a) Find $u'(t)$.

b) At what time is the voltage greatest, and how large is it then?

c) How fast is the voltage changing at $t = 1\,\text{s}$?

d) At what time is the voltage decreasing by $4\,\text{V/s}$?

Exercise 2.13 Interpreting the sign of the derivative

ANSWER P. 259

For a function f, $f'(1) = 3$, $f'(2) = 0$, $f'(3) = -2$ and $f''(2) = -5$.

a) Is f increasing or decreasing at $x = 1$?

b) Is f increasing or decreasing at $x = 3$?

c) What type of point is $x = 2$?

d) Describe in words what the graph looks like around $x = 2$.

Exercise 2.14 Find the error

ANSWER P. 259

Each line contains a common differentiation error. Explain the error and give the correct answer.

a) $f(x) = x^3 \quad \Rightarrow \quad f'(x) = 3x^2 \cdot x$

b) $f(x) = 5 \quad \Rightarrow \quad f'(x) = 5$

c) $f(x) = 4x \quad \Rightarrow \quad f'(x) = 4x$

d) $f(x) = 3x^2 + 2x \quad \Rightarrow \quad f'(x) = 6x + 2x$

e) If $f'(x_0) = 0$, then x_0 is always a maximum point.

f) $f''(x)$ is found by squaring $f'(x)$.

PART 3

Mathematics in C

The teacher shows the exercise below during the lesson, in VS Code and with `make`. You can also do it yourself in **OnlineGDB**, `https://www.onlinegdb.com/online_c_compiler`, with the language set to **C**.

Exercise 3.1 The definition of the derivative in C

ANSWER P. 259

The derivative $f'(x)$ is the limit of the slope between the points x and $x + h$ as h tends to zero:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

Let $f(x) = x^2$.

a) Find $f'(3)$ with the rules of differentiation.

b) Write a program that computes the slope $\frac{f(3+h) - f(3)}{h}$ for $h = 1;\ 0.1;\ 0.01;\ \ldots;\ 10^{-6}$ and prints h and the slope on each line. Write $f(x)$ as a function of its own, `double f(double x)`, and let h start at `1.0` and be divided by `10.0` in each pass of a loop.

c) How does the slope change as h decreases? Show algebraically that the slope is exactly $6 + h$.

CHAPTER 13

Derivatives, part II

The derivatives of sine, cosine, exponential functions and logarithms.

IN THIS CHAPTER

- The derivatives of trigonometric functions.
- The derivatives of exponential functions.
- The derivatives of logarithmic functions.
- The product rule and the chain rule.

In Chapter 12 we differentiated polynomials. Many quantities in electrical engineering are not polynomials, though: alternating currents and voltages are sinusoidal, a capacitor discharges exponentially and gain is measured with logarithms. This chapter gives the derivatives of trigonometric, exponential and logarithmic functions, and the product rule and the chain rule, which are needed when functions are multiplied together or substituted into each other. With them you can, for example, calculate the current through a capacitor when the voltage across it is sinusoidal, or how fast the voltage falls as the capacitor discharges.

§13.1 to §13.6 are the chapter's theory, with examples. §13.7 summarises the rules and what you should be able to do after the chapter, and §13.8 holds the exercises. Part 1 of them reviews the analysis of polynomials from Chapter 12.

13.1 Derivatives of trigonometric functions

Function	Derivative
$\sin(x)$	$\cos(x)$
$\cos(x)$	$-\sin(x)$
$\sin(kx)$	$k\cos(kx)$
$\cos(kx)$	$-k\sin(kx)$

EXAMPLE

$$f(x) = 3\sin(2x) \quad \Rightarrow \quad f'(x) = 3 \cdot 2\cos(2x) = 6\cos(2x)$$

13.2 Derivatives of exponential functions

Function	Derivative
e^x	e^x
e^{kx}	ke^{kx}
a^x	$a^x \ln a$

Derivation of $\frac{d}{dx}e^x = e^x$. With the definition of the derivative from §12.2,

$$\lim_{h\to 0} \frac{e^{x+h} - e^x}{h} = e^x \lim_{h\to 0} \frac{e^h - 1}{h} = e^x \cdot 1 = e^x.$$

Why e? Every exponential function has a derivative that is proportional to the function itself, $\frac{d}{dx}a^x = a^x \ln a$ (§13.4.2), but the factor $\ln a$ depends on the base:

Base a	Factor $\ln a$
2	≈ 0.69
$e \approx 2.718$	1
10	≈ 2.30

The number e is the only base for which the factor is exactly 1. That is precisely what the limit $\lim_{h\to 0} \frac{e^h-1}{h} = 1$ in the derivation above says: the graph $y = e^x$ has slope 1 where it crosses the y-axis. This is how the number e is defined.

For the same reason, ln is called the *natural* logarithm: with base e the derivative is the simplest, $\frac{d}{dx}\ln x = \frac{1}{x}$, while every other base adds a constant, for example $\frac{d}{dx}\log_{10} x = \frac{1}{x \ln 10}$ (§13.3).

In electrical engineering, e appears because many quantities change in step with their own value. When a capacitor discharges through a resistor, $\frac{du}{dt} = -\frac{u}{\tau}$, and the function that satisfies this is $u(t) = U_0 e^{-t/\tau}$. After the time τ, the voltage has fallen to $e^{-1} \approx 37\,\%$ of its initial value.

13.3 Derivatives of logarithmic functions

Function	Derivative
$\ln(x)$	$\dfrac{1}{x}$
$\ln(kx)$	$\dfrac{1}{x}$
$\log_{10}(x)$	$\dfrac{1}{x \ln 10}$

Note: $\ln(kx) = \ln k + \ln x$, whose derivative is $1/x$: the constant $\ln k$ disappears.

Why $\ln 10$? Every logarithm can be rewritten with the natural logarithm: $\log_{10} x = \dfrac{\ln x}{\ln 10}$. Since $\ln 10 \approx 2.303$ is a constant,

$$\frac{d}{dx} \log_{10} x = \frac{1}{\ln 10} \cdot \frac{d}{dx} \ln x = \frac{1}{\ln 10} \cdot \frac{1}{x} = \frac{1}{x \ln 10}.$$

In the same way, $\dfrac{d}{dx} \log_a x = \dfrac{1}{x \ln a}$ for every base $a > 0$, $a \neq 1$.

13.4 The product rule and the chain rule

The tables above give the derivative of a single function. When two functions are multiplied together, or when one function is substituted into another, two more rules are needed.

13.4.1 The product rule

The derivative of a product $f(x) = g(x) \cdot h(x)$ is

$$f'(x) = g'(x) \cdot h(x) + g(x) \cdot h'(x).$$

So differentiate one factor at a time while the other stays as it is, and add the two terms.

EXAMPLE $f(x) = x^2 \sin(x)$ with $g(x) = x^2$ and $h(x) = \sin(x)$ gives

$$f'(x) = 2x \cdot \sin(x) + x^2 \cdot \cos(x).$$

13.4.2 The chain rule

A composite function $f(x) = g(u(x))$ consists of an **outer function** g and an **inner function** $u(x)$. Its derivative is the derivative of the outer function, evaluated at the inner function, times the derivative of the inner function:

$$f'(x) = g'(u(x)) \cdot u'(x).$$

EXAMPLE $f(x) = \ln(2x+1)$ has the outer function $\ln u$ and the inner function $u = 2x+1$, with $u' = 2$:

$$f'(x) = \frac{1}{2x+1} \cdot 2 = \frac{2}{2x+1}.$$

The table rows for $\sin(kx)$, $\cos(kx)$, e^{kx} and $\ln(kx)$ above are special cases of the chain rule, with the inner function $u = kx$ and $u' = k$. For example,

$$\frac{d}{dx}\sin(kx) = \cos(kx) \cdot k = k\cos(kx), \qquad \frac{d}{dx}\ln(kx) = \frac{1}{kx} \cdot k = \frac{1}{x}.$$

EXAMPLE **The derivative of a^x.** Since $a = e^{\ln a}$, a^x can be written as $a^x = e^{x\ln a}$. The outer function is e^u and the inner function $u = x\ln a$, with $u' = \ln a$ since $\ln a$ is a constant:

$$\frac{d}{dx}a^x = e^{x\ln a} \cdot \ln a = a^x \ln a.$$

The factor $\ln a$ in §13.2 therefore comes from the chain rule. For $a = e$ it is $\ln e = 1$.

EXAMPLE **The product rule and the chain rule together.** A quotient such as $f(x) = \dfrac{2x}{x+5}$ can be written as a product, $f(x) = 2x \cdot (x+5)^{-1}$. The factor $(x+5)^{-1}$ is differentiated with the chain rule, with the outer function u^{-1} and the inner function $u = x+5$, where $u' = 1$:

$$\frac{d}{dx}(x+5)^{-1} = -(x+5)^{-2} \cdot 1 = -\frac{1}{(x+5)^2}.$$

The product rule then gives

$$f'(x) = 2 \cdot \frac{1}{x+5} + 2x \cdot \left(-\frac{1}{(x+5)^2}\right) = \frac{2(x+5) - 2x}{(x+5)^2} = \frac{10}{(x+5)^2}.$$

13.5 Table of derivatives

Function $f(x)$	Derivative $f'(x)$
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$\ln x$	$1/x$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\sin(kx)$	$k\cos(kx)$
$\cos(kx)$	$-k\sin(kx)$

13.6 Worked examples

EXAMPLE **Derivatives of common functions.** Differentiate **a)** $f(x) = 5e^{3x}$, **b)** $f(x) = \ln(4x)$ and **c)** $f(x) = 2\cos(x) - 3\sin(x)$.

a) $f'(x) = 5 \cdot 3e^{3x} = 15e^{3x}$

b) $f'(x) = 1/x$

c) $f'(x) = -2\sin(x) - 3\cos(x)$

EXAMPLE **The derivative at a point.** The current $i(t) = I_0 e^{-t/\tau}$ amperes describes the discharge of a capacitor. Calculate $i'(0)$.

$$i'(t) = -\frac{I_0}{\tau} e^{-t/\tau} \quad \Rightarrow \quad i'(0) = -\frac{I_0}{\tau}$$

$i'(0)$ is the instantaneous rate of change of the current at $t = 0$.

EXAMPLE **Stationary point of an exponential function.** Find the turning point of $f(x) = xe^{-x}$. The product rule (§13.4.1) with $g(x) = x$ and $h(x) = e^{-x}$ gives

$$f'(x) = e^{-x} + x \cdot (-e^{-x}) = e^{-x}(1 - x)$$

$f'(x) = 0$ gives $x = 1$, since $e^{-x} \neq 0$. The second derivative is

$$f''(x) = -e^{-x}(1 - x) + e^{-x}(-1) = e^{-x}(x - 2),$$

and

$$f''(1) = e^{-1}(1-2) = -e^{-1} < 0 \quad \Rightarrow \quad \textbf{maximum at } (1, e^{-1}) \approx (1, 0.37).$$

13.7 Summary

Function	Derivative
e^{kx}	ke^{kx}
$\ln x$	$1/x$
$\sin(kx)$	$k\cos(kx)$
$\cos(kx)$	$-k\sin(kx)$
$g(x) \cdot h(x)$	$g'(x) \cdot h(x) + g(x) \cdot h'(x)$ (the product rule)
$g(u(x))$	$g'(u(x)) \cdot u'(x)$ (the chain rule)

What you should be able to do

After this chapter, you should be able to:

- Know the derivatives of common functions.
- Use the product rule and the chain rule.

Test yourself

ANSWER P. 260

1. Differentiate $f(x) = e^{2x} + \ln(3x)$.
2. Calculate $f'(\pi/2)$ for $f(x) = \sin(x)$.

Next chapter

Chapter 14 is about integrals.

13.8 Exercises

How to work with the exercises. Parts 1 and 2 are done during the lesson: first on your own, a few minutes per exercise, then together. Part 3 is extra exercises for further practice after the lesson; most of them can be done in your head or with pencil and paper.
Answers and worked solutions. The answer to every exercise is in the answer key at the back of the book, and its page is given next to the exercise. An answer shows whether you got it right, but not how. Full worked solutions, in Swedish, are published in the course repository after the lesson, in `lectures/L13/appendix/c_solutions.md`. Always try it yourself first.

PART 1

Review exercises

Exercise 1.1 Analysing a cubic function

ANSWER P. 259

Consider the function

$$f(x) = -0.5x^3 + 1.5x^2 + 4.5x - 3.$$

a) Differentiate the function $f(x)$ and find the expression for $f'(x)$.

b) Find where the function is stationary, that is, solve the equation $f'(x) = 0$.

c) Use the second derivative to decide whether each stationary point is a maximum or a minimum.

d) Calculate the maximum and minimum values of the function, that is, the function values at the stationary points.

e) Plot the graph of $f(x)$ in GeoGebra, https://www.geogebra.org/graphing?lang=en, and check that your answers are right.

Exercise 1.2 Analysing the voltage across a capacitor

ANSWER P. 259

A capacitor in an RC circuit is charged and discharged repeatedly. The voltage across the capacitor can be described by the function

$$u(t) = -0.2t^3 + 1.2t^2 - 1.4625t + 3,$$

where $u(t)$ is the voltage across the capacitor in V and t the time in seconds, $0 \leq t \leq 6$.

a) Differentiate the function and find the expression for the rate of change of the voltage $u'(t)$.

b) Find the time (or times) at which the voltage is stationary, that is, when $u'(t) = 0$.

c) Use the second derivative to decide whether the points are maximum or minimum points.

d) Calculate the maximum and minimum values of the voltage, that is, the voltage at the stationary points.

e) Plot the graph of $u(t)$ in GeoGebra, https://www.geogebra.org/graphing?lang=en, and check that your answers are right.

PART 2
New material

Exercise 2.1 Current through an inductor

ANSWER P. 259

The current through an inductor varies over time according to the function

$$i(t) = 4\sin(50t),$$

where $i(t)$ is the current through the inductor in A and t the time in seconds.

a) Differentiate the function and find the expression for $i'(t)$.

b) Find the time (or times) at which the current is stationary, that is, when $i'(t) = 0$.

c) Use the second derivative to decide whether the points are maximum or minimum points.

d) Calculate the greatest and least values of the current. Consider whether this matches what you expected.

Exercise 2.2 Discharging a capacitor

ANSWER P. 260

The voltage across a capacitor that is discharging is given by the function

$$u(t) = 12e^{-0.4t},$$

where $u(t)$ is the voltage across the capacitor in V and t the time in seconds.

a) Differentiate the function and find the expression for $u'(t)$.

b) Calculate the rate of change of the voltage at $t = 2\,\text{s}$.

c) State whether the rate of change of the voltage is increasing or decreasing at this point, that is, find $u''(2)$.

Exercise 2.3 Gain in dB

ANSWER P. 260

The logarithmic gain of an amplifier system is described by the function

$$G(x) = 10\log_{10}(2x + 1),$$

where $G(x)$ is the gain in dB and x the input level (a dimensionless factor).

a) Calculate the value of x for which the gain is $G(x) = 2\,\text{dB}$.

b) Calculate the rate of change of the gain $G'(x)$ at this point.

> **Note:** The rate of change $G'(x)$ states how fast the gain changes when the input x changes.

PART 3

Extra exercises

The exercises below are extra practice, best done on your own after the lesson.

Exercise 3.1 Differentiating trigonometric functions

ANSWER P. 260

Differentiate.

a) $f(x) = \sin(3x)$ **b)** $f(x) = \cos(5x)$ **c)** $f(x) = 4\sin(x)$

d) $f(x) = -2\cos(2x)$ **e)** $f(x) = \sin(x) + \cos(x)$

Exercise 3.2 Differentiating exponential functions

ANSWER P. 260

Differentiate.

a) $f(x) = e^{5x}$ **b)** $f(x) = 3e^{-2x}$ **c)** $f(x) = e^x + x^2$

d) $f(x) = 2^x$ **e)** $f(x) = 10e^{-x/4}$

Exercise 3.3 Differentiating logarithmic functions

ANSWER P. 260

Differentiate.

a) $f(x) = \ln x$ **b)** $f(x) = \ln(7x)$ **c)** $f(x) = 3\ln x$

d) $f(x) = \log_{10} x$ **e)** $f(x) = \ln x + e^x$

Exercise 3.4 Mixed differentiation

ANSWER P. 260

Differentiate.

a) $f(x) = 2e^{3x} - \sin(4x)$ **b)** $f(x) = x^2 + \ln(2x)$

c) $f(x) = 5\cos(x) - 3e^{-x}$ **d)** $f(x) = 4\sin(2x) + 4\cos(2x)$

Exercise 3.5 The derivative at a point

ANSWER P. 260

Calculate the derivative at the given point.

a) $f(x) = e^{2x}$ at $x = 0$ **b)** $f(x) = \sin x$ at $x = \dfrac{\pi}{2}$ **c)** $f(x) = \ln x$ at $x = 2$

d) $f(x) = \cos(3x)$ at $x = 0$ **e)** $f(x) = 3e^{-x}$ at $x = 1$

Exercise 3.6 Current through a capacitor

ANSWER P. 260

The voltage across a capacitor with $C = 47\,\mu\text{F}$ is given by $u(t) = 10\sin(100\pi t)$ volts. The current is given by $i_C(t) = C\dfrac{du}{dt}$.

a) Find $u'(t)$.

b) Find an expression for $i_C(t)$.

c) What is the amplitude of the current?

d) At what time is the current greatest for the first time?

Exercise 3.7 Voltage across an inductor

ANSWER P. 260

The current through an inductor with $L = 25\,\text{mH}$ is given by $i(t) = 2\sin(200t)$ amperes. The voltage is given by $u_L(t) = L\dfrac{di}{dt}$.

a) Find $i'(t)$.

b) Find an expression for $u_L(t)$.

c) What is the amplitude of the voltage?

d) What is the phase difference between the voltage and the current?

Exercise 3.8 Discharging a capacitor

ANSWER P. 260

The voltage is given by $u(t) = 20e^{-t/0.5}$ volts.

a) Find $u'(t)$.

b) Calculate $u'(0)$.

c) Calculate $u'(0.5)$.

d) What does the sign of the derivative tell you?

e) Show that $u'(t) = -\dfrac{u(t)}{\tau}$ with $\tau = 0.5\,\text{s}$.

Exercise 3.9 Stationary point of an exponential function

ANSWER P. 260

The function $f(x) = xe^{-2x}$ is given.

a) Find $f'(x)$ with the product rule and factor out e^{-2x}.

b) Solve $f'(x) = 0$.

c) Use $f''(x)$ to decide whether the point is a maximum or a minimum.

d) Calculate the extreme value of the function.

Exercise 3.10 Maximum power over time

ANSWER P. 260

The power in a component is given by $p(t) = 100te^{-t}$ watts for $t \geq 0$.

a) Find $p'(t)$.

b) At what time is the power stationary?

c) Use $p''(t)$ to decide whether it is a maximum or a minimum.

d) Calculate the maximum power.

Exercise 3.11 The derivative of a dB function

ANSWER P. 260

The gain is given by $G(x) = 20\log_{10}(x)$ dB.

a) Find $G'(x)$. **b)** Calculate $G'(1)$. **c)** Calculate $G'(10)$.

d) Interpret the difference between the answers to **b)** and **c)**.

Exercise 3.12 Tangent to an exponential curve

ANSWER P. 260

The function $f(x) = e^{-x}$ is given. Consider the point where $x_0 = 0$.

a) Calculate $f(0)$.

b) Calculate $f'(0)$.

c) Find the equation of the tangent.

d) Where does the tangent cross the x-axis? Compare with the time constant τ of a discharge.

Exercise 3.13 The derivative of an exponential discharge

ANSWER P. 260

The voltage is given by $u(t) = U_0 e^{-t/\tau}$.

a) Show that $u'(t) = -\dfrac{u(t)}{\tau}$.

b) What does this relationship mean physically?

c) Calculate $u'(0)$ and $u'(\tau)$ when $U_0 = 10\,\text{V}$ and $\tau = 2\,\text{s}$.

d) What fraction of the initial voltage remains after the time τ?

Exercise 3.14 True or false

ANSWER P. 260

Decide whether each statement is true or false, and justify your answer briefly.

a) The derivative of $\sin x$ is $-\cos x$.

b) The derivative of e^{3x} is e^{3x}.

c) The derivative of $\ln(5x)$ is $\dfrac{1}{5x}$.

d) The derivative of $\cos(2x)$ is $-2\sin(2x)$.

e) e^x is its own derivative.

f) The derivative of a constant times a function is the constant times the derivative of the function.

Exercise 3.15 Find the error

ANSWER P. 260

Each line contains a common error. Explain the error and give the correct answer.

a) $\frac{d}{dx}\sin(3x) = \cos(3x)$

b) $\frac{d}{dx}e^{-2x} = -2e^{-2x-1}$

c) $\frac{d}{dx}\ln(3x) = \frac{1}{3x}$

d) $\frac{d}{dx}4\cos(x) = 4\sin(x)$

e) $\frac{d}{dx}2^x = x \cdot 2^{x-1}$

PART 4

Mathematics in C

The teacher shows the exercise below during the lesson, in VS Code and with `make`. You can also do it yourself in **OnlineGDB**, `https://www.onlinegdb.com/online_c_compiler`, with the language set to **C**.

Exercise 4.1 The derivative of e^x and $\sin v$ in C

ANSWER P. 260

The derivative can be computed numerically with

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}, \qquad h = 10^{-6}.$$

Here a single derivative function is written that works for any function. This is possible because a function can be passed as an argument to another function, using a **function pointer**.

a) Include `math.h` and `stdio.h` and define the macro `DERIVATIVE_H` with the value `1e-6`. Then define the type `math_func_t`, a pointer to a function that takes a `double` and returns a `double`, with `typedef double (*math_func_t)(double);`. Write the function `double derive(math_func_t f, double x)`, which returns the approximation above for the function `f()` at the point `x`.

b) Write the function `double e_x(double x)`, which returns e^x using `pow(M_E, x)`. Then let x run from 0 to 4 in a loop, call `derive(e_x, x)` and print x, the slope and e^x on each line. What does the output say about the derivative of e^x?

c) `sin()` in `math.h` also takes a `double` and returns a `double`, so it can be passed directly to `derive()`, without a function of your own: `derive(sin, v)`. Let v run from -1 to 1 radian in steps of 0.2, and print v, the slope, $\sin v$ and $\cos v$ with two decimals on

each line. Let an integer variable `i` run from -5 to 5 and compute `v = 0.2 * i`, so that rounding errors do not build up in the loop. Which of $\sin v$ and $\cos v$ matches the slope?

d) `pow()` takes two arguments and therefore does not match `math_func_t`, so it cannot be passed directly to `derive()`. Write the function `double two_x(double x)`, which returns 2^x using `pow(2.0, x)`. Let x run from 0 to 4, call `derive(two_x, x)` and print x, the slope, 2^x, the ratio of the slope to 2^x and $\ln 2$ using `log(2.0)` on each line. What happens to the ratio when x changes, and what separates 2^x from e^x? What had to change in `derive()` to differentiate this function?

e) `log10()` in `math.h` gives the common logarithm $\log_{10} x$ and matches `math_func_t`, so it can be passed directly to `derive()`, like `sin()`. Note that `log()` gives the natural logarithm $\ln x$. Let x run from 1 to 5, call `derive(log10, x)` and print x, the slope and $\dfrac{1}{x \ln 10}$, computed with `1.0 / (x * log(10.0))`, on each line. Does the slope match $f'(x) = \dfrac{1}{x \ln 10}$? Explain the factor $\ln 10$ using $\log_{10} x = \dfrac{\ln x}{\ln 10}$. Why can x not start at 0?

CHAPTER 14

Integrals

The derivative in reverse: antiderivatives, the area under a curve, and the charge that a current carries.

IN THIS CHAPTER

- What an integral is and how it is calculated.
- Antiderivatives and the constant of integration.
- Rules of integration for the most common functions.
- Definite integrals and area.
- Charge from current and energy from power.

In Chapter 12 and Chapter 13 we differentiated functions to find how fast they change. This chapter goes the other way: given a function, we look for a function whose derivative it is, an antiderivative. The chapter goes through the rules of integration for the most common functions, how the definite integral gives the area under a curve, and how the constant of integration is found from an initial condition. The application in electrical engineering is direct: charge is the integral of current, and the energy stored in an inductor is the integral of power.

§14.1 to §14.5 are the chapter's theory, and §14.6 works through four examples. §14.7 summarises the formulas and what you should be able to do after the chapter, and §14.8 holds the exercises. As preparation, you can watch a short video about integrals as a concept: https://www.youtube.com/watch?v=__Uw1SXPW7s.

14.1 Antiderivatives

An **antiderivative** $F(x)$ of $f(x)$ satisfies

$$F'(x) = f(x).$$

There are infinitely many antiderivatives, and they differ by a constant C:

$$\int f(x)\,dx = F(x) + C$$

C is called the **constant of integration**.

14.2 Basic rules of integration

The table shows the antiderivatives of the most common functions.

Function $f(x)$	**Antiderivative** $F(x)$
$x^n \quad (n \neq -1)$	$\dfrac{x^{n+1}}{n+1} + C$
e^x	$e^x + C$
e^{kx}	$\dfrac{1}{k}e^{kx} + C$
$\dfrac{1}{x}$	$\ln\lvert x\rvert + C$
$\sin(x)$	$-\cos(x) + C$
$\cos(x)$	$\sin(x) + C$
$\sin(kx)$	$-\dfrac{1}{k}\cos(kx) + C$
$\cos(kx)$	$\dfrac{1}{k}\sin(kx) + C$

The **linearity property** also holds:

$$\int [af(x) + bg(x)]\,dx = a\int f(x)\,dx + b\int g(x)\,dx$$

14.3 Definite integrals and area

The **definite integral** gives the area under the curve $y = f(x)$ from a to b:

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

> **Note:** If $f(x) < 0$ in parts of the interval, those parts contribute negative area.

14.4 The constant of integration and initial conditions

For an antiderivative with a known initial condition $F(x_0) = y_0$, C is given by

$$C = y_0 - F_{\text{without } C}(x_0),$$

where $F_{\text{without } C}$ is the antiderivative written without the constant.

14.5 Application: charge from current

The charge $q(t)$ is given by the integral of the current $i(t)$:

$$q(t) = \int_0^t i(\tau)\,d\tau + q(0)$$

14.6 Worked examples

EXAMPLE **Calculating an area.** Calculate the area under $f(x) = 2x + 1$ for $0 \le x \le 2$.

$$\int_0^2 (2x+1)\,dx = \Big[x^2 + x\Big]_0^2 = (4+2) - 0 = 6$$

EXAMPLE **Area under a parabola.** Calculate the area under $f(x) = x^2$ for $0 \le x \le 3$.

$$\int_0^3 x^2\,dx = \left[\frac{x^3}{3}\right]_0^3 = \frac{27}{3} - 0 = 9$$

EXAMPLE **Charge by integration.** During the time $0 \le t \le 2\,\text{s}$, the current $i(t) = (2t+4)\,\text{A}$ flows. Calculate the charge $q(t)$ with $q(0) = 0$.

$$q(t) = \int_0^t (2\tau + 4)\,d\tau = \Big[\tau^2 + 4\tau\Big]_0^t = t^2 + 4t$$

The antiderivative $I(t) = t^2 + 4t$ satisfies the initial condition, since $I(0) = 0$. After two seconds, the charge is

$$q(2) = 4 + 8 = 12\,\text{C}.$$

EXAMPLE **A definite integral with negative values.** Calculate $\int_{-1}^{2} (x^2 - 1)\,dx$.

$$\begin{aligned}\left[\frac{x^3}{3} - x\right]_{-1}^{2} &= \left(\frac{8}{3} - 2\right) - \left(-\frac{1}{3} + 1\right) \\ &= \frac{8}{3} - 2 + \frac{1}{3} - 1 = 3 - 3 = 0\end{aligned}$$

The positive and negative areas cancel out.

14.7 Summary

Concept	Formula
Antiderivative	$F'(x) = f(x)$
Indefinite integral	$\int f(x)\,dx = F(x) + C$
Definite integral	$\int_a^b f(x)\,dx = F(b) - F(a)$
$\int x^n\,dx$	$\frac{x^{n+1}}{n+1} + C$
$\int e^{kx}\,dx$	$\frac{1}{k}e^{kx} + C$
Charge	$q(t) = \int i(t)\,dt$

What you should be able to do

After this chapter, you should be able to:

- Understand what an integral is.
- Understand the concept of an antiderivative and find one for simple functions.
- Carry out simple integrations.

Test yourself

ANSWER P. 262

1. Calculate $\int_0^2 (3x^2 + 1)\,dx$.
2. Find the antiderivative $F(x)$ of $f(x) = 4x + e^x$. Include the constant of integration C in your answer.

Next chapter

Chapter 15 is about complex numbers: what they are, how to calculate with them and how to convert them from polar to rectangular form.

14.8 Exercises

How to work with the exercises. Parts 1 and 2 are done during the lesson: first on your own, a few minutes per exercise, then together. Part 3 is extra exercises for further practice after the lesson. Most of them can be done in your head or with pencil and paper.
Answers and worked solutions. The answer to every exercise is in the answer key at the back of the book, and its page is given next to the exercise. An answer shows whether you got it right, but not how. Full worked solutions, in Swedish, are published in the course repository after the lesson, in `lectures/L14/appendix/c_solutions.md`. Always try it yourself first.

PART 1

Review exercises

Exercise 1.1 Discharging a capacitor

ANSWER P. 261

The voltage across a discharging capacitor is given by the function

$$u(t) = 24e^{-0.25t},$$

where $u(t)$ is the voltage across the capacitor in V and t the time in seconds. The definition of the derivative is

$$u'(t) = \lim_{h \to 0} \frac{u(t+h) - u(t)}{h}.$$

a) Find the derivative $u'(t)$ using the definition of the derivative.

b) Calculate the rate of change of the voltage at $t = 4\,\text{s}$.

c) Find $u''(4)$. Is the rate of change increasing or decreasing at this point, and what does that mean for the discharge?

PART 2

New material

Exercise 2.1 Energy stored in an inductor

ANSWER P. 261

The current through an inductor depends on time and is given by the function

$$i(t) = 5t,$$

where t is the time in seconds. The current is measured in amperes (A). The inductance of the inductor is $L = 0.20\,\text{H}$ (henry).

The power $p(t)$ developed in the inductor is given by the function

$$p(t) = L \cdot i(t) \cdot i'(t),$$

where

- $p(t)$ is the power developed in watts (W),
- L is the inductance of the inductor in henries (H),
- $i(t)$ is the current through the inductor in amperes (A),
- $i'(t)$ is the rate of change of the current through the inductor in A/s.

The energy $w(t)$ stored in the inductor between time 0 and time t is given by the function

$$w(t) = \int_0^t p(t)\,dt$$

and is measured in joules (J).

a) Find $p(t)$.

b) Find the antiderivative $P(t)$ of $p(t)$.

c) Find an expression for the energy $w(t)$ in the inductor as a function of time.

d) The inductor is uncharged at the start, that is, $w(0) = 0$. Find the constant of integration C.

e) Find how much energy has been stored in the inductor during the first 3 seconds $(0 \le t \le 3)$.

PART 3

Extra exercises

The exercises below are extra practice, best done on your own after the lesson.

Exercise 3.1 Antiderivatives of polynomials

ANSWER P. 261

Find the antiderivative.

a) $\int x^3\,dx$ **b)** $\int 4x\,dx$ **c)** $\int 5\,dx$

d) $\int (2x+3)\,dx$ **e)** $\int (x^2-4x+1)\,dx$

Exercise 3.2 Antiderivatives of standard functions

ANSWER P. 261

Find the antiderivative.

a) $\int e^x\,dx$ **b)** $\int e^{3x}\,dx$ **c)** $\int \frac{1}{x}\,dx$

d) $\int \sin(x)\,dx$ **e)** $\int \cos(2x)\,dx$ **f)** $\int \sin(4x)\,dx$

Exercise 3.3 Checking by differentiation

ANSWER P. 261

a) Show that $F(x) = \dfrac{x^3}{3}$ is an antiderivative of $f(x) = x^2$.

b) Show that $F(x) = -\dfrac{\cos(2x)}{2}$ is an antiderivative of $f(x) = \sin(2x)$.

c) Show that $F(t) = -2e^{-t/2}$ is an antiderivative of $f(t) = e^{-t/2}$.

d) Why is the answer to an indefinite integral never unique?

Exercise 3.4 Definite integrals: polynomials

ANSWER P. 261

Calculate.

a) $\int_0^2 3x^2\,dx$ **b)** $\int_1^3 2x\,dx$ **c)** $\int_0^4 (x+1)\,dx$

d) $\int_{-1}^1 x^2\,dx$ **e)** $\int_0^3 (6-2x)\,dx$

Exercise 3.5 Definite integrals: other functions

ANSWER P. 261

Calculate.

a) $\int_0^1 e^x\,dx$ **b)** $\int_0^\pi \sin(x)\,dx$ **c)** $\int_0^{\pi/2} \cos(x)\,dx$

d) $\int_1^e \frac{1}{x}\,dx$ **e)** $\int_0^2 e^{-t}\,dt$

Exercise 3.6 Area under a curve

ANSWER P. 261

a) Calculate the area under $f(x) = x$ for $0 \le x \le 4$.

b) Check your answer to a) with the formula for the area of a triangle.

c) Calculate the area under $f(x) = 3$ for $1 \le x \le 5$ and check it with a rectangle.

d) Calculate the area under $f(x) = x^2$ for $0 \le x \le 2$.

Exercise 3.7 Negative areas

ANSWER P. 261

The function $f(x) = x - 2$ is given.

a) Calculate $\int_0^2 (x-2)\,dx$. **b)** Calculate $\int_2^4 (x-2)\,dx$. **c)** Calculate $\int_0^4 (x-2)\,dx$.

d) Explain why the integral in c) is zero even though the curve does not lie on the x-axis.

Exercise 3.8 The constant of integration from an initial condition

ANSWER P. 261

It is known that $f'(x) = 4x - 3$ and that $f(0) = 5$.

a) Find the general antiderivative.

b) Find the constant of integration C.

c) Write down $f(x)$.

d) Calculate $f(2)$.

Exercise 3.9 Voltage from its rate of change

ANSWER P. 261

The voltage across a capacitor changes at the rate $u'(t) = 6t\,\mathrm{V/s}$, and at the start $u(0) = 2\,\mathrm{V}$.

a) Find the general antiderivative.

b) Find the constant of integration.

c) Write down $u(t)$.

d) Calculate $u(4)$.

Exercise 3.10 Charge from current

ANSWER P. 261

The current through a conductor is given by $i(t) = 4t + 2$ amperes, and $q(0) = 0$.

a) Find an expression for the charge $q(t)$.

b) Calculate $q(3)$.

c) Calculate $q(5)$.

d) How much charge passes between $t = 3\,\mathrm{s}$ and $t = 5\,\mathrm{s}$?

e) Check your answer to d) by calculating $\int_3^5 i(t)\,dt$.

Exercise 3.11 Charge from a sinusoidal current

ANSWER P. 261

The current is given by $i(t) = 2\sin(100\pi t)$ amperes.

a) Find the antiderivative of $i(t)$.

b) Calculate $\int_0^{0.01} i(t)\,dt$.

c) Calculate $\int_0^{0.02} i(t)\,dt$.

d) Explain the result in c).

Exercise 3.12 Energy stored in an inductor

ANSWER P. 261

The current through an inductor with $L = 0.4\,\mathrm{H}$ is given by $i(t) = 3t$ amperes. The power is given by the relationship $p(t) = L\,i(t)\,i'(t)$.

a) Find $i'(t)$.

b) Find $p(t)$.

c) Find the energy $w(t) = \int_0^t p(\tau)\,d\tau$.

d) Calculate the energy after 2 seconds.

e) Check your answer with the formula $w = \dfrac{Li^2}{2}$.

Exercise 3.13 Energy dissipated in a resistor

ANSWER P. 261

The power in a resistor is given by $p(t) = 5 + 2t$ watts for $0 \le t \le 10\,\text{s}$.

a) Find the energy $w(t) = \int_0^t p(\tau)\,d\tau$.

b) Calculate the total energy after 10 seconds.

c) How much energy is dissipated during the last 5 seconds?

d) Calculate the average power over the whole interval.

Exercise 3.14 Mean value of a function

ANSWER P. 261

The mean value of f over the interval $[a, b]$ is given by

$$\frac{1}{b-a}\int_a^b f(x)\,dx.$$

a) Calculate the mean value of $f(x) = x^2$ over $[0, 3]$.

b) Calculate the mean value of $f(x) = 4$ over $[1, 5]$.

c) Calculate the mean value of $i(t) = 10\sin(100\pi t)$ over half a period, $[0, 0.01]$.

d) What fraction of the amplitude is the mean value in c)?

Exercise 3.15 Rules for integrals

ANSWER P. 261

It is known that $\int_0^2 f(x)\,dx = 5$ and $\int_0^2 g(x)\,dx = 3$. Calculate:

a) $\int_0^2 [f(x) + g(x)]\,dx$

b) $\int_0^2 2f(x)\,dx$

c) $\int_0^2 [3f(x) - 2g(x)]\,dx$

d) $\int_2^0 f(x)\,dx$

Exercise 3.16 Differentiation and integration as opposites

ANSWER P. 261

a) Differentiate $F(x) = x^3 - 2x$.

b) Integrate the result from a).

c) Why does a constant appear in b) that was not in $F(x)$?

d) Calculate $\frac{d}{dx}\left[\int 5x^4\,dx\right]$.

Exercise 3.17 Finding an unknown limit or constant

ANSWER P. 261

a) Find $a > 0$ such that $\int_0^a 2x\,dx = 25$.

b) Find a such that $\int_0^a 3\,dx = 12$.

c) Find k such that $\int_0^2 kx\,dx = 10$.

d) A capacitor is charged with the constant current $i = 0.5\,\text{A}$. After how long has a charge of $2\,\text{C}$ passed?

Exercise 3.18 Find the error

ANSWER P. 261

Each line contains a common error. Explain the error and give the correct answer.

a) $\int x^2\,dx = 2x + C$

b) $\int e^{2x}\,dx = 2e^{2x} + C$

c) $\int \sin(x)\,dx = \cos(x) + C$

d) $\int_1^2 3x^2\,dx = \left[x^3\right]_1^2 = 8$

e) A definite integral needs a constant of integration.

f) $\int \frac{1}{x}\,dx = \ln x + C$ holds for all $x \neq 0$.

PART 4

Mathematics in C

The teacher shows the exercise below during the lesson, in VS Code and with `make`. You can also do it yourself in **OnlineGDB**, `https://www.onlinegdb.com/online_c_compiler`, with the language set to **C**.

Exercise 4.1 The integral as a sum in C

ANSWER P. 261

The definite integral is the area under the curve, and it can be approximated by a sum of n narrow rectangles. With the width $\Delta x = \frac{b-a}{n}$ and the height taken at the left edge of each rectangle,

$$\int_a^b f(x)\,dx \approx \sum_{k=0}^{n-1} f(a + k\Delta x) \cdot \Delta x.$$

a) Evaluate $\int_0^\pi \sin(x)\,dx$ with an antiderivative.

b) Define the type `math_func_t` as in Chapter 13, Exercise 4.1, with `typedef double (*math_func_t)(double);`. Write the function `double integrate(math_func_t f, double a, double b, size_t n)`, which returns the sum above. The type `size_t` is in `stddef.h`. Then call `integrate(sin, 0.0, M_PI, n)` for $f(x) = \sin(x)$ on the interval

$[0, \pi]$ with $n = 4$, 10, 100 and 1000 rectangles, and print n, the sum and the error compared with the answer to a). The error is the distance between the sum and the expected answer 2.0 from a). If the sum is stored in the variable `area`, the error is computed with `fabs(area - 2.0)`, where `fabs()` gives the absolute value, so the error is never negative. Use `sin()`, `fabs()` and the constant `M_PI` from `math.h`.

c) How does the error change when n is multiplied by ten?

d) Evaluate $\int_0^1 e^{2x}\,dx$ with an antiderivative. Then write the function `double e_2x(double x)`, which returns e^{2x} using `pow(M_E, 2.0 * x)`, and call `integrate(e_2x, 0.0, 1.0, n)` with the same values of n. Print n, the sum and the error, which is now computed with `fabs(area - exact)`, where `exact` is the answer you calculated. How does the error change when n is multiplied by ten, compared with $\sin(x)$ in c)?

CHAPTER 15

Complex numbers, part I

A number whose square is negative, and how it describes voltages and currents.

IN THIS CHAPTER

- What complex numbers are and how to calculate with them.
- Rectangular form and the complex plane.
- Magnitude and phase angle, with quadrant correction.
- Conversion between rectangular and polar form.
- Applications in electrical engineering: voltage, current and impedance.

The equation $x^2 = -1$ has no solution among the real numbers. Introducing the imaginary unit j, with $j^2 = -1$, extends the numbers to the complex numbers. This chapter shows how a complex number is written in rectangular form and drawn in the complex plane, how its magnitude and phase angle are calculated, and how it is converted between rectangular and polar form. In electrical engineering, complex numbers are the natural way to calculate with AC voltages and alternating currents, which have both a size and a phase angle.

§15.1 to §15.5 are the chapter's theory, and §15.6 gives three worked examples. §15.7 summarises the formulas and what you should be able to do after the chapter, and §15.8 holds the exercises. You will need a calculator for the phase angles.

15.1 Introduction

Complex numbers are needed when we want to solve equations such as $x^2 = -1$. We introduce the **imaginary unit** j:

$$j^2 = -1 \quad \Rightarrow \quad j = \sqrt{-1}$$

In electrical engineering, j is used instead of i, which is reserved for current.

15.2 Rectangular form

A complex number z is written in **rectangular form** as

$$z = x + jy,$$

where

- $x = \mathrm{Re}(z)$ is the real part,
- $y = \mathrm{Im}(z)$ is the imaginary part.

Complex numbers are drawn in the **complex plane**, with the Re axis horizontal and the Im axis vertical.

15.3 Magnitude and phase angle

The **magnitude**, or absolute value, is the distance from the origin:

$$|z| = \sqrt{x^2 + y^2}$$

The **phase angle** is the angle to the positive Re axis:

$$\delta = \arctan\left(\frac{y}{x}\right) + \text{quadrant correction}$$

The correction depends on which quadrant the number lies in:

Quadrant	x	y	Correction
I	$+$	$+$	none
II	$-$	$+$	$+180^\circ$ $(+\pi)$
III	$-$	$-$	$+180^\circ$ $(+\pi)$
IV	$+$	$-$	none (negative δ)

15.4 Polar form

In **polar form**, a complex number is written with its magnitude and its phase angle:

$$z = |z| \angle \delta$$

The conversion from **rectangular to polar form** is given by

$$|z| = \sqrt{x^2 + y^2}, \qquad \delta = \arctan(y/x) + \text{correction},$$

and the conversion from **polar to rectangular form** by

$$x = |z| \cos\delta, \qquad y = |z| \sin\delta.$$

15.5 Operations in rectangular form

Addition and subtraction: the real and imaginary parts are added or subtracted separately.

$$(x_1 + jy_1) \pm (x_2 + jy_2) = (x_1 \pm x_2) + j(y_1 \pm y_2)$$

Multiplication: multiply out and use $j^2 = -1$.

$$(x_1 + jy_1)(x_2 + jy_2) = (x_1x_2 - y_1y_2) + j(x_1y_2 + x_2y_1)$$

Conjugate: the conjugate of $z = x + jy$ is $\bar{z} = x - jy$. It is useful in division, since it makes the denominator real.

15.6 Worked examples

EXAMPLE **Complex numbers in the complex plane.** Mark $z_1 = 3 + j4$ and $z_2 = -2 - j3$ in the complex plane and calculate their magnitudes and phase angles.

The number $z_1 = 3 + j4$ lies in quadrant I:

$$|z_1| = \sqrt{9 + 16} = 5, \qquad \delta_1 = \arctan\left(\frac{4}{3}\right) \approx 53.1^\circ$$

In polar form, $z_1 = 5\angle 53.1^\circ$.

The number $z_2 = -2 - j3$ lies in quadrant III, so 180° is added:

$$|z_2| = \sqrt{4 + 9} = \sqrt{13} \approx 3.61,$$
$$\delta_2 = \arctan\left(\frac{-3}{-2}\right) + 180^\circ \approx 56.3^\circ + 180^\circ = 236.3^\circ$$

EXAMPLE **Converting from rectangular to polar form.** The voltage is $U = 3 + j4\,\text{V}$. Find its magnitude and phase angle.

$$|U| = \sqrt{3^2 + 4^2} = 5\,\text{V}, \qquad \delta = \arctan\left(\frac{4}{3}\right) \approx 0.927\,\text{rad} \approx 53.1^\circ$$

In polar form, $U = 5\angle 0.927\,\text{rad}$.

EXAMPLE **Converting from polar to rectangular form.** The current is $I = 10\angle\frac{\pi}{4}$ mA. Find its rectangular form.

$$x = 10\cos\left(\frac{\pi}{4}\right) = \frac{10}{\sqrt{2}} \approx 7.07\,\text{mA}$$
$$y = 10\sin\left(\frac{\pi}{4}\right) = \frac{10}{\sqrt{2}} \approx 7.07\,\text{mA}$$
$$I = (7.07 + j7.07)\,\text{mA}$$

15.7 Summary

Concept	Formula
Rectangular form	$z = x + jy$
Magnitude	$\lvert z\rvert = \sqrt{x^2 + y^2}$
Phase angle	$\delta = \arctan(y/x) + \text{correction}$
Polar form	$z = \lvert z\rvert\angle\delta$
Polar to rectangular form	$x = \lvert z\rvert\cos\delta,\ y = \lvert z\rvert\sin\delta$
Addition	$(x_1 + x_2) + j(y_1 + y_2)$

What you should be able to do

After this chapter, you should be able to:

- Understand what complex numbers are and how they can be represented in the complex plane.
- Carry out simple calculations with complex numbers in rectangular form.
- Convert between rectangular and polar form, and give the magnitude and phase angle.

Test yourself

ANSWER P. 263

1. Convert $z = 3 + j4$ to polar form. Give the magnitude and the phase angle in radians.
2. Calculate $(2 + j3)(1 - j)$ in rectangular form.

Next chapter

Chapter 16 continues with complex numbers: Euler's formula, and complex numbers for vector calculations.

15.8 Exercises

How to work with the exercises. Parts 1 and 2 are done during the lesson: first on your own, a few minutes per exercise, then together. Part 3 is extra exercises for further practice after the lesson. Most of them can be done in your head or with pencil and paper.
Answers and worked solutions. The answer to every exercise is in the answer key at the back of the book, and its page is given next to the exercise. An answer shows whether you got it right, but not how. Full worked solutions, in Swedish, are published in the course repository after the lesson, in `lectures/L15/appendix/c_solutions.md`. Always try it yourself first.

PART 1
Review exercises

Exercise 1.1 Recovering a function of time from its derivative

ANSWER P. 262

A sensor voltage $u(t)$ in a measuring circuit is not known directly, but its derivative has been measured as

$$u'(t) = (6t - 4)\ \mathrm{V/s}.$$

The initial voltage is also known to be $u(0) = 2\,\mathrm{V}$.

a) Find an expression for $u(t)$.

b) Find the voltage after 3 seconds.

c) Calculate the definite integral $\int_0^3 u'(t)\,dt$. Compare the value with $u(3)$ from b) and explain any difference.

d) Find the time at which the voltage is stationary, i.e. when $u'(t) = 0$.

e) Use the **second derivative** to decide whether the stationary point is a **maximum** or a **minimum**.

f) Find the coordinates of the stationary point, i.e. the time t and the voltage $u(t)$ at the point.

g) Plot the graph of $u(t)$ in GeoGebra, `https://www.geogebra.org/graphing?lang=en`, and check that your answers are right.

PART 2
New material

Exercise 2.1 Rectangular → polar form

ANSWER P. 262

Convert the voltage $U = 5 - j2\,\text{V}$ to polar form.

Exercise 2.2 Polar → rectangular form

ANSWER P. 262

Convert the current $I = 5\angle\frac{\pi}{6}\,\text{mA}$ to rectangular form.

Exercise 2.3 Calculating an impedance

ANSWER P. 262

A circuit is supplied with the voltage from Exercise 2.1, and the current from Exercise 2.2 flows through the circuit. Calculate the circuit's impedance Z with "Ohm's law":

$$Z = \frac{U}{I},$$

where

- Z is the circuit's impedance in Ω,
- U is the supply voltage in V,
- I is the current through the circuit in mA.

Give the answer in both polar and rectangular form.

PART 3
Extra exercises

The exercises below are extra practice, best done on your own after the lesson.

Exercise 3.1 The imaginary unit

ANSWER P. 262

Calculate.

a) j^2 **b)** j^3 **c)** j^4 **d)** j^5 **e)** $\frac{1}{j}$

f) Solve the equation $x^2 = -25$.

Exercise 3.2 Real and imaginary parts

ANSWER P. 262

The complex numbers $z_1 = 4 - j3$, $z_2 = -2 + j5$, $z_3 = j6$ and $z_4 = -7$ are given.

a) Give $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$ for each of the numbers.

b) Which number is purely imaginary?

c) Which number is purely real?

Exercise 3.3 Addition and subtraction

ANSWER P. 262

Calculate with $z_1 = 3 + j2$ and $z_2 = -1 + j5$:

a) $z_1 + z_2$ **b)** $z_1 - z_2$ **c)** $2z_1 + 3z_2$ **d)** $z_2 - 2z_1$

Exercise 3.4 Multiplication

ANSWER P. 262

Calculate and write the answer in rectangular form.

a) $(2 + j3)(1 + j4)$ **b)** $(3 - j2)(3 + j2)$ **c)** $j(5 - j2)$ **d)** $(1 + j)^2$

Exercise 3.5 Conjugates

ANSWER P. 262

a) Give the conjugate of $5 + j2$.

b) Give the conjugate of $-3 - j7$.

c) Calculate $z\bar{z}$ for $z = 4 + j3$.

d) Show in general that $z\bar{z} = |z|^2$.

Exercise 3.6 Division

ANSWER P. 262

Calculate and write the answer in rectangular form.

a) $\dfrac{4 + j2}{1 - j}$ **b)** $\dfrac{6 + j8}{j}$ **c)** $\dfrac{10}{3 + j4}$

d) Why do you multiply by the conjugate of the denominator when dividing?

Exercise 3.7 Magnitude

ANSWER P. 262

Calculate.

a) $|3 + j4|$ **b)** $|-5 + j12|$ **c)** $|j7|$ **d)** $|-8|$ **e)** $|1 - j|$

Exercise 3.8 Phase angle with quadrant correction

ANSWER P. 262

Find the phase angle. State which quadrant the number lies in.

a) $4 + j4$ **b)** $-4 + j4$ **c)** $-4 - j4$ **d)** $4 - j4$ **e)** $3 + j4$

Exercise 3.9 Rectangular to polar form

ANSWER P. 262

Write in polar form.

a) $z = 6 + j8$ **b)** $z = -3 + j4$ **c)** $z = -1 - j1$ **d)** $z = -j5$

Exercise 3.10 Polar to rectangular form

ANSWER P. 262

Write in rectangular form.

a) $z = 10\angle 30°$ **b)** $z = 4\angle 90°$ **c)** $z = 6\angle 180°$

d) $z = 2\angle -60°$ **e)** $z = 8\angle \frac{\pi}{4}$

Exercise 3.11 A voltage as a complex number

ANSWER P. 262

A voltage is given by $U = 12 + j5\,\text{V}$.

a) Calculate $|U|$.

b) Calculate the phase angle δ.

c) Write U in polar form.

d) What does the phase angle represent physically?

Exercise 3.12 Impedance in rectangular form

ANSWER P. 262

A series circuit has $Z_R = 30\,\Omega$ and $Z_L = j40\,\Omega$.

a) Find the total impedance Z in rectangular form.

b) Calculate $|Z|$.

c) Calculate the phase angle of the impedance.

d) Is the circuit inductive or capacitive?

Exercise 3.13 Ohm's law with complex numbers

ANSWER P. 262

A circuit with the impedance $Z = 3 + j4\,\Omega$ is supplied with the voltage $U = 10\,\text{V}$ (purely real).

a) Calculate $|U|$ and $|Z|$.

b) Calculate the current $I = \dfrac{U}{Z}$ in rectangular form.

c) Calculate $|I|$ and check it against $\dfrac{|U|}{|Z|}$.

d) Calculate the phase angle of the current and interpret the result.

Exercise 3.14 Quadratic equations with complex roots

ANSWER P. 262

Solve the equations.

a) $x^2 + 9 = 0$ **b)** $x^2 + 4x + 13 = 0$ **c)** $x^2 - 2x + 5 = 0$

d) What is always true of the two roots when the discriminant is negative?

Exercise 3.15 Series and parallel impedance

ANSWER P. 262

Two impedances $Z_1 = 4 + j3\,\Omega$ and $Z_2 = 4 - j3\,\Omega$ are given.

a) Calculate the series connection $Z_1 + Z_2$ and comment on the result.

b) Calculate $|Z_1|$ and $|Z_2|$.

c) Calculate the product $Z_1 Z_2$.

d) Calculate the parallel connection $\dfrac{Z_1 Z_2}{Z_1 + Z_2}$.

Exercise 3.16 True or false

ANSWER P. 262

Decide whether each statement is true or false, and justify your answer briefly.

a) $j^2 = 1$

b) The magnitude $|z|$ can be negative.

c) A purely imaginary number has the phase angle $90°$ or $-90°$.

d) The product $(a + jb)(a - jb)$ is always real.

e) The calculator's arctan always gives the right phase angle directly.

f) Adding complex numbers is easiest in polar form.

PART 4

Mathematics in C

The teacher shows the exercise below during the lesson if time allows, in VS Code and with `make`. You can also do it yourself in **OnlineGDB**, https://www.onlinegdb.com/online_c_compiler, with the language set to **C**.

In the exercise below, a complex number is written as a **struct** of its own. A struct gathers several variables that belong together under one name, here the real part and the imaginary part. The type `complex_t` is defined with `typedef struct` and has the fields `re` and `im`, both of type `double`:

```c
/**
 * @brief Complex number structure.
 */
typedef struct
{
    /** Real part (3 in z = 3 + j4). */
    double re;

    /** Imaginary part (4 in z = 3 + j4). */
    double im;
} complex_t;
```

- `typedef` gives the struct the alias `complex_t`, so that we do not have to write `struct` every time we use it. The suffix `_t` shows that the name is an alias, just as in `uint8_t` from `stdint.h`.
- The number $3 + j4$ is created with `const complex_t z = {3.0, 4.0};`, or more clearly with `const complex_t z = {.re = 3.0, .im = 4.0};`. The fields are reached with a dot: `z.re` and `z.im`.
- Structs are passed to functions through a **pointer** (`complex_t*`), that is, the memory address of the struct, so that the whole struct does not have to be copied on every call. The main reason is efficiency. Here it hardly matters, since two `doubles` take $2 \cdot 64$ bits while a pointer takes 64 bits, but for larger structs the difference becomes significant.
- The parameter type `const complex_t*` says that the function does not change the number.
- In the call, the address is given with `&`, for example `complex_magnitude(&z)`. Inside the function, the fields are reached with an arrow: `z->re` and `z->im`.
- A `complex_t` can, however, be returned directly, as in `complex_from_polar()`. A new number has to be created anyway, so it is simpler to do that inside the function than to have the caller create a variable and pass its address.

The functions `sqrt()`, `sin()`, `cos()`, `atan()` and `atan2()` and the constant `M_PI` for π are in `math.h`. The trigonometric functions work in radians.

Exercise 4.1 Polar form in C

ANSWER P. 262

The numbers $z_1 = 3 + j4$ and $z_2 = -2 - j3$ from §15.6 and $z_3 = 10 \angle 45°$ are given.

a) Give z_1 and z_2 in polar form, with the argument in degrees, and z_3 in rectangular form. Calculate by hand.

b) Define `complex_t` and write the functions `double complex_magnitude(const complex_t* z)`, which returns the modulus $|z|$, `double complex_angle_deg(const complex_t* z)`, which returns the argument in degrees, computed with `atan2(z->im, z->re)`, `complex_t complex_from_polar(double r, double deg)`, which returns the number $r \angle \theta$ in rectangular form, and `void complex_print(const complex_t* z)`, which prints the number in both rectangular and polar form, with the argument in degrees. Write z_3 as `complex_from_polar(10.0, 45.0)` and call `complex_print()` once per number.

c) Also compute `atan(z2.im / z2.re)` for z_2 and convert the answer to degrees. Why is it wrong, and why does `atan2()` give the right quadrant?

CHAPTER 16

Complex numbers, part II

A formula that unites the exponential function with sine and cosine, and complex numbers that carry sinusoidal signals.

IN THIS CHAPTER

- Euler's formula and Euler form.
- Rectangular form, polar form and Euler form, and when each is easiest to calculate with.
- Phasors: sinusoidal signals as complex numbers.
- Converting between a phasor and a function of time.

This chapter builds on Chapter 15, where complex numbers were written in rectangular and polar form. Here you meet Euler's formula, which links the complex exponential function with cosine and sine and gives a third way of writing a complex number, Euler form. It makes multiplication and division as simple as in polar form, and it is the key to phasors: complex numbers that represent sinusoidal voltages and currents. You may like to read about Euler's formula at `https://en.wikipedia.org/wiki/Euler's_formula` before you start.

§16.1 to §16.6 are the chapter's theory, with worked examples. §16.7 summarises the formulas and what you should be able to do after the chapter, and §16.8 holds the exercises. The angles are sometimes given in degrees and sometimes in radians, so keep track of which mode your calculator is in.

16.1 Euler's formula

Euler's formula connects complex exponents and trigonometry:

$$e^{j\delta} = \cos\delta + j\sin\delta$$

This means that every complex number can be written in **Euler form**:

$$z = |z| \cdot e^{j\delta}$$

16.2 Three forms of complex numbers

A complex number can be written in three ways, and each form has its strengths.

Form	Notation	Comment
Rectangular	$x + jy$	Best for addition and subtraction
Polar	$\lvert z\rvert \angle \delta$	Clear geometric interpretation
Euler form	$\lvert z\rvert e^{j\delta}$	Best for multiplication and division

Multiplication in polar form:

$$z_1 \cdot z_2 = |z_1||z_2| \angle (\delta_1 + \delta_2)$$

Division in polar form:

$$\frac{z_1}{z_2} = \frac{|z_1|}{|z_2|} \angle (\delta_1 - \delta_2)$$

16.3 Phasors

A **phasor** is a complex number that represents a sinusoidal signal. A voltage $u(t) = |U| \sin(\omega t + \delta)$ is represented by the phasor

$$U = |U| \angle \delta = |U| e^{j\delta}.$$

Phasors simplify calculations with sine waves in electric circuits: adding sine waves reduces to adding complex numbers.

16.4 Converting between Euler form and the time domain

From phasor to time domain:

$$U = |U| e^{j\delta} \quad \Rightarrow \quad u(t) = |U| \sin(\omega t + \delta)$$

From time domain to phasor:

$$u(t) = |U| \sin(\omega t + \delta) \quad \Rightarrow \quad U = |U| \angle \delta$$

16.5 Why Euler form?

In calculations, the polar form $|U| \angle \delta$ is usually the simplest to write. Euler form explains why it works.

Multiplication and division follow from the laws of exponents. The exponents are added in multiplication and subtracted in division:

$$|z_1|e^{j\delta_1} \cdot |z_2|e^{j\delta_2} = |z_1||z_2|\, e^{j(\delta_1+\delta_2)}$$

This is why the angles are added when two numbers are multiplied in polar form.

The frequency can be left out. All voltages and currents in a circuit have the same frequency. The only things that differ between them are the amplitude and the phase, and these are contained in the phasor $U = |U|e^{j\delta}$. We therefore calculate with phasors and put ωt back only when the answer is to be written as $u(t) = |U|\sin(\omega t + \delta)$.

16.5.1 Differentiation with Euler's formula

This is beyond the course, but it shows why Euler's formula makes differentiation and integration so much simpler. We write the complex voltage in rectangular form using Euler's formula:

$$u(t) = |U|\cos(\omega t + \delta) + j|U|\sin(\omega t + \delta)$$

We differentiate with the chain rule, where the inner derivative is $(\omega t + \delta)' = \omega$:

$$\begin{aligned} u'(t) &= -|U|\sin(\omega t + \delta) \cdot (\omega t + \delta)' + j|U|\cos(\omega t + \delta) \cdot (\omega t + \delta)' \\ &= -\omega|U|\sin(\omega t + \delta) + j\omega|U|\cos(\omega t + \delta) \end{aligned}$$

Since $j^2 = -1$, $-\omega|U|\sin(\omega t + \delta)$ can be written as $j\omega \cdot j|U|\sin(\omega t + \delta)$, so $j\omega$ can be factored out:

$$u'(t) = j\omega(|U|\cos(\omega t + \delta) + j|U|\sin(\omega t + \delta)) = j\omega \cdot u(t)$$

In Euler form, the same differentiation takes a single step. With $u(t) = |U|e^{j(\omega t+\delta)}$, the inner derivative is $\left(j(\omega t + \delta)\right)' = j\omega$:

$$u'(t) = \left(j(\omega t + \delta)\right)' \cdot |U|e^{j(\omega t+\delta)} = j\omega \cdot |U|e^{j(\omega t+\delta)} = j\omega \cdot u(t)$$

The result is the same, but in Euler form it is enough to multiply by $j\omega$. Integration works the same way in reverse: integrating corresponds to dividing by $j\omega$.

16.5.2 Impedance of an inductor and a capacitor

With the result above, the impedances are easy to derive. The voltage across an inductor is $u = L\frac{di}{dt}$. With the current $i(t) = |I|e^{j(\omega t+\delta)}$, the differentiation above gives $\frac{di}{dt} = j\omega \cdot i(t)$, so

$$u(t) = L\frac{di}{dt} = L \cdot j\omega \cdot i(t) \quad \Rightarrow \quad Z_L = \frac{u(t)}{i(t)} = j\omega L$$

The current through a capacitor is $i = C\frac{du}{dt}$, and in the same way $\frac{du}{dt} = j\omega \cdot u(t)$, so

$$i(t) = C\frac{du}{dt} = C \cdot j\omega \cdot u(t) \quad \Rightarrow \quad Z_C = \frac{u(t)}{i(t)} = \frac{1}{j\omega C}$$

The impedances of the inductor and the capacitor thus follow directly from differentiation corresponding to multiplication by $j\omega$. Impedances are covered further in Chapter 17.

16.6 Worked examples

EXAMPLE **Multiplication in Euler form.** Calculate $z_1 \cdot z_2$ when $z_1 = 2e^{j\pi/6}$ and $z_2 = 3e^{j\pi/4}$. Give the answer in both Euler form and rectangular form.

The magnitudes are multiplied and the exponents added:

$$z_1 \cdot z_2 = 2 \cdot 3 \cdot e^{j(\pi/6+\pi/4)} = 6e^{j5\pi/12}$$

The angle is $\frac{\pi}{6} + \frac{\pi}{4} = \frac{2\pi}{12} + \frac{3\pi}{12} = \frac{5\pi}{12}$, that is, 75°. Euler's formula gives the rectangular form:

$$6e^{j5\pi/12} = 6\,(\cos 75^\circ + j \sin 75^\circ) \approx 6(0.259 + j0.966) \approx 1.55 + j5.80$$

EXAMPLE **Euler form to rectangular form.** Write $u(t) = 3e^{j(100\pi t - \pi/4)}$ V in rectangular form at $t = 0$.

$$u(0) = 3e^{-j\pi/4} = 3\left(\cos\left(-\frac{\pi}{4}\right) + j \sin\left(-\frac{\pi}{4}\right)\right)$$
$$= 3\left(\frac{\sqrt{2}}{2} - j\frac{\sqrt{2}}{2}\right) \approx 2.12 - j2.12\,\text{V}$$

EXAMPLE **Phasor to time domain.** The phasor $I = 5 - j4$ mA belongs to the angular frequency $\omega = 100\pi$ rad/s. Write $i(t)$ in Euler form.

$$|I| = \sqrt{25 + 16} = \sqrt{41} \approx 6.40\,\text{mA},$$
$$\delta = \arctan\left(\frac{-4}{5}\right) \approx -38.7^\circ \approx -0.675\,\text{rad},$$
$$i(t) = 6.40 \cdot e^{j(100\pi t - 0.675)}\,\text{mA}.$$

16.7 Summary

Concept	Formula
Euler's formula	$e^{j\delta} = \cos\delta + j\sin\delta$
Euler form	$z = \lvert z\rvert e^{j\delta}$
Multiplication	$z_1 z_2 = \lvert z_1\rvert\lvert z_2\rvert \angle (\delta_1 + \delta_2)$
Division	$z_1/z_2 = (\lvert z_1\rvert/\lvert z_2\rvert) \angle (\delta_1 - \delta_2)$
From phasor to time domain	$u(t) = \lvert U\rvert \sin(\omega t + \delta)$

What you should be able to do

After this chapter, you should be able to:

- Understand Euler's formula.
- Know the difference between a complex quantity and a phasor.
- Be able to multiply and divide complex numbers in Euler form.

Test yourself

ANSWER P. 264

1. Write $Z = 10 \angle \frac{\pi}{6}$ in rectangular form.

2. What happens geometrically to a complex number when it is multiplied by $2e^{j\pi/6}$?

Next chapter

Chapter 17 is about how complex numbers are used for calculations with sinusoidal signals.

16.8 Exercises

How to work with the exercises. Parts 1 and 2 are done during the lesson: first on your own, a few minutes per exercise, then together. Part 3 is extra exercises for further practice after the lesson.
Answers and worked solutions. The answer to every exercise is in the answer key at the back of the book, and its page is given next to the exercise. An answer shows whether you got it right, but not how. Full worked solutions, in Swedish, are published in the course repository after the lesson, in `lectures/L16/appendix/c_solutions.md`. Always try it yourself first.

PART 1

Review exercises

Exercise 1.1 Current in an RL circuit

ANSWER P. 263

A resistor $R = 100\,\Omega$ and an inductor $L = 1\,\text{mH}$ are connected in series and supplied with the voltage $U = 10\angle 0°\,\text{V}$, at the frequency $f = 50\,\text{kHz}$. The impedance of the inductor is

$$Z_L = j\omega L = j2\pi f L.$$

a) Calculate the impedance Z_L of the inductor. Keep π in the answer and round only at the end.

b) Calculate the total impedance $Z = R + Z_L$ of the circuit in both rectangular and polar form.

c) Calculate the current $I = \dfrac{U}{Z}$ in both polar and rectangular form.

Exercise 1.2 Current in an RC circuit

ANSWER P. 263

The inductor in Exercise 1.1 is replaced by a capacitor $C = 10\,\text{nF}$. The voltage $U = 10\angle 0°\,\text{V}$, the frequency $f = 50\,\text{kHz}$ and the resistor $R = 100\,\Omega$ are the same. The impedance of the capacitor is

$$Z_C = \frac{1}{j\omega C} = -\frac{j}{2\pi f C}.$$

a) Calculate the impedance Z_C of the capacitor. Keep π in the answer and round only at the end.

b) Calculate the total impedance $Z = R + Z_C$ of the circuit in both rectangular and polar form.

c) Calculate the current $I = \dfrac{U}{Z}$ in both polar and rectangular form.

d) Compare the current with the current in Exercise 1.1. What is nearly the same, and what differs?

PART 2

New material

Exercise 2.1 Rectangular form to Euler form

ANSWER P. 263

A voltage $u(t)$ in an AC circuit can be represented by a phasor U, which in rectangular form is written

$$U = -12 + j4\,\text{V}.$$

a) Draw the phasor U in the complex plane (the x-axis is the real part, the y-axis the imaginary part).

b) Express the phasor U in Euler form, that is, find the magnitude $|U|$ and the phase angle δ such that $U = |U|e^{j\delta}$.

c) Assume that the frequency of the voltage is $f = 100\,\text{Hz}$. Find the angular frequency ω.

d) Write the corresponding complex function of time for the voltage in Euler form, that is, $u(t) = |U|e^{j(\omega t+\delta)}$.

e) Draw the sine wave of the voltage $u(t) = |U|\sin(\omega t + \delta)$ over one period T.

Note: The phasor U is the complex voltage $u(t)$ without its time dependence.

Exercise 2.2 Euler form to rectangular form

ANSWER P. 263

A current $i(t)$ in an AC circuit is described by the complex function of time

$$i(t) = 10e^{j(100\pi t - 3\pi/4)}\,\text{mA}.$$

a) State the magnitude $|I|$, the phase angle δ (in degrees) and the angular frequency ω of the current.

b) Find the frequency f of the current.

c) Write the phasor $I = |I|e^{j\delta}$ in rectangular form.

d) Draw the phasor I in the complex plane (the x-axis is the real part, the y-axis the imaginary part).

e) Draw the sine wave of the current $i(t) = |I|\sin(\omega t + \delta)$ over one period T.

Note: The exercise goes the opposite way to Exercise 2.1, from Euler form to rectangular form.

PART 3

Extra exercises

The exercises below are extra practice, best done on your own after the lesson. Most of them can be done in your head or with pencil and paper.

Exercise 3.1 Euler's formula

ANSWER P. 263

Write in rectangular form.

a) $e^{j\pi/6}$ **b)** $e^{j\pi/4}$ **c)** $e^{j2\pi/3}$ **d)** $e^{-j\pi/3}$ **e)** $e^{-j3\pi/4}$

Exercise 3.2 Euler form to rectangular form

ANSWER P. 263

Write in rectangular form.

a) $z = 4e^{j\pi/6}$ **b)** $z = 5e^{j\pi/3}$ **c)** $z = 2e^{-j\pi/4}$ **d)** $z = 10e^{j2\pi/3}$

Exercise 3.3 Rectangular form to Euler form

ANSWER P. 263

Write in Euler form $|z|e^{j\delta}$.

a) $z = 3 + j3$ **b)** $z = -2 + j2$ **c)** $z = 1 - j\sqrt{3}$ **d)** $z = -4 - j3$

Exercise 3.4 Multiplication and division in polar form

ANSWER P. 263

Calculate with $z_1 = 4 \angle 30°$ and $z_2 = 3 \angle 45°$:

a) $z_1 z_2$ **b)** $\dfrac{z_1}{z_2}$ **c)** z_1^2

d) Why is polar form more convenient than rectangular form for multiplication?

Exercise 3.5 Comparing the methods

ANSWER P. 263

The complex numbers $z_1 = 3 + j4$ and $z_2 = 1 + j$ are given.

a) Calculate $z_1 z_2$ in rectangular form.

b) Write both numbers in polar form.

c) Multiply the numbers in polar form.

d) Check that the answers to a) and c) are the same number.

Exercise 3.6 Division in polar form

ANSWER P. 263

Calculate.

a) $\dfrac{10 \angle 80°}{2 \angle 20°}$ **b)** $\dfrac{12 \angle 0°}{4 \angle 90°}$ **c)** $\dfrac{6 \angle -30°}{3 \angle 60°}$

d) What happens to the phase angle when you divide by j?

Exercise 3.7 Powers of complex numbers

ANSWER P. 263

The number $z = 2\angle 30°$ is given.

a) Calculate z^2.

b) Calculate z^3, and write the answer in rectangular form as well.

c) Calculate z^4.

d) State the general rule for z^n in polar form.

Exercise 3.8 Phasor from a function of time

ANSWER P. 263

Give the phasor in polar form.

a) $u(t) = 10\sin\left(100\pi t + \frac{\pi}{6}\right)$ **b)** $u(t) = 4\sin\left(200\pi t - \frac{\pi}{2}\right)$

c) $i(t) = 6\sin(50t)$

d) What is required for two phasors to be added?

Exercise 3.9 Function of time from a phasor

ANSWER P. 263

The angular frequency is $\omega = 100\pi$ rad/s. Write the corresponding function of time.

a) $U = 8\angle 45°$ V **b)** $U = 5\angle -90°$ V **c)** $I = 3 + j4$ mA **d)** $I = -2 - j2$ mA

Exercise 3.10 Phasor addition

ANSWER P. 263

Two voltages with the same frequency are given by $U_1 = 6\angle 0°$ V and $U_2 = 8\angle 90°$ V.

a) Write both in rectangular form.

b) Calculate the sum $U = U_1 + U_2$.

c) Give the sum in polar form.

d) Write $u(t)$ if $\omega = 100\pi$ rad/s.

Exercise 3.11 Impedance from voltage and current

ANSWER P. 264

A circuit has $U = 20\angle 60°$ V and $I = 4\angle 15°$ A.

a) Calculate the impedance $Z = \dfrac{U}{I}$ in polar form.

b) Write Z in rectangular form.

c) Is the circuit inductive or capacitive?

d) What are the resistance and the reactance?

Exercise 3.12 Impedance of an inductor and a capacitor

ANSWER P. 264

At $\omega = 1000\,\text{rad/s}$, $L = 20\,\text{mH}$ and $C = 50\,\mu\text{F}$.

a) Calculate $Z_L = j\omega L$.

b) Calculate $Z_C = -\dfrac{j}{\omega C}$.

c) A resistor $R = 15\,\Omega$ is connected in series with both. Calculate the total impedance.

d) What is this condition called?

Exercise 3.13 Rotation by j

ANSWER P. 264

The number $z = 3 + j4$ is given.

a) Calculate jz.

b) Compare $|z|$ and $|jz|$.

c) Compare the phase angles of the two numbers.

d) What does multiplication by j do geometrically?

Exercise 3.14 True or false

ANSWER P. 264

Decide whether each statement is true or false, and justify your answer briefly.

a) Two complex numbers with the same magnitude are always equal.

b) $|z_1 z_2| = |z_1| + |z_2|$

c) In division, the phase angles are subtracted.

d) A phasor contains information about the frequency of the signal.

e) The phase angle of $z = a + jb$ is always $\tan^{-1}\frac{b}{a}$.

f) Multiplying complex numbers is easiest in rectangular form.

PART 4

Mathematics in C

The teacher shows the exercise below during the lesson if time allows, in VS Code and with `make`. You can also do it yourself in **OnlineGDB**, `https://www.onlinegdb.com/online_c_compiler`, with the language set to **C**. The struct `complex_t` and the functions from Exercise 4.1 of Chapter 15 are also used here.

Exercise 4.1 Rotation by j in C

ANSWER P. 264

The exercise builds on Exercise 3.13. The number $z = 3 + j4$ is given.

a) Compute jz, j^2z, j^3z and j^4z by hand.

b) Show that $j(a + jb) = -b + ja$. Then write the function `complex_t complex_rotate_j(const complex_t* z)`, which returns jz. Multiply z by j four times in a loop, with `z = complex_rotate_j(&z);`. Print the number in rectangular form and its argument in degrees before the first multiplication and after each multiplication.

c) How much does the argument change with each multiplication? Why is $j^4z = z$?

d) Change the call to `complex_rotate_j(&z);`, without `z =`. What is printed now? Explain why, given that the parameter is `const complex_t*`.

CHAPTER 17

Complex numbers, part III

Sinusoidal signals as phasors, and impedance as a complex number.

IN THIS CHAPTER

- Complex numbers for calculations with sinusoidal signals.
- Converting between a sinusoidal signal and a phasor.
- Phasor addition in five steps.
- Multiplication and division with phasors.
- Impedance of a resistor, an inductor and a capacitor.

This chapter concludes the three chapters on complex numbers. In Chapter 16 you learned that a sinusoidal signal can be represented by a phasor, a complex number that carries the signal's amplitude and phase angle. Here the phasors are put to the use they are made for: adding sinusoidal voltages and currents of the same frequency, and calculating with impedance, where voltage, current and impedance are related exactly as in Ohm's law, but with complex numbers. You will also see how the phase angle of the impedance decides whether the current leads or lags the voltage.

§17.1 to §17.5 are the chapter's theory, with worked examples. §17.6 summarises the formulas and what you should be able to do after the chapter, and §17.7 holds the exercises. The angles are sometimes given in degrees and sometimes in radians, so keep track of which mode your calculator is in.

17.1 Sinusoidal signals and phasors

A sinusoidal signal and its phasor are related by

$$u(t) = |U|\sin(\omega t + \delta) \quad \longleftrightarrow \quad U = |U| \angle \delta.$$

From sinusoidal signal to phasor: read the amplitude $|U|$ and the phase angle δ directly from the expression for $u(t)$.

From phasor to sinusoidal signal: write $u(t) = |U|\sin(\omega t+\delta)$ with the phasor $U = |U| \angle \delta$.

Note: All the phasors in a calculation must have the **same angular frequency** ω.

17.2 Phasor addition

Sinusoidal signals are most easily added in the phasor plane:

1. Convert each sinusoidal signal into a phasor in polar form.
2. Convert the phasors to rectangular form.
3. Add the real and imaginary parts.
4. Convert the sum back to polar form.
5. Write the sum as a sinusoidal signal.

17.3 Multiplication and division with phasors

Multiplication in polar form:

$$U_1 \cdot U_2 = |U_1||U_2| \angle (\delta_1 + \delta_2)$$

Division in polar form:

$$\frac{U_1}{U_2} = \frac{|U_1|}{|U_2|} \angle (\delta_1 - \delta_2)$$

These are used in impedance calculations, among other things: $U = Z \cdot I$.

17.4 Application: impedance

A voltage U and a current I, both complex, give the impedance Z:

$$Z = \frac{U}{I} = \frac{|U|}{|I|} \angle (\delta_U - \delta_I)$$

For the three basic components:

- Resistance: $Z_R = R$ (real, phase angle 0).
- Inductance: $Z_L = j\omega L$ (phase angle $+90°$).
- Capacitance: $Z_C = \dfrac{1}{j\omega C} = -\dfrac{j}{\omega C}$ (phase angle $-90°$).

The derivation of Z_L and Z_C is in §16.5.2.

17.5 Worked examples

EXAMPLE **Phasor addition.** Add $u_1(t) = 2\sin(\omega t + 45^\circ)$ V and $u_2(t) = 3\sin(\omega t - 60^\circ)$ V. *Step 1:* the phasors in polar form:

$$U_1 = 2\angle 45^\circ, \qquad U_2 = 3\angle -60^\circ$$

Step 2: convert to rectangular form:

$$U_1 = 2(\cos 45^\circ + j\sin 45^\circ) = \sqrt{2} + j\sqrt{2} \approx 1.414 + j1.414$$
$$U_2 = 3(\cos(-60^\circ) + j\sin(-60^\circ)) = 3(0.5 - j0.866) = 1.5 - j2.598$$

Step 3: add:

$$U_{\text{tot}} = (1.414 + 1.5) + j(1.414 - 2.598) = 2.914 - j1.184$$

Step 4: convert to polar form:

$$|U_{\text{tot}}| = \sqrt{2.914^2 + 1.184^2} \approx 3.15\,\text{V}$$
$$\delta_{\text{tot}} = \arctan\left(\frac{-1.184}{2.914}\right) \approx -22.1^\circ$$

Step 5: back to the time domain:

$$u_{\text{tot}}(t) = 3.15\sin(\omega t - 22.1^\circ)\,\text{V}$$

EXAMPLE **Finding the phase from an equation.** An AC voltage is given by $u(t) = 6\sin(80\pi t + \delta)$ V. At $t = 15$ ms, $u = 3$ V. Find δ.

$$6\sin(80\pi \cdot 0.015 + \delta) = 3 \quad \Rightarrow \quad \sin(1.2\pi + \delta) = 0.5$$
$$\delta_1 = \arcsin(0.5) - 1.2\pi \approx 0.5236 - 3.7699 \approx -3.25\,\text{rad}$$
$$\delta_2 = \pi - \arcsin(0.5) - 1.2\pi \approx -1.15\,\text{rad}$$

Checking both roots gives $u(0.015) = 3$ V.

EXAMPLE **Calculating an impedance.** A series RLC circuit has $R = 10\,\Omega$, $Z_L = j5\,\Omega$ and $Z_C = -j3\,\Omega$ at $\omega = 1000$ rad/s. The total impedance is

$$Z = R + Z_L + Z_C = 10 + j5 - j3 = 10 + j2\,\Omega,$$
$$|Z| = \sqrt{100 + 4} \approx 10.20\,\Omega, \qquad \delta_Z = \arctan\left(\frac{2}{10}\right) \approx 11.3^\circ.$$

17.6 Summary

Concept	Description
Phasor	$U = \lvert U\rvert \angle \delta$ represents $u(t) = \lvert U\rvert \sin(\omega t + \delta)$
Phasor addition	Convert to rectangular form, add, convert back
Multiplication	$\lvert z_1\rvert\lvert z_2\rvert \angle (\delta_1 + \delta_2)$
Division	$(\lvert z_1\rvert/\lvert z_2\rvert) \angle (\delta_1 - \delta_2)$
Impedance	$Z = U/I$

What you should be able to do

After this chapter, you should understand how complex numbers can be used for calculations with sinusoidal signals.

Test yourself

ANSWER P. 265

1. A current is given by $i(t) = 5\sin(100\pi t + \pi/3)$ A. Give the corresponding phasor I in polar form.
2. Add the phasors $U_1 = 4 \angle 0$ and $U_2 = 3 \angle \frac{\pi}{2}$. Give the sum in polar form.

Next chapter

Chapter 18 is a lab in mathematics in C with the standard library `math.h`.

17.7 Exercises

How to work with the exercises. Parts 1 and 2 are done during the lesson: first on your own, a few minutes per exercise, then together. Part 3 is extra exercises for further practice after the lesson.

Answers and worked solutions. The answer to every exercise is in the answer key at the back of the book, and its page is given next to the exercise. An answer shows whether you got it right, but not how. Full worked solutions, in Swedish, are published in the course repository after the lesson, in `lectures/L17/appendix/c_solutions.md`. Always try it yourself first.

PART 1

Review exercises

Exercise 1.1 Impedance of an RC circuit

ANSWER P. 264

A resistor R and a capacitor C are connected in series and supplied with the voltage $U = 10\angle 0°$ V. The current through the circuit is measured as $I = 50\angle 30°$ mA, and the frequency is $f = 1$ kHz.

a) Calculate the impedance $Z = \dfrac{U}{I}$ of the circuit in polar form.

b) Convert Z to rectangular form, $Z = R - jX_C$, and find the resistance R and the reactance X_C of the capacitor.

c) Calculate the capacitance C that gives the calculated reactance, where $X_C = \dfrac{1}{\omega C}$.

Exercise 1.2 Rectangular form to Euler form

ANSWER P. 264

A voltage $u(t)$ in an AC circuit can be represented by a phasor U, which in rectangular form is written

$$U = 3 - j6\,\text{V}.$$

a) Draw the phasor U in the complex plane (the x-axis is the real part, the y-axis the imaginary part).

b) Express the phasor U in Euler form, that is, find the magnitude $|U|$ and the phase angle δ such that $U = |U|e^{j\delta}$.

c) Assume that the frequency of the voltage is $f = 50$ Hz. Find the angular frequency ω.

d) Write $u(t)$ as a function of time, $u(t) = |U|e^{j(\omega t+\delta)}$.

PART 2
New material

Exercise 2.1 Adding three currents in an AC circuit

ANSWER P. 264

A node in an electric circuit is fed by three sinusoidal currents:

$$\begin{aligned} i_1(t) &= 5\sin(\omega t + 30^\circ)\ \text{mA}, \\ i_2(t) &= 3\sin(\omega t - 30^\circ)\ \text{mA}, \\ i_3(t) &= 4\sin(\omega t + 120^\circ)\ \text{mA}. \end{aligned}$$

The total current $i_{\text{tot}}(t)$ in the circuit is

$$i_{\text{tot}}(t) = i_1(t) + i_2(t) + i_3(t).$$

a) Rewrite the currents $i_1(t)$, $i_2(t)$ and $i_3(t)$ as phasors I_1, I_2 and I_3 in complex rectangular form.

b) Calculate the phasor sum $I_{\text{tot}} = I_1 + I_2 + I_3$.

c) Draw the phasors in the complex plane (the x-axis is the real part, the y-axis the imaginary part).

d) Convert the result back into a sinusoidal current in the time domain, of the form

$$i_{\text{tot}}(t) = |I_{\text{tot}}|\sin(\omega t + \delta).$$

PART 3
Extra exercises

The exercises below are extra practice, best done on your own after the lesson. Most of them can be done in your head or with pencil and paper.

Exercise 3.1 Phasor from a sinusoidal signal

ANSWER P. 264

Give the phasor in polar form.

a) $u(t) = 12\sin(\omega t + 60^\circ)$

b) $u(t) = 5\sin(\omega t - 45^\circ)$

c) $i(t) = 2\sin(\omega t)$

d) $i(t) = 7\sin(\omega t + 180^\circ)$

Exercise 3.2 Phasors in rectangular form

ANSWER P. 264

Write the phasors in rectangular form.

a) $10\angle 0^\circ$ **b)** $4\angle 90^\circ$ **c)** $6\angle 30^\circ$ **d)** $5\angle -53.1^\circ$

Exercise 3.3 Adding two voltages

ANSWER P. 264

Two voltages with the same frequency are $u_1(t) = 3\sin(\omega t)$ V and $u_2(t) = 4\sin(\omega t + 90°)$ V.

a) Give the phasors in polar form.

b) Write them in rectangular form.

c) Calculate the phasor sum and give it in polar form.

d) Write $u_{\text{tot}}(t)$.

Exercise 3.4 Adding three phasors

ANSWER P. 264

Three voltages are given by $U_1 = 10 \angle 0°$ V, $U_2 = 5 \angle 90°$ V and $U_3 = 5 \angle -90°$ V.

a) Write all of them in rectangular form.

b) Calculate the sum.

c) Give the sum in polar form.

d) Explain why U_2 and U_3 cancel.

Exercise 3.5 Sum and difference of two phasors

ANSWER P. 264

Two voltages are given by $U_1 = 8 \angle 30°$ V and $U_2 = 8 \angle -30°$ V.

a) Write both in rectangular form.

b) Calculate $U_1 + U_2$ and give the answer in polar form.

c) Calculate $U_1 - U_2$ and give the answer in polar form.

d) Comment on the results.

Exercise 3.6 Multiplying and dividing phasors

ANSWER P. 264

Calculate.

a) $(5 \angle 20°)(4 \angle 40°)$

b) $\dfrac{20 \angle 100°}{5 \angle 40°}$

c) $(6 \angle -30°)(2 \angle -30°)$

d) $\dfrac{9 \angle 0°}{3 \angle 90°}$

Exercise 3.7 Impedance of R, L and C

ANSWER P. 265

At $\omega = 2000$ rad/s, $R = 100\,\Omega$, $L = 50$ mH and $C = 5\,\mu$F.

a) Give Z_R and its phase angle.

b) Calculate $Z_L = j\omega L$ and give its phase angle.

c) Calculate $Z_C = -\dfrac{j}{\omega C}$ and give its phase angle.

d) What do the phase angles mean for the current through each component?

Exercise 3.8 Series RL circuit

ANSWER P. 265

A series circuit has $R = 40\,\Omega$ and $X_L = 30\,\Omega$.

a) Give the impedance Z in rectangular form.

b) Calculate $|Z|$ and the phase angle.

c) The circuit is supplied with $U = 100 \angle 0°$ V. Calculate the current.

d) Does the current lead or lag the voltage?

Exercise 3.9 Series RC circuit

ANSWER P. 265

A series circuit has $R = 60\,\Omega$ and $X_C = 80\,\Omega$.

a) Give the impedance Z in rectangular form.

b) Calculate $|Z|$ and the phase angle.

c) The circuit is supplied with $U = 200 \angle 0°$ V. Calculate the current.

d) Does the current lead or lag the voltage?

Exercise 3.10 Series RLC circuit

ANSWER P. 265

A series circuit has $R = 20\,\Omega$, $X_L = 60\,\Omega$ and $X_C = 45\,\Omega$.

a) Give the impedance Z in rectangular form.

b) Calculate $|Z|$ and the phase angle.

c) The circuit is supplied with $U = 50 \angle 0°$ V. Calculate the current.

d) Is the circuit inductive or capacitive?

Exercise 3.11 Resonance

ANSWER P. 265

A series circuit has $R = 10\,\Omega$, $L = 100\,\text{mH}$ and $C = 10\,\mu\text{F}$. Resonance occurs at $\omega_0 = \dfrac{1}{\sqrt{LC}}$.

a) Calculate ω_0.

b) Calculate the resonant frequency f_0.

c) Calculate X_L and X_C at ω_0.

d) What impedance does the circuit have at resonance?

Exercise 3.12 Voltage division in an AC circuit

ANSWER P. 265

A series circuit with $Z_R = 30\,\Omega$ and $Z_L = j40\,\Omega$ is supplied with $U = 100 \angle 0°\,\text{V}$.

a) Calculate Z_{tot} and $|Z_{\text{tot}}|$.

b) Calculate the current I.

c) Calculate the voltage U_R across the resistor.

d) Calculate the voltage U_L across the inductor.

e) Show that $U_R + U_L = U$ as phasors, even though $|U_R| + |U_L| \neq |U|$.

Exercise 3.13 Finding the phase from an equation

ANSWER P. 265

An AC voltage is given by $u(t) = 10\sin(100\pi t + \delta)\,\text{V}$. At $t = 5\,\text{ms}$, $u = 5\,\text{V}$.

a) Set up the equation for δ.

b) Solve the equation as two cases.

c) Check both solutions.

Exercise 3.14 Impedances in parallel

ANSWER P. 265

The impedances $Z_1 = 10\,\Omega$ and $Z_2 = j10\,\Omega$ are connected in parallel.

a) Calculate $Z = \dfrac{Z_1 Z_2}{Z_1 + Z_2}$ in rectangular form.

b) Calculate $|Z|$ and the phase angle.

c) Compare with the series connection of the same impedances.

d) What do the two phase angles have in common?

Exercise 3.15 From the phasor domain to the time domain

ANSWER P. 265

The current $i(t) = 4\sin(500t - 30°)\,\text{A}$ flows through a circuit. The circuit's impedance is $Z = 25 \angle 60°\,\Omega$.

a) Give the phasor of the current.

b) Calculate the phasor of the voltage, $U = ZI$.

c) Write $u(t)$.

d) What is the phase difference of the voltage relative to the current?

Exercise 3.16 Two currents at a node

ANSWER P. 265

A node is fed by $i_1(t) = 6\sin(\omega t + 45°)$ mA and $i_2(t) = 8\sin(\omega t - 45°)$ mA.

a) Give the phasors in polar form.

b) Write them in rectangular form.

c) Calculate the phasor sum and give it in polar form.

d) Write $i_{\text{tot}}(t)$.

Exercise 3.17 True or false

ANSWER P. 265

Decide whether each statement is true or false, and justify your answer briefly.

a) Phasors with different angular frequencies may be added.

b) An inductance has an impedance with the phase angle +90°.

c) $|Z| = R + X$

d) In a purely resistive circuit, current and voltage are in phase.

e) In a capacitive circuit, the current lags the voltage.

f) At resonance, the impedance of the circuit is purely resistive.

PART 4

Mathematics in C

The teacher shows the exercise below during the lesson if time allows, in VS Code and with `make`. You can also do it yourself in **OnlineGDB**, `https://www.onlinegdb.com/online_c_compiler`, with the language set to **C**. The struct `complex_t` and the functions from Exercise 4.1 of Chapter 15 are also used here.

Exercise 4.1 Phasor addition in C

ANSWER P. 265

The exercise checks the worked example in §17.5, where the sinusoidal voltages

$$u_1(t) = 2\sin(\omega t + 45°)\ \text{V}, \qquad u_2(t) = 3\sin(\omega t - 60°)\ \text{V}$$

are added with phasors.

a) Give the phasors U_1 and U_2 in rectangular form, and compute $U_{tot} = U_1 + U_2$ in both rectangular and polar form. Calculate by hand.

b) Write the function `complex_t complex_add(const complex_t* a, const complex_t* b)`, which returns the sum of two complex numbers. Create the phasors with `complex_from_polar(2.0, 45.0)` and `complex_from_polar(3.0, -60.0)`, store them in a variable each and compute U_{tot}. Print the sum in rectangular and polar form, with the argument in degrees.

c) Extend the program so that it also computes $u_1(0) + u_2(0)$ with `sin()`. Compare with the imaginary part of U_{tot}, and explain why they are equal.

CHAPTER 18

Mathematics in C

The mathematics of the course written as programs: from hand calculation to code that calculates the same way.

IN THIS CHAPTER

- The standard library `math.h`: functions, constants and linking with `-lm`.
- Floating-point numbers in C: `float` and `double`, and how they differ from the integer types you have already used.
- Powers, roots, logarithms and decibels in code.
- Trigonometric functions, and converting between degrees and radians.
- Numerical differentiation and integration.
- Complex numbers and phasors with `complex.h`.
- Sampling a sinusoidal signal – from formula to sequence of numbers.
- Machine learning: a linear regression model in C and a small neural network in C++.

This chapter is a lab. It gathers the mathematics of the earlier chapters – powers, logarithms and decibels, trigonometry, derivatives, integrals and complex numbers – and lets a computer do the same calculations. Each exercise is solved in two steps: first by hand, then with a C program that uses the standard libraries `math.h` and `complex.h`, after which the results are compared. The code is written and run in the web browser, in the OnlineGDB compiler, so no installation is needed. The chapter assumes basic C programming: variables, functions, `printf()` and loops. Floating-point numbers are covered in the chapter.

§18.1 to §18.11 are the theory: how each part of the mathematics is written in C and what pitfalls there are, and §18.12 works through five examples. §18.13 summarises what you should be able to do, and §18.14 holds the lab's exercises. Parts 1–3 are compulsory, as are the hand calculations in Exercises 6.1, 6.4 a, 7.1 and 7.4. The other exercises in Parts 4–7 are optional further study. In Parts 6 and 7 a linear regression model and a neural network are trained, using the mathematics of the earlier chapters.

Preparation: Read the theory and do the **hand calculations for all the exercises in Parts 1–3, and Exercises 6.1, 6.4 a, 7.1 and 7.4**, before the lesson. The lesson time is not enough for both calculating and programming, and the calculations are required to pass. Also open `https://www.onlinegdb.com/online_c_compiler`, check that the language is set to **C** and press **Run**: the example program should print `Hello World`.

18.1 The standard library `math.h`

`math.h` is the part of C's standard library that contains mathematical functions. The functions work with floating-point numbers of type `double`.

```
#include <math.h>
#include <stdio.h>
```

18.1.1 Development environment

During the lesson you use OnlineGDB, a `gcc` compiler that runs directly in the web browser: `https://www.onlinegdb.com/online_c_compiler`. No installation is needed, and no account.

1. Check that the language is set to **C** in the list at the top right.
2. Write the code in the editor.
3. Press **Run**. The program's output appears in the box below the editor.

Note: The code is not saved automatically. Copy it to a file of your own before you close the tab.

Programs that print sample values as CSV (§18.10.1) are run in the usual way. Select the output in the output box, copy it and paste it into a spreadsheet program to plot the signal.

18.1.2 Compiling on your own computer (recommended)

It is a good idea to set up **WSL** (Windows Subsystem for Linux) with `gcc` and `make` and run the code locally. The environment is used in later courses, so it pays to set it up now. It is required for the programming exercises in Parts 6 and 7 in §18.14, but not for Parts 1–5, where OnlineGDB is enough. WSL runs a Linux distribution, here Ubuntu, in a terminal directly in Windows, without a virtual machine. The installation requires administrator rights and a restart.

Open **Windows PowerShell** as administrator and run:

```
wsl --install
wsl --set-default-version 2
```

Restart the computer if the system asks you to. The command `wsl --status` should then show `Default Version: 2`. If Ubuntu was not installed, the latest LTS version is in the Microsoft Store. WSL requires Windows 11, or Windows 10 version 2004 (build 19041) or later, and virtualisation enabled in the computer's UEFI/BIOS. Without administrator rights, WSL cannot be installed; use OnlineGDB instead.

Then start **Ubuntu** from the Windows search box. On the first start you choose a user name and a password; the password is not shown while you type it, which is normal. Update the system and install the development tools, including `gcc` and `make`:

```
sudo apt -y update
sudo apt -y upgrade
sudo apt -y install build-essential
gcc --version
```

The Windows files are reached from Ubuntu under `/mnt/c/`, but work is faster if the files are in the Linux home directory, ~. **VS Code** with the **WSL** extension opens the files directly in the Ubuntu environment; `code .` in the terminal opens the current directory.

18.1.3 Linking with the maths library

The functions in `math.h` and `complex.h` are in a separate maths library, `libm`. On your own computer the program must therefore be linked with the flag `-lm`, which is placed **last** on the command line, after the source files:

```
gcc main.c -o main -lm
./main
```

Without `-lm` the program compiles, but linking fails with error messages such as `undefined reference to 'sqrt'`. In OnlineGDB the library is linked automatically, so no setting is needed there.

18.1.4 Constants

`M_PI` (π) and `M_E` (e) are in `math.h` on most systems, but they are not part of the C standard and are therefore missing in strict standard mode (`-std=c11`). OnlineGDB uses `gcc`'s default mode, so `M_PI` works there, as it does with `-std=gnu11`. For the code to work everywhere, the constant can be defined directly in the program:

```
#define PI 3.14159265358979323846
```

18.2 Floating-point numbers in C: `float` and `double`

The integer types you have already used, such as `int`, `uint8_t` and `uint16_t`, can only store whole numbers: `7 / 2` becomes 3, not 3.5. The mathematics in this course requires decimal numbers, and for those **floating-point numbers** are used.

Why are floating-point numbers avoided in microcontroller code? An AVR microcontroller has no floating-point unit (FPU). Floating-point calculations must therefore be done in software, which is slow and takes a lot of memory. On AVR, `double` is moreover often only 32 bits, the same precision as `float`. An ordinary computer has an FPU and calculates with floating-point numbers at no extra cost – which is why they are used freely in this chapter.

18.2.1 Types

Type	Size	Approximate precision	Use
`float`	32 bits	about 7 significant figures	When memory is limited
`double`	64 bits	about 15 significant figures	**The default in this course**

Every function in `math.h` takes and returns `double`. So use `double` throughout.

18.2.2 Literals

How a number is written decides which type it gets:

```
int    a = 2;      // Integer.
double b = 2.0;    // Floating point (double).
float  c = 2.0f;   // Floating point (float).
```

Note: If two integers are divided, integer division is performed, whatever type the result is stored in:

```
double x = 1 / 2;      // 0.0 - integer division!
double y = 1.0 / 2.0;  // 0.5
```

18.2.3 Printing

Type	Format specifier
`int`, `uint8_t`, `uint16_t`	`%d`
`float`, `double`	`%f`, for example `%.3f` for three decimals

`printf()` automatically converts `float` to `double`, so `%f` works for both. Never use `%d` for a floating-point number – the output is then meaningless.

18.2.4 Converting between integers and floating-point numbers

Integers are converted to floating-point numbers automatically. In the other direction the number is **truncated**, not rounded:

```
const double d = 7;           // 7.0
const int    n = 3.9;         // 3, not 4!
const int    m = round(3.9);  // 4
```

18.2.5 Precision

Floating-point numbers are stored in binary and cannot represent every decimal number exactly. The expression `0.1 + 0.2` gives `0.30000000000000004`, not exactly `0.3`.

Never compare floating-point numbers with ==, therefore. Check instead that the difference is small enough:

```
if (fabs(a - b) < 1e-9) { /* The numbers are considered equal. */ }
```

18.3 Basic functions

Mathematics	C function	Comment
$\sqrt{x}$	`sqrt(x)`	Square root, $x \geq 0$
x^y	`pow(x, y)`	Power
e^x	`exp(x)`	Exponential function
$\ln x$	`log(x)`	Natural logarithm, $x > 0$
$\lg x$	`log10(x)`	Base-10 logarithm, $x > 0$
$\|x\|$	`fabs(x)`	Absolute value of a floating-point number
$\sqrt{x^2 + y^2}$	`hypot(x, y)`	Magnitude of a vector
Remainder of a division	`fmod(x, y)`	Like `%`, but for floating-point numbers
Rounding	`round(x)`	`floor(x)` down, `ceil(x)` up

Note: `%` works only for integers. For floating-point numbers, `fmod()` is used.

```c
const double r = sqrt(pow(3.0, 2.0) + pow(4.0, 2.0)); // r = 5.0
printf("r = %.2f\n", r);
```

18.4 Trigonometric functions

Mathematics	C function	Return value
$\sin(x)$	`sin(x)`	–
$\cos(x)$	`cos(x)`	–
$\tan(x)$	`tan(x)`	–
$\arcsin(x)$	`asin(x)`	$[-\pi/2,\ \pi/2]$
$\arccos(x)$	`acos(x)`	$[0,\ \pi]$
$\arctan(x)$	`atan(x)`	$(-\pi/2,\ \pi/2)$
$\arctan(y/x)$	`atan2(y, x)`	$(-\pi,\ \pi]$

All the functions work in radians, never in degrees. `sin(30)` is read as $\sin(30\,\text{rad})$ and gives -0.988, not 0.5.

18.4.1 Converting between degrees and radians

$$\text{rad} = \text{degrees} \cdot \frac{\pi}{180}, \qquad \text{degrees} = \text{rad} \cdot \frac{180}{\pi}$$

```c
const double deg = 30.0;
const double rad = deg * PI / 180.0;
printf("sin(30 degrees) = %.3f\n", sin(rad)); // 0.500
```

18.4.2 `atan2()` instead of `atan()`

`atan(y / x)` cannot tell quadrant 1 from quadrant 3, or quadrant 2 from quadrant 4, because the signs disappear in the division. `atan2(y, x)` takes y and x separately and therefore gives the right angle in all four quadrants. Always use `atan2()` when converting from rectangular to polar form (Chapter 15).

18.5 Powers, logarithms and decibels

From Chapter 10, for voltage and power respectively:

$$A_{\text{dB}} = 20\log_{10}\left(\frac{U_{\text{out}}}{U_{\text{in}}}\right), \qquad A_{\text{dB}} = 10\log_{10}\left(\frac{P_{\text{out}}}{P_{\text{in}}}\right)$$

```
const double gain_db = 20.0 * log10(u_out / u_in);
```

The way back from decibels to a ratio:

$$\frac{U_{\text{out}}}{U_{\text{in}}} = 10^{A_{\text{dB}}/20}$$

```
const double ratio = pow(10.0, gain_db / 20.0);
```

Logarithms to other bases are calculated with the change-of-base rule $\log_b x = \frac{\ln x}{\ln b}$:

```
const double log2_x = log(x) / log(2.0);
```

18.6 Numerical differentiation

The derivative is defined as the limit

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

A computer cannot let $h \to 0$, but a small h gives an approximate value. The **central difference** gives better accuracy than the formula above, because the errors on either side of x cancel each other out:

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

```
double derivative(double (*f)(double), const double x, const double h)
{
    return (f(x + h) - f(x - h)) / (2.0 * h);
}
```

The call is made with a function pointer, that is, the name of the function to be differentiated:

```
const double slope = derivative(square, 3.0, 1e-6);
```

18.6.1 Choosing h

- An h that is too **large** gives a truncation error – the approximation lies too far from the limit.
- An h that is too **small** gives a rounding error – $f(x+h)$ and $f(x-h)$ become so alike that the significant figures vanish in the subtraction.

For `double`, $h \approx 10^{-6}$ is a good compromise.

18.7 Numerical integration

The definite integral corresponds to the area under the curve (Chapter 14). If the interval $[a, b]$ is divided into n equal parts of width $h = \frac{b-a}{n}$, the area can be approximated with the **trapezoidal rule**:

$$\int_a^b f(x)\,dx \approx h\left[\frac{f(a)+f(b)}{2} + \sum_{k=1}^{n-1} f(a+kh)\right]$$

Each subinterval is thus approximated by a trapezium instead of a rectangle, which follows the curve considerably better. The accuracy increases with the number of subintervals n.

```
double integral(double (*f)(double), const double a, const double b, const int n)
{
    const double h = (b - a) / n;
    double sum     = (f(a) + f(b)) / 2.0;

    for (int k = 1; k < n; ++k)
    {
        sum += f(a + k * h);
    }
    return sum * h;
}
```

For a first-degree polynomial the trapezoidal rule gives an **exact** result, since the curve is then a straight line. For other functions there is an error, which decreases as n increases.

18.8 Complex numbers with `complex.h`

In Chapters 15–17, complex numbers were written as a struct of their own, `complex_t`. The standard library `complex.h` instead has a ready-made type and is used together with `math.h`:

```
#include <complex.h>
#include <math.h>
```

The type is called `double complex` and the imaginary unit is called `I`:

```
const double complex z = 3.0 + 4.0 * I;
```

> **Note:** In electrical engineering the imaginary unit is written j, so that it is not confused with the current i. In C it is always called `I`.

18.8.1 Functions

Mathematics	C function	Description
$\mathrm{Re}(z)$	`creal(z)`	The real part
$\mathrm{Im}(z)$	`cimag(z)`	The imaginary part
$\lvert z\rvert$	`cabs(z)`	The magnitude
$\arg(z)$	`carg(z)`	The argument in radians, $(-\pi, \pi]$
$\bar{z}$	`conj(z)`	The conjugate
e^z	`cexp(z)`	The exponential function
$\sqrt{z}$	`csqrt(z)`	The square root

The four arithmetic operations are written as usual, with `+`, `-`, `*` and `/`.

18.8.2 Rectangular and polar form

From rectangular to polar form:

```
const double magnitude = cabs(z);        // |z|
const double angle     = carg(z);        // arg(z) in radians
```

From polar to rectangular form, Euler's formula $z = re^{j\theta}$ is used:

```
const double complex z = r * cexp(I * theta);
```

Note: `printf()` cannot print a complex number directly. The real and imaginary parts are printed separately:

```
printf("z = %.2f + %.2fj\n", creal(z), cimag(z));
```

18.9 Impedance and phasors

With complex numbers, impedances are calculated directly in code (Chapters 16–17). With the angular frequency $\omega = 2\pi f$:

$$Z_R = R, \qquad Z_L = j\omega L, \qquad Z_C = \frac{1}{j\omega C} = -\frac{j}{\omega C}$$

```
const double omega          = 2.0 * PI * f;
const double complex z_r    = r;
const double complex z_l    = I * omega * l;
const double complex z_c    = -I / (omega * c);
const double complex z_tot  = z_r + z_l; // Series connection.
```

A phasor $U = |U|\angle\delta$ is written as a complex number in polar form. Phasors are then added with ordinary addition:

```
const double complex u1 = a1 * cexp(I * d1);
const double complex u2 = a2 * cexp(I * d2);
const double complex u  = u1 + u2;
```

The amplitude and phase angle of the result are then given by `cabs()` and `carg()` respectively.

18.10 Sampling a sinusoidal signal

A sinusoidal voltage is described by

$$u(t) = |U| \sin(2\pi f t + \delta),$$

where $|U|$ is the amplitude in V, f the frequency in Hz, t the time in s and δ the phase angle in rad.

In **sampling**, the signal's value is calculated at evenly spaced times. With the sampling frequency f_s, the number of samples per period is

$$N = \frac{f_s}{f},$$

and the kth sample value $(k = 0, 1, \ldots, N-1)$ is given by

$$u_k = |U| \sin\left(\frac{2\pi k}{N} + \delta\right).$$

For the signal to be recoverable from its samples, the **sampling theorem** must be satisfied:

$$f_s > 2f$$

```
for (int k = 0; k < n; ++k)
{
    const double u_k = amplitude * sin(2.0 * PI * k / n + delta);
    printf("%d,%.6f\n", k, u_k);
```

```
}
```

18.10.1 Printing for further analysis

If the sample values are printed as **CSV** (comma-separated values), one line per sample, they can be read into another program to be plotted:

```
k,u
0,0.000000
1,3.535534
2,5.000000
```

In OnlineGDB, select the output in the output box and copy it into a spreadsheet program, where the signal can be plotted as a chart.

If the program is run on your own computer, the output can instead be saved directly to a file by redirection in the terminal:

```
./samples > samples.csv
```

The file can then be opened in a spreadsheet program, or fed into a simulator that plots the signal.

18.11 Common pitfalls

Pitfall	Consequence	Remedy
Forgotten `-lm`	`undefined reference to 'sqrt'`	Put the flag last when compiling
Degrees instead of radians	Completely wrong value from `sin()`, `cos()`, `tan()`	Multiply by $\pi/180$
Integer division, e.g. `1 / 2`	Gives `0`, not `0.5`	Write `1.0 / 2.0`
`atan(y / x)` for polar form	Wrong quadrant	Use `atan2(y, x)`
Comparing floating-point numbers with `==`	Rarely true	Compare with `fabs(a - b) < 1e-9`
`sqrt()` or `log()` with an invalid argument	Gives `nan` and `-inf` respectively	Check the argument first
`%d` for a `double`	Undefined behaviour	Use `%f`, for example `%.3f`

18.12 Worked examples

EXAMPLE **Gain and decibels.** An amplifier has the input voltage $U_{\text{in}} = 0.5\,\text{V}$ and the output voltage $U_{\text{out}} = 8.0\,\text{V}$. Calculate the gain in dB.

$$A_{\text{dB}} = 20\log_{10}\left(\frac{8.0}{0.5}\right) = 20\log_{10}(16) \approx 24.1\,\text{dB}$$

In code:

```
const double gain_db = 20.0 * log10(8.0 / 0.5); // 24.082 dB.
```

EXAMPLE **Numerical differentiation.** Differentiate $f(x) = x^2$ at the point $x = 3$ and compare with the analytical derivative.

Analytically, $f'(x) = 2x$, so $f'(3) = 6$. Numerically, with $h = 10^{-6}$:

$$f'(3) \approx \frac{(3.000001)^2 - (2.999999)^2}{2 \cdot 10^{-6}} = 6.000000$$

The deviation is of the order of 10^{-9} and is due to the finite precision of floating-point numbers.

EXAMPLE **Numerical integration.** The current $i(t) = 2t + 4\,\text{A}$ flows for $0 \le t \le 2\,\text{s}$. Calculate the charge q.

Analytically (Chapter 14):

$$q = \int_0^2 (2t + 4)\,dt = \Big[t^2 + 4t\Big]_0^2 = 4 + 8 = 12\,\text{C}$$

The trapezoidal rule with $n = 4$ and $h = 0.5$:

$$q \approx 0.5\left[\frac{4 + 8}{2} + (5 + 6 + 7)\right] = 0.5 \cdot 24 = 12\,\text{C}$$

The result is exact, because $i(t)$ is a straight line.

EXAMPLE **Impedance of an RL circuit.** A resistor $R = 100\,\Omega$ is connected in series with an inductor $L = 50\,\text{mH}$. Calculate the circuit's impedance at $f = 50\,\text{Hz}$.

$$\omega = 2\pi \cdot 50 \approx 314.16\,\text{rad/s}$$
$$Z = R + j\omega L = 100 + j15.71\,\Omega$$

$$|Z| = \sqrt{100^2 + 15.71^2} \approx 101.23\,\Omega, \qquad \arg(Z) = \arctan\left(\frac{15.71}{100}\right) \approx 0.156\,\text{rad} \approx 8.93°$$

In code, the magnitude and the argument are calculated with `cabs(z)` and `carg(z)` respectively.

EXAMPLE **Sample values.** A signal $u(t) = 5\sin(2\pi \cdot 50t)\,\text{V}$ is sampled at $f_s = 400\,\text{Hz}$. Calculate the first three sample values.

$$N = \frac{400}{50} = 8 \text{ samples per period}$$
$$u_0 = 5\sin(0) = 0\,\text{V}$$
$$u_1 = 5\sin\left(\frac{2\pi}{8}\right) = 5\sin\left(\frac{\pi}{4}\right) \approx 3.54\,\text{V}$$
$$u_2 = 5\sin\left(\frac{2\pi \cdot 2}{8}\right) = 5\sin\left(\frac{\pi}{2}\right) = 5.00\,\text{V}$$

The sampling theorem is satisfied, since $400 > 2 \cdot 50$.

18.13 Summary

Concept	Mathematics	In C
Root, power	$\sqrt{x}$, x^y	`sqrt(x)`, `pow(x, y)`
Logarithms	$\ln x$, $\lg x$	`log(x)`, `log10(x)`
Decibels	$20\log_{10}(U_{\text{out}}/U_{\text{in}})$	`20.0 * log10(u_out / u_in)`
Degrees to radians	degrees $\cdot\, \pi/180$	`deg * PI / 180.0`
Angle in the right quadrant	$\arg(x + jy)$	`atan2(y, x)`
Central difference	$(f(x+h) - f(x-h))/2h$	`derivative(f, x, 1e-6)`
Trapezoidal rule	$\int_a^b f(x)\,dx$, n subintervals	`integral(f, a, b, n)`
Complex number	$3 + 4j$	`3.0 + 4.0 * I`
Magnitude, argument	$\lvert z\rvert$, $\arg(z)$	`cabs(z)`, `carg(z)`
Polar form	$re^{j\theta}$	`r * cexp(I * theta)`
Sample	$\lvert U\rvert\sin(2\pi k/N + \delta)$, $N = f_s/f$	`amplitude * sin(2.0 * PI * k / n + delta)`

What you should be able to do

After this chapter, you should be able to:

- Understand the difference between integers and floating-point numbers in C, and choose the right type and format specifier.
- Use `math.h` to calculate powers, roots, logarithms and decibels.
- Calculate with angles in radians and convert to and from degrees.
- Calculate a derivative numerically and compare it with the analytical derivative.
- Calculate a definite integral numerically with the trapezoidal rule.
- Represent complex numbers and phasors with `complex.h`, and calculate magnitude and argument.
- Calculate sample values of a sinusoidal signal and print them for further analysis.

Test yourself

ANSWER P. 268

1. What is the result of `1 / 2` in C, and why?
2. What does the flag `-lm` do, and what error message do you get if the maths library is not linked?
3. Why does `sin(30)` not give 0.5?
4. How is the derivative of $f(x)$ at the point x calculated numerically, and what controls the accuracy?
5. How are the magnitude and the argument of a complex number calculated in C?

Next chapter

Chapter 19 closes the book with a review of the course and preparation for the exam.

18.14 Exercises

How to work with the exercises. Each exercise is solved in two steps. First you calculate **by hand** and write down the intermediate steps; this is done **before the lesson**. Then, during the lesson, you write **a C program** that does the same calculation, and compare the results. Parts 1–3 are compulsory and are assessed U/G, the Swedish fail and pass. The hand calculations in Exercises 6.1, 6.4 a, 7.1 and 7.4 are compulsory too. The other exercises in Parts 4–7 are optional further study for those who have time, and do not affect the assessment.

Answers and worked solutions. The answer to every exercise is in the answer key at the back of the book, and its page is given next to the exercise. An answer shows whether you got it right, but not how. Full worked solutions, in Swedish, are published in the course repository after the lesson, in `lectures/L18/appendix/c_solutions.md`. Always try it yourself first.

> **Note:** The hand calculations for Parts 1–3, and Exercises 6.1, 6.4 a, 7.1 and 7.4, must be done before the lesson and brought to it. The lesson time is not enough for both calculating and programming.

The code is written and run in OnlineGDB, https://www.onlinegdb.com/online_c_compiler, as in §18.1.1. Check that the language is set to **C** and press **Run**. Print floating-point numbers with three decimals, for example `printf("R = %.3f ohm\n", r);`.

How to work: one file per exercise

All the exercises are collected in **one and the same OnlineGDB project**, with one file per exercise:

```
assignment1_1.c   assignment2_1.c   assignment3_1.c
assignment1_2.c   assignment2_2.c   assignment3_2.c
assignment1_3.c   assignment2_3.c   assignment3_3.c
main.c
```

New files are created with the **New file** button in the toolbar, or with `Ctrl + M`.

OnlineGDB compiles and links all the files in the project together. Therefore only **one** file may contain `main()`. Instead, each exercise is written as a function of its own, with the same name as the file:

```c
/* assignment1_1.c */
#include <stdio.h>

void assignment1_1(void)
{
    printf("--- Exercise 1.1 ---\n");
    /* Your code here. */
}
```

In `main.c` the exercises are declared at the top and then called in turn. Add one line per exercise as you finish it – a function that is declared but does not exist gives a linker error:

```c
/* main.c */
void assignment1_1(void);
void assignment1_2(void);

int main(void)
{
    assignment1_1();
    assignment1_2();
```

```
    return 0;
}
```

At the presentation it is then enough to press **Run** once: all the exercises run and print their results in order.

> **Note:** The project is not saved automatically. Copy the files to your own computer before you close the tab.

Working in WSL (recommended)

The directory `lectures/L18/code/math/` in the course repository is an example of what a project can look like in WSL (§18.1.2). It holds two files: `main.c`, a small example program that divides two numbers and prints the quotient, and `Makefile`, which builds and runs the program with `make`.

Get the code by cloning the course repository to your Linux home directory. Then go into the directory, and build and run the example:

```
cd ~
git clone https://github.com/Yrgo-26/electrical-engineering-math.git
cd electrical-engineering-math/lectures/L18/code/math
make
```

The clone also holds the code for Parts 6 and 7, so it only has to be fetched once.

Then use the directory as the starting point for the exercises. Replace the example in `main.c` with the declarations and calls of the exercises as above, and add each new file under `SRC_FILES` in the makefile, one file per line:

```
SRC_FILES := assignment1_1.c \
             assignment1_2.c \
             main.c \
```

A file that is not listed under `SRC_FILES` is not compiled, and calling its function then gives a linker error.

Presentation and assessment

The exercises are presented to the teacher during the lesson:

- **The hand calculations** for each exercise, done before the lesson and brought to it, hand-written or digital and with clear steps.
- **The code**: the OnlineGDB project with one file per exercise, which must be runnable and commented. All the exercises must run with a single press of **Run**.

- A **comparison** between the hand-calculated and the computed result, with an explanation of any discrepancies.

The lab is assessed U/G and does not affect the points towards the course grade. It is part of the requirement that all the course's learning outcomes are achieved for a pass grade in the course.

Requirement	Assessment
Parts 1–3 and the hand calculations in 6.1, 6.4 a, 7.1 and 7.4 completed and correctly presented	G
Parts 1–3 or the hand calculations in 6.1, 6.4 a, 7.1 and 7.4 not completed or not presented	U

The other exercises in Parts 4–7 are optional further study and do not affect the assessment.

PART 1

Basics of `math.h`

Exercise 1.1 Resistance and power

ANSWER P. 265

Two resistors, $R_1 = 220\,\Omega$ and $R_2 = 330\,\Omega$, are connected in parallel across the voltage $U = 12\,\text{V}$.

a) Calculate the parallel resistance R_p.

b) Calculate the current I and the power dissipated, $P = \dfrac{U^2}{R_\text{p}}$.

c) Write a program that calculates R_p, I and P. Use `pow()` for the squaring.

d) Extend the program so that it also calculates the voltage needed for the power to be 2.0 W, from $U = \sqrt{PR_\text{p}}$.

Exercise 1.2 Decibels

ANSWER P. 265

An amplifier has the input voltage $U_\text{in} = 25\,\text{mV}$ and the output voltage $U_\text{out} = 1.6\,\text{V}$.

a) Calculate the gain in dB.

b) A filter attenuates the signal by $-3\,\text{dB}$. Calculate the ratio U_out/U_in for the filter.

c) Write a program that calculates both values with `log10()` and `pow()` respectively.

Exercise 1.3 Angles and trigonometry

ANSWER P. 265

An AC voltage is given by $u(t) = 325\sin(2\pi \cdot 50t)$ V.

a) Calculate $u(t)$ at $t = 2.0$ ms.

b) Give the angle $2\pi \cdot 50t$ in both radians and degrees at this time.

c) Write a program that calculates $u(t)$ for $t = 0$, 2.0 ms, 5.0 ms and 10.0 ms. Print the time, the angle in degrees and the voltage on each line.

d) Find the angle of the vector $\vec{v} = (-3, 4)$ with `atan2()`, in radians and degrees. Explain why `atan(4.0 / -3.0)` gives a different answer.

PART 2

Derivatives and integrals

Exercise 2.1 Numerical differentiation

ANSWER P. 266

Let $f(x) = x^3 - 2x + 1$.

a) Differentiate $f(x)$ analytically and calculate $f'(2)$.

b) Write a function `derivative()` that calculates the derivative numerically with the central difference

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}.$$

c) Calculate $f'(2)$ numerically with $h = 10^{-2}$, $h = 10^{-6}$ and $h = 10^{-12}$. Print the absolute error against the analytical value for each h.

d) Which h gives the smallest error? Explain why values of h that are too large and too small both give worse results.

Exercise 2.2 Voltage across an inductor

ANSWER P. 266

The current through an inductor is given by $i(t) = 5t^2$ A. The inductor's inductance is $L = 0.20$ H. The voltage across the inductor is given by

$$u(t) = L \cdot i'(t).$$

a) Find $i'(t)$ analytically, and hence an expression for $u(t)$.

b) Calculate $u(3)$ by hand.

c) Write a program that calculates $u(3)$ by differentiating $i(t)$ numerically, and compare with the hand calculation.

Exercise 2.3 Numerical integration

ANSWER P. 266

Let $f(x) = x^2$.

a) Calculate $\int_0^3 x^2\,dx$ analytically.

b) Write a function `integral()` that calculates the definite integral with the trapezoidal rule:

$$\int_a^b f(x)\,dx \approx h\left[\frac{f(a)+f(b)}{2} + \sum_{k=1}^{n-1} f(a+kh)\right], \qquad h = \frac{b-a}{n}.$$

c) Calculate the integral with $n = 10$, $n = 100$ and $n = 1000$. Print the absolute error for each n.

d) How does the error change when n is multiplied by ten?

PART 3

Complex numbers and phasors

Exercise 3.1 Calculating with complex numbers

ANSWER P. 266

Let $z_1 = 4 + 3j$ and $z_2 = 2 - 5j$.

a) Calculate $z_1 + z_2$, $z_1 \cdot z_2$ and $\dfrac{z_1}{z_2}$ by hand.

b) Calculate $|z_1|$ and $\arg(z_1)$ in radians and degrees.

c) Write a program that does all the calculations with `complex.h`. Print each result in the form `a + bj` using `creal()` and `cimag()`.

Exercise 3.2 Polar and rectangular form

ANSWER P. 266

A complex number is given in polar form as $z = 10\angle\frac{\pi}{6}$.

a) Convert z to rectangular form by hand.

b) Write a program that does the conversion with `cexp()`, and converts back with `cabs()` and `carg()`.

c) Check that you get the original number back. Why should the results be compared with `fabs(a - b) < 1e-9` rather than with ==?

Exercise 3.3 Impedance of an RLC circuit

ANSWER P. 266

A series circuit consists of $R = 47\,\Omega$, $L = 100\,\text{mH}$ and $C = 10\,\mu\text{F}$. The circuit is supplied with $U = 230\,\text{V}$ at $f = 50\,\text{Hz}$.

a) Calculate ω, $X_L = \omega L$ and $X_C = \dfrac{1}{\omega C}$.

b) Give the total impedance $Z = R + j(X_L - X_C)$ in rectangular form.

c) Calculate $|Z|$ and $\arg(Z)$ in degrees. Is the circuit inductive or capacitive?

d) Calculate the magnitude of the current, $|I| = \dfrac{|U|}{|Z|}$.

e) Write a program that calculates all of the above with `complex.h`. Let f be a variable, and run the program for $f = 500\,\text{Hz}$ as well. What happens to the character of the circuit?

PART 4

Further study (optional)

Exercise 4.1 Sampling a sinusoidal signal

ANSWER P. 266

A signal is given by $u(t) = 5\sin(2\pi \cdot 50t + \frac{\pi}{4})\,\text{V}$ and is sampled at $f_s = 800\,\text{Hz}$.

a) Check that the sampling theorem is satisfied.

b) Calculate the number of samples per period, $N = \dfrac{f_s}{f}$.

c) Calculate the sample values u_0, u_1 and u_2 by hand from

$$u_k = |U| \sin\left(\frac{2\pi k}{N} + \delta\right).$$

d) Write a program that calculates and prints all N sample values for one period, as CSV with one line per sample:

```
k,u
0,3.536
1,4.619
```

e) Copy the output into a spreadsheet program and plot u_k as a function of k. Check that the curve is sinusoidal and that the values repeat after N samples.

Exercise 4.2 Adding phasors

ANSWER P. 266

Three voltages with the same frequency are given by the phasors

$$U_1 = 10\angle 0, \qquad U_2 = 6\angle \frac{\pi}{3}, \qquad U_3 = 4\angle -\frac{\pi}{2}.$$

a) Convert all the phasors to rectangular form by hand.

b) Calculate the sum $U = U_1 + U_2 + U_3$ in rectangular form.

c) Give the sum in polar form, $|U|\angle\delta$, with δ in both radians and degrees.

d) Write a program that does the addition with `complex.h` and prints the sum in both rectangular and polar form.

e) Draw the phasors and their sum in the complex plane.

Exercise 4.3 The composite signal

ANSWER P. 266

a) Write a program that samples the composite signal

$$u(t) = u_1(t) + u_2(t) + u_3(t),$$

where each component signal is given by the corresponding phasor in Exercise 4.2 and all of them have the frequency $f = 50\,\text{Hz}$.

b) Print the sample values as CSV with $f_s = 1600\,\text{Hz}$.

c) Check that the largest sample value agrees with the $|U|$ calculated in Exercise 4.2. Explain any discrepancy.

PART 5

Extra exercises (optional)

The exercises below are extra practice for those who want to go further. They do not affect the assessment. As in Parts 1–3, the hand calculation comes first and the program after.

Exercise 5.1 Rounding and absolute value

ANSWER P. 266

Consider the numbers -2.7, 3.2 and 5.5.

a) Give by hand what `floor()`, `ceil()` and `round()` return for each of the numbers.

b) Write a program that prints all nine values in a table.

c) Why does `round(-2.5)` give -3 and not -2?

d) What is the difference between `fabs()` and `abs()`, and which should be used in this course?

Exercise 5.2 Tolerance check of resistors

ANSWER P. 266

A resistor is marked $4.7\,\text{k}\Omega \pm 5\,\%$.

a) Calculate the permitted range by hand.

b) Write the function

```
int within_tolerance(double nominal, double tol_percent, double measured);
```

which returns 1 if the measured value is within the tolerance, and 0 otherwise.

c) Test the function with the measured values $4600\,\Omega$, $4950\,\Omega$ and $4465\,\Omega$.

d) Why should the comparison be written as

```
fabs(measured - nominal) <= nominal * tol_percent / 100.0
```

rather than as a comparison with two separate limits?

Exercise 5.3 Series and parallel resistance

ANSWER P. 266

a) Calculate R_s and R_p by hand for $R_1 = 100\,\Omega$ and $R_2 = 400\,\Omega$.

b) Write the functions

```
double series(double r1, double r2)
double parallel(double r1, double r2)
```

c) Write a program that prints both values.

d) Extend the program with a function that calculates the parallel resistance of n resistors in an array, via the sum of $1/R$. Test with $\{100, 400, 200\}\,\Omega$.

Exercise 5.4 Table of an RC discharge

ANSWER P. 267

The voltage is given by $u(t) = 12e^{-t/\tau}\,\text{V}$, where $\tau = RC$ with $R = 10\,\text{k}\Omega$ and $C = 100\,\mu\text{F}$.

a) Calculate τ by hand.

b) Calculate $u(0)$, $u(1)$ and $u(2)$ by hand.

c) Write a program that uses `exp()` to print $u(t)$ for $t = 0$ to $5\,\text{s}$ in steps of $0.5\,\text{s}$.

d) After how many time constants is the voltage below $1\,\%$ of its initial value? Let the program find the answer.

Exercise 5.5 A quadratic equation with all three cases

ANSWER P. 267

a) Write a program that solves $ax^2 + bx + c = 0$ using the discriminant.

b) The program must handle the three cases $D > 0$, $D = 0$ and $D < 0$ separately.

c) Test with the coefficients $(1, -5, 6)$, $(1, -4, 4)$ and $(1, 2, 5)$. Work out the expected answers by hand first.

d) In the case $D < 0$, print the roots in the form `a + bj` using `sqrt(-D)`.

Exercise 5.6 Solving an equation numerically by bisection

ANSWER P. 267

Find the root of $f(x) = e^{-x} - x$ in the interval $[0, 1]$.

a) Calculate $f(0)$ and $f(1)$ by hand and show that they have opposite signs.

b) Write a function `bisect()` that repeatedly halves the interval until its width is less than 10^{-9}.

c) Run the program. What value of the root do you get?

d) Why must $f(a)$ and $f(b)$ have opposite signs for the method to work?

Exercise 5.7 RMS value by numerical integration

ANSWER P. 267

For a sinusoidal voltage $u(t) = |U| \sin(\omega t)$,

$$U_{\text{RMS}} = \sqrt{\frac{1}{T} \int_0^T u(t)^2 \, dt}.$$

a) Give the theoretical value of U_{RMS} when $|U| = 325\,\text{V}$.

b) Write a program that calculates the integral numerically with the trapezoidal rule and $n = 1000$ subintervals over one period.

c) Compare with the theoretical value and print the relative error.

d) What happens to the error if n is reduced to 10? Explain why.

Exercise 5.8 Charge through a conductor

ANSWER P. 267

The current is given by $i(t) = 3t + 2\,\text{A}$.

a) Calculate the charge $q = \int_0^4 i(t) \, dt$ analytically.

b) Calculate the same charge numerically with the trapezoidal rule and $n = 4$.

c) Why is the numerical result exact in this case, but not in Exercise 2.3?

PART 6

Linear regression

Here a simple machine learning model, a **linear regression model**, is trained to find the straight line $y = kx + m$ from a number of measured points. In machine learning, k is called the model's **weight** and m its **bias**.

The Ee class later takes a course of its own in machine learning, and for you Parts 6 and 7 give a first taste. For the Eh class this is the only chance to learn how machine learning works, so make the most of it! Whichever class you are in, the parts show that the mathematics of the course is the foundation there too.

> **Note:** The hand calculations in Exercise 6.1 and Exercise 6.4 a) are compulsory and are done before the lesson, as in Parts 1–3. They need no computer. The other exercises in Part 6 are optional.

The code is in `lectures/L18/code/lin_reg/` and consists of four files:

File	Contents
`include/ml/lin_reg.h`	The structs for the model, the training data and the training parameters, and the function declarations.
`source/ml/lin_reg.c`	The implementation of the model. The function `lin_reg_optimize()` is empty, and you are to implement it.
`source/main.c`	The main program, where the training data and the training parameters are defined.
`include/ml/utils.h`	Helper macros, among them one that computes the number of elements in an array.

> **Note:** The programming exercises in Part 6 require **WSL** with `gcc` and `make`, and cannot be done in OnlineGDB. The program consists of several files in subdirectories and is built with a makefile. Set up WSL as in §18.1.2 before you start.

Go into the directory in your clone of the course repository, see *Working in WSL* above, and build and run the program with `make`:

```
cd ~/electrical-engineering-math/lectures/L18/code/lin_reg
make
```

Without an implementation in `lin_reg_optimize()` the model is not trained at all, so every prediction is 0 until Exercise 6.2 is done.

Structs in C

A **struct**, like `complex_t` in Chapters 15–17, gathers several variables that belong together under one name. The model, for example, consists of two floating-point numbers:

```
typedef struct
{
    double bias;   // The m value.
    double weight; // The k value.
} lin_reg_t;
```

A variable of type `lin_reg_t` is created and initialised like an ordinary variable, and its fields are reached with a dot:

```
lin_reg_t model = {.bias = 0.0, .weight = 0.0};
printf("k = %.3f, m = %.3f\n", model.weight, model.bias);
```

When the struct is instead reached through a **pointer**, as in the functions in `lin_reg.c`, an arrow is used: `self->bias` and `self->weight`.

The training algorithm

The model is trained by going through the measured points, the so-called **training sets** (x, y_{ref}), one at a time. One pass through all the training sets is called an **epoch**. Both parameters are zero at the start. For each training set:

1. Compute the model's prediction y_p for the input x:

$$y_p = k \cdot x + m.$$

2. Compute the deviation δ between the reference value and the prediction:

$$\delta = y_{\text{ref}} - y_p.$$

   If $\delta > 0$ the prediction is too low, and k and m should be increased. If $\delta < 0$ it is too high, and k and m should be decreased.

3. Compute the adjustment Δe, where LR is the **learning rate**, a small number that sets how strongly the parameters are adjusted:

$$\Delta e = \delta \cdot LR.$$

4. Adjust the parameters:

$$m = m + \Delta e, \qquad k = k + \Delta e \cdot x.$$

Note that k is not adjusted when $x = 0$, since k does not affect the prediction then. The larger x is, the more k is adjusted.

Exercise 6.1 Training by hand (compulsory)

ANSWER P. 267

The training data in `main.c` follow the line $y = 2x + 2$:

x	0	1	2	3	4
y_{ref}	2	4	6	8	10

a) Carry out one epoch by hand with the learning rate $LR = 0.1$, starting from $k = 0$ and $m = 0$. Show y_p, δ, Δe, k and m after each training set. Round to four decimals.

b) Have k and m moved towards the correct values $k = 2$ and $m = 2$?

Exercise 6.2 Implement the optimisation (optional)

ANSWER P. 267

a) Implement the function `lin_reg_optimize()` in `lin_reg.c` according to the training algorithm above. The comments in the function describe the steps.

b) Set the number of epochs to 1 and the learning rate to 0.1 in `main.c`. Print `model.weight` and `model.bias` after the training and compare with your hand calculation in Exercise 6.1.

c) Restore the number of epochs to 50 and the learning rate to 0.25. Does the model now predict $y = 2x + 2$?

Exercise 6.3 Learning rate and epochs (optional)

ANSWER P. 267

a) Train the model with the learning rate 0.01 for 50 epochs. How close to the correct values do k and m end up? How many epochs are needed to get as close as in Exercise 6.2 c)?

b) Train the model with the learning rate 0.5. What happens? Explain why a learning rate that is too high keeps the model from finding the line.

c) Multiply all the x values by 10 and compute the y values from $y = 2x + 2$, that is $x = 0, 10, \ldots, 40$. Train with the learning rate 0.1. What happens, and why? Hint: see step 4 of the training algorithm.

Exercise 6.4 Your own training data

ANSWER P. 267

In Chapter 8, Exercise 2.6, the voltage $U(t)$ was measured during a discharge:

t (s)	0	1	2	3	4
$U(t)$ (V)	12	9	6	3	0

a) (*Compulsory.*) Find k and m in $U(t) = kt + m$ by hand, or take them from your solution in Chapter 8.

b) (*Optional, as are c)–e).*) Replace the training data in `main.c` with the table above. Choose the number of epochs and the learning rate so that the model finds k and m with at least three correct decimals.

c) Let the model predict $U(5)$. Does the answer agree with your calculation in Chapter 8?

d) Change the measured values slightly, for example to 12.2, 8.9, 6.1, 2.8 and 0.1, and retrain the model with the learning rate 0.25 for 50 epochs. Why are k and m no longer exactly -3 and 12?

e) Retrain the model with the learning rate 0.01 for 1000 epochs and compare k and m with d). Why does the learning rate matter when the measured values do not lie on a straight line? Hint: the model is adjusted after each training set, not after the whole epoch.

Exercise 6.5 Precision (optional)

ANSWER P. 267

How well the model fits the training data can be given by the mean absolute error (MAE), the mean of the absolute values of the deviations:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_{\mathrm{ref},i} - y_{p,i}|.$$

The model's **precision** is defined here as $1 - \mathrm{MAE}$. A precision close to 1 means that the model predicts very close to the reference values.

a) Write a function that computes the precision of a trained model. Use `fabs()` from `math.h`.

b) Print the precision after each epoch. How does it change during the training?

c) Stop the training as soon as the precision exceeds 0.9999. After how many epochs does that happen?

PART 7

A neural network in C++

The linear regression model of Part 6 is in fact a **node** with a single input. In a **neural network** each node has several inputs, each with a weight of its own, and the nodes are arranged in **layers**. In a so-called **dense layer** every node is connected to all the inputs.

In Part 7 a neural network is trained to learn a pattern that no straight line can describe. The network is written in **C++**, which the Ee class studies in a later course. You do not need to understand all of the code: the aim is that you implement a central part of the network yourself and then watch it learn.

> **Note:** The hand calculations in Exercise 7.1 and Exercise 7.4 are compulsory and are done before the lesson. They need no computer. The other exercises in Part 7 are optional.

> **Note:** The programming exercises in Part 7, like those in Part 6, require **WSL** with `g++` and `make`. Both are in the package `build-essential`, see §18.1.2.

The code is in `lectures/L18/code/neural_network/`. Go into the directory in your clone of the course repository, and build and run the program with `make`:

```
cd ~/electrical-engineering-math/lectures/L18/code/neural_network
make
```

The training data

The network has two inputs x_1 and x_2 and one output y. The output is to be 1 when the inputs differ, and 0 when they are equal:

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	0

The network consists of two layers:

- A **hidden layer** with 15 nodes, which takes the inputs x_1 and x_2.
- An **output layer** with one node, which takes the 15 outputs of the hidden layer.

The weights and biases are given random starting values and are then adjusted during the training, in the same way as k and m in Part 6.

Feedforward

Computing a layer's outputs from its inputs is called **feedforward**. For node i, first the weighted sum z_i of the inputs plus the node's bias b_i is computed:

$$z_i = w_{i1}x_1 + w_{i2}x_2 + \cdots + b_i = \sum_j w_{ij}x_j + b_i,$$

where w_{ij} is the weight between input j and node i. With a single input this becomes $z = kx + m$, the same thing as in Part 6.

The sum is then passed through an **activation function** f, so that the node's output is $y_i = f(z_i)$. The network uses two activation functions:

- The hidden layer uses **ReLU**, which sets negative sums to zero: $f(z) = \max(0, z)$.
- The output layer uses the **hyperbolic tangent**, which squeezes all real numbers into the interval $(-1, 1)$:

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}.$$

The code in brief

File	Contents
`source/main.cpp`	The training data, the size of the network, and the training and printing.
`source/ml/dense_layer/dense.cpp`	The dense layer. This is where the method `feedforward()` is, which you are to implement.
`source/ml/neural_network/shallow.cpp`	The network, which connects the layers and trains them.
`source/ml/utils.cpp`	The activation functions and random starting values.

A few things in C++ that differ from C:

- `ml::Matrix1d` is a `std::vector<double>`, an array that keeps track of its own size and can change it. The size is given by `.size()`, and the elements are reached with square brackets just as in C, for example `input[j]`.
- `ml::Matrix2d` is a vector of vectors, that is a two-dimensional array. The weights are reached with `myWeights[i][j]`.
- A **class** works like a struct that also holds its own functions, called **methods**. The layer's variables, for example `myWeights`, `myBias` and `myOutput`, are therefore reached directly inside the methods, without `self->`.
- The size of the layer is given by the methods `nodeCount()` and `weightCount()`, and the activation function is computed with `actFuncOutput(myActFunc, sum)`.

Feel free to read the rest of the code, but the exercises do not require it.

Exercise 7.1 Feedforward by hand (compulsory)

ANSWER P. 267

A node has the weights $w_1 = 0.5$ and $w_2 = -1$ and the bias $b = 0.2$.

a) Compute the sum z for the inputs $(x_1, x_2) = (1, 0)$ and $(x_1, x_2) = (1, 1)$.

b) Compute the node's output with ReLU for both cases.

c) Compute the node's output with tanh for the case $(1, 0)$. Round to four decimals.

Exercise 7.2 Implement feedforward (optional)

ANSWER P. 267

a) Implement the method `feedforward()` in `dense.cpp`. For each node i, compute the sum z_i with an inner loop over the inputs, and then store both the sum z_i in `myPreActOutput[i]` and the output $f(z_i)$ in `myOutput[i]`. The sum is needed later, when the network is trained. The comments in the method describe the steps.

b) Build and run the program. It prints the network's predictions before and after the training. Compare them with the table of training data. Has the network learnt the pattern?

> **Note:** Without a working feedforward every prediction is 0, and the training fails after several attempts.

Exercise 7.3 Experiment (optional)

ANSWER P. 267

a) Run the program several times. Why does the number of epochs needed vary from run to run?

b) Reduce the number of nodes in the hidden layer, `hiddenCount` in `main.cpp`, to 5, 2 and 1. What happens to the training?

Exercise 7.4 Why is one straight line not enough? (compulsory)

ANSWER P. 268

A single node without an activation function gives $z = w_1 x_1 + w_2 x_2 + b$, and the relation $z = 0$ is a straight line in the $x_1 x_2$ plane. The node can therefore only separate points on different sides of one line.

a) Plot the four points $(0, 0)$, $(0, 1)$, $(1, 0)$ and $(1, 1)$ in a coordinate system, and mark the ones that are to give the output 1.

b) Can a single straight line be drawn that separates the points that are to give 1 from the points that are to give 0? What does that say about why the network needs a hidden layer?

CHAPTER 19

Review and exam preparation

The whole course once more, and two exams to practise on.

IN THIS CHAPTER

- Review of the course content before the exam.
- How the exam is organised: format, permitted aids, points and grade thresholds.
- Two practice exams covering the whole course, with answers in the answer key.

The course ends with a written exam, and this chapter prepares you for it. There is no new material here. Instead, the chapter describes how the exam works, and then come two practice exams that test the central parts of the course: arithmetic, algebra, equations, trigonometry, logarithms, derivatives, integrals and complex numbers, that is, more or less everything you have calculated with in Chapters 1 to 17.

§19.1 goes through the exam's format, the permitted aids, the points and the grade thresholds, and how to prepare. §19.2 holds practice exam 1 and §19.3 practice exam 2. Sit them on your own first, under the same conditions as the exam: with only the permitted aids and without looking at the answer key. Then mark it against the answer key at the back of the book, and after that go through the worked solutions published in the course repository, for the questions where you went wrong or got stuck. §19.4 closes the chapter, and the course, with a few questions to reflect on.

19.1 Before the exam

In the exam you show that you have achieved the course's learning outcomes. This section describes how the exam works, what it contains and how to prepare for it.

19.1.1 Format

The exam is a written exam, sat in an exam hall. Every question requires a solution and a justification, with a stated answer, including the unit where relevant. In other words, you must explain how you arrived at your results.

19.1.2 Permitted aids

- Writing materials and a calculator of any kind.
- One A4 sheet (both sides), handwritten or typed, with formulas, examples and notes of your own choice.

> **Note:** Mobile phones may not be used during the exam and must be left in the place indicated.

19.1.3 Points and grade thresholds

The exam gives 16 points in total. On top of that, up to 1.0 bonus point can be earned from the course's two tests: 0.5 points per test for anyone who scores at least 3.5 of the test's 5 points. The course uses the Swedish grades IG (fail), G (pass) and VG (pass with distinction), with the thresholds in the table below; a pass grade also requires that all the course's learning outcomes are achieved.

Grade	Threshold
IG	< 6.5 points
G	≥ 6.5 points
VG	≥ 11.0 points

19.1.4 Content

The exam is divided into two parts:

- **Part I**: arithmetic, algebra, equations, trigonometry and logarithms (Chapters 1–10).
- **Part II**: derivatives, integrals and complex numbers (Chapters 12–17).

19.1.5 Preparation

- Sit practice exam 1 in §19.2 on your own. It is gone through in the review lesson.
- Then compare with the answer key and with the worked solutions published in the course repository in `exam/en/practice_exam_1_solutions.md`.
- Sit practice exam 2 in §19.3 as extra practice, with worked solutions in `exam/en/practice_exam_2_solutions.md`.
- Review the central parts of the course, especially the areas where you went wrong in the practice exams.
- Prepare the A4 sheet (§19.1.2) in good time.

19.2 Practice exam 1

Practice exam 1 has the same aids, points and grade thresholds as the exam.

> **How to work with the practice exam.** Sit it on your own and under the same conditions as the exam: with only the permitted aids, without a mobile phone and without looking at the answer key. Work each question as far as you can before you move on.
> **Answers and worked solutions.** The answer to every question is in the answer key at the back of the book, and its page is given next to the question. An answer shows whether you got it right, but not how. Full worked solutions are published in the course repository, in `exam/en/practice_exam_1_solutions.md`. Always try it yourself first.

Aids	Writing materials and a calculator of any kind. One A4 sheet (both sides), handwritten or typed, with formulas, examples and notes of your own choice.
Points	16 points in total. On top of that, up to 1.0 bonus point can be earned from tests 1–2.
Grades	IG $<$ 6.5 points, G $\geq$ 6.5 points, VG $\geq$ 11 points.
Solutions	Every question requires a solution and a justification with a stated answer, including the unit where relevant. In other words, you must explain how you arrived at your results. Good luck!

> **Note:** Mobile phones may not be used while the exam is in progress and must be left in the place indicated.

Question 1

ANSWER P. 268 1.0 POINTS

Solve the system of equations below to calculate the currents I_1 and I_2, in mA:

$$\begin{cases} 3I_1 - I_2 = 7 \\ 3I_2 = I_1 + 3 \end{cases}$$

Question 2

ANSWER P. 268 2.0 POINTS

A battery with the open-circuit voltage $E = 24\,\text{V}$ and the internal resistance $R_\text{i} = 6\,\Omega$ is connected to a load with the resistance R. The power P developed in the load is given by

$$P = \frac{E^2 R}{(R + R_\text{i})^2}.$$

a) Find the load resistance R such that the power in the load is $P = 18\,\text{W}$. The equation has two solutions.

b) Calculate the current $I = \dfrac{E}{R + R_\text{i}}$ for each solution. Which solution is preferable if the battery is to last as long as possible? Justify your answer.

Question 3

ANSWER P. 268 1.5 POINTS

The voltage U in volts across a component depends on the current I in amperes as

$$U(I) = \frac{4I^2 - 36}{I^2 - I - 6}.$$

a) Simplify the expression and state which values of I it is not defined for.

b) Calculate the voltage U when the current is $I = 2\,\text{A}$.

Question 4

ANSWER P. 268 1.5 POINTS

When a capacitor discharges through a resistor, the voltage across the capacitor decreases exponentially with time. The voltage $u(t)$ can be described by the function

$$u(t) = U_0 e^{-t/(RC)},$$

where

- $u(t)$ is the voltage across the capacitor at time t,
- U_0 is the initial voltage in volts,
- RC is the circuit's time constant in seconds,
- t is the time in seconds.

A capacitor with the capacitance 680 µF is connected to a 10 kΩ resistor. It is initially charged to 10 V and starts to discharge at time $t = 0$.

a) Calculate the voltage after three seconds.

b) Find the function's domain and range for the time interval 0–10 seconds.

c) Calculate how long it takes for the voltage to fall to 40 % of its original value.

Question 5

ANSWER P. 268 1.0 POINTS

An AC voltage has the amplitude 5 V, the frequency 100 Hz and the phase 90°.

a) Find the equation $u(t)$ of the AC voltage. Give the phase in radians.

b) Draw the sine wave of the AC voltage over one period T.

Question 6

ANSWER P. 268 2.0 POINTS

The battery in Question 2, with the open-circuit voltage $E = 24\,\text{V}$ and the internal resistance $R_\text{i} = 6\,\Omega$, delivers the current I to a load. The terminal voltage is $U = E - R_\text{i}I$, so the power in the load as a function of the current is

$$P(I) = U \cdot I = (24 - 6I) \cdot I = 24I - 6I^2.$$

a) Differentiate the function $P(I)$ and find the expression for $P'(I)$.

b) Find where the function is stationary, i.e. solve $P'(I) = 0$.

c) Use the second derivative to decide whether the point is a maximum or a minimum.

d) Calculate the largest or smallest value of the function, i.e. find $P(I)$ at the point where the function is stationary.

e) Draw the graph of $P(I)$ for the interval $0 \le I \le 4\,\text{A}$. Mark the turning point and the points where the graph crosses the axes.

f) Calculate the load resistance $R = \dfrac{U}{I}$ at the stationary point and compare it with R_i.

Question 7

ANSWER P. 268 1.5 POINTS

The charge $q(t)$ that passes through a conductor during the time interval $(0, t)$ can be calculated with

$$q(t) = \int_0^t i(t)\,dt = \big[I(t)\big]_0^t,$$

where

- $q(t)$ is the charge through the conductor in coulombs (C),
- $i(t)$ is the current through the conductor in amperes (A),
- $I(t)$ is the antiderivative (indefinite integral) of $i(t)$.

The current $i(t)$ in a capacitor is given by

$$i(t) = 0.03t^2 + 0.1t + 0.5,$$

where t is the time in seconds. The capacitor already has the charge $q_0 = 1\,\text{C}$ at the start.

a) Find an expression for the antiderivative $I(t) = \int i(t)\,dt$, i.e. integrate without any limits.

b) Find an expression for the charge $q(t)$ of the capacitor by integrating over the interval $[0, t]$.

c) Find a formula for the total charge $q_\text{tot}(t)$ in the capacitor, including the initial charge q_0.

d) How much total charge (including the initial charge) has flowed into the capacitor after 4 seconds?

Question 8

ANSWER P. 268 2.0 POINTS

A circuit is supplied with the voltage $U = 3 - j4\,\text{V}$. The circuit has the impedance $Z = 2 + j2\,\Omega$. The current I through the circuit can be calculated with "Ohm's law", that is, Ohm's law adapted to complex numbers:

$$I = \frac{U}{Z}$$

a) Calculate the current I and give it in both rectangular and polar form.

b) The impedance can be written $Z = R + jX$. State the resistance R and the reactance X, and decide whether the circuit is inductive or capacitive.

c) The frequency is $f = 50\,\text{Hz}$. Calculate the inductance or capacitance that gives the circuit's reactance.

Question 9

ANSWER P. 268 1.5 POINTS

A current $i(t)$ in an AC circuit can be represented by a phasor I, which in rectangular form is written

$$I = 10 - j5\,\text{mA}.$$

a) Draw the phasor I in the complex plane (the x-axis is the real part, the y-axis the imaginary part).

b) Express the phasor I in Euler form, i.e. find the magnitude $|I|$ and the phase angle δ such that $I = |I|e^{j\delta}$.

c) Assume that the frequency of the current is $f = 10\,\text{Hz}$. Find the angular frequency w.

d) Write $i(t)$ as a function of time $i(t) = |I| \cdot e^{j(wt+\delta)}$.

Question 10

ANSWER P. 268 2.0 POINTS

In a series connection of three components, the AC voltage across each component is measured as

$$\begin{aligned} u_1(t) &= 2\sin(wt + 25°)\,\text{V}, \\ u_2(t) &= 5\sin(wt - 45°)\,\text{V}, \\ u_3(t) &= 1.5\sin(wt + 36°)\,\text{V}. \end{aligned}$$

The total voltage in the circuit is

$$u_{\text{tot}}(t) = u_1(t) + u_2(t) + u_3(t).$$

a) Rewrite the voltages $u_1(t)$, $u_2(t)$ and $u_3(t)$ as phasors U_1, U_2 and U_3 in complex rectangular form.

b) Calculate the phasor sum $U_{\text{tot}} = U_1 + U_2 + U_3$.

c) Draw the phasors in the complex plane (the x-axis is the real part, the y-axis the imaginary part).

d) Convert the result back to a sinusoidal voltage in the time domain, in the form $u_{\text{tot}}(t) = |U_{\text{tot}}| \sin(\omega t + \delta_{\text{tot}})$.

19.3 Practice exam 2

Practice exam 2 has the same layout, aids, points and grade thresholds as practice exam 1, but new numbers and partly different questions. Use it as extra practice once you have worked through and marked practice exam 1. The answers are in the answer key, and full worked solutions in `exam/en/practice_exam_2_solutions.md`.

Question 1

ANSWER P. 269 1.0 POINTS

Solve the system of equations below to calculate the currents I_1 and I_2, in mA:

$$\begin{cases} 2I_1 + 3I_2 = 16 \\ 4I_1 = I_2 + 4 \end{cases}$$

Question 2

ANSWER P. 269 2.0 POINTS

A battery with the open-circuit voltage $E = 20\,\text{V}$ and the internal resistance $R_\text{i} = 4\,\Omega$ is connected to a load. The current is $I = \dfrac{E - U}{R_\text{i}}$, where U is the terminal voltage, so the power in the load is given by

$$P(U) = U \cdot I = \frac{U(20 - U)}{4}.$$

a) Find the terminal voltage U such that the power in the load is $P = 16\,\text{W}$. The equation has two solutions.

b) Calculate the current I and the load resistance $R = \dfrac{U}{I}$ for each solution. Which solution is preferable if the battery is to last as long as possible? Justify your answer.

Question 3

ANSWER P. 269 1.5 POINTS

The voltage U in volts across a component depends on the current I in amperes as

$$U(I) = \frac{3I^2 - 3I - 18}{I^2 - 9}.$$

a) Simplify the expression and state which values of I it is not defined for.

b) Find the current I when the voltage is $U = 2.7\,\text{V}$.

Question 4

ANSWER P. 269 1.5 POINTS

When a capacitor discharges through a resistor, the voltage across the capacitor decreases exponentially with time. The voltage $u(t)$ can be described by the function

$$u(t) = U_0 e^{-t/(RC)},$$

where $u(t)$ is the voltage across the capacitor at time t, U_0 the initial voltage in volts, RC the circuit's time constant in seconds and t the time in seconds. A capacitor with the capacitance $100\,\mu\text{F}$ discharges through a $100\,\text{k}\Omega$ resistor from time $t = 0$. After 5 seconds the voltage is measured as 6.0 V.

a) Calculate the time constant RC of the circuit.

b) Find the initial voltage U_0.

c) Calculate how long it takes for the voltage to fall to half of U_0. Does the answer depend on U_0?

Question 5

ANSWER P. 269 1.0 POINTS

An AC voltage is described by the function

$$u(t) = 325 \sin\left(100\pi t + \frac{\pi}{6}\right) \text{ V}.$$

a) State the voltage's amplitude, RMS value, frequency, period and phase. The RMS value is given by $U_{\text{RMS}} = \dfrac{|\hat{U}|}{\sqrt{2}}$. Give the phase in degrees.

b) Draw the sine wave of the AC voltage over one period T.

Question 6

ANSWER P. 269 2.0 POINTS

The current through an inductor varies during a measurement as

$$i(t) = t^3 - 9t^2 + 24t\,\text{mA},$$

where t is the time in seconds and $0 \leq t \leq 5$.

a) Differentiate the function $i(t)$ and find the expression for $i'(t)$.

b) Find the times at which the function is stationary, i.e. solve $i'(t) = 0$.

c) Use the second derivative to decide whether each point is a maximum or a minimum.

d) Calculate the current at the stationary points.

e) Draw the graph of $i(t)$ for the interval $0 \leq t \leq 5\,\text{s}$. Mark the turning points.

f) The voltage across the inductor is $u(t) = L \cdot i'(t)$. Calculate the voltage at time $t = 1\,\text{s}$ if the inductance is $L = 0.5\,\text{H}$.

Question 7

ANSWER P. 269 1.5 POINTS

The charge $q(t)$ that passes through a conductor during the time interval $(0, t)$ can be calculated with

$$q(t) = \int_0^t i(t)\,dt = [I(t)]_0^t,$$

where $q(t)$ is the charge in coulombs (C), $i(t)$ the current in amperes (A) and $I(t)$ the antiderivative of $i(t)$. The current into a battery that is being charged is given by

$$i(t) = 0.06t^2 + 0.4t + 1,$$

where t is the time in seconds.

a) Find an expression for the antiderivative $I(t) = \int i(t)\,dt$, i.e. integrate without any limits.

b) Find an expression for the charge $q(t)$ by integrating over the interval $[0, t]$.

c) Calculate the charge that flows into the battery between $t = 2\,\text{s}$ and $t = 5\,\text{s}$.

d) Calculate the average current over the same interval, i.e. the charge divided by the length of the interval.

Question 8

ANSWER P. 269 2.0 POINTS

A resistor $R = 30\,\Omega$ and an inductor with the reactance $X_L = 40\,\Omega$ are connected in series and supplied with the voltage $U = 100\angle 0°\,\text{V}$. The current I through the circuit can be calculated with "Ohm's law", that is, Ohm's law adapted to complex numbers:

$$I = \frac{U}{Z}$$

a) Find the impedance $Z = R + jX_L$ of the circuit and give it in both rectangular and polar form.

b) Calculate the current I and give it in both polar and rectangular form.

c) The frequency is $f = 50\,\text{Hz}$. Calculate the inductance L of the inductor, where $X_L = \omega L$.

Question 9

ANSWER P. 269 1.5 POINTS

A current $i(t)$ in an AC circuit can be represented by a phasor I, which in rectangular form is written

$$I = -8 - j6\,\text{mA}.$$

a) Draw the phasor I in the complex plane (the x-axis is the real part, the y-axis the imaginary part).

b) Express the phasor I in Euler form, i.e. find the magnitude $|I|$ and the phase angle δ such that $I = |I|e^{j\delta}$.

c) Assume that the frequency of the current is $f = 50\,\text{Hz}$. Find the angular frequency ω.

d) Write the current as a sinusoidal function of time, $i(t) = |I| \sin(\omega t + \delta)$.

Question 10

ANSWER P. 269 2.0 POINTS

In a series connection of three components, the AC voltage across each component is measured as

$$u_1(t) = 4 \sin(wt + 60°)\,\text{V},$$
$$u_2(t) = 3 \sin(wt - 30°)\,\text{V},$$
$$u_3(t) = 2 \sin(wt + 150°)\,\text{V}.$$

The total voltage in the circuit is

$$u_{\text{tot}}(t) = u_1(t) + u_2(t) + u_3(t).$$

a) Rewrite the voltages $u_1(t)$, $u_2(t)$ and $u_3(t)$ as phasors U_1, U_2 and U_3 in complex rectangular form.

b) Calculate the phasor sum $U_{\text{tot}} = U_1 + U_2 + U_3$.

c) Draw the phasors in the complex plane (the x-axis is the real part, the y-axis the imaginary part).

d) Convert the result back to a sinusoidal voltage in the time domain, in the form $u_{\text{tot}}(t) = |U_{\text{tot}}| \sin(wt + \delta_{\text{tot}})$.

19.4 Summary

What you should be able to do

After this chapter, you should:

- Be prepared for the exam.
- Have an overall understanding of the central parts of the course.

Reflect

1. Which area of the course do you feel least sure of? Write down a specific question about it and find out the answer before the exam.
2. Solve a question from the practice exam without aids, and then compare with the worked solution.
3. Go through the question you found hardest in the practice exam, and explain the method of solution to a fellow student.

End of the course

The course ends with the written exam described in §19.1. Thank you for following the course, and good luck!

APPENDIX A

Answers

This appendix holds the answers to all the exercises in the book, to the questions under *Test yourself* in each chapter, and to the practice tests and the practice exam. An exercise's number is a link back to the exercise, and every exercise gives the page its answer is on.

The answer key holds only the answers. An answer shows whether you got it right, but not how to get there; the full worked solutions, in Swedish, are published in the course repository after each lesson. Where an exercise asks you to show, justify or explain something, the answer key gives the conclusion and the decisive reason, not the whole argument.

The answers are rounded as in the worked solutions. An answer that differs in the last decimal place is most often because you rounded earlier in the calculation; calculate with unrounded values and round only in the answer.

CHAPTER 1

Number sets and basic arithmetic

1.1 **a)** 13 **b)** 25 **c)** 5 **d)** 15

1.2 **a)** $\frac{5}{12}$ **b)** $\frac{7}{20}$ **c)** $\frac{3}{2} = 1.5$ **d)** 2

1.3 **a)** 100 **b)** 25 % **c)** Lowest 95 Ω, highest 105 Ω.

2.1 **a)** $\frac{1}{4}\,\Omega^{-1}$ **b)** 4 Ω

2.2 **a)** $R_\text{p} = 6\,\text{k}\Omega$ with both formulas. **b)** 10 kΩ **c)** The parallel connection works like a bracket: its value must be known before R_1 can be added. Adding all three resistors gives the wrong answer, 36 kΩ.

2.3 **a)** 2 mA **b)** 8 V **c)** 12 V **d)** $I_2 = 0.5\,\text{mA}$, $I_3 = 1.5\,\text{mA}$; the sum is $2\,\text{mA} = I$.

2.4 **a)** 8 V **b)** 5.3 V

2.5 **a)** 75 % **b)** 21.6 W

2.6 **a)** $\mathbb{N}$ **b)** $\mathbb{Z}$ **c)** $\mathbb{Q}$ **d)** Irrational **e)** $\mathbb{Q}$ **f)** Irrational

3.1 **a)** −13 **b)** 0 **c)** −6 **d)** 16 **e)** 1

3.2 **a)** $-|4| < -2.5 < 0 < |-3| < \frac{7}{2}$ **b)** 4 **c)** Both are 5: the absolute value of a difference is the distance between the numbers on the number line. **d)** $|U_1 - U_2| = 0.3\,\text{V}$

3.3 $\frac{1}{4} = 0.25 = 25\,\%$; $\frac{3}{5} = 0.6 = 60\,\%$; $\frac{1}{8} = 0.125 = 12.5\,\%$; $\frac{3}{8} = 0.375 = 37.5\,\%$

3.4 **a)** $\frac{1}{3}$ **b)** 1 **c)** 1 **d)** 2 **e)** The parallel resistance of two 4 Ω resistors, which is 2 Ω.

3.5 **a)** 25 % **b)** 15 % **c)** No. $1.10 \times 0.90 = 0.99$, so the value ends up 1 % lower than at the start. **d)** 75 W

3.6 **a)** 4.23 kΩ to 5.17 kΩ **b)** Yes. **c)** About 4.3 % **d)** No. The permitted range is 209–231 Ω, and 208 Ω deviates by about 5.5 %.

3.7 **a)** True: $n = \frac{n}{1}$. **b)** False, for example $\frac{1}{2}$. **c)** False: $\sqrt{9} = 3$. **d)** False, for example $\sqrt{2} + (-\sqrt{2}) = 0$. **e)** True: $0.333\ldots = \frac{1}{3}$. **f)** False, for example π and $\sqrt{2}$.

3.8 **a)** 50 Ω **b)** 25 Ω **c)** n equal terms give $\frac{1}{R_p} = \frac{n}{R}$. **d)** 5 resistors

3.9 **a)** 20 Ω **b)** 280 Ω **c)** Yes, 12 Ω < 20 Ω. **d)** No. A parallel connection is always smaller than the smallest of the resistors in it, here 30 Ω.

3.10 From the top:
$I = 3\,\text{mA}$; $U = 11\,\text{V}$; $R = 3\,\text{k}\Omega$; $I = 5\,\text{mA}$

3.11 **a)** 190 W **b)** 152 W **c)** 76 % **d)** 48 W

Test yourself **1)** The irrational numbers, and so $\mathbb{R}$: 5 is not the square of an integer, so $\sqrt{5}$ cannot be written as a fraction. **2)** 3 Ω

CHAPTER 2

Algebra

1.1 $R_{\text{TOT}} = 1\,\Omega$

1.2 **a)** 460 W **b)** 40 W

1.3 $U_{\text{out}} = 4\,\text{V}$

2.1 **a)** $3I_1 + 10I_2$ **b)** $3R_1 + 2R_2$ **c)** $7I^2 + 3I - 1$ **d)** Both expressions give 33.

2.2 **a)** $U_R = RI_1 + RI_2 + RI_3$; every term has the unit $\Omega \cdot \text{A} = \text{V}$. **b)** 10 V **c)** $2 + 3 + 5 = 10\,\text{V}$, the same answer.

2.3 **a)** $3I(2 + 3R)$ **b)** $(P_1 + P_2)(P_1 - P_2)$ **c)** $(U - 5)^2$ **d)** $(2R + 1)(2R - 1)$

2.4 **a)** $P_{\text{TOT}} = (R_1 + R_2 + R_3)I^2$ **b)** 1.25 W **c)** $P_1 = 0.30\,\text{W}$, $P_2 = 0.45\,\text{W}$, $P_3 = 0.50\,\text{W}$, sum 1.25 W. The factorised form took the least work: one multiplication by I^2 instead of three.

2.5 **a)** Take out $\frac{1}{R}$; what is left is $U_1^2 - U_2^2$, a difference of two squares, which factorises by that rule. **b)** 22 W **c)** $P_1 = 72\,\text{W}$ and $P_2 = 50\,\text{W}$; the difference is 22 W.

2.6 **a)** $P = RI_0^2 + 2RI_0\,\Delta I + R\,\Delta I^2$ **b)** 44.1 W **c)** $\Delta P = 4.1\,\text{W}$ **d)** 0.1 W, about 2.4 % of ΔP.

2.7 **a)** $RI^2 = RI \cdot I$, and $RI = U$ is the voltage across the resistor by Ohm's law, so $P = UI$. **b)** $R \cdot \frac{U^2}{R^2} = \frac{U^2}{R}$ **c)** $I = 2\,\text{A}$; all three forms give 24 W.

3.1 **a)** Three **b)** 7 and −4 respectively **c)** 9 **d)** 7 and R^2 (or 7, R and R) **e)** 29

3.2 **a)** $3I + 14$ **b)** $U + 7$ **c)** $4R_1 + 3$ **d)** 5

3.3 **a)** $R^2 + 8R + 15$ **b)** $2I^2 + 7I - 4$ **c)** $U^2 - 4$ **d)** $9R^2 - 12R + 4$ **e)** $7I$

3.4 **a)** $I^2 + 8I + 16$ **b)** $R^2 - 12R + 36$ **c)** $4U^2 + 12U + 9$ **d)** $25 - 10I + I^2$ **e)** $(3 + 4)^2 = 49$ but $3^2 + 4^2 = 25$; the difference is twice the product, $2ab = 24$.

3.5 **a)** 9996 **b)** 1575 **c)** 3.99 **d)** 200 **e)** 80 W

3.6 **a)** $4(2R + 3)$ **b)** $RI(I + 1)$ **c)** $(U + 7)(U - 7)$ **d)** $(I + 6)^2$ **e)** $(3R + 4)(3R - 4)$ **f)** $2(U + 2)(U - 2)$

3.7 **a)** $2I$ **b)** $R - 3$ **c)** $U + 2$ **d)** $I - 5$ **e)** $2R + 1$

3.8 **a)** 12 V **b)** 18 W **c)** 18 W, the same answer. **d)** The power quadruples, to 72 W: $R(2I)^2 = 4RI^2$.

3.9 **a)** The fractions have the same denominator, and the numerators add up to the denominator $R_1 + R_2$, so $U_1 + U_2 = U_{\text{in}}$. **b)** In the quotient, U_{in} and the denominator $R_1 + R_2$ cancel, leaving $\frac{R_1}{R_2}$. **c)** $U_1 = \frac{U_{\text{in}}}{2}$ **d)** $U_1 = 3\,\text{V}$, $U_2 = 12\,\text{V}$

3.10 **a)** Twice the product is missing: $R^2 + 4R + 4$. **b)** The factor must multiply both terms: $3I - 6$. **c)** A fraction can only be cancelled by factors: $\frac{R}{4} + 1$. **d)** The minus sign changes the sign of every term: $-U + 3$. **e)** The variables are multiplied too: $6R^2$. **f)** The exponent applies to the whole bracket: $4I^2$.

Test yourself **1)** $(3R + 2)(3R - 2)$ **2)** $5I + 7$ **3)** 20 W

CHAPTER 3

Linear equations and inequalities

1.1 **a)** $4R + 14$ **b)** $U^2 + 6U = U(U + 6)$ **c)** $3I + 2$

1.2 **a)** $(R + 5)(R - 5)$ **b)** $3U(U + 2)$ **c)** $(I - 4)^2$

2.1 **a)** $15 = R \cdot 0.3$ gives $R = 50\,\Omega$. **b)** $50 \times 0.3 = 15\,\text{V}$, which is correct.

2.2 **a)** $24 = (R_1 + 40) \times 0.4$ gives $R_1 + 40 = 60$, so $R_1 = 20\,\Omega$. **b)** The same answer, $R_1 = 20\,\Omega$. **c)** $I = \frac{24}{20+40} = 0.4\,\text{A}$, which is correct.

2.3 **a)** $3 = \frac{3}{R_1+3} \times 9$ gives $R_1 = 6\,\Omega$. **b)** $\frac{3}{6+3} \times 9 = 3\,\text{V}$, which is correct.

2.4 $R = \frac{\tau}{C} = 500\,\Omega$

2.5 **a)** $U_{CE} \cdot I_C \leq P_{max}$, that is, $U_{CE} \cdot 0.05 \leq 0.5$ **b)** $U_{CE} \leq 10\,\text{V}$

2.6 **a)** 10.4 V **b)** $12 - 0.8I \geq 9$ gives $I \leq 3.75\,\text{A}$; the sign reverses on dividing by -0.8. **c)** $12 - 0.8I = 13 - 1.0I$ gives $I = 5\,\text{A}$, where both give 8 V. **d)** The first battery: 5.6 V against 5.0 V.

2.7 **a)** $-0.5\,\text{V} \leq \delta \leq 0.5\,\text{V}$ **b)** $4.5\,\text{V} \leq u \leq 5.5\,\text{V}$

2.8 **a)** 5 A **b)** $I = 2\,\text{A}$ and $I = 8\,\text{A}$ **c)** $2\,\text{A} < I < 8\,\text{A}$

3.1 **a)** $x = 5$ **b)** $x = 9$ **c)** $x = 5$ **d)** $x = 24$ **e)** $x = 4$ **f)** $x = -6$

3.2 **a)** $x = 4$ **b)** $x = 2$ **c)** $x = 5$ **d)** Every x: the two sides are identical, so the equation is an identity. **e)** No solution: x cancels and what is left is $8 = 5$, which is false.

3.3 **a)** $x = 6$ **b)** $x = 8$ **c)** $x = 10$ **d)** $x = 3$

3.4 **a)** $x = 6$ **b)** $x = 15$ **c)** $R_1 = 1\,\text{k}\Omega$ **d)** $\frac{6}{0.2} = \frac{U}{0.5}$ gives $U = 15\,\text{V}$.

3.5 **a)** $I = \frac{U}{R}$ **b)** $R = \frac{P}{I^2}$ **c)** $R_2 = R_{TOT} - R_1$ **d)** $C = \frac{\tau}{R}$ **e)** $R_i = \frac{E-U}{I}$ **f)** $P_{in} = \frac{P_{out}}{\eta}$

3.6 **a)** $x < 5$: open circle at 5, arrow to the left. **b)** $x \geq 4$: filled circle at 4, arrow to the right. **c)** $x < -4$: open circle at -4, arrow to the left. **d)** $x \geq 3$: filled circle at 3, arrow to the right. **e)** $x < -3$: open circle at -3, arrow to the left.

3.7 **a)** $1 < x < 6$ **b)** $-1 \leq x \leq 3$ **c)** $5I \leq 0.25$ gives $I \leq 0.05\,\text{A} = 50\,\text{mA}$. **d)** $4.0 \leq x \leq 6.6$

3.8 **a)** $x = 6$ or $x = -6$ **b)** $x = 8$ or $x = -2$ **c)** $x = 3$ or $x = -4$ **d)** $x = -4$, a single solution. **e)** No solution: an absolute value is never negative.

3.9 **a)** $-2 \leq x - 5 \leq 2$, so $3 \leq x \leq 7$. **b)** $1 < x < 5$ **c)** $|u - 5| \leq 0.1$, so $4.9\,\text{V} \leq u \leq 5.1\,\text{V}$. **d)** No: $|1015 - 1000| = 15 > 10$.

3.10 **a)** $120\,\Omega$ **b)** 45 °C **c)** At most 95 °C, since $\Delta T \leq 75\,°\text{C}$. **d)** $\Delta T = \frac{R-R_0}{R_0\,\alpha}$

Test yourself **1)** $x = 8$; both sides are then 24. **2)** $I = 0.03\,\text{A} = 30\,\text{mA}$ **3)** $I \leq 6\,\text{A}$

CHAPTER 4

Systems of equations

1.1 **a)** 9.4 V **b)** 1.88 W

1.2 **a)** $R = \frac{\tau}{C} = 2\,000\,\Omega$ **b)** 2 kΩ

1.3 $5U_{\text{in}} \leq 15$, so $U_{\text{in}} \leq 3$ V.

2.1 $x = \frac{20}{7} \approx 2.86$, $y = \frac{17}{7} \approx 2.43$

2.2 **a)** $I_1 + I_2 = 5$ and $I_1 = 2I_2$
b) $I_1 = \frac{10}{3} \approx 3.33$ A, $I_2 = \frac{5}{3} \approx 1.67$ A

2.3 **a)** $U_1 + U_2 = 12$ and $U_2 = 3U_1$
b) $U_1 = 3$ V, $U_2 = 9$ V **c)** $R_1 = 1.5\,\Omega$, $R_2 = 4.5\,\Omega$

3.1 **a)** $(2,5)$ **b)** $(9,3)$ **c)** $(7,3)$

3.2 **a)** $(3,1)$ **b)** $(4,2)$ **c)** $(3,2)$ **d)** $(2,1)$

3.3 **a)** Infinitely many: equation (2) is equation (1) times 2, so the lines coincide. **b)** None: the same left-hand side but different right-hand sides, so the lines are parallel. **c)** Exactly one, $(2,2)$: the lines have different slopes and intersect. **d)** Infinitely many: equation (2) times −2 is equation (1), so the lines coincide.

3.4 **a)** For $x = 0,1,2,3,4$, $y = 2x - 1$ gives the values $-1,1,3,5,7$ and $y = -x + 5$ the values $5,4,3,2,1$. **b)** At $x = 2$; the point of intersection is $(2,3)$. **c)** $(x,y) = (2,3)$, the same as the reading. **d)** The rows would never give the same y: the difference between them would be constant, as for two parallel lines.

3.5 **a)** $(4,2)$ **b)** $(6,3)$

3.6 **a)** $I_1 + I_2 = 12$ and $I_1 - I_2 = 4$ **b)** $I_1 = 8$ A, $I_2 = 4$ A **c)** $8 + 4 = 12$ and $8 - 4 = 4$ both hold.

3.7 **a)** $I_1 + I_2 = 3$ and $I_1 = 2I_2$ **b)** $I_1 = 2$ A, $I_2 = 1$ A **c)** $R_1 = 12\,\Omega$, $R_2 = 24\,\Omega$
d) $R_1 \,/\!/\, R_2 = \frac{12\times24}{12+24} = 8\,\Omega = \frac{24}{3}\,\Omega$

3.8 **a)** $U_1 + U_2 = 15$ and $U_1 - U_2 = 3$
b) $U_1 = 9$ V, $U_2 = 6$ V **c)** $R_1 = 30\,\Omega$, $R_2 = 20\,\Omega$ **d)** $R_1 + R_2 = 50\,\Omega = \frac{15}{0.3}\,\Omega$

3.9 **a)** $U = 12 - 2I$ and $U = 4I$ **b)** $I = 2$ A, $U = 8$ V **c)** $12 - 2\times2 = 8$ and $4\times2 = 8$ both hold. **d)** The operating point.

3.10 **a)** $\frac{6}{9} = \frac{R_1}{900}$, which gives $R_1 = 600\,\Omega$.
b) $R_1 = 600\,\Omega$, $R_2 = 300\,\Omega$
c) $U_1 = 9 \times \frac{600}{900} = 6$ V, which is correct.

3.11 **a)** $I_1 = 3$ A, $I_2 = 2$ A, $I_3 = 4$ A
b) $3 + 2 + 4 = 9$, $3 - 2 = 1$ and $I_3 = 4$ all hold.
c) It gives one unknown directly; substituted into the others, it leaves a system with two unknowns.

3.12 **a)** $5R_A + 3R_B = 4\,390$ and $2R_A + 4R_B = 3\,660$ **b)** $R_A = 470\,\Omega$, $R_B = 680\,\Omega$ **c)** $2\,350 + 2\,040 = 4\,390$ and $940 + 2\,720 = 3\,660$ both hold.

3.13 **a)** Yes. **b)** Yes. **c)** No: $5 - 2\times1 = 3$ holds, but $5 + 1 = 6 \neq 7$. **d)** The solution must lie on both lines at once. A pair of numbers that satisfies only one of the equations lies on only one of the lines, as in c).

3.14 **a)** $(2,3)$ **b)** $(4,-1)$ **c)** $(2,-1)$
d) Substitution in a) and b), where one unknown is easy to solve for; elimination in c), where the y-terms easily become opposite. Both methods give the same answer.

Test yourself **1)** $(x,y) = (3,2)$ **2)** $I_1 = 3$ A, $I_2 = 1$ A

CHAPTER 5

Powers, roots and number systems

1.1 $I_1 = 2\,\text{A}, I_2 = 3\,\text{A}$

1.2 $U_A = \frac{54}{11} \approx 4.91\,\text{V}, U_B = \frac{30}{11} \approx 2.73\,\text{V}$

2.1 **a)** x^8 **b)** R^4 **c)** I^6 **d)** U^3 **e)** $\frac{8}{R^3}$

2.2 **a)** $4.7 \times 10^4\,\Omega$ **b)** $2.2 \times 10^{-5}\,\text{F}$ **c)** $3.3 \times 10^6\,\text{Hz}$ **d)** $1.5 \times 10^{-8}\,\text{A}$

2.3 **a)** 250 mW **b)** 80 mW

2.4 **a)** 10 **b)** 25 **c)** 58 **d)** 200

2.5 **a)** 10110_2 **b)** 1100100_2 **c)** 64_{16} **d)** $0101\,1010_2$

3.1 **a)** 81 **b)** −32 **c)** 0.01 **d)** 1 **e)** $\frac{1}{8} = 0.125$ **f)** 0.001

3.2 **a)** a^4 **b)** $\frac{1}{U^3}$ **c)** $4R^6$ **d)** I^5 **e)** $\frac{R^6}{27}$ **f)** $3x^3$

3.3 **a)** 10^{-3} **b)** 10^3 **c)** 10^{-6} **d)** 10^7 **e)** 10^{-3}

3.4 **a)** 9 **b)** 0.5 **c)** 3 **d)** 0.04 **e)** $5\sqrt{2}$ **f)** $\sqrt{9+16} = \sqrt{25} = 5$, but $\sqrt{9} + \sqrt{16} = 3 + 4 = 7$.

3.5 **a)** 4 **b)** 3 **c)** 27 **d)** $\frac{1}{2}$ **e)** 2

3.6 **a)** $2.2 \times 10^3\,\Omega$ **b)** $4.7 \times 10^{-4}\,\text{F}$ **c)** $1.5 \times 10^{-11}\,\text{F}$ **d)** $2.5 \times 10^{-1}\,\text{A}$ **e)** $1.8 \times 10^6\,\text{Hz}$ **f)** $6.8 \times 10^{-9}\,\text{H}$

3.7 **a)** 3.3 µF **b)** 470 kΩ **c)** 250 µA **d)** 80 MHz **e)** 12 pF

3.8 **a)** 8×10^{-3} **b)** 3×10^{-5} **c)** 2.5×10^7 **d)** 3.25×10^{-3} **e)** 9×10^1

3.9 **a)** Three. **b)** 0.025 **c)** $1.23 \times 10^4\,\Omega$ **d)** Two significant figures, like the voltage: $I = 0.050\,\text{A} = 50\,\text{mA}$.

3.10 **a)** 15 mA **b)** 11.75 V **c)** 3 kΩ

3.11 **a)** 0.6 W **b)** ≈ 11.4 mW **c)** ≈ 23 mA **d)** ≈ 10.8 V

3.12 **a)** 47 ms **b)** 10 kΩ **c)** 500 nF

3.13 **a)** ≈ 7.07 V **b)** ≈ 325 V **c)** $U_{\text{RMS}}^2 = \frac{|U|^2}{(\sqrt{2})^2} = \frac{|U|^2}{2}$, so $P = \frac{|U|^2}{2R}$. **d)** 9 W

3.14 **a)** The exponents should be added: $2^7 = 128$. **b)** The exponents should be multiplied: a^6. **c)** The exponents should be subtracted: 10^4. **d)** The exponent applies to both factors: $8a^3$. **e)** A root cannot be split over a sum: $\sqrt{25} = 5$. **f)** A negative exponent gives a reciprocal, not a negative number: $\frac{1}{a^2}$.

3.15 **a)** 7 **b)** 16 **c)** 42 **d)** 100101_2 **e)** 1000000_2 **f)** $1111\,1111_2$

3.16 **a)** 31 **b)** 128 **c)** 2 748 **d)** AF_{16} **e)** 1000_{16} **f)** $3\text{E}8_{16}$

3.17 **a)** 9E_{16} **b)** $\text{F}0_{16}$ **c)** 2B_{16} **d)** $0011\,1100_2$ **e)** $1010\,0101_2$ **f)** $0111\,1111_2$

3.18 **a)** 16 values; the largest is $1111_2 = 15$. **b)** 1 024 values; the largest is 1 023. **c)** Bit 2 and bit 5. **d)** $1000\,0001_2 = 81_{16} = 129$

Test yourself **1)** $3x^2$ **2)** $2.2 \times 10^{-7}\,\text{F}$ **3)** $\text{D}2_{16} = 210$

CHAPTER 6

Quadratic equations

1.1 **a)** 10 **b)** $4 \cdot 10^6$ **c)** $2 \times 10^3 = 2\,000$

1.2 **a)** $I \approx 7.02 \times 10^{-1}\,\text{mA} = 0.702\,\text{mA}$
b) $P \approx 2.32\,\text{mW}$

1.3 **a)** 12 **b)** 0.07 **c)** 20 V

1.4 **a)** $0100\,0100_2 = 44_{16}$ **b)** $0000\,0100_2 = 04_{16}$ **c)** $x = 0100\,0000_2 = 40_{16}$, and PORTD becomes $0100\,0100_2 = 44_{16}$. **d)** $R = 300\,\Omega$ gives exactly 10 mA; the nearest E12 values are $330\,\Omega$ ($I \approx 9.1\,\text{mA}$) and $270\,\Omega$ ($I \approx 11.1\,\text{mA}$). **e)** About 45 mW per circuit with $330\,\Omega$ (about 56 mW with $270\,\Omega$).

2.1 **a)** $I = \pm 0.4\,\text{A}$ **b)** The negative root only says that the current flows the other way; the power RI^2 is the same, so the magnitude of the current is $I = 0.4\,\text{A}$.

2.2 **a)** $x_1 = 3, x_2 = 2$ **b)** $x_1 = 2, x_2 = -5$ **c)** $x_1 = x_2 = 3$ (a double root)

2.3 **a)** $x_1 = 3, x_2 = 0.5$ **b)** $x_1 = \frac{1}{3}, x_2 = -2$

2.4 **a)** $R_1^2 - 13R_1 + 36 = 0$ **b)** $9\,\Omega$ and $4\,\Omega$

3.1 **a)** $x = \pm 7$ **b)** $x = \pm 3$ **c)** $x = \pm 5$ **d)** No real solutions, since $x^2 = -4$. **e)** $x = \pm 6$

3.2 **a)** $x_1 = 0, x_2 = 7$ **b)** $x = \pm 4$ **c)** $x_1 = -2, x_2 = -3$ **d)** $x_1 = 0, x_2 = 4$ **e)** $x_1 = x_2 = 2$ (a double root)

3.3 **a)** $x_1 = 2, x_2 = -4$ **b)** $x_1 = 6, x_2 = 2$ **c)** $x_1 = 3, x_2 = -4$ **d)** $x_1 = 5, x_2 = -3$

3.4 **a)** $x_1 = 3, x_2 = 2$ **b)** $x_1 = 1, x_2 = -5$ **c)** $x_1 = 3, x_2 = 1$

3.5 The expression under the root sign, $\left(\frac{p}{2}\right)^2 - q$, is: **a)** $4 > 0$, two real roots **b)** 0, a double root **c)** $-2.75 < 0$, no real roots **d)** 0, a double root **e)** $\frac{4}{9} > 0$, two real roots

3.6 **a)** $x_1 = 1, x_2 = -5$ **b)** $x_1 = 7, x_2 = 3$ **c)** $x = -3 \pm \sqrt{5}$, so $x_1 \approx -0.76$ and $x_2 \approx -5.24$

3.7 **a)** 0.5 A **b)** 30 mA **c)** About 33 mA **d)** The negative root only means the opposite direction of current; the power is the same because I is squared, and it is the magnitude of the current that is asked for.

3.8 **a)** 10 V **b)** About 15.8 V **c)** $52.9\,\Omega$

3.9 **a)** $12\,\Omega$ and $8\,\Omega$ **b)** $8\,\Omega$ and $7\,\Omega$ **c)** No. For $R_1^2 - 10R_1 + 30 = 0$ the expression under the root sign is $25 - 30 = -5 < 0$, so there are no real roots.

3.10 **a)** Product over sum gives $R_1 R_2 = (R_1 /\!/ R_2)(R_1 + R_2) = 2.4 \times 10 = 24\,\Omega^2$. **b)** $R_1^2 - 10R_1 + 24 = 0$ gives $6\,\Omega$ and $4\,\Omega$. **c)** $6 + 4 = 10\,\Omega$ and $\frac{6\times4}{6+4} = 2.4\,\Omega$.

3.11 **a)** $P = R\left(\frac{12}{4+R}\right)^2 = 8$ gives $18R = (4+R)^2$, so $R^2 - 10R + 16 = 0$. **b)** $R = 2\,\Omega$ or $R = 8\,\Omega$ **c)** $R = 2\,\Omega$: $I = 2\,\text{A}$ and $P = 8\,\text{W}$. $R = 8\,\Omega$: $I = 1\,\text{A}$ and $P = 8\,\text{W}$. **d)** $I = 1.5\,\text{A}$ and $P = 9\,\text{W}$, more than the two solutions' 8 W: the power in R is greatest when $R = R_1$ (maximum power transfer).

3.12 **a)** False: the expression under the root sign in the PQ-formula can give two, one or no real roots. **b)** False: $x = -4$ is also a solution. **c)** False: the equation must first be normalised, $x^2 + 2x - 3 = 0$. **d)** True: the square-root term becomes zero, and both roots become $-\frac{p}{2}$. **e)** True: that is the zero-product rule. **f)** True: $x^2 = -4$, and no real square is negative.

Test yourself **1)** $2x(x-4) = 0$, so $x_1 = 0$ and $x_2 = 4$. **2)** $I = 0.5$ (amperes). The negative root is rejected: it only indicates the opposite direction of current, and the power RI^2 is the same.

PRACTICE TEST 1

Arithmetic, algebra, powers, number systems and equations

1 $R_2 \approx 8\,\text{k}\Omega$ (unrounded $8.18\,\text{k}\Omega$)

2 $c(c^2+1) = c^3 + c$

3 $x = -\frac{17}{7}, y = -\frac{19}{7}$

4 $1001\,1100_2 = 156$

5 $P \approx 2.9 \cdot 10^{-2}\,\text{W}$ (about 29 mW)

6 $x_1 \approx 0.21, x_2 \approx -27.54$

CHAPTER 7

Vectors

1.1 **a)** $P = (12 - 4I) \cdot I = 5$ gives $4I^2 - 12I + 5 = 0$. **b)** $I = 0.5\,\text{mA}$ or $I = 2.5\,\text{mA}$ **c)** At 0.5 mA there are 10 V across the module and the loss in the resistor is 1 mW. At 2.5 mA there are 2 V across the module and the loss is 25 mW. The first operating point is the most reasonable: the module gets a usable supply voltage and the losses are small.

2.1 **a)** $(7, 8)\,\text{V}$ **b)** $(18, 22)\,\text{V}$ **c)** $(-1, -4)\,\text{V}$ **d)** $(-22, -28)\,\text{V}$

2.2 **a)** $(0, 2)\,\text{A}$ **b)** $(2, 6)\,\text{A}$ **c)** $(-14, 16)\,\text{A}$ **d)** $(18, 0)\,\text{A}$

2.3 **a)** $\sqrt{5}\,\text{V}$ **b)** $\sqrt{50} = 5\sqrt{2}\,\text{V}$ **c)** $\sqrt{10}\,\text{V}$ **d)** $\sqrt{125} = 5\sqrt{5}\,\text{V}$

2.4 **a)** $x = 23, y = -5$ **b)** $x = -8, y = 2$

2.5 a) and d): $(6, -9) = 3\mathbf{I}$ (in phase) and $(-12, 18) = -6\mathbf{I}$ (in antiphase).

2.6 $a = 1.5, b = 3.5$

2.7 **a)** $|\mathbf{U}| = \sqrt{32} \approx 5.66\,\text{V}$, $|\mathbf{I}| = \sqrt{5} \approx 2.24\,\text{A}$ **b)** $v_U = 45°$, $v_I \approx 116.6°$ **c)** About 71.6°

3.1 **a)** 5 **b)** 5 **c)** 5 **d)** 8 **e)** 4

3.2 **a)** $\overrightarrow{AB} = (4, 3)$, $|\overrightarrow{AB}| = 5$ **b)** $\overrightarrow{AB} = (3, -4)$, $|\overrightarrow{AB}| = 5$ **c)** $\overrightarrow{BA} = (-4, -3)$: the same length, the opposite direction. **d)** $\overrightarrow{BA} = -\overrightarrow{AB}$

3.3 **a)** 10 V **b)** 13 V **c)** $\sqrt{2} \approx 1.41\,\text{V}$ **d)** $3\sqrt{2} \approx 4.24\,\text{V}$ **e)** 7 V

3.4 **a)** 45°, quadrant I, no correction **b)** 135°, quadrant II, +180° **c)** 225° (or −135°), quadrant III, +180° **d)** −45° (or 315°), quadrant IV, no correction **e)** 36.9°, quadrant I, no correction

3.5 **a)** $(8.66, 5.00)\,\text{V}$ **b)** $(0, 5)\,\text{V}$ **c)** $(-4, 0)\,\text{V}$ **d)** $(-1.41, -1.41)\,\text{V}$ **e)** $\sqrt{75.0 + 25} = \sqrt{100} = 10\,\text{V}$

3.6 **a)** 45° **b)** 90° **c)** About 81.9° **d)** 180° (in antiphase)

3.7 **a)** $(12, 5)\,\text{V}$ **b)** 13 V **c)** About 22.6° **d)** The voltages are perpendicular and add geometrically, like the two shorter sides of a right-angled triangle: $\sqrt{12^2 + 5^2} = 13\,\text{V}$.

3.8 **a)** $(30, 40)\,\text{V}$ **b)** 50 V **c)** About 53.1° **d)** $(30, 0)\,\text{V}$: the inductor and capacitor voltages cancel, so the whole voltage lies across the resistor.

3.9 **a)** $(0.6, 0.8)$ **b)** $\sqrt{0.36 + 0.64} = 1$ **c)** $(-0.6, 0.8)$ **d)** $(9, 12)\,\text{A}$ **e)** $(-9, -12)\,\text{A}$

3.10 **a)** $(0, 0)\,\text{V}$ **b)** The phase voltages cancel: their sum is zero. **c)** About 230 V and about 240° (or −120°) **d)** $(345, 199)\,\text{V}$, magnitude about 398 V

3.11 **a)** $(2, -3)\,\text{V}$, $(-8, 12)\,\text{V}$ and $(-2, 3)\,\text{V}$ **b)** $(-8, 12)\,\text{V}$ and $(-2, 3)\,\text{V}$ **c)** $t = -6$ **d)** Yes: $\mathbf{U} = 1 \cdot \mathbf{U}$.

3.12 **a)** $(4, 3)\,\text{A}$ **b)** 5 A **c)** About 36.9° **d)** 4 A, that is, $\mathbf{I}_C = (0, 4)\,\text{A}$

3.13 **a)** 1000 VA **b)** About 36.9° **c)** 0.8 **d)** $\mathbf{S} = (800, 0)$ and 800 VA: the reactive power is fully compensated.

3.14 **a)** The length is given by Pythagoras' theorem, not by the sum: $|(3,4)| = 5$. **b)** The number multiplies both components: $(6, -2)$. **c)** The angle between the phasors is the difference between the angles, not the sum: 70°. **d)** The vector lies in quadrant II, so 180° must be added: 126.9°. **e)** A length is never negative: $|-2\mathbf{u}| = 2|\mathbf{u}|$.

4.1 **a)** $|\mathbf{u}| = |\mathbf{v}| = 5$, $v_u \approx 53.13°$, $v_v \approx 143.13°$ (quadrant II), $\mathbf{u} + \mathbf{v} = (-1, 7)$ with magnitude $\sqrt{50} \approx 7.071$ and angle 98.13°, $\alpha = 90°$: the vectors are perpendicular. **b)** The same values as in a). **c)** `atan(3.0 / -4.0)` gives −36.87°, the angle of $(4, -3)$ in quadrant IV: the ratio does not show which component was negative. `atan2(y, x)` gets the components separately and sees their signs.

Test yourself **1)** $|\mathbf{u}| = 5$; the angle is about −53.1° (or 306.9°), quadrant IV. **2)** $\mathbf{w} = (5, -1)$

CHAPTER 8

Functions

1.1 **a)** $\mathbf{U} = (4, -1)$ V, $\mathbf{I} = (-3, 4)$ A **b)** $|\mathbf{U}| = \sqrt{17} \approx 4.12$ V, $|\mathbf{I}| = 5$ A **c)** $v_U \approx -14.0°$, $v_I \approx 126.9°$ **d)** $\alpha \approx 140.9°$: the current leads the voltage. **e)** $\mathbf{I}_{\text{tot}} = (4, 3)$ A, $|\mathbf{I}_{\text{tot}}| = 5$ A **f)** $\mathbf{I}_a = -2\mathbf{I} = (6, -8)$ A

2.1 **a)** A function: each U gives exactly one I. **b)** A function: all the U values are different ($I = 250$ occurring twice is allowed). **c)** Not a function: $U = 1$ gives both $I = 2$ and $I = 5$.

2.2 **a)** $D_U = [-40, 125]$ °C, $V_U = [0.1, 1.75]$ V **b)** $D_U = [0, \infty)$ W, $V_U = [0, \infty)$ V **c)** $D_P = [-20, 20]$ V, $V_P = [0, 40]$ W

2.3 **a)** $k = 1.5\,\Omega/°\text{C}$, $m = 100\,\Omega$ **b)** 137.5 Ω

2.4 **a)** $C(10) \approx 2950$ mAh: after 10 years about 74 % of the capacity remains. **b)** After about 7.3 years.

2.5 **a)** A power function, $P(I) = 50I^2$ (a quadratic function). **b)** $P(0.5) = 12.5$ W, $P(2) = 200$ W **c)** $D_P = [0, 3]$ A, $V_P = [0, 450]$ W

2.6 **a)** The voltage falls by 3 V every second; a constant change gives a linear relationship. **b)** $U(t) = -3t + 12$ **c)** $t = 5$ s. No: the voltage across a discharging capacitor does not become negative but levels out towards 0 V, so the model holds only within the measured interval $0 \le t \le 4$ s.

3.1 **a)** 10 V **b)** 4 V **c)** 0 W **d)** 45 W **e)** 3 A **f)** 2 A

3.2 **a)** 1.5 A **b)** $I = 6$ A: the terminal voltage is zero, the battery is short-circuited. **c)** 2 A (mathematically also −2 A, the same current in the opposite direction) **d)** No: $I^2 \ge 0$, so $P(I) \ge 0$. **e)** 48 Ω

3.3 **a)** $R \neq 0$; reasonable $R > 0$ **b)** $R \neq -2$; reasonable $R \ge 0$ **c)** $P \ge 0$; the same **d)** $f \neq 0$; reasonable $f > 0$

3.4 **a)** $[0, 40]$ W **b)** $[0, 12]$ V **c)** $[0, 10]$ V **d)** $[-40, 125]$ °C

3.5 **a)** $U(I) = -0.5I + 9$ **b)** $U(T) = 0.01T + 0.5$ **c)** $R(T) = 0.385T + 100$ **d)** $I(t) = 2$, a constant function.

3.6 **a)** The slope: how much U changes when I increases by 1 A. **b)** The intercept with the y-axis, $U(0) = m$. **c)** Decreasing, since $k = -0.5 < 0$. **d)** $I = 24$ A, the short-circuit

current. **e)** $m = 12\,\text{V}$ is the open-circuit voltage, the terminal voltage at $I = 0$. The slope $k = -0.5$ is minus the internal resistance, $R_\text{i} = 0.5\,\Omega$.

3.7 **a)** Both are 90 W: the square removes the minus sign. **b)** 9 times larger. **c)** It becomes a quarter. **d)** 2.5 W and 10 W: the power becomes 4 times larger.

3.8 **a)** 5, 10 and 40 mV **b)** 100, 50 and 12.5 mV **c)** u is increasing and g decreasing. The base decides: $a > 1$ gives an increasing function, $0 < a < 1$ a decreasing one. **d)** $C = 5\,\text{mV}$ and $C = 100\,\text{mV}$; C is the initial value, the signal before the first stage.

3.9 **a)** 1.05 **b)** 0.80 **c)** $f(n) = 2 \cdot 0.7^n\,\text{V}$ **d)** 0.686 V **e)** After 2 stages, when it is 0.98 V.

3.10 **a)** f: the difference between adjacent values is constant, 2. **b)** g: the ratio between adjacent values is constant, 2. **c)** $f(x) = 2x + 3$ **d)** $g(x) = 3 \cdot 2^x$ **e)** $f(10) = 23$, $g(10) = 3072$

3.11 **a)** $I = 3\,\text{A}$: the short-circuit current. **b)** $I = 0$ and $I = 6\,\text{A}$: no power is delivered without current, nor at a short circuit, when the whole voltage lies across the internal resistance. **c)** $t = 12\,\text{s}$: then the voltage is down to 0 V.

3.12 **a)** 0 V, 6 V and 10.8 V **b)** $V_U = [0, 10.8]\,\text{V}$ **c)** $x = 2\,\text{k}\Omega$ **d)** No: the denominator $1 + x$ is always greater than the numerator x, so $U < 12\,\text{V}$. The voltage approaches 12 V but never reaches it.

3.13 **a)** 10 V **b)** $u(2) \approx 3.68\,\text{V}$, $u(4) \approx 1.35\,\text{V}$ **c)** Decreasing. **d)** $V_u = (0, 10]\,\text{V}$ **e)** No: $e^{-t/2} > 0$ for all t, so the voltage approaches 0 V but never becomes exactly zero.

3.14 **a)** False: $I^2 \geq 0$, so the power is never negative. **b)** False: $R = 0$ gives division by zero. **c)** False: a zero requires $U = 0$, but $U(2) = 5$. **d)** True: a function gives exactly one I for each U. **e)** True: $u(0) = 3 \cdot 2^0 = 3\,\text{V}$.

4.1 **a)** $f(0) = -50\,^\circ\text{C}$, $f(0.8) = 30\,^\circ\text{C}$, $f(5) = 450\,^\circ\text{C}$ **b)** −50, 0, 50, ..., 450 °C: the temperature rises by 50 °C per half volt, the slope $k = 100$. **c)** $x = \dfrac{f + 50}{100}$. 30 °C gives 0.8 V, and `voltage(temperature(0.8))` gives 0.8 back, since the functions are each other's inverses.

Test yourself **1)** For example $f(x) = \sqrt{x}$ with $D_f = [0, \infty)$: the square root of a negative number is not defined. (Or $f(x) = \frac{1}{x}$, where $x = 0$ gives division by zero.) **2)** $f(2) = 9$, $f(-1) = 6$

CHAPTER 9

Trigonometric functions

1.1 **a)** 0 V **b)** $0 \leq u_\text{c}(t) \leq 5(1 - e^{-5}) \approx 4.97\,\text{V}$ **c)** $t = -2 \ln 0.2 \approx 3.22\,\text{s}$

1.2 **a)** 0°, 6°, 8°, 6° and 0° **b)** A downward-opening parabola through the points in a), symmetric about $t = 4\,\text{s}$; the largest angle, 8°, is reached at $t = 4\,\text{s}$. **c)** $t = 1\,\text{s}$ and $t = 7\,\text{s}$

1.3 **a)** $x^2 + x - 12 = (x + 4)(x - 3)$, $x^2 - 16 = (x + 4)(x - 4)$ **b)** $x = 4$ and $x = -4$ **c)** $G(x) = \frac{x-3}{x-4}$, $x \neq \pm 4$

2.1 **a)** $-\frac{\pi}{6}$ **b)** $\frac{\pi}{9}$ **c)** $\frac{5\pi}{4}$ **d)** $-\frac{\pi}{4}$ **e)** $\frac{5\pi}{3}$

2.2 **a)** 90° **b)** 135° **c)** −150° **d)** ≈ −57.3° **e)** ≈ 126.1°

2.3 $|U| = 8\,\text{V}$; $f = 10\,\text{Hz}$; $\omega = 20\pi \approx 62.8\,\text{rad/s}$; $\delta = \frac{\pi}{3}\,\text{rad}$; $u(t) = 8\sin(20\pi t + \frac{\pi}{3})\,\text{V}$

2.4 $u(t) = 3\sin(50\pi t + \frac{\pi}{4})$ V. The curve has the amplitude 3 V and the period 40 ms, and lies $T/8 = 5$ ms ahead of a sine curve with no phase shift: the peak comes at $t = 5$ ms.

2.5 $\delta \approx -1.63$ rad or $\delta \approx 1.00$ rad

3.1 **a)** $\frac{\pi}{3}$ **b)** $\frac{2\pi}{3}$ **c)** $-\frac{\pi}{2}$ **d)** $\frac{7\pi}{6}$ **e)** $\frac{\pi}{12}$ **f)** 2π

3.2 **a)** 60° **b)** 225° **c)** −90° **d)** 210° **e)** ≈ 171.9° **f)** ≈ 28.6°

3.3 **a)** 0 **b)** 1 **c)** 0 **d)** −1 **e)** $\frac{1}{2}$ **f)** $\frac{\sqrt{2}}{2} \approx 0.707$

3.4 From the top: $T = 20$ ms, $\omega = 100\pi \approx 314$ rad/s; $f = 1$ kHz, $\omega = 2000\pi \approx 6283$ rad/s; $f = 200$ Hz, $T = 5$ ms; $T = 2.5$ ms, $\omega = 800\pi \approx 2513$ rad/s

3.5 In the order $|U|$, ω, f, T, δ: **a)** 5 V, 100π rad/s, 50 Hz, 20 ms, 0 **b)** 12 V, 200π rad/s, 100 Hz, 10 ms, $\frac{\pi}{2}$ **c)** 3 V, 50π rad/s, 25 Hz, 40 ms, $-\frac{\pi}{6}$ **d)** 8 V, 2000π rad/s, 1000 Hz, 1 ms, 0

3.6 **a)** $u(t) = 10\sin(100\pi t)$ V
b) $u(t) = 6\sin(200\pi t + \frac{\pi}{2})$ V
c) $u(t) = 2\sin(400\pi t - \frac{\pi}{3})$ V
d) $u(t) = 325\sin(100\pi t + \frac{\pi}{4})$ V

3.7 **a)** 0 V **b)** 10 V **c)** 0 V **d)** −10 V **e)** $t = 0$, 10 and 20 ms

3.8 **a)** ≈ 8.49 V **b)** $u(t) = 325\sin(100\pi t)$ V **c)** 0.72 W

3.9 **a)** 5 ms **b)** ≈ 1.67 ms **c)** $\delta = \frac{\pi}{5} \approx 0.628$ rad **d)** $\delta = -\frac{\pi}{2}$ rad

3.10 **a)** u_2 leads u_1 by 60°. **b)** ≈ 3.33 ms **c)** $u_1 = 4$ V, $u_2 = 2$ V **d)** Yes: both have the amplitude 4 V and the frequency 50 Hz; only the phase differs.

3.11 **a)** $t = \frac{1}{600}$ s ≈ 1.67 ms **b)** $t = \frac{5}{600}$ s ≈ 8.33 ms **c)** $t = 5$ ms **d)** Sine is periodic: every solution repeats once per period, that is, every 20 milliseconds.

3.12 **a)** False: the range is $[-1, 1]$. **b)** True: $T = \frac{1}{f}$. **c)** False: ω is measured in rad/s; it is the frequency f that is measured in hertz. **d)** True: the curve reaches its values earlier. **e)** True: half a turn is π radians. **f)** False: the amplitude is measured from zero to the peak value; from the lowest to the highest value is $2|U|$.

4.1 **a)** $\sin 30° = 0.5$, $30° = \frac{\pi}{6} \approx 0.5236$ rad **b)** `sin(30.0)` gives −0.988, `sin(deg_to_rad(30.0))` gives 0.5: `sin()` takes the angle as radians, so `sin(30.0)` is the sine of 30 rad. **c)** $t = 0$: 5.000; 2: 9.135; 4: 9.781; 6: 6.691; 8: 1.045; 10: −5.000, and then the same values with the opposite sign, until $u(20\text{ ms}) = 5.000$ V. **d)** $T = 20$ ms, and the table repeats after 20 ms. The largest value is 9.781 V, since the peak falls between two times in the table: $u(t) = 10$ V at $t = \frac{1}{300}$ s ≈ 3.33 ms.

Test yourself **1)** $150° = \frac{5\pi}{6}$ and $\frac{5\pi}{6} = 150°$ **2)** $u(t) = 5\sin(100\pi t + \frac{\pi}{4})$ V

CHAPTER 10

Exponential functions, more logarithms and decibels

1.1 **a)** $\frac{\pi}{12}$ **b)** $-\frac{5\pi}{12}$

1.2 **a)** −30° **b)** 270°

1.3 $u(t) = 4\sin\left(50\pi t - \frac{\pi}{4}\right)$ V

1.4 $u(t) = 4\sin\left(100\pi t - \frac{\pi}{6}\right)$ V. The curve has the period $T = 20$ ms and starts at $u(0) = -2$ V. It passes upwards through zero at $t \approx 1.67$ ms (a twelfth of a period late), reaches the peak 4 V at 6.67 ms and the trough −4 V at 16.67 ms.

1.5 $\delta \approx -3.25\,\text{rad}$ or $\delta \approx -1.15\,\text{rad}$, exactly $-\frac{31\pi}{30}$ and $-\frac{11\pi}{30}$.

2.1 **a)** 3 **b)** 10 **c)** ≈ 3.79

2.2 $t \approx 132.88\,\text{h}$, about 132 hours and 53 minutes.

2.3 $G_{\text{lin}} \approx 50.12$, that is, about 50.

2.4 $|U| \approx 10.0\,\text{V}$ ($U_{\text{RMS}} \approx 7.08\,\text{V}$).

3.1 **a)** 2 **b)** 3 **c)** 0 **d)** -2 **e)** 1 **f)** 0

3.2 **a)** $\log 12$ **b)** $\log 5$ **c)** $\log 25$ **d)** 1 **e)** $\log 2$

3.3 **a)** 5 **b)** -3 **c)** ≈ 2.86 **d)** ≈ 3.00 **e)** 2

3.4 **a)** 100 **b)** 1 **c)** 5 **d)** 100 **e)** $\frac{e}{2} \approx 1.36$

3.5 **a)** 40 dB **b)** 0 dB **c)** -20 dB **d)** Attenuation: the output signal is smaller than the input signal, so the ratio is less than 1 and its logarithm negative.

3.6 **a)** 10 **b)** 1000 **c)** ≈ 2.00 **d)** ≈ 0.708 **e)** $\approx 1.0\,\text{V}$

3.7 **a)** 20 dB **b)** ≈ -3.01 dB **c)** The power is proportional to the square of the voltage, $P = U^2/R$, and the exponent 2 becomes a factor in front of the logarithm: $10 \cdot 2 = 20$. **d)** The power: ≈ 3 dB (3.01). The voltage: ≈ 6 dB (6.02).

3.8 **a)** 26 dB **b)** ≈ 20.0 **c)** $3.98 \times 7.94 \times 0.631 \approx 20.0$ **d)** Linear gains are multiplied but dB values are added, since $\log(ab) = \log a + \log b$.

3.9 **a)** 0 dBV **b)** 20 dBV **c)** -20 dBV **d)** $\approx 3.98\,\text{V}$ **e)** $\approx 5.63\,\text{V}$

3.10 **a)** $a = 0.98$ **b)** $\approx 34.3\,\text{h}$ **c)** ≈ 7.3 days **d)** $C \cdot a^t = 2C$ gives $a^t = 2$, and taking logarithms gives $t \log a = \log 2$.

3.11 **a)** It is $\log(ab)$ that equals $\log a + \log b$; $\log(a + b)$ cannot be simplified. **b)** The quotient is a change of base: $\frac{\log 8}{\log 2} = \log_2 8 = 3$. **c)** The exponent becomes a factor: $\log(a^n) = n \log a$. **d)** 0 dB gives $G_{\text{lin}} = 1$: the signal passes unchanged. **e)** A power of ten and the base-10 logarithm cancel each other: $10^{\log x} = x$. **f)** $\ln 0$ is undefined; it is $\ln 1$ that is 0.

4.1 **a)** 0, 20 and 40 dB; $10^{-3/20} \approx 0.708$ **b)** 0.5: -6.02 dB; 1: 0; 2: 6.02; 10: 20; 100: 40; 1000: 60 dB. `ratio_from_db()` gives the same ratios back. **c)** Doubling adds 6.02 dB and multiplying by ten adds 20 dB, wherever you start. A ratio less than 1 gives negative dB, that is attenuation.

Test yourself **1)** $x = 3$ **2)** $G_{\text{lin}} = 100$

CHAPTER 11

Review: vectors, functions, trigonometry and logarithms

1.1 **a)** $(-1, 6)$ **b)** $(5, 4)$ **c)** $(6, 15)$ **d)** $(13, 7)$

1.2 **a)** 10 **b)** 13 **c)** $\approx 36.9°$ **d)** $\approx 247.4°$

1.3 **a)** $\approx 63.4°$ and $\approx 18.4°$ **b)** $45°$ **c)** Yes: the angles are $\approx 18.4°$ and $\approx -71.6°$, so the angle between the vectors is $90°$.

1.4 **a)** $(-0.8, 0.6)$ **b)** $(-20, 15)$ **c)** $(4, -3)$

1.5 **a)** $(9, -12)\,\text{V}$ **b)** $15\,\text{V}$ **c)** $\approx -53.1°$ **d)** The voltages are perpendicular and add geometrically: $|\mathbf{U}|$ is the hypotenuse of a right-angled triangle, which is shorter than the sum of the other two sides.

2.1 **a)** $D_f = [2, \infty)$, $V_f = [0, \infty)$ **b)** $D_f = \mathbb{R} \setminus \{-1\}$, $V_f = \mathbb{R} \setminus \{0\}$ **c)** $D_f = \mathbb{R}$, $V_f = (-\infty, 4]$

2.2 **a)** $k = -2\,\Omega/°\text{C}$, $m = 200\,\Omega$ **b)** $150\,\Omega$ **c)** 50°C **d)** Decreasing, since $k < 0$.

2.3 **a)** $f(x) = 80 \cdot 0.85^x$ **b)** ≈ 30 faults **c)** After 9 whole months ($x \approx 8.5$).

2.4 **a)** 4 W **b)** 100 W **c)** $V_P = [0, 100]$ W **d)** 1.2 A

2.5 **a)** $(x+3)(x-3)$ **b)** $x = 3$ **c)** $x + 3$, where $x \neq 3$ **d)** $x = -3$

3.1 **a)** $\frac{3\pi}{4}$ **b)** $-\frac{2\pi}{3}$ **c)** 150° **d)** ≈ 114.6°

3.2 **a)** 6 V **b)** $\omega = 400\pi$ rad/s, $f = 200$ Hz **c)** 5 ms **d)** ≈ 0.83 ms; the phase is negative, so the curve is late.

3.3 $u(t) = 12 \sin\left(1000\pi t + \frac{\pi}{6}\right)$ V

3.4 **a)** 8 V **b)** ≈ 5.66 V **c)** 0.64 W

3.5 **a)** $t = 1/1200$ s ≈ 0.83 ms **b)** $t = 5/1200$ s ≈ 4.17 ms **c)** $t = 2.5$ ms

4.1 **a)** 7 **b)** ≈ 2.70 **c)** ≈ 0.805 **d)** 101

4.2 **a)** 2 **b)** 2 **c)** $\log 8$ **d)** 3

4.3 **a)** 40 dB **b)** 28 dB **c)** ≈ 25.1 **d)** ≈ 0.50 V

4.4 **a)** ≈ 14.0 dB **b)** 100 mW **c)** 30 dBm

4.5 **a)** 2.2 s **b)** ≈ 3.07 V **c)** ≈ 1.53 s **d)** ≈ 5.47 s

PRACTICE TEST 2

Vectors, functions, trigonometry and logarithms

1 **a)** **u** is an arrow from the origin to $(1, 2)$ and **v** an arrow from the origin to $(-3, 4)$. **b)** $|\mathbf{u}| = \sqrt{5} \approx 2.24$, $|\mathbf{v}| = 5$ **c)** $v_u \approx 63.4°$, $v_v \approx 126.9°$ **d)** $\mathbf{w} = (5, 0)$

2 $f(x) = x + 1$, where $x \neq 2$.

3 **a)** ≈ 4.14 V **b)** D_f: $0 \leq t \leq 20$ s; V_f: $0.17\,\text{V} \leq u(t) \leq 12\,\text{V}$ **c)** ≈ 3.26 s

4 **a)** $u(t) = 5 \sin\left(200\pi t + \frac{\pi}{2}\right)$ V **b)** A sine curve with the amplitude 5 V and the period $T = 10$ ms, shifted 2.5 ms to the left: peak 5 V at $t = 0$, zero at 2.5 ms, trough −5 V at 5 ms, zero at 7.5 ms and peak again at 10 ms.

5 $G_{\text{lin}} = 10^{1.6} \approx 39.8$, that is, about 40 times.

CHAPTER 12

Derivatives, part I

1.1 **a)** $f'(x) = -6x + 5$ **b)** $f'(x) = 6x^2 - 10x + \frac{2}{3}$ **c)** $f'(x) = 8x^3 - 3x^2 + \frac{6x}{5} + 4$

1.2 **a)** $f'(x) = -2x + 8$ **b)** $x = 4$ **c)** Maximum, since $f''(4) = -2 < 0$. **d)** The greatest value is $f(4) = 9$; the maximum point is $(4, 9)$. **e)** A downward-opening parabola with its peak at $(4, 9)$ and zeros at $x = 1$ and $x = 7$.

1.3 **a)** $i'(t) = -0.6t^2 + 3.6t - 3.6$ **b)** $t = 3 \pm \sqrt{3}$, so $t_1 \approx 1.27$ s and $t_2 \approx 4.73$ s. **c)** $i''(t_1) \approx 2.08 > 0$: minimum at t_1; $i''(t_2) \approx -2.08 < 0$: maximum at t_2. **d)** The minimum value is $i(t_1) \approx 0.42$ A and the maximum value $i(t_2) \approx 4.58$ A. **e)** The curve starts at $i(0) = 2.5$ A, has a valley at $(1.27, 0.42)$ and a peak at $(4.73, 4.58)$, and then falls.

2.1 **a)** $5x^4$ **b)** $12x^2$ **c)** 0 **d)** $4x - 5$ **e)** $-12x^3 + 1$ **f)** x^2

2.2 **a)** The quotient simplifies to $4 + h$. **b)** 4 **c)** $f'(2) = 2 \times 2 = 4$, the same answer. **d)** 4.1 and 4.01 respectively

2.3 **a)** $f'(x) = 3x^2 - 3$ **b)** −3 **c)** 9 **d)** Where $x = -1$ and $x = 1$, that is, at $(-1, 2)$ and $(1, -2)$.

2.4 **a)** 2 **b)** 2 **c)** $y = 2x - 4$ **d)** At the point $(2, 0)$.

2.5 **a)** $f'(x) = 2x - 6$ **b)** $x = 3$ **c)** Minimum, since $f''(x) = 2 > 0$. **d)** $f(3) = -1$; the minimum point is $(3, -1)$.

2.6 **a)** $f'(x) = 3x^2 - 6x - 9 = 3(x - 3)(x + 1)$ **b)** $x = -1$ and $x = 3$ **c)** Maximum at $x = -1$ ($f''(-1) = -12 < 0$), minimum at $x = 3$ ($f''(3) = 12 > 0$). **d)** $f(-1) = 10$ and $f(3) = -22$

2.7 **a)** $f'(x) = 2x - 4$ **b)** $x > 2$ **c)** $x < 2$ **d)** $f'(2) = 0$: the function goes from decreasing to increasing, so $x = 2$ is a minimum point.

2.8 **a)** $f'(x) = 4x^3 - 4x, f''(x) = 12x^2 - 4$ **b)** $f'(x) = 15x^2 + 2, f''(x) = 30x$ **c)** $f'(x) = -2x + 7, f''(x) = -2$ **d)** $f''(x) > 0$: the curve is concave up, like a valley; $f''(x) < 0$: concave down, like a hill.

2.9 **a)** $u'(t) = 8t + 2$ **b)** $i_C(t) = (0.8t + 0.2)\,\text{mA}$ **c)** 1.0 mA **d)** $t = 2.25\,\text{s}$

2.10 **a)** $i'(t) = 4t - 4$ **b)** $u_L(t) = (2t - 2)\,\text{V}$ **c)** $t = 1\,\text{s}$ **d)** The current is stationary, a minimum of $i(1) = 1\,\text{A}$: for an instant it stops changing.

2.11 **a)** $P'(I) = 24 - 6I$ **b)** $I = 4\,\text{A}$ **c)** Maximum, since $P''(I) = -6 < 0$. **d)** $P(4) = 48\,\text{W}$

2.12 **a)** $u'(t) = -4t + 12$ **b)** At $t = 3\,\text{s}$, when $u = 18\,\text{V}$. **c)** It is increasing by $8\,\text{V/s}$. **d)** $t = 4\,\text{s}$

2.13 **a)** Increasing, since $f'(1) > 0$. **b)** Decreasing, since $f'(3) < 0$. **c)** A maximum point: $f'(2) = 0$ and $f''(2) < 0$. **d)** The graph rises up to $x = 2$, has a horizontal tangent at the peak, and then falls.

2.14 **a)** No extra x should remain: $f'(x) = 3x^2$. **b)** A constant has the derivative zero: $f'(x) = 0$. **c)** The derivative of $4x$ is the slope: $f'(x) = 4$. **d)** The term $2x$ must be differentiated too: $f'(x) = 6x + 2$. **e)** False: a stationary point can also be a minimum or a stationary point of inflection. **f)** False: $f''(x)$ is found by differentiating $f'(x)$ once more. For example, $f(x) = x^3$ gives $f''(x) = 6x$, not $9x^4$.

3.1 **a)** $f'(3) = 6$ **b)** $h = 1$: 7; 0.1: 6.1; 0.01: 6.01; ...; 10^{-6}: 6.000001 **c)** The slope approaches 6, off by exactly h:

$$\frac{(3+h)^2 - 9}{h} = \frac{6h + h^2}{h} = 6 + h.$$

Test yourself **1)** $f'(x) = 12x^2 - 4x$ **2)** $(1, -9)$, a minimum, since $f''(x) = 2 > 0$.

CHAPTER 13

Derivatives, part II

1.1 **a)** $f'(x) = -1.5x^2 + 3x + 4.5$ **b)** $x = -1$ and $x = 3$ **c)** Maximum at $x = 3$ ($f''(3) = -6 < 0$), minimum at $x = -1$ ($f''(-1) = 6 > 0$). **d)** The maximum value is $f(3) = 10.5$ and the minimum value $f(-1) = -5.5$. **e)** The curve has a valley at $(-1, -5.5)$ and a peak at $(3, 10.5)$, and crosses the y-axis at -3.

1.2 **a)** $u'(t) = -0.6t^2 + 2.4t - 1.4625$ **b)** $t_1 = 0.75\,\text{s}$ and $t_2 = 3.25\,\text{s}$ **c)** $u''(0.75) = 1.5 > 0$: minimum; $u''(3.25) = -1.5 < 0$: maximum. **d)** The minimum value is $u(0.75) \approx 2.49\,\text{V}$ and the maximum value $u(3.25) \approx 4.06\,\text{V}$. **e)** The curve starts at 3 V, has a valley at $(0.75, 2.49)$ and a peak at $(3.25, 4.06)$, and then falls.

2.1 **a)** $i'(t) = 200\cos(50t)$ **b)** $50t = \pi/2 + k\pi$, so $t = \pi/100 + k\pi/50$, $k = 0, 1, 2, \ldots$; the first two are $t_1 \approx 0.031\,\text{s}$ and $t_2 \approx 0.094\,\text{s}$. **c)** $i''(t) = -10\,000\sin(50t)$:

$i''(t_1) = -10\,000 < 0$ gives a maximum, $i''(t_2) = 10\,000 > 0$ a minimum. **d)** 4 A and −4 A, which matches the amplitude 4 A.

2.2 **a)** $u'(t) = -4.8e^{-0.4t}$ **b)** $u'(2) \approx -2.16\,\text{V/s}$ **c)** $u''(2) \approx 0.86\,\text{V/s}^2 > 0$: the rate of change is increasing, that is, the voltage is falling more and more slowly.

2.3 **a)** $x = (10^{0.2} - 1)/2 \approx 0.29$ **b)** $G'(x) = 20/((2x + 1)\ln 10) \approx 5.48$ dB per unit

3.1 **a)** $3\cos(3x)$ **b)** $-5\sin(5x)$ **c)** $4\cos(x)$ **d)** $4\sin(2x)$ **e)** $\cos(x) - \sin(x)$

3.2 **a)** $5e^{5x}$ **b)** $-6e^{-2x}$ **c)** $e^x + 2x$ **d)** $2^x \ln 2$ **e)** $-2.5e^{-x/4}$

3.3 **a)** $\frac{1}{x}$ **b)** $\frac{1}{x}$ **c)** $\frac{3}{x}$ **d)** $\frac{1}{x\ln 10}$ **e)** $\frac{1}{x} + e^x$

3.4 **a)** $6e^{3x} - 4\cos(4x)$ **b)** $2x + \frac{1}{x}$ **c)** $-5\sin(x) + 3e^{-x}$ **d)** $8\cos(2x) - 8\sin(2x)$

3.5 **a)** 2 **b)** 0 **c)** 0.5 **d)** 0 **e)** $-3e^{-1} \approx -1.10$

3.6 **a)** $u'(t) = 1000\pi\cos(100\pi t)$ **b)** $i_C(t) \approx 0.148\cos(100\pi t)$ A **c)** About 148 mA **d)** At $t = 0$, when the voltage passes through zero and changes fastest.

3.7 **a)** $i'(t) = 400\cos(200t)$ **b)** $u_L(t) = 10\cos(200t)$ V **c)** 10 V **d)** The voltage leads the current by 90°.

3.8 **a)** $u'(t) = -40e^{-2t}\,\text{V/s}$ **b)** $-40\,\text{V/s}$ **c)** $-40e^{-1} \approx -14.7\,\text{V/s}$ **d)** The derivative is negative, so the voltage is decreasing, fastest at the start. **e)** $-u(t)/0.5 = -20e^{-2t}/0.5 = -40e^{-2t}$, the same expression as $u'(t)$ in a).

3.9 **a)** $f'(x) = e^{-2x}(1 - 2x)$ **b)** $x = 0.5$ **c)** Maximum: $f''(x) = e^{-2x}(4x - 4)$ gives $f''(0.5) = -2e^{-1} \approx -0.736 < 0$. **d)** $f(0.5) = 0.5e^{-1} \approx 0.184$

3.10 **a)** $p'(t) = 100e^{-t}(1 - t)$ **b)** $t = 1$ s **c)** Maximum, since $p''(1) = -100e^{-1} < 0$. **d)** $p(1) = 100e^{-1} \approx 36.8$ W

3.11 **a)** $G'(x) = \frac{20}{x\ln 10}$ **b)** ≈ 8.69 dB per unit **c)** ≈ 0.869 dB per unit **d)** The derivative is ten times smaller at $x = 10$: the gain in dB grows more and more slowly, since the logarithmic scale compresses large values.

3.12 **a)** 1 **b)** −1 **c)** $y = 1 - x$ **d)** At $x = 1$, that is, after one time constant ($\tau = 1$): had the discharge continued as fast as at the start, it would have been complete after the time τ.

3.13 **a)** With $k = -1/\tau$, $u'(t) = -U_0e^{-t/\tau}/\tau = -u(t)/\tau$. **b)** The rate of change is proportional to the present voltage: the lower the voltage, the more slowly it falls. **c)** $u'(0) = -5\,\text{V/s}$, $u'(\tau) \approx -1.84\,\text{V/s}$ **d)** $e^{-1} \approx 37\,\%$

3.14 **a)** False: the derivative is $\cos x$. **b)** False: it is $3e^{3x}$. **c)** False: $\ln(5x) = \ln 5 + \ln x$, so the derivative is $\frac{1}{x}$. **d)** True: $\cos(kx)$ has the derivative $-k\sin(kx)$, here with $k = 2$. **e)** True: e^{kx} has the derivative ke^{kx}, and here $k = 1$. **f)** True: the constant can be moved outside, $(Cf(x))' = Cf'(x)$.

3.15 **a)** The inner factor 3 must be included: $3\cos(3x)$. **b)** The exponent does not change; only a factor is added: $-2e^{-2x}$. **c)** The constant inside the logarithm disappears: $\frac{1}{x}$. **d)** The derivative of cosine is minus sine: $-4\sin(x)$. **e)** The power rule does not apply when x is in the exponent: $2^x \ln 2$.

4.1 **a)** `derive()` returns `(f(x + DERIVATIVE_H) - f(x)) / DERIVATIVE_H`. **b)** The slope equals e^x at every point, for example 2.71828 for $x = 1$ and 54.5982 for $x = 4$, so $\frac{d}{dx}e^x = e^x$. **c)** The slope matches $\cos v$, for example 0.54 for $v = \pm 1$ and 1.00 for $v = 0$, so $\frac{d}{dv}\sin v = \cos v$. **d)** The ratio is 0.693147 = $\ln 2$ for all x, since $\frac{d}{dx}2^x = 2^x \ln 2$. For e^x the factor is $\ln e = 1$, so e^x is its own derivative. Nothing had to change in `derive()`, only the argument in the call. **e)** The slope matches $\frac{1}{x\ln 10}$, for example 0.434294 for $x = 1$ and 0.0868589 for $x = 5$. Since $\log_{10} x = \frac{\ln x}{\ln 10}$, the derivative is $\frac{1}{\ln 10} \cdot \frac{1}{x}$. At $x = 0$, $\log_{10} x$ is not defined.

Test yourself **1)** $f'(x) = 2e^{2x} + \frac{1}{x}$ **2)** $f'(\pi/2) = \cos(\pi/2) = 0$

CHAPTER 14

Integrals

1.1 **a)** $\dfrac{24e^{-0.25t-0.25h} - 24e^{-0.25t}}{h} = 24e^{-0.25t} \cdot \dfrac{e^{-0.25h} - 1}{h} \to 24e^{-0.25t} \cdot (-0.25)$, so $u'(t) = -6e^{-0.25t}$ **b)** $u'(4) = -6e^{-1} \approx -2.21\,\text{V/s}$ **c)** $u''(t) = 1.5e^{-0.25t}$, so $u''(4) \approx 0.552\,\text{V/s}^2 > 0$. The rate of change is increasing, that is, the voltage falls more and more slowly.

2.1 **a)** $p(t) = 5t\,\text{W}$ **b)** $P(t) = \frac{5t^2}{2} + C$ **c)** $w(t) = \frac{5t^2}{2} + C$, the same expression as $P(t)$. **d)** $C = 0$, so $w(t) = \frac{5t^2}{2}\,\text{J}$. **e)** 22.5 J

3.1 **a)** $\frac{x^4}{4} + C$ **b)** $2x^2 + C$ **c)** $5x + C$ **d)** $x^2 + 3x + C$ **e)** $\frac{x^3}{3} - 2x^2 + x + C$

3.2 **a)** $e^x + C$ **b)** $\frac{e^{3x}}{3} + C$ **c)** $\ln|x| + C$ **d)** $-\cos(x) + C$ **e)** $\frac{\sin(2x)}{2} + C$ **f)** $-\frac{\cos(4x)}{4} + C$

3.3 **a)** $F'(x) = \frac{3x^2}{3} = x^2 = f(x)$ **b)** $F'(x) = -\frac{1}{2} \cdot (-2\sin(2x)) = \sin(2x) = f(x)$ **c)** $F'(t) = -2 \cdot (-\frac{1}{2})e^{-t/2} = e^{-t/2} = f(t)$ **d)** The derivative of a constant is zero, so $F(x) + C$ has the same derivative for every value of C.

3.4 **a)** 8 **b)** 8 **c)** 12 **d)** $\frac{2}{3}$ **e)** 9

3.5 **a)** $e - 1 \approx 1.72$ **b)** 2 **c)** 1 **d)** 1 **e)** $1 - e^{-2} \approx 0.865$

3.6 **a)** 8 **b)** A triangle with base 4 and height 4: $\frac{4 \cdot 4}{2} = 8$. **c)** 12, like the rectangle with width 4 and height 3. **d)** $\frac{8}{3} \approx 2.67$

3.7 **a)** -2 **b)** 2 **c)** 0 **d)** The integral counts area with a sign: the part below the x-axis and the equally large part above it cancel out. The total area is $|-2| + |2| = 4$.

3.8 **a)** $f(x) = 2x^2 - 3x + C$ **b)** $C = 5$ **c)** $f(x) = 2x^2 - 3x + 5$ **d)** 7

3.9 **a)** $u(t) = 3t^2 + C$ **b)** $C = 2\,\text{V}$ **c)** $u(t) = (3t^2 + 2)\,\text{V}$ **d)** 50 V

3.10 **a)** $q(t) = (2t^2 + 2t)\,\text{C}$ **b)** 24 C **c)** 60 C **d)** 36 C **e)** $\left[2t^2 + 2t\right]_3^5 = 60 - 24 = 36\,\text{C}$, the same answer.

3.11 **a)** $-\frac{2\cos(100\pi t)}{100\pi} + C$ **b)** $\frac{4}{100\pi} \approx 0.0127\,\text{C}$ **c)** 0 **d)** The current is positive during the first half-period and equally large but negative during the second, so no net charge passes during a whole period.

3.12 **a)** $i'(t) = 3\,\text{A/s}$ **b)** $p(t) = 3.6t\,\text{W}$ **c)** $w(t) = 1.8t^2\,\text{J}$ **d)** 7.2 J **e)** $i(2) = 6\,\text{A}$ gives $w = \frac{0.4 \cdot 6^2}{2} = 7.2\,\text{J}$, the same answer.

3.13 **a)** $w(t) = (5t + t^2)\,\text{J}$ **b)** 150 J **c)** 100 J **d)** 15 W

3.14 **a)** 3 **b)** 4 **c)** $\frac{20}{\pi} \approx 6.37\,\text{A}$ **d)** $\frac{2}{\pi} \approx 0.637$, so about 64 % of the amplitude.

3.15 **a)** 8 **b)** 10 **c)** 9 **d)** -5

3.16 **a)** $F'(x) = 3x^2 - 2$ **b)** $x^3 - 2x + C$ **c)** A constant term disappears on differentiation, since its derivative is zero, so on integration there is no way of knowing which constant was there, and a general C is written. **d)** $5x^4$

3.17 **a)** $a = 5$ **b)** $a = 4$ **c)** $k = 5$ **d)** 4 s

3.18 **a)** It has been differentiated instead of integrated: $\frac{x^3}{3} + C$. **b)** The inner factor belongs in the denominator: $\frac{e^{2x}}{2} + C$. **c)** The sign is wrong: $-\cos(x) + C$. **d)** The lower limit has been forgotten: $8 - 1 = 7$. **e)** False. The constant cancels itself: $(F(b) + C) - (F(a) + C) = F(b) - F(a)$. **f)** The absolute value is missing; $\ln x$ is undefined for $x < 0$. The correct form is $\ln|x| + C$.

4.1 **a)** $\left[-\cos x\right]_0^{\pi} = 2$ **b)** $n = 4$: 1.896, error 0.10; $n = 10$: 1.984, error 0.017; $n = 100$: 1.99984, error $1.6 \cdot 10^{-4}$; $n = 1000$: 1.999998,

error $1.6 \cdot 10^{-6}$ **c)** The error becomes about 100 times smaller. That is unusually good for rectangles: since $\sin 0 = \sin \pi = 0$ and the curve is symmetric, the errors on the rising and the falling side almost cancel. For most functions the error only becomes about 10 times smaller. **d)** $[\frac{e^{2x}}{2}]_0^1 = \frac{e^2-1}{2} \approx 3.1945$. The error is 0.73, 0.31, 0.032 and 0.0032, so only about 10 times smaller per tenfold increase. Since e^{2x} is increasing, all rectangles lie below the curve, so the errors do not cancel.

Test yourself **1)** 10 **2)** $F(x) = 2x^2 + e^x + C$

CHAPTER 15
Complex numbers, part I

1.1 **a)** $u(t) = (3t^2 - 4t + 2)$ V **b)** 17 V **c)** 15 V, which is 2 V less than $u(3)$. The integral only gives the change $u(3) - u(0)$, since the constant $C = u(0) = 2$ V cancels between the limits. **d)** $t = \frac{2}{3} \approx 0.67$ s **e)** Minimum, since $u''(t) = 6 > 0$. **f)** $(\frac{2}{3}, \frac{2}{3}) \approx (0.67, 0.67)$, so $u \approx 0.67$ V at $t \approx 0.67$ s. **g)** An upward-opening parabola that starts at $u(0) = 2$ V, has its minimum at $(0.67, 0.67)$ and then rises.

2.1 $U = \sqrt{29}\angle -0.38 \text{ rad} \approx 5.39\angle -0.38 \text{ rad}$ V, that is, $\delta_u \approx -21.8°$.

2.2 $I = \frac{5\sqrt{3}}{2} + j2.5 \approx (4.33 + j2.5)$ mA

2.3 $Z \approx 1.08\angle -0.9 \text{ rad} \approx (0.67 - j0.85)$ kΩ

3.1 **a)** -1 **b)** $-j$ **c)** 1 **d)** j **e)** $-j$ **f)** $x = \pm j5$

3.2 **a)** Real part and imaginary part: z_1: 4 and -3; z_2: -2 and 5; z_3: 0 and 6; z_4: -7 and 0. **b)** $z_3 = j6$ **c)** $z_4 = -7$

3.3 **a)** $2 + j7$ **b)** $4 - j3$ **c)** $3 + j19$ **d)** $-7 + j$

3.4 **a)** $-10 + j11$ **b)** 13 **c)** $2 + j5$ **d)** $j2$

3.5 **a)** $5 - j2$ **b)** $-3 + j7$ **c)** 25 **d)** $(x+jy)(x-jy) = x^2 - j^2y^2 = x^2 + y^2 = |z|^2$

3.6 **a)** $1 + j3$ **b)** $8 - j6$ **c)** $1.2 - j1.6$ **d)** The product of the denominator and its conjugate is real, and with a real denominator the real and imaginary parts can be divided separately.

3.7 **a)** 5 **b)** 13 **c)** 7 **d)** 8 **e)** $\sqrt{2} \approx 1.41$

3.8 **a)** 45°, quadrant I **b)** 135°, quadrant II **c)** 225° (or $-135°$), quadrant III **d)** $-45°$, quadrant IV **e)** 53.1°, quadrant I

3.9 **a)** $10\angle 53.1°$ **b)** $5\angle 126.9°$ **c)** $\sqrt{2}\angle 225° \approx 1.41\angle 225°$ **d)** $5\angle -90°$

3.10 **a)** $8.66 + j5$ **b)** $j4$ **c)** -6 **d)** $1 - j1.73$ **e)** $5.66 + j5.66$

3.11 **a)** 13 V **b)** 22.6° **c)** $13\angle 22.6°$ V **d)** How far the voltage is shifted in time relative to the reference, for example the current.

3.12 **a)** $Z = (30 + j40)\ \Omega$ **b)** 50 Ω **c)** 53.1° **d)** Inductive: the imaginary part is positive.

3.13 **a)** $|U| = 10$ V, $|Z| = 5\ \Omega$ **b)** $I = (1.2 - j1.6)$ A **c)** $|I| = \frac{10}{5} = 2$ A **d)** $-53.1°$: the current lags the voltage by 53.1°, as in an inductive circuit.

3.14 **a)** $x = \pm j3$ **b)** $x = -2 \pm j3$ **c)** $x = 1 \pm j2$ **d)** They are each other's conjugates: the same real part and opposite imaginary parts.

3.15 **a)** 8 Ω. The imaginary parts cancel, so the series connection is purely resistive. **b)** Both are 5 Ω. **c)** $25\ \Omega^2$ **d)** 3.125 Ω

3.16 **a)** False: $j^2 = -1$. **b)** False: $|z|$ is a length, always ≥ 0. **c)** True: the number lies on the imaginary axis. **d)** True: the product is $a^2 + b^2$. **e)** False: arctan only gives angles between $-90°$ and 90°; in quadrants II and III, 180° is added. **f)** False: addition is easiest in rectangular form.

4.1 **a)** $z_1 = 5\angle 53.13°$, $z_2 \approx 3.606\angle -123.69°$ (or $236.31°$), $z_3 \approx 7.071 + j7.071$ **b)** The same values. `atan2()` gives the argument between $-180°$ and $180°$, so z_2 gets $-123.69°$. **c)** $56.31°$, the angle of $2 + j3$. The ratio $\frac{-3}{-2}$ has lost the signs and with them the quadrant. `atan2()` gets the imaginary and real parts separately and sees their signs.

Test yourself **1)** $|z| = 5$ and $\delta \approx 0.927$ rad, so $z = 5\angle 0.927$. **2)** $5 + j$

CHAPTER 16
Complex numbers, part II

1.1 **a)** $Z_L = j100\pi \approx j314.2\,\Omega$
b) $Z \approx 100 + j314.2 \approx 329.7\angle 72.3°\,\Omega$
c) $I \approx 30.3\angle -72.3°$ mA $\approx 9.20 - j28.90$ mA

1.2 **a)** $Z_C = -j\frac{1000}{\pi} \approx -j318.3\,\Omega$
b) $Z \approx 100 - j318.3 \approx 333.6\angle -72.6°\,\Omega$
c) $I \approx 30.0\angle 72.6°$ mA $\approx 8.98 + j28.59$ mA
d) Nearly the same magnitude, since $|Z_L| \approx |Z_C|$, but opposite phase angles: the current lags the voltage in the RL circuit and leads it in the RC circuit.

2.1 **a)** An arrow from the origin to $(-12, 4)$, in the second quadrant.
b) $|U| = \sqrt{160} \approx 12.6$ V, $\delta \approx 2.8$ rad $\approx 161.6°$, so $U \approx 12.6\,e^{j2.8}$ V.
c) $\omega = 200\pi \approx 628$ rad/s
d) $u(t) \approx 12.6\,e^{j(200\pi t + 2.8)}$ V
e) $u(t) \approx 12.6\sin(200\pi t + 2.82)$ V with $T = 10$ ms: starts at 4 V, zero at 0.51 and 5.51 ms, minimum at 3.01 ms and maximum at 8.01 ms.

2.2 **a)** $|I| = 10$ mA, $\delta = -\frac{3\pi}{4} = -135°$, $\omega = 100\pi$ rad/s **b)** $f = 50$ Hz
c) $I = -5\sqrt{2} - j5\sqrt{2} \approx -7.07 - j7.07$ mA
d) An arrow from the origin to $(-7.07, -7.07)$, in the third quadrant.
e) $i(t) = 10\sin(100\pi t - 3\pi/4)$ mA with $T = 20$ ms: starts at -7.07 mA, minimum at 2.5 ms, zero at 7.5 and 17.5 ms and maximum at 12.5 ms.

3.1 **a)** $\frac{\sqrt{3}}{2} + j\frac{1}{2} \approx 0.866 + j0.5$
b) $\frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2} \approx 0.707 + j0.707$
c) $-\frac{1}{2} + j\frac{\sqrt{3}}{2} \approx -0.5 + j0.866$
d) $\frac{1}{2} - j\frac{\sqrt{3}}{2} \approx 0.5 - j0.866$
e) $-\frac{\sqrt{2}}{2} - j\frac{\sqrt{2}}{2} \approx -0.707 - j0.707$

3.2 **a)** $2\sqrt{3} + j2 \approx 3.46 + j2$ **b)** $2.5 + j4.33$
c) $\approx 1.41 - j1.41$ **d)** $-5 + j8.66$

3.3 **a)** $3\sqrt{2}\,e^{j\pi/4} \approx 4.24\,e^{j\pi/4}$
b) $2\sqrt{2}\,e^{j3\pi/4} \approx 2.83\,e^{j3\pi/4}$ **c)** $2\,e^{-j\pi/3}$
d) $5\,e^{-j2.50}$ $(\delta \approx -143.1°)$

3.4 **a)** $12\angle 75°$ **b)** $\frac{4}{3}\angle -15° \approx 1.33\angle -15°$
c) $16\angle 60°$ **d)** The magnitudes are multiplied and the angles added; rectangular form needs four partial products and the handling of j^2.

3.5 **a)** $-1 + j7$ **b)** $z_1 \approx 5\angle 53.1°$, $z_2 = \sqrt{2}\angle 45° \approx 1.41\angle 45°$ **c)** $\approx 7.07\angle 98.1°$
d) Yes: $|-1 + j7| = \sqrt{50} \approx 7.07$ and the angle is $\approx 98.1°$.

3.6 **a)** $5\angle 60°$ **b)** $3\angle -90°$ **c)** $2\angle -90°$ **d)** It decreases by $90°$ and the magnitude is unchanged, since $j = 1\angle 90°$: the number is rotated a quarter turn clockwise.

3.7 **a)** $4\angle 60°$ **b)** $8\angle 90° = j8$ **c)** $16\angle 120°$
d) $z^n = |z|^n\angle n\delta$ (de Moivre's formula)

3.8 **a)** $10\angle\frac{\pi}{6} = 10\angle 30°$
b) $4\angle -\frac{\pi}{2} = 4\angle -90°$ **c)** $6\angle 0°$ **d)** The signals must have the same angular frequency ω.

3.9 **a)** $u(t) = 8\sin\left(100\pi t + \frac{\pi}{4}\right)$ V
b) $u(t) = 5\sin\left(100\pi t - \frac{\pi}{2}\right)$ V
c) $i(t) \approx 5\sin(100\pi t + 0.927)$ mA
d) $i(t) \approx 2.83\sin(100\pi t - 2.36)$ mA

3.10 **a)** $U_1 = 6 + j0\,\text{V}$, $U_2 = 0 + j8\,\text{V}$ **b)** $6 + j8\,\text{V}$ **c)** $10 \angle 53.1^\circ\,\text{V}$ **d)** $u(t) \approx 10\sin(100\pi t + 0.927)\,\text{V}$

3.11 **a)** $Z = 5 \angle 45^\circ\,\Omega$ **b)** $Z \approx 3.54 + j3.54\,\Omega$ **c)** Inductive: the imaginary part is positive. **d)** $R \approx 3.54\,\Omega$ and $X \approx 3.54\,\Omega$

3.12 **a)** $Z_L = j20\,\Omega$ **b)** $Z_C = -j20\,\Omega$ **c)** $Z = 15\,\Omega$ **d)** Resonance: the reactances cancel, $\omega L = 1/(\omega C)$, and the impedance is purely resistive.

3.13 **a)** $jz = -4 + j3$ **b)** Equal: $|z| = |jz| = 5$. **c)** 53.1° and 143.1°; the difference is 90°. **d)** It rotates the number 90° anticlockwise without changing its length.

3.14 **a)** False: the phase angles can differ, for example $3 + j4$ and 5. **b)** False: the magnitudes are multiplied, $|z_1 z_2| = |z_1||z_2|$. **c)** True: $z_1/z_2 = (|z_1|/|z_2|) \angle (\delta_1 - \delta_2)$. **d)** False: a phasor carries only amplitude and phase; ω is given separately. **e)** False: in quadrants II and III, π must be added or subtracted. **f)** False: multiplication is easiest in polar form or Euler form.

4.1 **a)** $jz = -4 + j3$, $j^2 z = -3 - j4$, $j^3 z = 4 - j3$, $j^4 z = 3 + j4$ **b)** The argument is 53.13°, 143.13°, −126.87°, −36.87° and 53.13°. `atan2()` gives angles between −180° and 180°, so 233.13° is printed as −126.87°. **c)** The argument increases by 90°, while the modulus stays 5. $j^4 = (j^2)^2 = 1$, so four quarter turns make a whole turn. **d)** The number does not change: $3 + j4$ is printed every time. The parameter is `const complex_t*`, so `complex_rotate_j()` cannot change z but returns the result as a new number, and without `z =` the return value is never used.

Test yourself **1)** $Z = 5\sqrt{3} + j5 \approx 8.66 + j5$ **2)** The magnitude is doubled and the number is rotated 30° anticlockwise, since the magnitudes are multiplied and the angles added.

CHAPTER 17

Complex numbers, part III

1.1 **a)** $Z = 200 \angle -30^\circ\,\Omega$ **b)** $Z \approx 173.2 - j100\,\Omega$, so $R \approx 173.2\,\Omega$ and $X_C = 100\,\Omega$ **c)** $C = \frac{1}{200\,000\pi}\,\text{F} \approx 1.59\,\mu\text{F}$

1.2 **a)** An arrow from the origin to $(3, -6)$, in the fourth quadrant. **b)** $|U| = \sqrt{45} \approx 6.7\,\text{V}$, $\delta \approx -1.1\,\text{rad} \approx -63.4^\circ$, so $U \approx 6.7\,e^{-j1.1}\,\text{V}$. **c)** $\omega = 100\pi \approx 314\,\text{rad/s}$ **d)** $u(t) \approx 6.7\,e^{j(100\pi t - 1.1)}\,\text{V}$

2.1 **a)** $I_1 \approx 4.3 + j2.5\,\text{mA}$, $I_2 \approx 2.6 - j1.5\,\text{mA}$, $I_3 \approx -2 + j3.5\,\text{mA}$ **b)** $I_{\text{tot}} \approx 4.9 + j4.5\,\text{mA}$ **c)** Arrows from the origin: I_1 in the first quadrant (30°), I_2 in the fourth (−30°), I_3 in the second (120°) and I_{tot} in the first (42°). **d)** $i_{\text{tot}}(t) \approx 6.6\sin(\omega t + 42.2^\circ)\,\text{mA}$

3.1 **a)** $12 \angle 60^\circ$ **b)** $5 \angle -45^\circ$ **c)** $2 \angle 0^\circ$ **d)** $7 \angle 180^\circ$, that is, -7

3.2 **a)** $10 + j0$ **b)** $0 + j4$ **c)** $\approx 5.20 + j3.00$ **d)** $\approx 3 - j4$

3.3 **a)** $U_1 = 3 \angle 0^\circ\,\text{V}$, $U_2 = 4 \angle 90^\circ\,\text{V}$ **b)** $U_1 = 3 + j0$, $U_2 = 0 + j4$ **c)** $3 + j4 \approx 5 \angle 53.1^\circ\,\text{V}$ **d)** $u_{\text{tot}}(t) \approx 5\sin(\omega t + 53.1^\circ)\,\text{V}$

3.4 **a)** $U_1 = 10 + j0$, $U_2 = 0 + j5$, $U_3 = 0 - j5$ **b)** $10 + j0\,\text{V}$ **c)** $10 \angle 0^\circ\,\text{V}$ **d)** They are equal in size but opposite in direction (their phase angles differ by 180°), so the imaginary parts $+j5$ and $-j5$ cancel.

3.5 **a)** $U_1 \approx 6.93 + j4\,\text{V}$, $U_2 \approx 6.93 - j4\,\text{V}$ **b)** $\approx 13.86 \angle 0^\circ\,\text{V}$ **c)** $j8 = 8 \angle 90^\circ\,\text{V}$ **d)** The phasors are each other's conjugates, so the sum is purely real and the difference purely imaginary.

3.6 **a)** $20\angle 60°$ **b)** $4\angle 60°$ **c)** $12\angle -60°$ **d)** $3\angle -90°$

3.7 **a)** $Z_R = 100\,\Omega$, phase angle 0° **b)** $Z_L = j100\,\Omega$, +90° **c)** $Z_C = -j100\,\Omega$, −90° **d)** Resistor: current and voltage in phase. Inductor: the current lags the voltage by 90°. Capacitor: the current leads the voltage by 90°.

3.8 **a)** $Z = 40 + j30\,\Omega$ **b)** $|Z| = 50\,\Omega$, $\delta_Z \approx 36.9°$ **c)** $I \approx 2\angle -36.9°$ A **d)** It lags the voltage, as in an inductive circuit.

3.9 **a)** $Z = 60 - j80\,\Omega$ **b)** $|Z| = 100\,\Omega$, $\delta_Z \approx -53.1°$ **c)** $I \approx 2\angle 53.1°$ A **d)** It leads the voltage, as in a capacitive circuit.

3.10 **a)** $Z = 20 + j15\,\Omega$ **b)** $|Z| = 25\,\Omega$, $\delta_Z \approx 36.9°$ **c)** $I \approx 2\angle -36.9°$ A **d)** Inductive, since $X_L > X_C$ gives a positive imaginary part.

3.11 **a)** $\omega_0 = 1000\,\text{rad/s}$ **b)** $f_0 \approx 159\,\text{Hz}$ **c)** $X_L = X_C = 100\,\Omega$ **d)** $Z = 10\,\Omega$, purely resistive and equal to R.

3.12 **a)** $Z_{\text{tot}} = 30 + j40\,\Omega$, $|Z_{\text{tot}}| = 50\,\Omega$ **b)** $I \approx 2\angle -53.1°$ A **c)** $U_R \approx 60\angle -53.1°$ V **d)** $U_L \approx 80\angle 36.9°$ V **e)** $U_R = 36 - j48$ V and $U_L = 64 + j48$ V give $100 + j0$ V $= U$, whereas the magnitudes give $60 + 80 = 140$ V: the two voltages are 90° apart.

3.13 **a)** $10\sin(0.5\pi + \delta) = 5$, that is, $\sin\left(\frac{\pi}{2} + \delta\right) = 0.5$ **b)** $\delta_1 = -\frac{\pi}{3} \approx -1.05$ rad, $\delta_2 = \frac{\pi}{3} \approx 1.05$ rad **c)** $10\sin\frac{\pi}{6} = 10\sin\frac{5\pi}{6} = 5$ V, so both are correct.

3.14 **a)** $Z = 5 + j5\,\Omega$ **b)** $|Z| = 5\sqrt{2} \approx 7.07\,\Omega$, 45° **c)** Series: $10 + j10\,\Omega$ with $|Z| = 10\sqrt{2} \approx 14.1\,\Omega$, twice the magnitude. **d)** Both have the phase angle 45°.

3.15 **a)** $I = 4\angle -30°$ A **b)** $U = 100\angle 30°$ V **c)** $u(t) = 100\sin(500t + 30°)$ V **d)** The voltage leads the current by 60°, the same as the phase angle of the impedance.

3.16 **a)** $I_1 = 6\angle 45°$ mA, $I_2 = 8\angle -45°$ mA **b)** $I_1 \approx 4.24 + j4.24$ mA, $I_2 \approx 5.66 - j5.66$ mA **c)** $I_{\text{tot}} \approx 9.90 - j1.41 \approx 10.0\angle -8.1°$ mA **d)** $i_{\text{tot}}(t) \approx 10.0\sin(\omega t - 8.1°)$ mA

3.17 **a)** False: all the phasors in a calculation must have the same angular frequency. **b)** True: $Z_L = j\omega L$ is purely imaginary and positive. **c)** False: $|Z| = \sqrt{R^2 + X^2}$. **d)** True: the impedance has the phase angle 0°. **e)** False: the current leads the voltage. **f)** True: $X_L = X_C$, the reactances cancel and $Z = R$.

4.1 **a)** $U_1 \approx 1.414 + j1.414$, $U_2 \approx 1.500 - j2.598$, $U_{tot} \approx 2.914 - j1.184 \approx 3.146\angle -22.11°$ **b)** The same values as in a). **c)** $u_1(0) + u_2(0) \approx 1.414 - 2.598 = -1.184$ V, the imaginary part of U_{tot}. At $t = 0$, $|U|\sin(\omega t + \delta) = |U|\sin\delta$, which is the imaginary part of the phasor $|U|\angle\delta$, and the imaginary parts add when the phasors are added.

Test yourself **1)** $I = 5\angle\frac{\pi}{3}$ A, that is, $5\angle 60°$ A **2)** $U_1 + U_2 = 4 + j3 \approx 5\angle 0.64$ (the angle in radians, $\approx 36.9°$)

CHAPTER 18

Mathematics in C

1.1 **a)** $R_{\text{p}} = 132\,\Omega$ **b)** $I \approx 90.9$ mA, $P \approx 1.09$ W **c)** $R_{\text{p}} = 132.000\,\Omega$, $I = 0.091$ A, $P = 1.091$ W **d)** $U \approx 16.248$ V

1.2 **a)** $20\log_{10} 64 \approx 36.12$ dB **b)** $10^{-3/20} \approx 0.708$ **c)** 36.124 dB and 0.708

1.3 **a)** $u \approx 191.03$ V **b)** $0.2\pi \approx 0.628\,\text{rad} = 36°$ **c)** $t = 0$: 0°, 0.000 V; 2.0 ms: 36°, 191.030 V; 5.0 ms: 90°, 325.000 V; 10.0 ms: 180°, 0.000 V **d)** 2.214 rad ≈ 126.870°. `atan(4.0 / -3.0)` gives

$-0.927\,\text{rad} \approx -53.130°$: the quotient $-4/3$ does not show which component was negative, so the angle lands in quadrant 4 instead of 2, 180° out.

2.1 **a)** $f'(x) = 3x^2 - 2, f'(2) = 10$ **b)** With $x = 2$ the function should give about 10; see c). **c)** $h = 10^{-2}$: 10.0001, error $1.0 \cdot 10^{-4}$; $h = 10^{-6}$: error about $5 \cdot 10^{-10}$; $h = 10^{-12}$: error about $9 \cdot 10^{-4}$. The last two errors are rounding errors and can vary somewhat with how f is written. **d)** $h = 10^{-6}$. A large h gives a truncation error, since the approximation lies far from the limit; a small h gives a rounding error, since $f(x+h)$ and $f(x-h)$ become so alike that the significant figures vanish in the subtraction.

2.2 **a)** $i'(t) = 10t$, $u(t) = 2t\,\text{V}$ **b)** 6 V **c)** 6.000 V, the same as by hand.

2.3 **a)** 9 **b)** With $n = 10$ the function should give 9.045; see c). **c)** $n = 10$: 9.045, error $4.5 \cdot 10^{-2}$; $n = 100$: 9.00045, error $4.5 \cdot 10^{-4}$; $n = 1000$: 9.0000045, error $4.5 \cdot 10^{-6}$ **d)** The error decreases by a factor of 100: it is proportional to $1/n^2$.

3.1 **a)** $z_1 + z_2 = 6 - 2j$, $z_1 z_2 = 23 - 14j$, $z_1/z_2 = (-7 + 26j)/29 \approx -0.241 + 0.897j$ **b)** $|z_1| = 5$, $\arg(z_1) \approx 0.644\,\text{rad} \approx 36.87°$ **c)** $6.000 - 2.000j$, $23.000 - 14.000j$, $-0.241 + 0.897j$; 5.000, 0.644 rad, 36.870°

3.2 **a)** $z = 5\sqrt{3} + 5j \approx 8.660 + 5.000j$ **b)** $8.660 + 5.000j$, and back $|z| = 10.000$, $\arg(z) = 0.524\,\text{rad} = 30.000°$ **c)** Yes, to three decimals. Floating-point numbers are rounded at every step of a calculation, so the number that comes back can differ from the original in the last binary digits, and then == gives false.

3.3 **a)** $\omega \approx 314.16\,\text{rad/s}$, $X_L \approx 31.42\,\Omega$, $X_C \approx 318.31\,\Omega$ **b)** $Z \approx 47 - j286.89\,\Omega$ **c)** $|Z| \approx 290.72\,\Omega$, $\arg(Z) \approx -80.70°$. Capacitive, since $X_C > X_L$. **d)** $|I| \approx 0.791$ A **e)** 50 Hz: the values above. 500 Hz: $X_L \approx 314.159\,\Omega$, $X_C \approx 31.831\,\Omega$, $Z \approx 47.000 + j282.328\,\Omega$, $|Z| \approx 286.214\,\Omega$, $\arg(Z) \approx 80.548°$, $|I| \approx 0.804$ A. The circuit becomes inductive.

4.1 **a)** Yes, 800 Hz > $2 \cdot 50\,\text{Hz} = 100\,\text{Hz}$. **b)** $N = 16$ **c)** $u_0 \approx 3.536$ V, $u_1 \approx 4.619$ V, $u_2 = 5.000$ V **d)** k = 0–15: 3.536, 4.619, 5.000, 4.619, 3.536, 1.913, 0.000, −1.913, −3.536, −4.619, −5.000, −4.619, −3.536, −1.913, 0.000, 1.913. The zero at $k = 14$ may be printed as `-0.000`. **e)** One whole sine period, with the peak 5 V at $k = 2$ and the trough −5 V at $k = 10$; sample 16 is the same as sample 0.

4.2 **a)** $U_1 = 10$, $U_2 = 3 + j5.196$, $U_3 = -j4$ **b)** $U \approx 13 + j1.196$ **c)** $U \approx 13.055\angle 0.0918$, so $\delta \approx 0.0918\,\text{rad} \approx 5.26°$ **d)** $13.000 + 1.196j$ and $13.055\angle 0.092\,\text{rad}$ (5.257°) **e)** U_1 along the positive real axis, U_2 at the angle 60° and U_3 straight down. Placed end to end, they finish at $13 + j1.196$, just above the real axis.

4.3 **a)** $u(t) = 10\sin(\omega t) + 6\sin(\omega t + \pi/3) + 4\sin(\omega t - \pi/2)$ with $\omega = 2\pi \cdot 50\,\text{rad/s}$ **b)** 32 samples per period, starting $u_0 \approx 1.196$, $u_1 \approx 3.709$, $u_2 \approx 6.080$, ... **c)** The largest sample value is 13.000 (at $k = 8$), against $|U| \approx 13.055$. The peak lies at $\omega t \approx 84.7°$, between two samples (78.75° and 90°), so no sample hits it exactly.

5.1 **a)** −2.7: −3, −2, −3; 3.2: 3, 4, 3; 5.5: 5, 6, 6 (`floor`, `ceil`, `round`) **b)** The same nine values. **c)** `round()` rounds halfway cases away from zero, not upwards. **d)** `fabs()` (`math.h`) is for `double`, `abs()` (`stdlib.h`) for `int`, and it truncates a floating-point number, so `abs(-2.7)` gives 2. Use `fabs()`.

5.2 **a)** 4465–4935 Ω **b)** See c). **c)** 4600 Ω: within (1); 4950 Ω: outside (0); 4465 Ω: within (1), exactly on the limit. **d)** The absolute value catches the deviation in both directions in a single condition, with fewer limits to calculate and fewer places where a sign can go wrong.

5.3 **a)** $R_s = 500\,\Omega$, $R_p = 80\,\Omega$ **b)** See c). **c)** 500.000 Ω and 80.000 Ω **d)** $R_p \approx 57.143\,\Omega$

5.4 **a)** $\tau = 1.0\,\text{s}$ **b)** 12 V, 4.415 V, 1.624 V **c)** $t = 0, 0.5, \ldots, 5.0$ s: 12.000, 7.278, 4.415, 2.678, 1.624, 0.985, 0.597, 0.362, 0.220, 0.133, 0.081 V **d)** 5 time constants, since the condition is $t > \tau \ln 100 \approx 4.61\tau$.

5.5 **a)** See c) and d). **b)** See c) and d). **c)** $(1, -5, 6)$: $D = 1$, $x_1 = 3$, $x_2 = 2$; $(1, -4, 4)$: $D = 0$, the double root $x = 2$; $(1, 2, 5)$: $D = -16$, $x = -1 \pm 2j$ **d)** $-1.000 + 2.000j$ and $-1.000 - 2.000j$

5.6 **a)** $f(0) = 1 > 0$ and $f(1) = e^{-1} - 1 \approx -0.632 < 0$ **b)** See c). **c)** $x \approx 0.567143$ **d)** A continuous function that changes sign between a and b must pass through zero there (the intermediate value theorem); the change of sign shows which half the root is in.

5.7 **a)** $325/\sqrt{2} \approx 229.81$ V **b)** 229.810 V **c)** The relative error is practically zero, of the order of 10^{-14} %. **d)** Nothing: $n = 10$ also gives 229.810 V. $u(t)^2$ is periodic, and the trapezoidal rule with evenly spaced points over a whole period is then exact.

5.8 **a)** 32 C **b)** 32 C **c)** $i(t)$ is a straight line, which the straight top edge of the trapezium follows exactly; x^2 is curved, so the trapezium ends up above the curve.

6.1 **a)** $x = 0$: $y_p = 0$, $\delta = 2$, $\Delta e = 0.2$, $k = 0$, $m = 0.2$; $x = 1$: $y_p = 0.2$, $\delta = 3.8$, $\Delta e = 0.38$, $k = 0.38$, $m = 0.58$; $x = 2$: $y_p = 1.34$, $\delta = 4.66$, $\Delta e = 0.466$, $k = 1.312$, $m = 1.046$; $x = 3$: $y_p = 4.982$, $\delta = 3.018$, $\Delta e = 0.3018$, $k = 2.2174$, $m = 1.3478$; $x = 4$: $y_p = 10.2174$, $\delta = -0.2174$, $\Delta e = -0.0217$, $k = 2.1304$, $m = 1.3261$ **b)** Yes. k has already gone slightly past 2, and m has gone from 0 to 1.33.

6.2 **a)** See the worked solution. **b)** $k \approx 2.1304$ and $m \approx 1.3261$, the same as in Exercise 6.1. **c)** Yes: $k = 2.000$ and $m = 2.000$, with an error of about $3 \cdot 10^{-6}$.

6.3 **a)** $k \approx 2.18$ and $m \approx 1.46$, still far off. About 840 epochs are needed. **b)** k and m grow without limit and change sign; after 50 epochs they are of the order of 10^{38}. Each adjustment is larger than the deviation it is meant to correct, so the prediction ends up ever further away on the other side. **c)** k and m become `nan`. The adjustment of k is $\Delta e \cdot x$, and with x up to 40 it is 40 times larger, so the learning rate 0.1 is far too high. With 0.001, for example, the model finds the line.

6.4 **a)** $k = -3$, $m = 12$ **b)** For example the learning rate 0.25 for 50 epochs, which gives $k = -3.000$ and $m = 12.000$. With 0.1 about 60 epochs are needed. **c)** $U(5) \approx -3.000$ V, the same as in Chapter 8. **d)** $k \approx -2.01$ and $m \approx 11.21$. No straight line passes through all the points, so the deviation is never zero for all the training sets, and the parameters never stop being adjusted. **e)** $k \approx -3.02$ and $m \approx 12.08$, close to the line that fits the points best, $k = -3.03$ and $m = 12.08$. The model is adjusted after each point. With a high learning rate each point pulls hard on k and m, so the result depends mostly on the last points of the epoch. With a low learning rate each point moves the parameters only a little, and the model ends up at an average of all the points.

6.5 **a)** See the worked solution. **b)** With the learning rate 0.25 the precision is negative for the first epochs, about −2.4 after one epoch, and then approaches 1: 0.97 after 10 epochs and 0.997 after 20. **c)** After 34 epochs.

7.1 **a)** $(1, 0)$: $z = 0.7$; $(1, 1)$: $z = -0.3$ **b)** 0.7 and 0 **c)** $\tanh(0.7) \approx 0.6044$

7.2 **a)** See the worked solution. **b)** Before the training the predictions are random and do not follow the pattern. After the training they are 0.00, 1.00, 1.00 and 0.00, exactly as in the table.

7.3 **a)** The weights and biases are given random starting values, so the training starts

from a new place each time. With 15 nodes, 150 000 to 300 000 epochs are usually needed. **b)** With 5 nodes the training succeeds, sometimes after a retry. With 2 nodes several attempts are often needed, and sometimes all of them fail. With 1 node the training always fails: one node gives only one straight line, see Exercise 7.4.

7.4 **a)** $(0,1)$ and $(1,0)$ are to give 1. They lie in two opposite corners of the square, and $(0,0)$ and $(1,1)$ in the other two. **b)** No. A line that has both the points that are to give 1 on the same side always takes in at least one of the points that are to give 0. Each node in the hidden layer gives a line of its own, and the output layer combines them, so that the network can separate points that no single line can.

Test yourself **1)** 0, since both operands are integers and integer division truncates the decimal part. `1.0 / 2.0` gives 0.5. **2)** It links in the maths library `libm`. Without it, linking fails with `undefined reference to 'sqrt'`. **3)** `sin()` works in radians, so `sin(30)` is $\sin(30\,\text{rad}) \approx -0.988$. Convert first: $30 \cdot \pi/180$. **4)** With the central difference $(f(x+h) - f(x-h))/2h$. The step h controls the accuracy: too large gives a truncation error, too small a rounding error; $h \approx 10^{-6}$ is a good compromise. **5)** With `cabs(z)` and `carg(z)`, where the argument is in radians.

PRACTICE EXAM 1

The whole course

1 $I_1 = 3\,\text{mA}$, $I_2 = 2\,\text{mA}$

2 **a)** $R = 2\,\Omega$ or $R = 18\,\Omega$ **b)** $I = 3\,\text{A}$ and $1\,\text{A}$ respectively. $R = 18\,\Omega$ is preferable: the same power with a third of the current, and the loss in the internal resistance is 6 W instead of 54 W.

3 **a)** $U(I) = \frac{4(I+3)}{I+2}$, where $I \neq -2$ and $I \neq 3$ **b)** $U(2) = 5\,\text{V}$

4 **a)** $u(3) \approx 6.43\,\text{V}$ **b)** $0 \le t \le 10\,\text{s}$ and $2.30\,\text{V} \le u(t) \le 10\,\text{V}$ **c)** After about 6.23 s.

5 **a)** $u(t) = 5\sin(200\pi t + \frac{\pi}{2})\,\text{V}$ **b)** The period is 10 ms. The curve starts at 5 V, is zero at 2.5 and 7.5 ms and reaches $-5\,\text{V}$ at 5 ms.

6 **a)** $P'(I) = 24 - 12I$ **b)** $I = 2\,\text{A}$ **c)** Maximum, since $P''(2) = -12 < 0$. **d)** The largest value is $P(2) = 24\,\text{W}$. **e)** A downward-opening parabola with its maximum at $(2, 24)$. It crosses the I-axis at $(0,0)$ and $(4,0)$. **f)** $U = 12\,\text{V}$ and $R = 6\,\Omega$, so $R = R_\text{i}$ (maximum power transfer).

7 **a)** $I(t) = 0.01t^3 + 0.05t^2 + 0.5t + C$
b) $q(t) = 0.01t^3 + 0.05t^2 + 0.5t$
c) $q_\text{tot}(t) = 0.01t^3 + 0.05t^2 + 0.5t + 1$
d) Exactly 4.44 C

8 **a)** $I = -0.25 - j1.75\,\text{A} \approx 1.8\angle{-1.7}\,\text{A}$, with the angle in radians (-98.1°). **b)** $R = 2\,\Omega$ and $X = 2\,\Omega$, so inductive.
c) $L = \frac{1}{50\pi}\,\text{H} \approx 6.37\,\text{mH}$

9 **a)** An arrow from the origin to $10 - j5$, in the fourth quadrant. **b)** $I \approx 11.2e^{-j0.46}\,\text{mA}$: $|I| = \sqrt{125} \approx 11.2\,\text{mA}$ and $\delta \approx -0.46\,\text{rad}$ (-26.6°) **c)** $w = 20\pi \approx 62.8\,\text{rad/s}$
d) $i(t) \approx 11.2e^{j(20\pi t - 0.46)}\,\text{mA}$

10 **a)** $U_1 \approx 1.81 + j0.85\,\text{V}$, $U_2 \approx 3.54 - j3.54\,\text{V}$, $U_3 \approx 1.21 + j0.88\,\text{V}$ **b)** $U_\text{tot} \approx 6.56 - j1.81\,\text{V}$
c) U_1 and U_3 in the first quadrant, U_2 and U_tot in the fourth; U_tot lies just below the real axis. **d)** $u_\text{tot}(t) \approx 6.81\sin(wt - 15.4^\circ)\,\text{V}$

PRACTICE EXAM 2

The whole course

1 $I_1 = 2\,\text{mA}$, $I_2 = 4\,\text{mA}$

2 **a)** $U = 4\,\text{V}$ or $U = 16\,\text{V}$ **b)** $U = 16\,\text{V}$: $I = 1\,\text{A}$, $R = 16\,\Omega$. $U = 4\,\text{V}$: $I = 4\,\text{A}$, $R = 1\,\Omega$. $U = 16\,\text{V}$ is preferable: the same power with a quarter of the current.

3 **a)** $U(I) = \frac{3(I+2)}{I+3}$, where $I \neq \pm 3$ **b)** $I = 7\,\text{A}$

4 **a)** $RC = 10\,\text{s}$ **b)** $U_0 = 6.0 \cdot e^{0.5} \approx 9.89\,\text{V}$ **c)** $t = 10 \ln 2 \approx 6.93\,\text{s}$, independent of U_0.

5 **a)** Amplitude 325 V, RMS value $\approx 230\,\text{V}$, $f = 50\,\text{Hz}$, $T = 20\,\text{ms}$ and phase $30°$. **b)** The curve starts at 162.5 V, has its maximum at 3.33 ms, zeros at 8.33 and 18.33 ms and its minimum at 13.33 ms.

6 **a)** $i'(t) = 3t^2 - 18t + 24$ **b)** $t = 2\,\text{s}$ and $t = 4\,\text{s}$ **c)** Maximum at $t = 2$ ($i''(2) = -6$), minimum at $t = 4$ ($i''(4) = 6$). **d)** $i(2) = 20\,\text{mA}$, $i(4) = 16\,\text{mA}$ **e)** The graph starts at the origin, rises to $(2, 20)$, falls to $(4, 16)$ and rises to $(5, 20)$. **f)** $u(1) = 0.5 \cdot 0.009 = 4.5\,\text{mV}$

7 **a)** $I(t) = 0.02t^3 + 0.2t^2 + t + C$ **b)** $q(t) = 0.02t^3 + 0.2t^2 + t$ **c)** 9.54 C **d)** 3.18 A

8 **a)** $Z = 30 + j40 \approx 50\angle 53.1° \,\Omega$ **b)** $I \approx 2\angle -53.1° \,\text{A} = 1.2 - j1.6\,\text{A}$ **c)** $L = \frac{2}{5\pi}\,\text{H} \approx 127\,\text{mH}$

9 **a)** An arrow from the origin to $(-8, -6)$, in the third quadrant. **b)** $I \approx 10e^{-j2.50}\,\text{mA}$ ($\delta \approx -143.1°$) **c)** $\omega = 100\pi\,\text{rad/s}$ **d)** $i(t) \approx 10\sin(100\pi t - 2.50)\,\text{mA}$

10 **a)** $U_1 \approx 2.00 + j3.46\,\text{V}$, $U_2 \approx 2.60 - j1.50\,\text{V}$, $U_3 \approx -1.73 + j1.00\,\text{V}$ **b)** $U_{\text{tot}} \approx 2.87 + j2.96\,\text{V}$ **c)** U_1 in the first, U_2 in the fourth and U_3 in the second quadrant; U_{tot} in the first. **d)** $u_{\text{tot}}(t) \approx 4.12\sin(wt + 46.0°)\,\text{V}$